

Aberration Correction in STEM

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*Tom described how Lancashire housewives cleaned their carpets by sprinkling them with damp tea-leaves, which were subsequently swept up after they had absorbed the dust and dirt. ‘But,’ observed Tom after a pause, ‘they didn’t start by **filling** the room with tea-leaves!’*

T. Mulvey’s comment on O. Sherzer’s suggestions for aberration correction, as related by P.W. Hawkes (1995).

Preface

This dissertation is the account of work carried out at the Cavendish Laboratory, Cambridge from October 1997 to January 2001, under the supervision of Dr A.L. Bleloch, and originally Prof. L.M. Brown. Except where explicitly stated in the text, this work is entirely my own and includes nothing which is the outcome of work done in collaboration. It is not substantially the same as any other dissertation submitted for a degree or other qualification at this or any other university.

Accounts of some of this work have appeared in the following publications:

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Glossary and Notation

The most commonly used notation in this report is listed below. Vectors are listed in bold, **V**, and matrices in bold underlined, **M**.

ρ	The axial angle of a ray (see Chapter 2), with components (u,v) .
(u,v)	The ray tangents to the axes, such that $\rho^2=u^2+v^2$.
θ	The azimuthal angle, where $\tan(\theta)=u/v$.
ρ	Spatial frequency.
R	A 2-dimensional vector in the sample plane.
K_i	The incident 2-dimensional wave-vector ($ \mathbf{K}_i ^2=1/\lambda$).
K_f	The scattered 2-dimensional wave-vector.
K or Q	General 2-dimensional vectors, normally wave-vectors.
$\psi(\mathbf{R})$	The specimen transmission function in real space.
$\Psi(\mathbf{K})$	The specimen transmission function in reciprocal space.
χ	The aberration function (with units of length, see Chapter 2).
λ	The electron wavelength (3.7 pm at 100kV).
$2\pi\chi/\lambda$	The phase change due to the aberration function.
B, E	Magnetic and electric fields respectively.
B_z	Component of the B field in the z direction.
C_s	The (third order) spherical aberration.
C_c	The (first order) chromatic aberration.
f	The focal length of a lens.
X	A 2-dimensional vector at the sample plane.
Y	A 2-dimensional vector at the detector plane.
<u>H</u>	The 2x2 Hessian matrix of second derivatives of χ (Chapter 6).
$M(\mathbf{K}_f, \mathbf{R}_0)$	The complex amplitude at the microdiffraction plane.
∇	The gradient operator (in 2-dimensions), normally: $\left(\frac{\partial}{\partial u}, \frac{\partial}{\partial v} \right)$

When an aperture size is given, unless otherwise stated this will refer to the semi-angle. The apertures used are the objective (OA), the virtual objective (VOA), the selected area (SAA), the collector, and the differential pumping aperture (DPA).

The sign assigned to free-space waves (and thus the choice of when to use forward and reverse Fourier transforms) will use the so-called crystallographic convention (Spence 1977, section 5.12) to give a free-space wave as:

$$\phi(\mathbf{R}) = e^{-2\pi i \mathbf{K} \cdot \mathbf{R}}$$

The following operations are used in the main text:

The forward Fourier transform (from real to reciprocal space) is defined as:

$$\Psi(\mathbf{K}) = \int \psi(\mathbf{R}) e^{2\pi i \mathbf{R} \cdot \mathbf{K}} d\mathbf{R}$$

The reverse Fourier transform (from reciprocal to real space) is defined as:

$$\psi(\mathbf{R}) = \int \Psi(\mathbf{K}) e^{-2\pi i \mathbf{R} \cdot \mathbf{K}} d\mathbf{K}$$

The convolution operator, $\otimes$, is defined by:

$$f(\mathbf{X}) \otimes_{\mathbf{x}} g(\mathbf{X}) = \int f(\mathbf{X}) g(\mathbf{Y} - \mathbf{X}) d\mathbf{X}$$

The similar correlation operation is also used, and is defined by:

$$\text{Correlation}_{\mathbf{x}}(f(\mathbf{X}), g(\mathbf{X})) = \int f(\mathbf{X}) g(\mathbf{X} - \mathbf{Y}) d\mathbf{X}$$

The Dirac delta function, $\delta(\mathbf{X} - \mathbf{Y})$, is zero unless $\mathbf{X} = \mathbf{Y}$, defined by:

$$\delta(\mathbf{X} - \mathbf{Y}) = \int e^{2\pi i \mathbf{K} \cdot (\mathbf{X} - \mathbf{Y})} d\mathbf{K}.$$

The following acronyms are used often in the main text:

ADF	Annular Dark Field (normally high angle).
BF	Bright Field (axial unless specified otherwise).
CCD	Charge Coupled Device.
CTEM	Conventional Transmission Electron Microscope.
EELS	Electron Energy Loss Spectroscopy.
FEG	Field Emission Gun.
FFT	Fast Fourier Transform.
OL	Objective Lens.
STEM	Scanning Transmission Electron Microscope.
TIS	Tilt Induced Shift (method for measuring aberrations).
VG	Vacuum Generators.

Chapter 1

Introduction

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1.1. Introduction

The Transmission Electron Microscope (TEM) is one of the foremost tools in materials science. Following its invention by Knoll and Ruska in 1932, there has been much effort devoted to improving the resolution at which materials can be studied. In a modern microscope, the performance is limited by the aberrations of the electron optical lenses. Therefore much of the work to develop such instruments has concentrated on eliminating or reducing these aberrations.

Since the work of Abbe at the end of the last century, light optical microscopes have been able to achieve a resolution approaching the wavelength (λ) of the illuminating radiation, typically a few hundred nanometres. The De Broglie wavelength of electrons in a conventional transmission electron microscope (CTEM) is far smaller than this, normally several picometers. So it might reasonably be expected that the achievable resolution should be correspondingly improved. However, instruments currently available are limited to resolutions two orders of magnitude poorer than this wavelength because of the lens aberrations. In particular the spherical aberration (C_s) of the objective lens limits the resolution of a state-of-the-art instrument to several Ångströms at best. It is interesting to note that this resolution is of the order of the lattice spacings in many solids and so any improvement in this figure would have direct applications in many realms of materials research.

The key to the operation of an electron microscope is that electrons, being charged particles, can be focussed by certain configurations of electric or magnetic fields. The lenses used in most high performance TEMs are round magnetic lenses. In their simplest form these consist of a coil around the electron beam path. In practice they will also include a magnetic circuit and carefully designed pole-pieces to shape the magnetic flux into the desired geometry. These initially had many advantages over non-round lenses and it was only after their introduction that the first electron microscopes were developed (Knoll and Ruska, 1932).

Most aberrations can be reduced by suitable column design and alignment, and many even eliminated with simple correcting elements. However, in 1936, Scherzer

proved that any electron optical system will always suffer from spherical aberration and chromatic aberration (C_c), if simultaneously:

- (a) The optical system is rotationally symmetric,
- (b) The system produces a real image of the object,
- (c) The fields of the system do not vary with time,
- (d) There is no charge on the axis.

No attempts to find exploitable loopholes in this proof, most notably those of Glaser, have met with any success (Hawkes and Kasper, 1994). Spherical aberration impairs the performance of the optical system by causing rays travelling at different angles to be focused to different planes. Thus, once the design of the microscope had improved to the stage where the performance was limited by the spherical aberration, it became necessary to seek some method to overcome this effect. The techniques to eliminate the effects of spherical aberration can be classed into two main categories.

The first class of method to overcome spherical aberration is to obtain aberrated images and remove the effects of the aberrations, normally by some sort of image processing. It is interesting to note that holograms were first invented for this purpose (Gabor, 1948). This route has some interesting possibilities; most excitingly the super-resolution techniques (Rodenburg and Bates, 1992), or holography (Van Dyck et al, 1996, and Wang and Cowley, 1995). However, these all have some experimental drawbacks and have not demonstrated a useful increase in resolution. In any case, even where these methods have exhibited some success, correction of the spherical aberration is likely to improve the effectiveness still further.

The second class of methods arises directly from Scherzer's proof that all conventional round lenses suffer from spherical aberration. Although the magnitude of spherical and chromatic aberration can be reduced by further refinement of the lens design they can never be completely eliminated in such systems. The way out of this situation was given by Scherzer in 1947, when it was proposed that one of the above conditions must therefore be broken to obtain a device that is free of spherical aberration. In that paper several methods of accomplishing this were given and so,

much of the subsequent work has consisted of constructing suitable devices to achieve this effect.

It will be seen that many of the difficulties in implementing such devices are technological, with severe stability and tolerance requirements involved. Also some problems are less obvious, such as the complexity of adjusting and aligning such a system. Computer control and image analysis will be seen to be not just useful tools, but essential components of a practical aberration corrector. Thus despite the length of time for which solutions have been proposed, it is only recently that practical devices that improve the resolution of a state-of-the-art microscope have been implemented (Krivanek et al, 1999a,b and Rose et al, 1999).

The relevance of spherical aberration in conventional optics has recently been highlighted by the launch of the Hubble space telescope. Due to a small manufacturing defect, this device suffered from spherical aberration. A separate mission was necessary to fit an aberration correction system in order to improve the resolution. This illustrates the importance of being able to measure and then correct lens aberrations. The correction of spherical aberration in electron microscopy is expected to have a similar impact in materials research to that of the aberration corrected Hubble telescope in astronomy.

1.2. TEM imaging and limitations

We begin by considering the factors that might be expected to limit the resolution of a general microscope. A simple model applicable to both light and electron transmission microscopes is shown in Fig. 1.1. A parallel beam of incident radiation illuminates a thin specimen. The radiation is scattered by the specimen and then collected and focussed by the objective lens. Propagation of waves to the far field is represented by Fourier transforms. Hence, a Fraunhofer diffraction pattern is formed in the back focal plane and a real image formed at the image plane.

The objective lens will be finite in extent and normally limited by an aperture in the back focal plane. Thus, the microscope will always have a limit on the amount of information that it transfers. In electron optics it is only possible to record the intensity and not the phase of the beam. This is known as the **phase problem** and once again means that information is lost.

The wavelike nature of the electron, first proposed by Louis de Broglie, will limit the resolution obtainable. However, in a typical TEM, operating at 100 keV, the electrons will have a wavelength of 3.7 pm. This is significantly smaller than any inter-atomic spacings and so would not be expected to limit the resolution. Similarly, it is possible to choose large enough apertures that diffraction does not limit the information attainable.

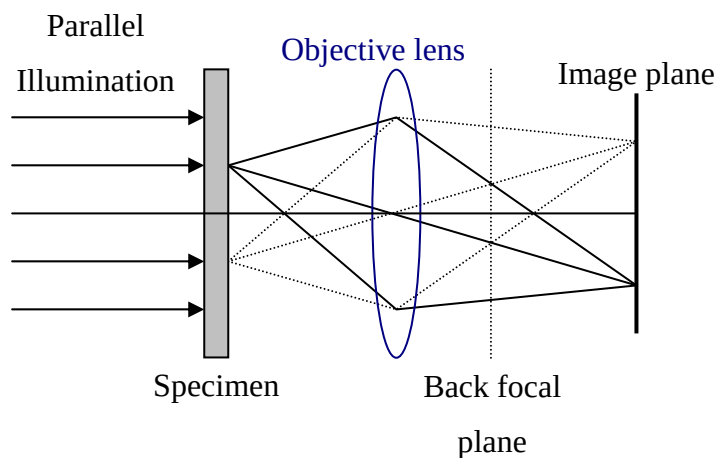


Figure 1.1. Schematic diagram of conventional bright field imaging in a CTEM. All of the rays that leave the sample at a given angle meet at the same point in the back focal plane.

It is the unavoidable spherical aberration of the objective lens that limits the resolution in all modern TEMs. The effects of positive spherical aberration are illustrated in Fig. 1.2, which shows the imaging of a point source by a spherically aberrated lens. Rays travelling at higher angles to the optic axis are focussed too strongly relative to those closer to the axis (called Gaussian or paraxial rays). Thus the paraxial rays are focussed at the Gaussian focal plane while the aberrated rays are focused at earlier planes. This results in a blurring of the image formed at the Gaussian plane. It can be seen that the smallest diameter image (the disk of least confusion) will be at a plane positioned before the Gaussian focal plane. This illustrates why some defocus is normally used to compensate for the effects of spherical aberration to obtain the sharpest possible image. The choice of aperture is conventionally determined by balancing the need to block out these aberrated rays against the resultant diffraction limit.

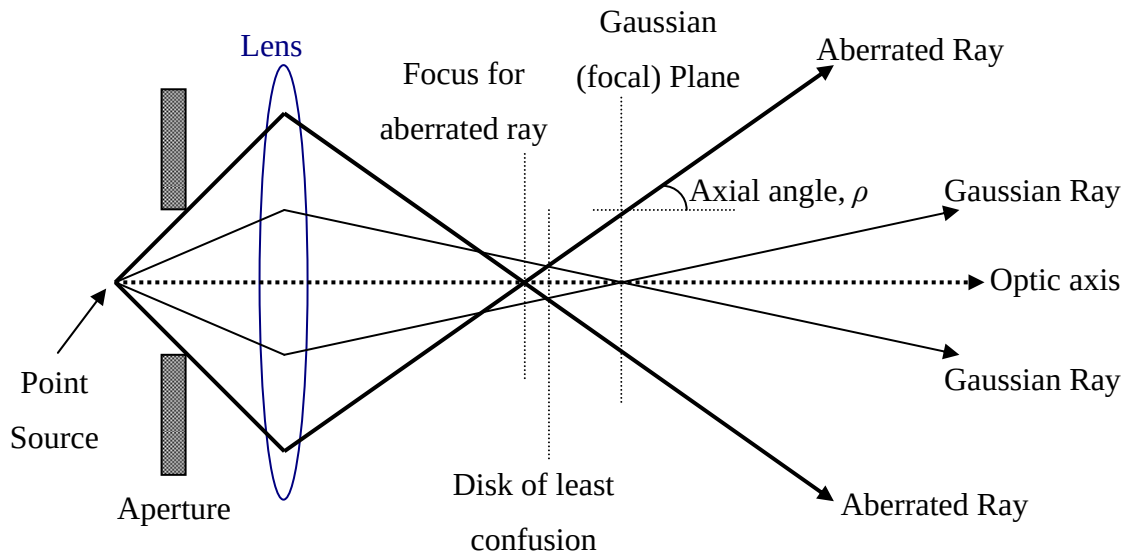


Figure 1.2. Schematic diagram of the effects of positive spherical aberration in an aberrated lens. The rays travelling at higher angles to the axis are focussed more strongly than the Gaussian rays. The disk of least confusion, where the beam has its smallest diameter, is shown.

The resolution of an optical system as defined by Rayleigh (1896) is based on the shortest distance between two points that can still be resolved. This resolution limit in a TEM is caused by the spherical aberration, and is known as the ***point resolution***. This will be more rigorously defined later, but can be described as the highest spatial frequency in the image that can be directly interpreted. Higher spatial frequencies may also be transferred, but these will not be directly interpretable due to the lens aberrations, although they may be recovered by other means. One practical implementation is to use a through focal series (for example, Van Dyck et al, 1996), although this is restricted by dynamical effects, drifts, and detector limitations. A method of correcting spherical aberration is thus expected to offer significant benefits for all sorts of transmission electron microscope.

The microscope will also be limited by the lack of perfect spatial and temporal coherence. This will be due to the instabilities present in any real microscope, the finite spatial extent of the source and the spread of electron energies. The effects of these will depend on the chromatic aberration of the electron lenses, described in Chapter 3. Thus there also exists another resolution limit known as the ***information limit***, beyond which no information is transferred to the image. This is the ultimate limit of the instrument and gives the highest resolution detail that could be extracted by simple image processing techniques. In bright-field imaging, this can be thought of as a damping envelope in frequency space, as discussed by Spence (1977). In annular dark-field modes this effect can be shown to be less severe (Nellist and Pennycook, 1998). The super-resolution schemes (Rodenburg and Bates, 1992) may offer a way around this limit, but are ultimately limited by dynamical effects and experimental restrictions.

1.3. STEM, reciprocity and microdiffraction

Most of the work on C_s -correction has concentrated on improving the resolution of either a conventional or a Scanning Transmission Electron Microscope (STEM). Although examples exist of other types of electron microscope that have been aberration corrected (Zach and Haider, 1995a,b), it is in these two forms of microscope, where the spherical aberration is the primary resolution limit, that the most dramatic benefits are expected.

In many respects a simple STEM can be considered as a CTEM with the source and detector interchanged. In CTEM the specimen is illuminated with a parallel beam (normally adjustable with several condenser lenses) and most of the electron optics are after the specimen. The objective lens forms an image of the illuminated area of the specimen that subsequent lenses then project into the detector plane.

In STEM, most of the electron optics are before the specimen. The condenser lenses demagnify the source, which is then imaged by the objective lens onto the sample, forming a **probe**. This small spot of illumination is scanned across the specimen, and the intensity is recorded as a function of probe position, at a detector in the far field, to form the image. The main detectors that are used in a STEM are bright field (BF) and annular dark field (ADF). The bright field detector is assumed to be on the optic axis unless otherwise stated. The signal that this records will be composed of the interference between the bright field disk and Bragg diffracted disks. The only type of dark field detector that will be considered here is the (high-angle) annular dark field detector. This is an extended annular detector that records the scattering to high angles. These detectors are schematically illustrated in Fig. 1.4. The choice of detector will be seen to have profound consequences for the details of the image recorded.

The images obtained in a CTEM and a STEM are related by the reciprocity theorem of Helmholtz. The principle of reciprocity states that if a particular intensity is recorded at point *A* for a source at point *B*, then in the same system, an identical intensity will be recorded at point *B* for the same point source at *A*. This theorem also

applies to electron microscopy, for elastic scattering, and has been described by Pogany and Turner (1968).

The simple picture of a STEM as a CTEM with the source and detector interchanged suggests immediately that the same image will be recorded in both cases. This is substantially true, although the relative efficiency of the two modes is not equal, because the STEM image is recorded in series, while all pixels in the CTEM image are recorded in parallel. Conversely one advantage of a STEM is that the optics are before the specimen and so there are no extra chromatic effects due to energy losses in the sample, unlike a CTEM. A relevant description of STEM imaging is given by Spence and Cowley (1978).

In addition to the real-space image in a STEM, the intensity across the far field can be recorded. In the limit of a small objective aperture, a relatively large part of the specimen will be illuminated with essentially parallel illumination and a diffraction pattern will be formed. As the objective aperture becomes larger the convergence of the incident beam causes the diffracted spots to be spread out into disks. The image formed is called a microdiffraction pattern and is illustrated in Fig. 1.4. This sort of image is extensively described by Cowley (1978), and this description is further extended in Chapter 3.

Electron microdiffraction is a convenient technique for providing information from very small parts of a specimen, as reviewed by Spence and Zuo (1992). In the limit of a very large (or no) objective aperture the disks will become large enough to overlap and interfere, forming a shadow image. This image is also known as a Ronchigram after the pioneer of the equivalent image in light optics, and is discussed in detail by Cowley (for example, 1986). Fig. 1.2 illustrates that the position of the cross-over of the beams relative to the sample will vary as a function of incident angle due to lens aberrations. Thus the Ronchigram can be seen to have a magnification that also varies as a function of incident angle. This geometrical approach provides some insight into why the Ronchigram is so sensitive to the aberrations and thus useful for alignment purposes. This image will be examined further in subsequent chapters.

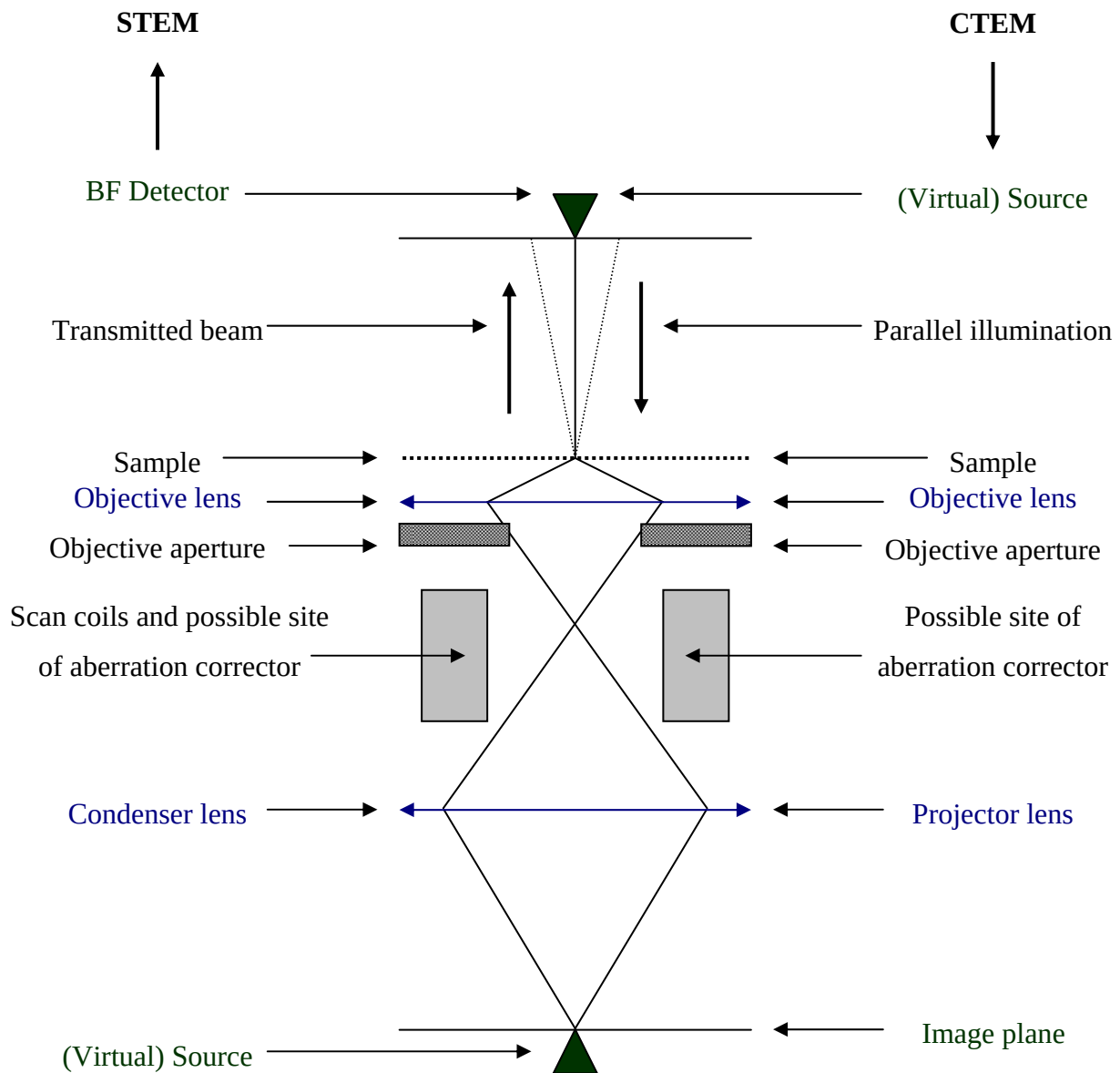


Figure 1.3. Schematic contrasting STEM (left column) and CTEM (right column) imaging in bright field modes. By the principle of reciprocity a STEM can be pictured as a CTEM in reverse.

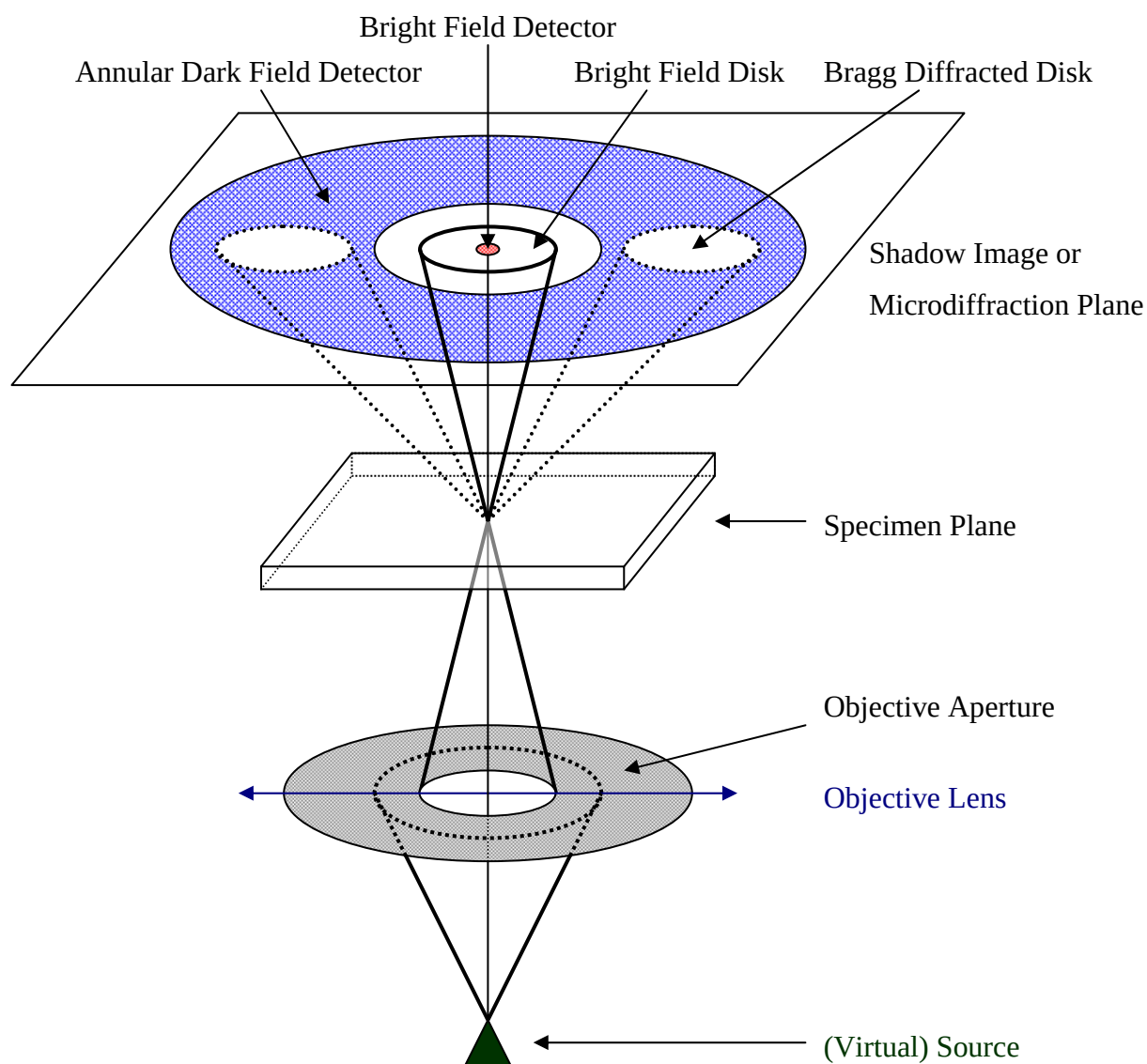


Figure 1.4. Schematic showing the formation of a microdiffraction pattern in a STEM. Representations are also shown of a bright field detector and an annular dark field detector (here spanning only quite low angles). Because there is no overlap between disks, the condition illustrated would not give any contrast in the usual imaging modes. If a larger aperture is used, these disks overlap and interfere to form a shadow image or Ronchigram at the microdiffraction plane.

1.4. Aberration correction

Aberration correction relies on constructing a device that produces negative spherical aberration. This then combines with the positive aberration of the objective lens to give a total of zero spherical aberration. The alternative route of just producing systems that do not have any round lenses has not achieved performance equivalent to that of conventional round lenses. This is because the stable, round objective lens, with its short focal length, provides an essential demagnification of any instabilities and misalignments in preceding optical systems.

A system to produce negative spherical aberration must break one of the conditions necessary for Scherzer's proof to hold (listed in Section 1.1). It is clearly necessary to form a real image, and so the other three methods must be examined. Proposals for accomplishing each of these were given by Scherzer in 1947, and have been investigated extensively since.

A thorough review of each of these possibilities is given by Hawkes and Kasper (1994). All of these methods have some associated experimental drawbacks. The problem with putting a charge on axis is that any structure on axis will also affect the image in other ways. This includes axial conductors or charged foils. Electron mirrors can achieve negative spherical aberration, but it is not clear how such a system could be integrated into a transmission electron microscope. Time varying fields must change in the time it takes an electron to travel through the system. A high-energy electron will thus require very high frequency fields with precise distributions. Even if this could be achieved, introducing a high frequency perturbation seems incompatible with a microscope that must be optimised for stability. None of these methods has displayed much experimental success compatible with high-resolution imaging. It is also not clear how to align such systems or adjust parasitic aberrations. Thus most work on aberration correction has concentrated on breaking the rotational symmetry of the system. This requires non-round optical elements, which are defined in Chapter 2. Non-round elements are already used in stigmators and spectrometers, and so have been proved to be compatible with high-resolution imaging.

There are two main ways of imparting negative spherical aberration to the beam using non-round elements. Firstly, octupoles acting on a round beam are known to exhibit negative spherical aberration in some directions, while making it worse in others. It is therefore necessary to deform the beam in order to obtain only the desired phase changes. This is achieved by using a series of quadrupole lenses to generate an appropriately shaped beam. The main difficulties are due to the stability and alignment precisions required. This thesis will concentrate on the analysis of quadrupole-octupole correctors. All of the experimental data has been obtained on these systems, which are further described in Chapters 2 and 4.

The second method relies on hexapole elements, which are also known as sextupoles. These have a primary effect of causing 3-fold astigmatism. However they also have a secondary effect of imparting negative spherical aberration. Such a system has a little less flexibility than an octupole corrector and is similarly difficult to align. More importantly these systems are not directly compatible with the correction of chromatic aberration.

Both of these methods to correct the spherical aberration rely on significantly distorting the beam shape away from rotational symmetry in order to correct the round spherical aberration. The construction and alignment tolerances of such a device in a high-resolution microscope will be severe. The C_s -correction system will also need adjustment, which requires a method for describing and quantifying the aberrations of the beam. It is therefore clear that an aberration corrector will be a complicated system consisting of several components.

1.5. Dissertation outline

This dissertation consists of eight chapters and three appendices. A brief glossary of the most commonly used terms is included in the preface. In this chapter, the

transmission electron microscope, the principle of reciprocity, and the resolution limits have been introduced. The notation necessary to describe the significant aberrations in a STEM is introduced in Chapter 2, with a review of the system used for the experimental work. This includes a consideration of the uncorrected system and some of the requirements for a successfully C_s -corrected microscope. The necessary software, which was largely developed during this work, is also reviewed.

In Chapter 3 the different sort of images that can be obtained are examined in the presence of non-round aberrations. This provides a summary of how the existing methods of measuring the aberration function were implemented, and a new description of the electron Ronchigram. The methods outlined there are used later in Chapter 5 to calculate the effects of aberrations on the probe size. This is used to calculate the factors that limit the expected resolution in a C_s -corrected system, and to give stability requirements for the different aberration coefficients. Chapter 4 begins with a historical review of the previous attempts to implement such systems, which is essential to avoid repeating some of the earlier failed attempts. Then more detailed descriptions of the aberration correctors used and their effects on the beam are presented. The conclusions from this work led to the construction of a second aberration corrector and some discussion of the development process is given.

In Chapter 6 the description of the Ronchigram given earlier is extended and simplified to a purely geometrical argument. By describing the magnification in such images in two dimensions, this is used to explain the significant features in the Ronchigrams of amorphous materials, and to produce an entirely new method for measuring the aberration function of a STEM. Chapter 7 gives some details of the experimental procedures that form an important part of using the C_s -correctors. Finally some experimental results from the systems examined are presented. These show that for the first time, the aberration correctors described here are finally able to improve the resolution of a C_s -corrected instrument, beyond that of an uncorrected system at the same voltage.

In Chapter 8, the achievements that have been presented are reviewed, and some of the future prospects are examined. Appendix 1 lists further experimental details of the Ronchigram analysis. Appendix 2 describes the method used to convert between the different descriptions of the aberration function that are encountered.

This relates the form of the aberration function used here to the output of the trajectory calculations and experimental measurements that are made. Appendix 3 gives some details of the image simulations used.

1.6. Conclusions

The imaging process in the scanning transmission microscope has been introduced. It has been seen that the aberrations of the electron optical lenses limit the resolution that can be obtained. In particular the spherical aberration of the objective lens provides the practical resolution limit. Aberration correction provides one possible route to overcome these effects. Other methods to enhance the resolution, such as holography and super-resolution exist, but these have experimental limitations, and would anyway benefit from reduced aberrations.

It is clear that using non-round optical elements is the only method of aberration correction that is likely to produce a device that can be included in a practical high-resolution instrument. Such a device will need high stability and accurate alignment, as well as extensive computer control and aberration analysis.

The benefits of C_s -correction will include enhanced resolution in both bright and ADF imaging, and higher resolution electron energy loss spectra with more current in a smaller probe. In this thesis, the first successful aberration corrector in a STEM is described, the development of an improved design is discussed, and some of the first results from both systems are presented. The newly developed aberration correctors are expected to offer a wide range of benefits for many aspects of materials analysis.

Chapter 2

Instrumentation and Aberrations

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2.1. Introduction

In this chapter the instrumentation used to perform the experiments is described. The notation used to describe the aberrations, and the relationship between geometrical and wave aberrations will be presented. To be of practical use an aberration corrector must be installed in a working electron microscope. The STEM is particularly well suited to this purpose, because of the variety of imaging modes available and the lack of sensitivity to off-axial and chromatic aberrations. The instrumentation for STEM imaging will be reviewed, and the aberration corrected Vacuum Generators (VG) HB5 used for most of the experimental work will be introduced. The significant features of the HB5 aberration corrector are then described. Finally, round lenses are numerically investigated with regard to improvements in a spherical aberration free system.

For some of the later experimental work a more modern VG HB501 was used. A detailed description of that instrument is given by Nellist (1995). This is considered to be similar to the HB5 other than where indicated. The primary differences are the lack of a gun lens in the HB5, and the second condenser lens and improved objective lens in the HB501.

2.2. Aberrations

In this section the nomenclature employed to describe the aberration function will be introduced. This is primarily designed for the axial aberrations that are the only significant terms in STEM. The notation used is extendible to arbitrary orders and symmetries. Off axial aberrations will not be considered here, because they can (at least in principle) be neglected in a STEM. This is because the field of view is limited and the beam should be scanned such that it always passes through the centre of the objective lens (for example, Thomson, 1971).

2.2.1. Classification of Aberrations

The aberrations of a transmission electron microscope are described by a scalar aberration function, χ . This can be defined in a number of slightly different ways; the form used here is similar to that described by Krivanek et al (1999b). It is possible to consider the aberration function either in terms of the wave aberration or the ray deviation, which will be related because the rays propagate normally to the wave-fronts. We define the aberration function as the distance between the aberrated wave-front and the un-aberrated (or “Gaussian”) wave-front along the aberrated ray, as illustrated in Fig. 2.1. The wave aberration, which represents the phase change across the back focal plane of the objective lens, is thus given by $2\pi\chi/\lambda$. The ray deviation is defined as the vector from the Gaussian image point to the intersection of the aberrated ray and the image plane. This depends on the gradient of the aberration surface because the rays are perpendicular to the wave-fronts. A derivation of this result is given by Born and Wolf (1970, section 5.1).

In a more general optical system, a ray is defined by its intersection with two planes, as described by Hawkes and Kasper (1994). However, in STEM, we are primarily concerned only with axial aberrations, where the Gaussian image points (those points which un-aberrated rays are imaged to) are always on-axis. In this case, it is possible to assume that the aberration function is a function only of the angle that a ray makes to the optic axis. This simplifies the following discussion and is sufficient for consideration of STEM instruments. Off-axial aberrations cannot be neglected in CTEM and will be encountered in Chapter 4 where extra notation is introduced.

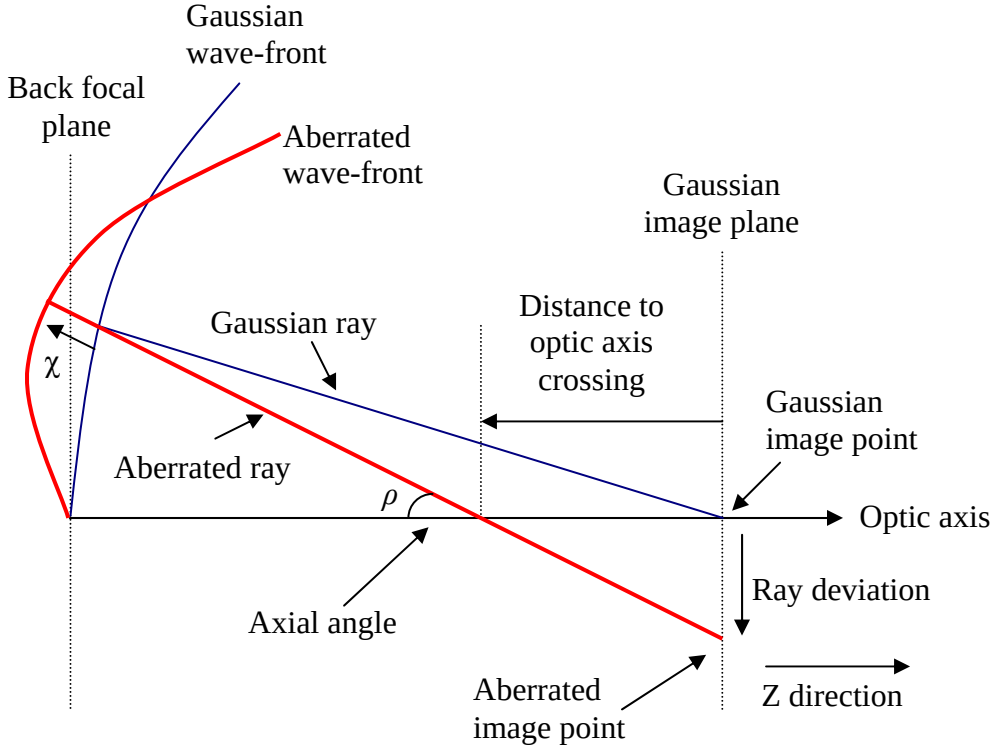


Figure 2.1. Effect of aberrations on rays in an aberrated system. The aberration function gives the distance from the un-aberrated Gaussian wave-front to the aberrated wave-surface as a function of angle to the optic axis.

The coordinates used to define the aberration function are shown in figures 2.1 and 2.2. The angles to the optic axis, the axial angle, ρ , and the azimuthal angle, θ , are defined at the sample plane. These are extrapolated so that ρ and θ also give the coordinates in the back focal plane. Strictly, this extrapolation gives a small error because these two definitions are not quite the same, however the error in this assumption is small for a large demagnification. In a magnetic lens, the trajectories will also be spiraling continuously. The optic axis is always defined as the Z-direction.

The aberration function in polar coordinates is then written as $\chi(\rho, \theta)$. The slope can also be written in Cartesian coordinates, in terms of the inclinations, u and v , to axes defined perpendicular to the optic axis as shown in Fig. 2.2. Thus if the coordinates along the ray at a general plane are $(x(z), y(z), z)$ then (u, v) are the

slopes $\left(\frac{\partial x(z)}{\partial z}, \frac{\partial y(z)}{\partial z} \right)$ evaluated at the specimen plane.

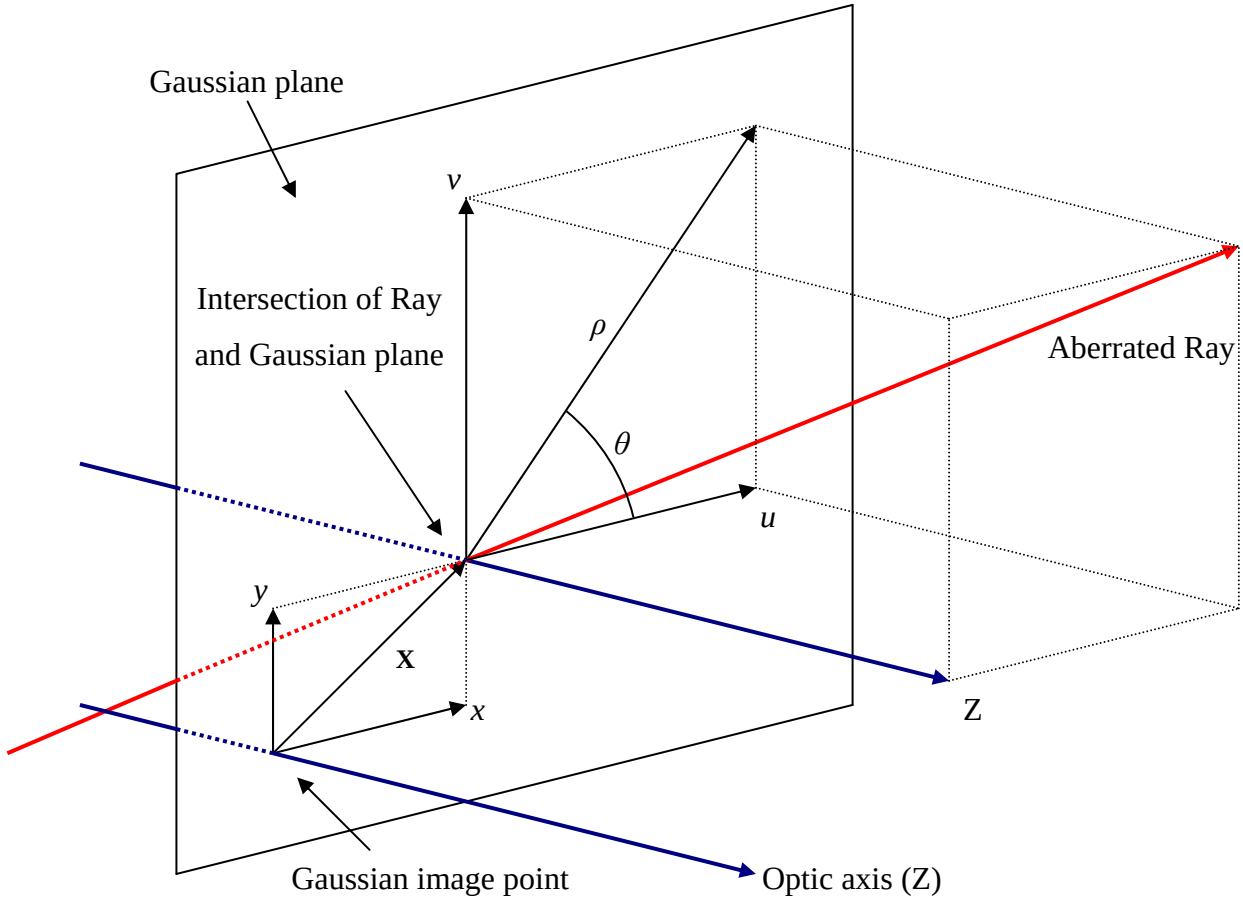


Figure 2.2. Schematic showing the coordinates used to define the aberration function. (ρ, θ) and (u, v) are defined so that they give the slopes of a ray to the optic axis at the sample. It is assumed that the demagnification is sufficient that these will also give the coordinates in the back focal plane, which implies that X is small. The intersection of the ray with the image plane is given by the vector $\mathbf{X}=(x, y)$. The optic axis defines the Z -direction.

The ray deviation is defined as the vector, $\mathbf{X}$, from the optic axis to the position of the aberrated ray in the image plane (which is of course the specimen plane in a STEM). Born and Wolf (1970, section 5.1) give this as:

$$\mathbf{X}(u, v) = \left(\frac{\partial \chi(u, v)}{\partial u}, \frac{\partial \chi(u, v)}{\partial v} \right) = M \nabla \chi(u, v) \quad (2.1)$$

Where the refractive index and additional factors have been included in the magnification, M , which is taken as unity in a STEM. This form could be considered as an alternative definition of the aberration function.

Even where interest is confined to the axial aberrations, there is still some choice as to what constants are introduced when the coefficients are defined. Several examples are given by Saxton (1995). To be consistent with the usual definitions of C_s

and defocus, the convention used here is that the additional constants should vanish in the expression for the radius of the ray deviation in polar coordinates. Other notations for aberration coefficients can be found (for example Hawkes and Kapser, 1994), but this form is particularly convenient for algebraic manipulation.

Using the notation of Krivanek et al (1999b), the aberration coefficients will be denoted by a capital C followed by up to two numbers and then a letter in subscript: C_{NSA} . The first number, N indicates the order of the ray aberration, which is referred to as ‘the order’. The second number, S indicates the symmetry of the aberration, and is the number of times that the function repeats on being rotated about the optic axis. The final letter indicates the orientation, which is specified as A or B type. These represent the cosine and sine terms respectively in the expansion below. Using these two components any orientation can be specified (equivalently, it is also possible to specify a magnitude and rotation). This form is well suited for use with an aberration corrector, because ideal multipoles correspond closely to particular aberrations (for example quadrupoles and C_{12} type aberrations). The orientation is redundant for rotationally symmetric aberrations, and there the associated orientation and symmetry subscripts can be omitted (hence C_3 for spherical aberration).

This expansion takes its simplest form in polar coordinates. Each aberration will be represented in the aberration function by terms:

$$(C_{NSA} \cos(S\theta) + C_{NSA} \sin(S\theta)) \frac{\rho^{N+1}}{N+1} \quad (2.2)$$

The corresponding terms in the radius of the ray intersection are:

$$(C_{NSA} \cos(S\theta) + C_{NSA} \sin(S\theta)) \rho^N \quad (2.3)$$

The angle between the A and B orientations will depend on the rotational symmetry and is given by: $\pi/2S$. The phase shift repeats every $2\pi/S$ radians. Similar results will apply for all orders of the same symmetry (so C_{12} and C_{32} will have the same axes) and for multipoles of the same symmetry. The angle between A and B type axes will be: $\pi/2$ for dipoles and probe shifts (C_{10}), $\pi/4$ for quadrupoles and astigmatism (C_{12}), $\pi/6$ for hexapoles and 3-fold astigmatism (C_{23}), and so on. All of the allowed terms up to fifth order are listed in Table 2.1.

Notation	Order of ray aberration	Order of wave aberration	Name	Symmetry
C_{01a}	0	1	Probe shift in x direction	1-fold
C_{01b}	0	1	Probe shift in y direction	1-fold
C_{12}	1	2	Astigmatism	2-fold
C_1	1	2	Defocus	Rotational
C_{23}	2	3	3-fold astig	3-fold
C_{21}	2	3	Coma	1-fold
C_{34}	3	4	4-fold astig of C_s	4-fold
C_{32}	3	4	2-fold astig of C_s	2-fold
C_3 or C_s	3	4	Spherical aberration	Rotational
C_{45}	4	5	5-fold astig	5-fold
C_{43}	4	5	4 th order 3-fold astig	3-fold
C_{41}	4	5	4 th order coma	1-fold
C_{56}	5	6	6-fold astig of C_5	6-fold
C_{54}	5	6	4-fold astig of C_5	4-fold
C_{52}	5	6	2-fold astig of C_5	2-fold
C_5	5	6	5 th order spherical aberration	Rotational

Table 2.1. Names and symmetries of all axial aberrations up to 5th order

Hence the expression for the aberration function up to fifth order is given by:

$$\begin{aligned}
\chi(\rho, \theta) = & \rho(C_{01a}\cos(\theta) + C_{01b}\sin(\theta)) \\
& + \frac{\rho^2}{2}(C_{10} + C_{12a}\cos(2\theta) + C_{12b}\sin(2\theta)) \\
& + \frac{\rho^3}{3}(C_{23a}\cos(3\theta) + C_{23b}\sin(3\theta) + C_{21a}\cos(\theta) + C_{21b}\sin(\theta)) \\
& + \frac{\rho^4}{4}(C_{30} + C_{34a}\cos(4\theta) + C_{34b}\sin(4\theta) + C_{32a}\cos(2\theta) + C_{32b}\sin(2\theta)) \\
& + \frac{\rho^5}{5}(C_{45a}\cos(5\theta) + C_{45b}\sin(5\theta) + C_{43a}\cos(3\theta) + C_{43b}\sin(3\theta) + C_{41a}\cos(\theta) + C_{41b}\sin(\theta)) \\
& + \frac{\rho^6}{6}(C_{50} + C_{56a}\cos(6\theta) + C_{56b}\sin(6\theta) + C_{54a}\cos(4\theta) + C_{54b}\sin(4\theta) + C_{52a}\cos(2\theta) + C_{52b}\sin(2\theta))
\end{aligned}
\tag{2.4}$$

In the discussion that follows, retaining higher order terms would make the notation unnecessarily cumbersome and so only terms up to third order will be shown. The discussion and all of the calculations within this thesis will consider terms up to the fifth order unless otherwise stated. Similarly higher order terms could be considered, but it is likely that other effects, such as off-axial aberrations, would also begin to be significant at such a level, even in the STEM, so they would also have to be included.

In Cartesian coordinates the aberration function is given by:

$$\begin{aligned}
\chi(u, v) = & C_{01a}u + C_{01b}v \\
& + \frac{1}{2} \left(C_{10} (u^2 + v^2) + C_{12a} (u^2 - v^2) + 2C_{12b}uv \right) \\
& + \frac{1}{3} \left(C_{23a} (u^3 - 3uv^2) + C_{23b} (3u^2v - v^3) + C_{21a} (u^3 + uv^2) + C_{21b} (v^3 + u^2v) \right) \\
& + \frac{1}{4} \left(C_{30} (u^2 + v^2)^2 + C_{34a} (u^4 - 6u^2v^2 + v^4) + C_{34b} (4u^3v - 4uv^3) + C_{32a} (u^4 - v^4) + C_{32b} (2u^3v + 2uv^3) \right) \\
& + \dots
\end{aligned} \tag{2.5}$$

Thus the gradient of the aberration function, which relates to the ray deviation at the specimen plane, is given by:

$$\begin{aligned}
\frac{\partial}{\partial u} \chi(u, v) = & C_{01a} + C_{10}u + C_{12a}u + C_{12b}v \\
& + C_{23a} (u^2 - v^2) + C_{23b} 2uv + C_{21a} (u^2 + \frac{1}{3}v^2) + C_{21b} \frac{2}{3}uv \\
& + C_{30} (u^3 + uv^2) + C_{34a} (u^3 - 3uv^2) + C_{34b} (3u^2v - v^3) + C_{32a} u^3 + C_{32b} (\frac{3}{2}u^2v + \frac{1}{2}v^3)
\end{aligned} \tag{2.6}$$

$$\begin{aligned}
\frac{\partial}{\partial v} \chi(u, v) = & C_{01b} + C_{10}v - C_{12a}v + C_{12b}u \\
& + C_{23a} (-2uv) + C_{23b} (u^2 - v^2) + C_{21a} \frac{2}{3}uv + C_{21b} (\frac{1}{3}u^2 + v^2) \\
& + C_{30} (u^2v + v^3) + C_{34a} (v^3 - 3u^2v) + C_{34b} (u^3 - 3uv^2) - C_{32a} v^3 + C_{32b} (\frac{1}{2}u^3 + \frac{3}{2}uv^2)
\end{aligned} \tag{2.7}$$

It is useful to also list the second derivatives; these will be collectively referred to as the curvature of the aberration function:

$$\begin{aligned}
\frac{\partial^2}{\partial u^2} \chi(u, v) = & C_{10} + C_{12a} + 2C_{23a}u + 2C_{23b}v + 2C_{21a}u + \frac{2}{3}C_{21b}v \\
& + C_{30} (3u^2 + v^2) + C_{34a} (3u^2 - 3v^2) + 6C_{34b}uv + 3C_{32a}u^2 + 3C_{32b}uv \\
\frac{\partial^2}{\partial u \partial v} \chi(u, v) = & C_{12b} - 2C_{23a}v + 2C_{23a}u + \frac{2}{3}C_{21a}v + \frac{2}{3}C_{21b}u - 6C_{34a}uv \\
& + 2C_{30}uv + C_{34b} (3u^2 - 3v^2) + C_{32b} (\frac{3}{2}u^2 + \frac{3}{2}v^2)
\end{aligned} \tag{2.8}$$

$$\begin{aligned}
\frac{\partial^2}{\partial v^2} \chi(u, v) = & C_{10} - C_{12a} - 2C_{23a}u - 2C_{23a}v + \frac{2}{3}C_{21a}u + 2C_{21b}v \\
& + C_{30} (u^2 + 3v^2) + C_{34a} (3v^2 - 3u^2) + 6C_{34b}uv - 3C_{32a}v^2 + 3C_{32b}uv
\end{aligned}$$

The simple but rather lengthy algebraic manipulation necessary for this work was performed using the computer program MAPLE (Waterloo Maple Inc, 2000). The fourth and fifth order terms follow in exactly the same way. The C_{01} type aberrations represent a shift of the probe and so are not included in the aberration function unless explicitly stated.

2.2.2. Spherical Aberration

The aberration that limits the point resolution in a modern transmission electron microscope is spherical aberration, C_s , which will often be referred to as C_3 or C_{30} in our notation. This rotationally symmetric aberration takes its name from light optics, where it was caused by the spherical shape of early lenses, and is unavoidable for all round electron lenses. The effect of positive spherical aberration is to focus rays at higher angles to the optic axis more strongly than those traveling at smaller angles as illustrated in Fig. 2.3. The wave aberration term associated with spherical aberration is $C_s \rho^4 / 4$. Thus the ray deviation from the Gaussian image point is $C_s \rho^3$.

Positive spherical aberration will cause higher angle rays to cross the axis **before** the Gaussian image plane, as indicated in Fig. 2.3. For pure spherical aberration this is at a distance $C_s \rho^2$ along the Z-axis. Continuing this geometrical optics view, it can be seen that the narrowest cross-section of the beam, the ‘disk of least confusion’ will be at some distance before the Gaussian focal plane. For C_s only, this would be at a distance from the Gaussian plane (or equivalently an underfocus value) of $3C_s \rho_{\max}^4 / 4$. This gives a disk radius of $C_s \rho_{\max}^4 / 4$, where $\rho_{\max}$ is the highest angle admitted by the aperture. This point, like the others on the caustic can be found by considering the extreme values of the intersection between the rays. Thus with C_s present, forming the smallest probe will require some underfocus. However this has neglected diffraction effects and so the wave-optical methods outlined in Chapter 5 are preferred for detailed calculations. The location of the beam caustic, the surface in which neighboring rays intersect, is discussed in more detail by Hawkes and Kasper (1994, chapter 42).

Scherzer's demonstration that the spherical aberration is unavoidable in conventional round lenses means that this aberration limits the ultimate performance of an electron lens (Scherzer, 1936). The point resolution is conveniently obtained by assuming that the diffraction and spherical aberration effects add in quadrature, such that the effective probe radius, r_{Eff} , is given by:

$$r_{\text{Eff}}^2 = \left(\frac{0.61\lambda}{\rho_{\text{max}}} \right)^2 + \left(\frac{1}{4} C_s \rho_{\text{max}}^3 \right)^2 \quad (2.9)$$

Optimizing this with respect to angle gives the ideal size of the objective aperture, ρ_{max} , and the optimal probe radius as:

$$\rho_{\text{max}} \propto 1.1 \left(\frac{\lambda}{C_s} \right)^{\frac{1}{4}}, r_{\text{Eff}} \propto 0.65 (C_s \lambda^3)^{\frac{1}{4}} \quad (2.10)$$

Where the constants can take several different values depending on which other effects are included and how much defocus is used to balance the spherical aberration. Similarly a wave-optical approach gives slightly different constants, but all of these limits scale with the wavelength and C_s in the same way (Chapter 5).

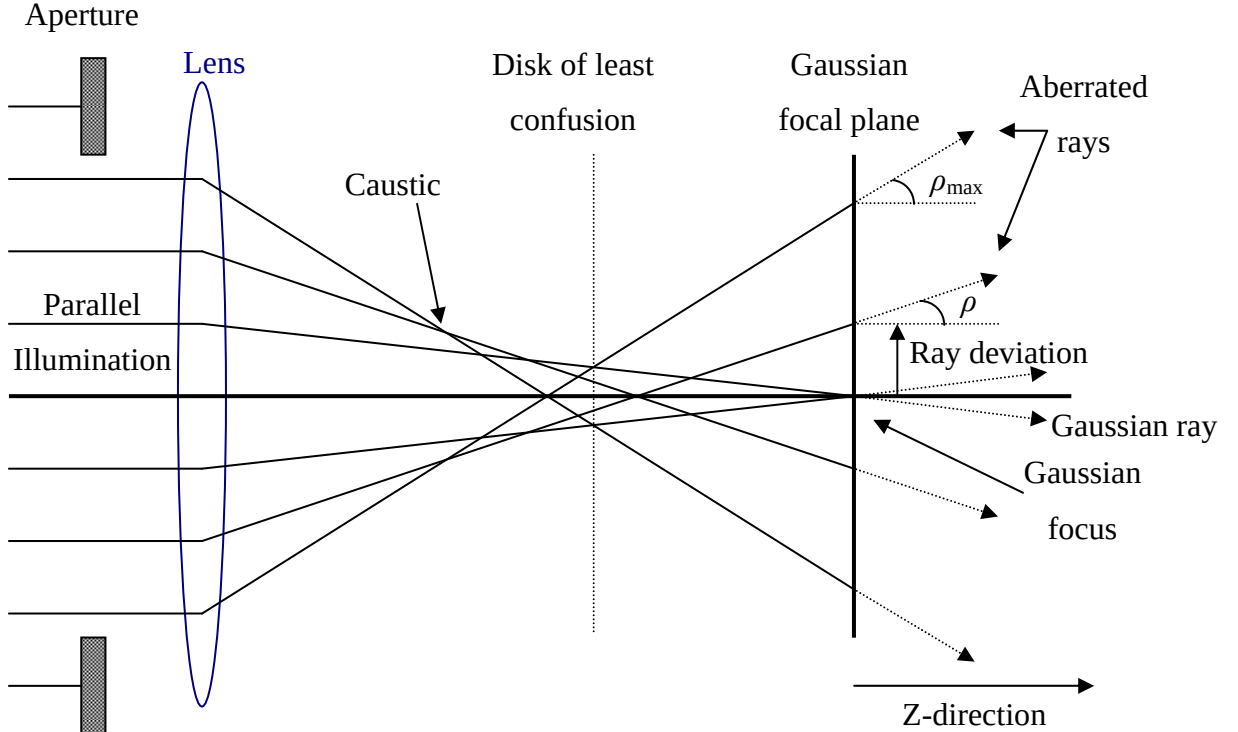


Figure 2.3. The effects of positive spherical aberration. Rays traveling at higher angles to the axis are overfocused. The rays then intersect the Gaussian focal plane at different points and the minimum probe diameter occurs in underfocus. This aberration is rotationally symmetric.

The phase changes caused by defocus and C_s are plotted in figures 2.4 and 2.5. In Fig. 2.4 it can be seen that underfocus (scaling as ρ^2) is used to balance the extra phase change introduced by the spherical aberration (scaling as ρ^4). In Fig. 2.5 the same aberration function is plotted, with the first and second radial derivatives indicated.

In an aberration corrected system, spherical aberration can also be negative. This just corresponds to the opposite phase change, or equivalently to higher angle rays being under-focused.

The probe intensity distributions, plotted later (Fig. 2.8), illustrate how the probe size is limited by the spherical aberration. The probe consists of a large central maximum surrounded by the so-called probe tails. The existence of these probe tails shows that in EELS, as well as imaging, the signal at any point will have contributions from a relatively large area of the specimen. A C_s -corrected system is therefore expected to offer significant advantages.

Chromatic aberration is described in Chapter 3.

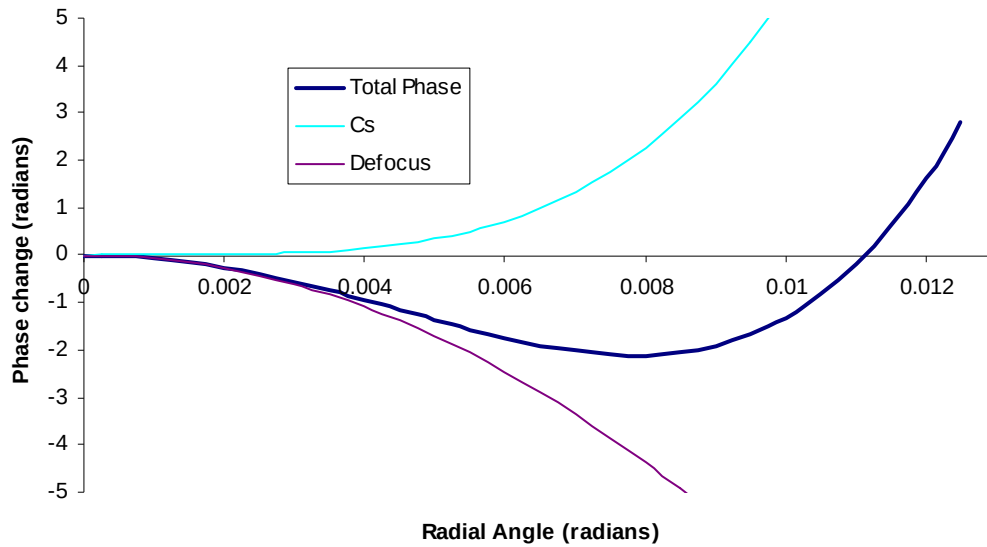


Figure 2.4. The phase change associated with spherical aberration. The contributions from C_s and defocus are shown and added to produce a small total phase across a range of usable angles. The physically realistic (uncorrected) values of $C_s=1.3$ mm defocus=-80 nm are used.

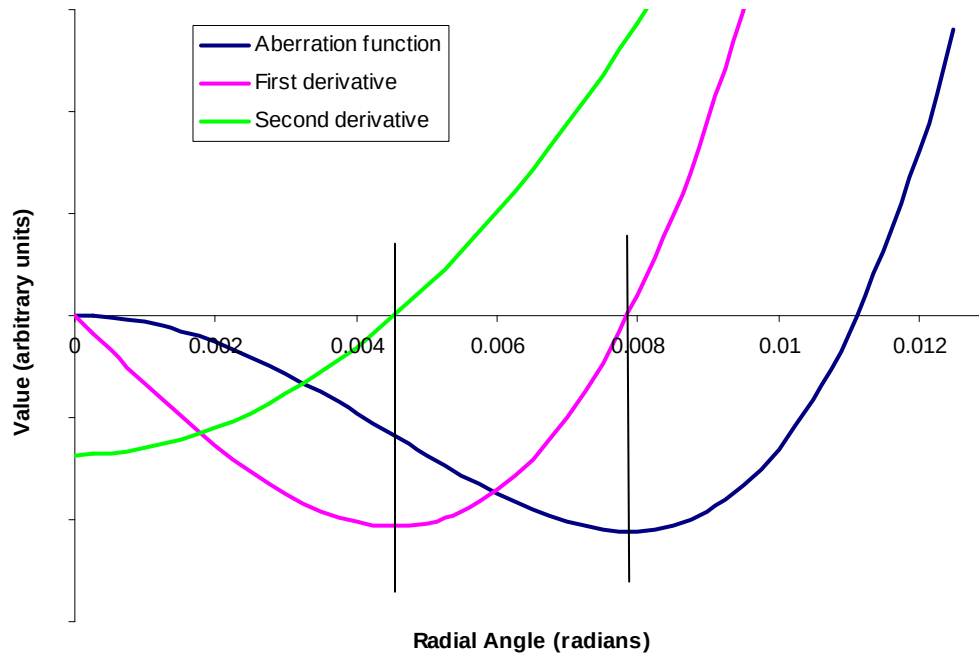


Figure 2.5. The same aberration function as for Fig. 2.4. The first two radial derivatives of the aberration function are shown (using arbitrary scales). In the rotationally symmetric case, the two points where these are zero (marked with lines) relate to the loci of infinite magnification in the Ronchigram encountered later.

2.2.3. Non-symmetric Aberrations

The rotationally symmetric description of aberrations is normally adequate for a correctly aligned and stigmated system consisting of round elements. In the case where strong non-round elements are being used, the other terms will also be significant. This means that the aberration function now has to be considered as a two dimensional surface rather than the simple function of one coordinate only. The contributions from all of the terms up to third order are shown in Fig. 2.6. The contours of these functions, which clearly show the symmetry of the aberrations, are plotted out in Fig. 2.7.

Some details of the effects of non-round aberrations on imaging are considered by Krivanek (1994) and Ishizuka (1994). In the Ronchigram these aberrations have particularly interesting effects, distorting the symmetry of the entire pattern (Chapter 6). In conventional images the effect is less obvious, as the resultant images will be related to the convolution of the probe with the object function. The symmetry of the probe will reflect the symmetry of the aberration function and, using the methods outlined in Chapter 5, it is possible to calculate the expected intensity distribution in the probe. The intensity distribution expected in the optimal case for the VG STEM is shown in Fig. 2.8(A). It can be seen that the probe consists of a narrow central maximum surrounded by symmetric probe tails, caused by the spherical aberration. In Fig. 2.8(B) the same aperture size and aberration values are used, along with the addition of significant amounts of 3-fold astigmatism. It can be seen that the probe becomes 3-fold, with a changed current distribution degrading the expected resolution.

The remaining terms in equation (2.5) include aberrations of higher symmetry, such as 5 and 6-fold astigmatisms. These terms also have a higher order dependence on the radial angle, and in the uncorrected case the coefficients are of the order of the focal length, so they are not normally significant at the typical aperture angles employed (much less than 1 radian). However, these will become significant in C_s -corrected systems.

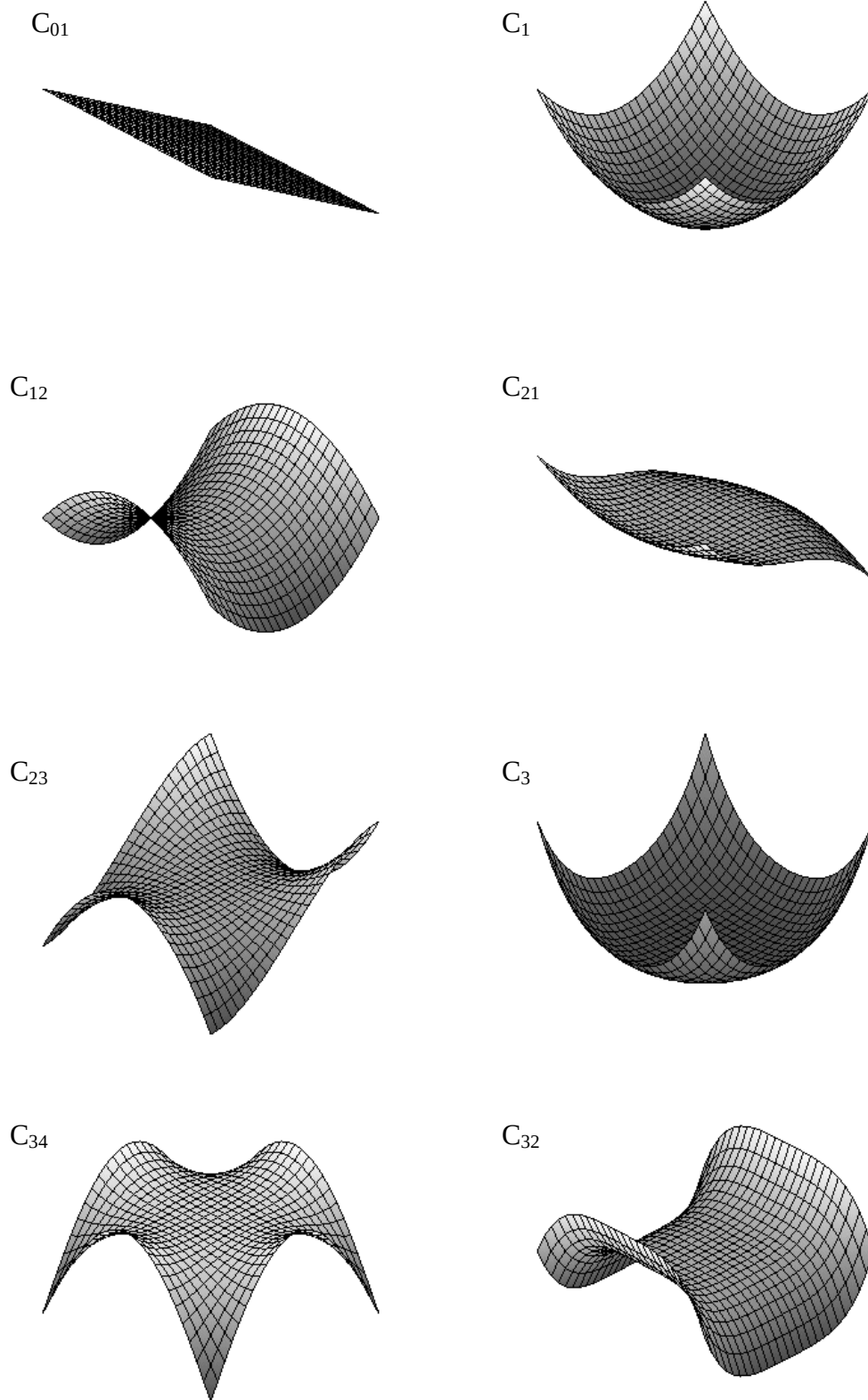


Figure 2.6. 3D projections of the contributions from terms in the aberration function up to third order. The vertical direction corresponds to the magnitude of the phase change imparted to the beam on an arbitrary scale.

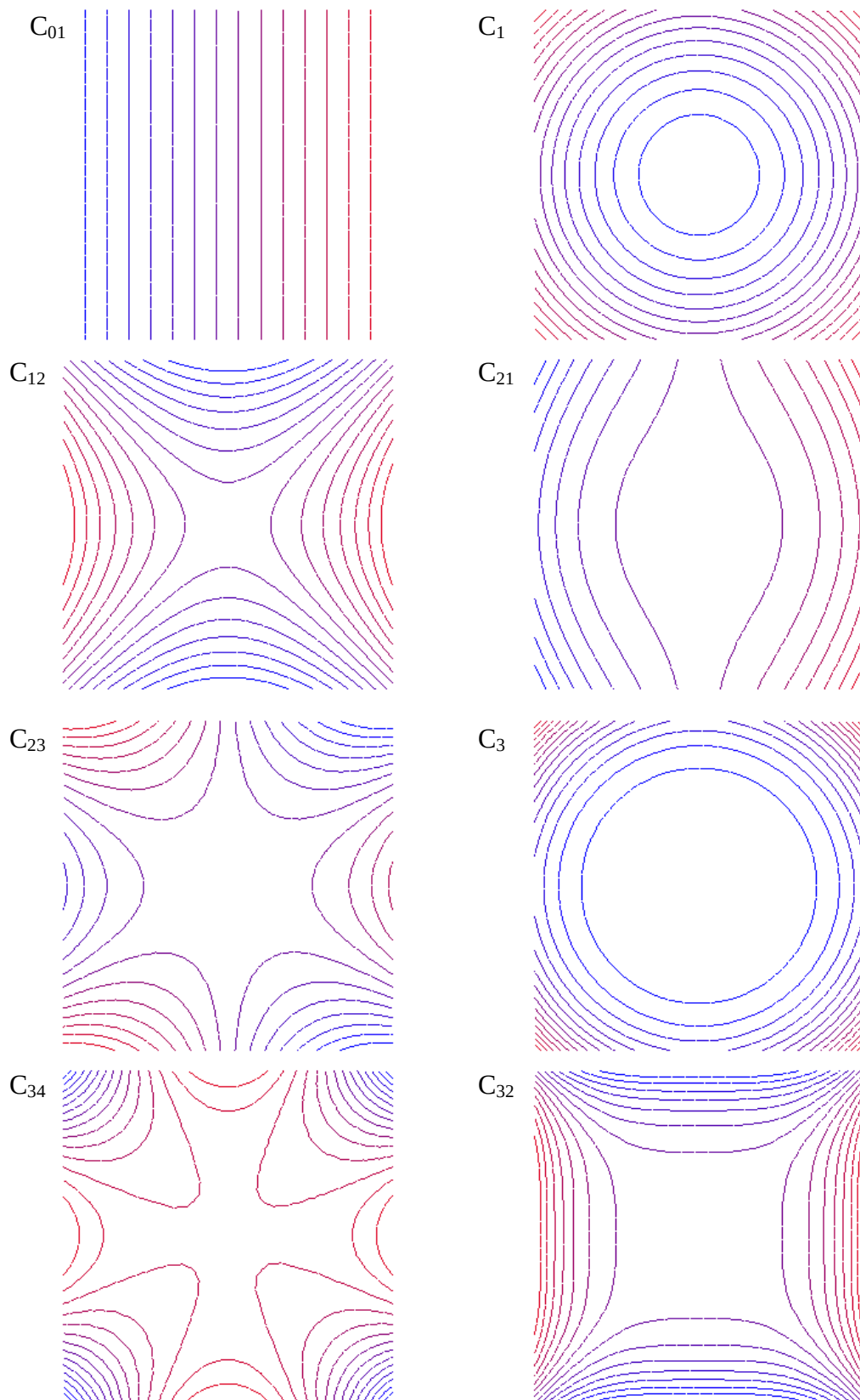


Figure 2.7. Contours of the contributions from terms in the aberration function. Color has been used to indicate the sign of the phase change, with increasing red corresponding to the vertical component plotted in Fig. 2.6.

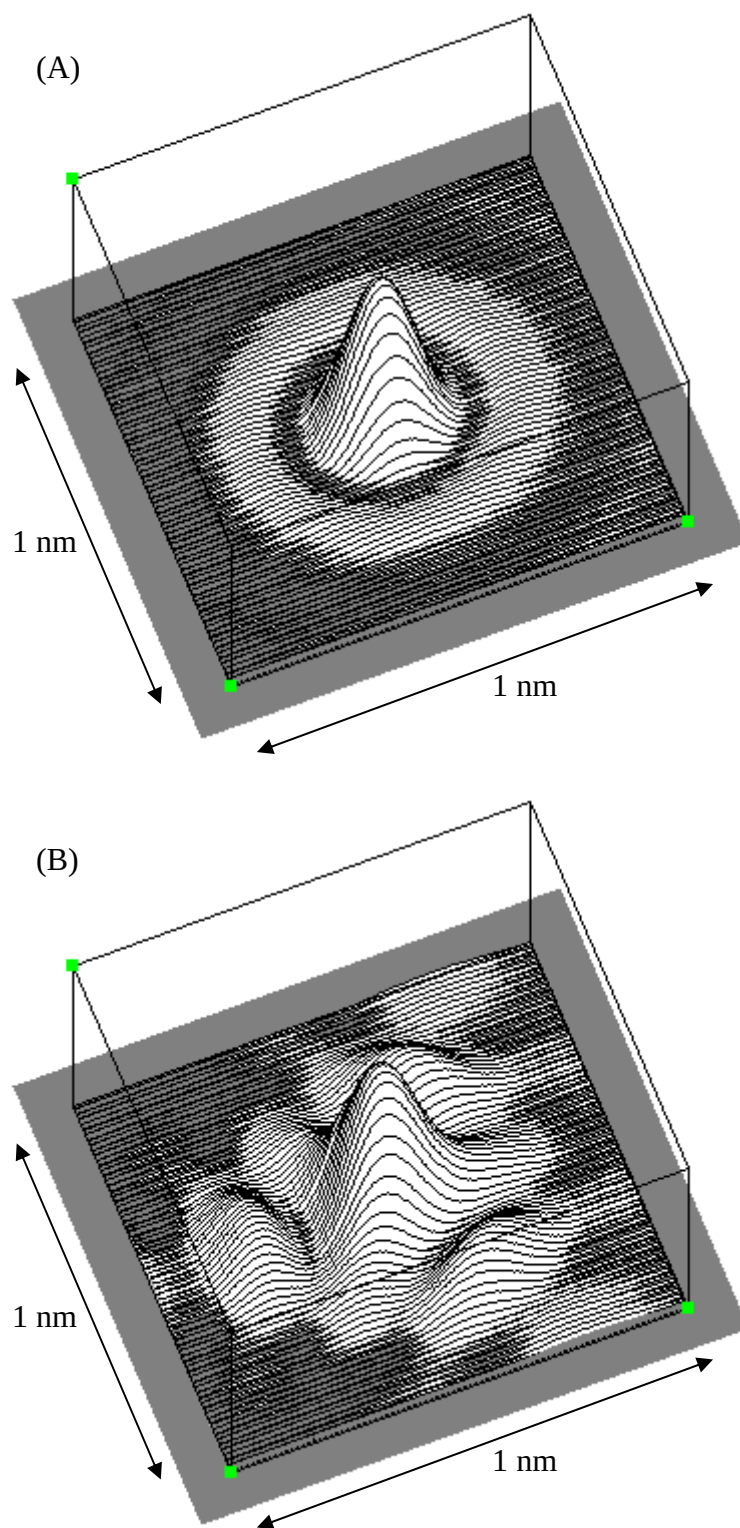


Figure 2.8. Probe intensity distributions in the VG STEM (with no chromatic aberration).
(A) The distribution for 1.3 mm C_s using defocus and apertures optimized according to the table in Chapter 5. Symmetric probe tails are visible outside the central maximum.
(B) The distribution for the same values but with the addition of 3 μm of 3-fold astigmatism.

2.3. Introduction to STEM

The formation of an electron microscope image can be summarised in 3 stages: A beam of electrons is produced and focused onto a specimen, the beam interacts with the specimen, and finally it is imaged onto a detector and the intensity recorded. The essential feature of STEM imaging is that the beam is focused into a cross-over (referred to as the **probe**) at the specimen plane and scanned across the sample. The transmitted signal is then recorded on a variety of detectors, as a function of probe position. The principles of STEM operation are shown in Fig. 2.9. The first successful STEM instrument was built by Crewe and co-workers in the late sixties (Crewe et al, 1968). Reviews of STEM operation are given by Cowley (1979c) and Colliex and Mory (1983).

In the instruments used here, the electron beam is formed by emission from a cold field emission gun (FEG). This is preferred to the alternatives such as thermionic emitters because of the high brightness and coherence, and low range of emitted energies (typically 0.3 eV). A FEG operates by applying a strong electric field to the tip, which is an orientated single crystal of tungsten. The field is sufficiently strong so that the electrons are able to tunnel out of the tip, and are accelerated up the column by suitably positioned anodes. Details of field emission, brightness, and gun design are given by James (1997).

Because the FEG is emitting from a very small area, it will be very coherent, but the total current will be quite small (typically limited to $<5 \mu\text{A}$). Brightness, B , is usually defined by the current, dF , passing through a solid angle element, $d\Omega$, emitted from an area element, dS , at an angle, φ :

$$dF = B dS d\Omega \cos(\varphi) \quad (2.11)$$

Thus the current in the probe, and therefore the signal at the detectors, will depend on the brightness and also the geometrical demagnification of the source. As the image of the source is demagnified to form a smaller and more coherent probe the detected intensity (and signal to noise ratio) will decrease. Obtaining a high quality STEM image will involve balancing these two considerations (for example, Nellist 1995).

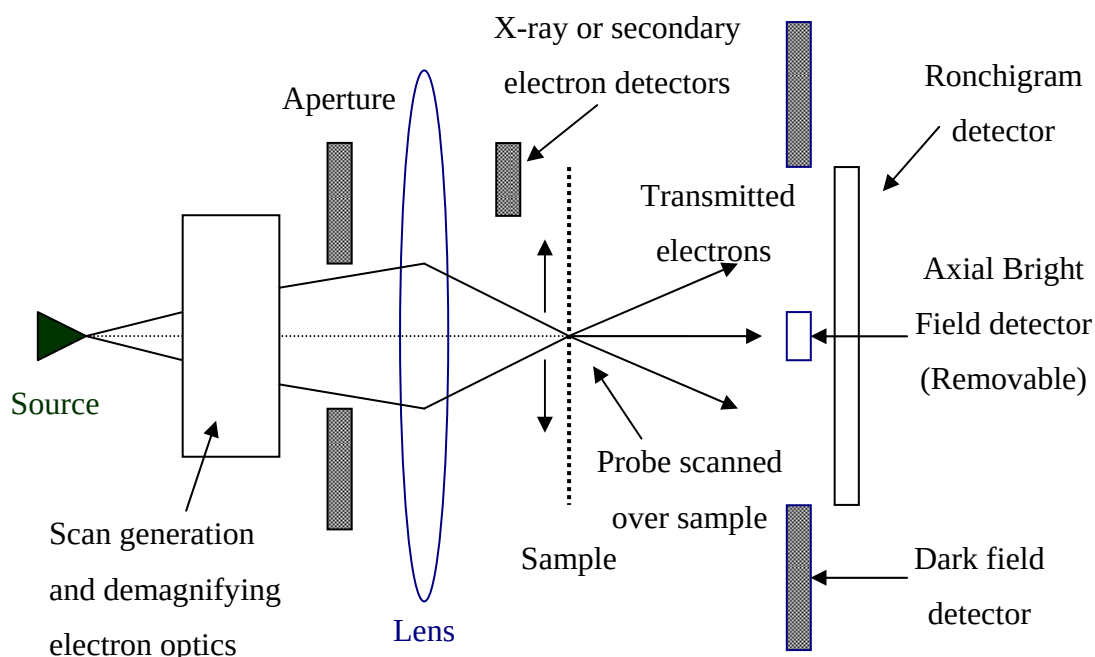


Figure 2.9. Simplified schematic of STEM operation. Electrons are emitted from the source and focussed into a probe which is scanned across the sample. The transmitted signal is recorded by detectors in the far-field.

In practice, it is necessary to have a number of other elements besides the ones shown in the simple system of Fig. 2.9, and a complete column is illustrated in Fig. 2.10. The beam of electrons is focused onto the sample by a series of round lenses to provide a flexible range of illumination conditions. The condenser lenses (C1 and C2) are used to demagnify the image of the source, which is then imaged onto the sample by the objective lens (OL). In practice a STEM will also include alignment coils to optically centre the lenses, and stigmators to correct astigmatism of the beam caused by imperfections and misalignments. Removable apertures exclude unwanted parts of the beam (particularly aberrated high-angle rays) and minimise scattering from other components in the column. Details of the pre-specimen optics in STEM are given by Mory (1985).

In an aberration-corrected STEM, the additional elements for correction and alignment will be situated before the objective lens. This contrasts to the situation in a CTEM where the corrector would be situated after the objective lens. So a fully corrected microscope able to operate in both of these modes would require correctors both before and after the objective lens.

Various detector geometries are employed in STEM in the far field. A point detector on axis at the centre of the zero-order disk provides bright field (BF) imaging. An annular dark field (ADF) detector is used to collect high scattering angles. A shadow image is recorded using a charge coupled device (CCD) detecting a finite range of angles around the axis. The different imaging modes are further described by Nellist (1995). A spectrometer can be used to record electron energy loss spectra (EELS). This process is described in detail by Egerton (1986) and more recent developments are given by Stolojan (2000). Other signals can also be collected with different detectors, which are summarised by Williams and Carter (1996). One advantage of STEM is that all the signals from the very small area of the sample illuminated by the probe are obtained simultaneously. The STEM thus provides a unique method for microanalysis of a wide variety of materials, as reviewed, for instance, by Brown (1981).

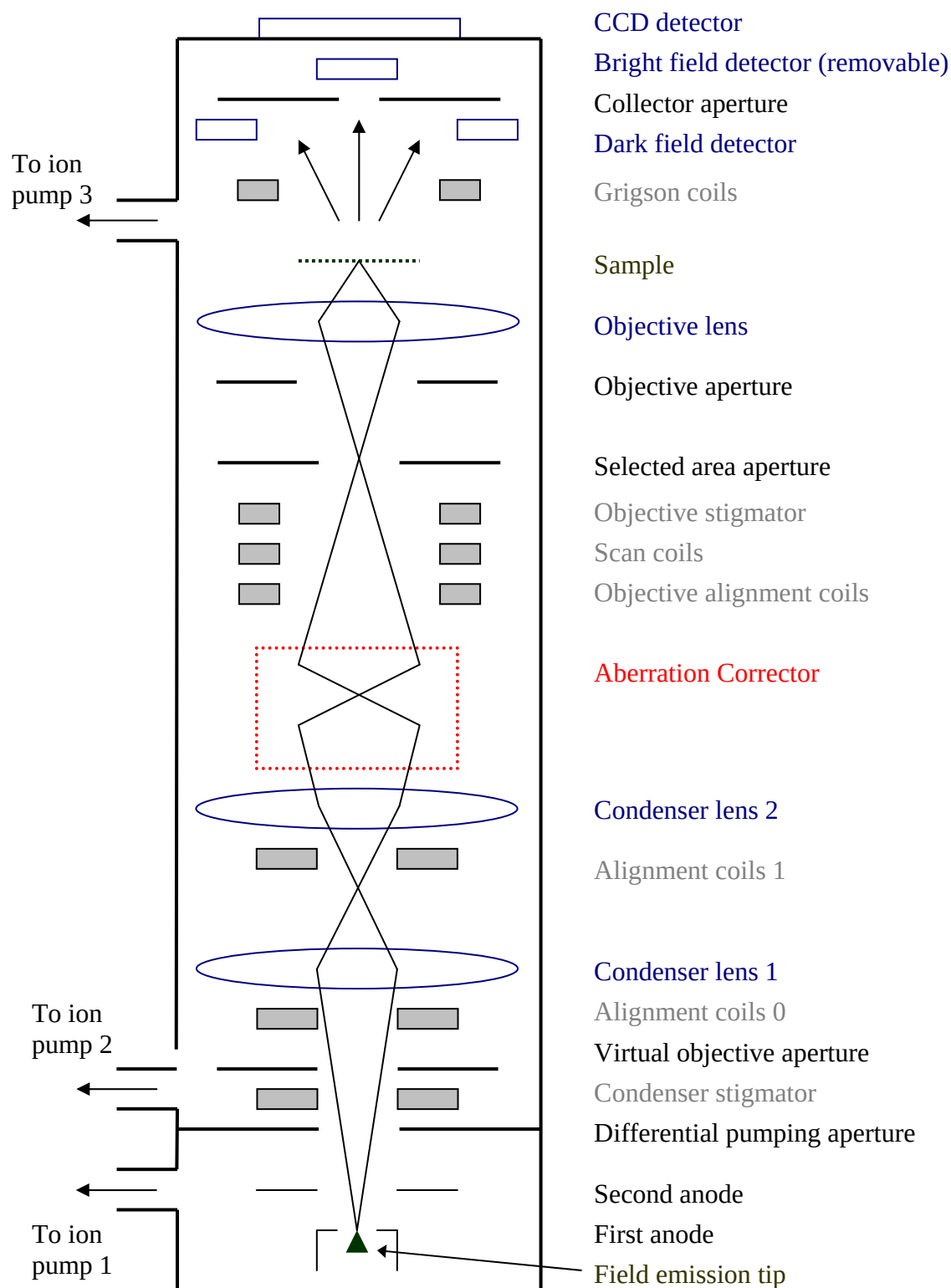


Figure 2.10. Schematic diagram of the modified VG HB5, used experimentally, indicating the approximate ray paths through the system in a typical imaging mode and the position of the aberration corrector. (The ray paths through the corrector are only representations of the trajectories in one plane). The functions of some of the normal VG stigmators and alignment coils have been taken over by the computer controlled controls on the corrector.

2.3.1. The Aberration Corrected VG HB5

The instrument used for most of the experimental work was a Vacuum Generators (VG) HB5 STEM. The microscope was modified by Krivanek et al (1997b) to include a quadrupole-octupole aberration corrector, to have a second condenser lens, and to have a completely ion pumped vacuum system. A schematic of the modified instrument is shown in Fig. 2.10. Some experimental work was also done on the closely related VG HB501, which is described in detail by Nellist (1995). A second aberration corrector was later installed in a modified HB501 (Krivanek et al, 1999a,b).

Both of these machines operate at up to 100 kV and employ cold field emission guns. The pole-pieces used in the HB5 have spherical aberration values quoted as approximately 7 mm for the “analytical” and 3.5 mm for the “high-resolution” pole-pieces. In the HB501, the spherical aberration values of the similarly named pole-pieces were given as 3.1 mm and 1.7 mm, respectively. Comparing schematics suggests that these were the same two designs of pole-pieces. The difference in the performance of similar pole-pieces in the HB5 was numerically confirmed to be due to the lower current rating of the objective lens power supply (Lupini and Krivanek, 1998). Using a cartridge in which the specimen height in the lens could be adjusted, it was possible to optimise the value of C_s for the analytical pole-piece in the HB5. This resulted in a measured value of approximately 4 mm of C_s , with a slightly modified power supply in the HB5.

The entire microscope operates under high vacuum conditions, with the chamber vacuum better than 10^{-8} mbar and the gun chamber vacuum of order 10^{-11} mbar. To improve the vacuum in the lower part of the column, an additional sublimation pump chamber was constructed and installed just above the differential pumping aperture.

2.3.2. Electron optics in the STEM

In an uncorrected STEM, the tip and the specimen plane are separated by an electron optical system designed to provide a probe that is both bright and coherent. A detailed review is given by Mory (1985).

The tip is often assumed to be a perfectly coherent point source. However, in any real gun the tip has a finite size and limited coherence, as described by Hawkes and Kasper (1994). In an uncorrected instrument, the two condenser lenses are adjusted to balance the competing needs of coherence and current required. The condenser lenses are modelled as thin lenses (James, 1997), with a focal length, f , such that:

$$f = \frac{k}{I^2} \quad (2.12)$$

Where I is the lens current, and k is a constant fitted to measurements. The values of k fitted experimentally were:

Condenser 1: $k=42.5$ mm,

Condenser 2: $k=42.2$ mm.

Where hysteresis means that the uncertainty is at least a millimetre on C1 and more on C2. These values are within this experimental error of the values measured by James (1997) for the HB501. The values for the lens current were determined from the voltage measured across a sense resistor in the power supply, and the focal length was measured by locating the beam cross-over. This is normally done by adjusting the lens until part of an aperture is at infinite magnification. However in an aberration corrected microscope, the cross-overs can also be located by wobbling the quadrupole elements and identifying the points at which they have minimum effect on the beam. Similarly, more fixed points can be obtained by using the additional differential pumping apertures inside the corrector.

For first-order calculations, the lenses are taken to be thin, in which case, the deflection imparted to the beam, ρ' , at an off-axial distance, x , is given by:

$$\rho' = -x / f \quad (2.13)$$

The magnification is then given by the ratio of slopes before and after the lens, or can be calculated with the thin lens formula. This allows the demagnification of the source and corrector to be calculated. It is also possible to include higher order aberrations, for example spherical aberration is included as:

$$\rho' = -\frac{x}{f} - \frac{C_s}{f} \left(\frac{x}{f} \right)^3 \quad (2.14)$$

In a C_s -corrected STEM, the additional role of the condenser lenses is to adjust the beam to the correct diameter going into the corrector. In practice, the way that the amount of C_s -correction was adjusted was to use the condenser lenses to alter the ratio of the beam diameter in the corrector to the objective lens. (As opposed to adjusting the strength of every octupole element in the corrector and correcting every single parasitic aberration.)

Ray diagrams of the trajectories in typical operating modes for a VG STEM are shown in Fig. 2.11. First the standard imaging mode suitable for collection of bright-field, ADF or Ronchigram images is illustrated. In this mode, a cross-over is positioned at the selected area aperture (SAD) and the angles that contribute to the probe are limited by either the objective aperture (OA) or the optically conjugate virtual objective aperture (VOA). The scan coils scan the beam such that the beam still passes through the centre of the objective lens (to minimise any off-axial aberrations) and the probe moves across the sample. In the selected area mode a cross-over is positioned at the OA so that the sample is illuminated by a nearly parallel beam. In this mode the scan coils now rock the beam, through a range of angles rather than positions. In both of these modes on the uncorrected HB501 it is common practice to use only one condenser lens to minimise the instabilities imparted to the beam. In the corrected system the VOA is used as the beam limiting aperture so that the beam does not hit the drift tube or any insulating parts when distorted by the corrector elements.

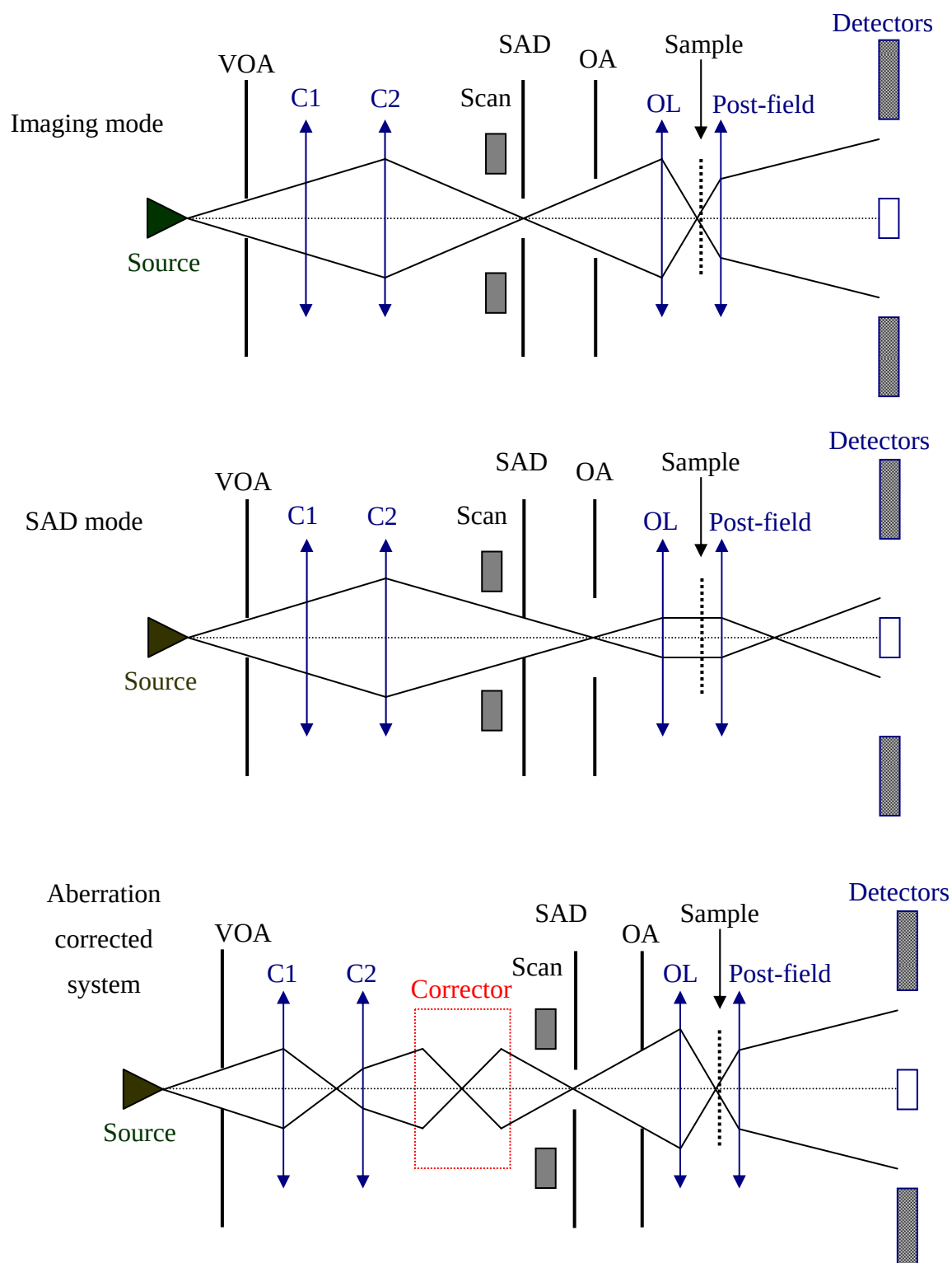


Figure 2.11. Schematic ray diagrams for the VG STEM in typical modes. In imaging mode a cross-over is scanned across the sample. In SAD mode the sample is illuminated by a nearly parallel beam which is rocked rather than scanned. The final ray diagram shows typical trajectories in the aberration corrected HB5 STEM in imaging mode.

2.4. The HB5 Aberration Corrector

The C_s -corrector installed in the VG HB5 is a quadrupole octupole (QO) corrector, positioned between the second condenser and the objective lens, as illustrated in Fig. 2.10. This is further described by Krivanek et al (1997b, 1998) and in Chapter 4. The central principle of operation of such a corrector is that the octupoles produce a field with the correct radial field dependence to correct spherical aberration, but have an unwanted azimuthal variation. By using suitable combinations of quadrupoles and octupoles it is possible to obtain only the desired C_s -correction without the azimuthal variation.

This corrector consists of six identical stages, each one consisting of 12 magnetic poles. On these poles, the power supplies and windings are arranged to provide a strong octupole and quadrupole. The remaining windings act as auxiliary adjustments to provide alignment and other multipole fields to remove any parasitic aberrations. These supplies are weaker than the main quadrupole and octupole elements in order to minimize the instabilities. The power supply for the corrector is able to provide a total of 84 adjustable outputs with the stability on each better than one part per million. The arrangement of elements is shown in Fig. 2.12, with the cross-section of the beam at the center of each stage.

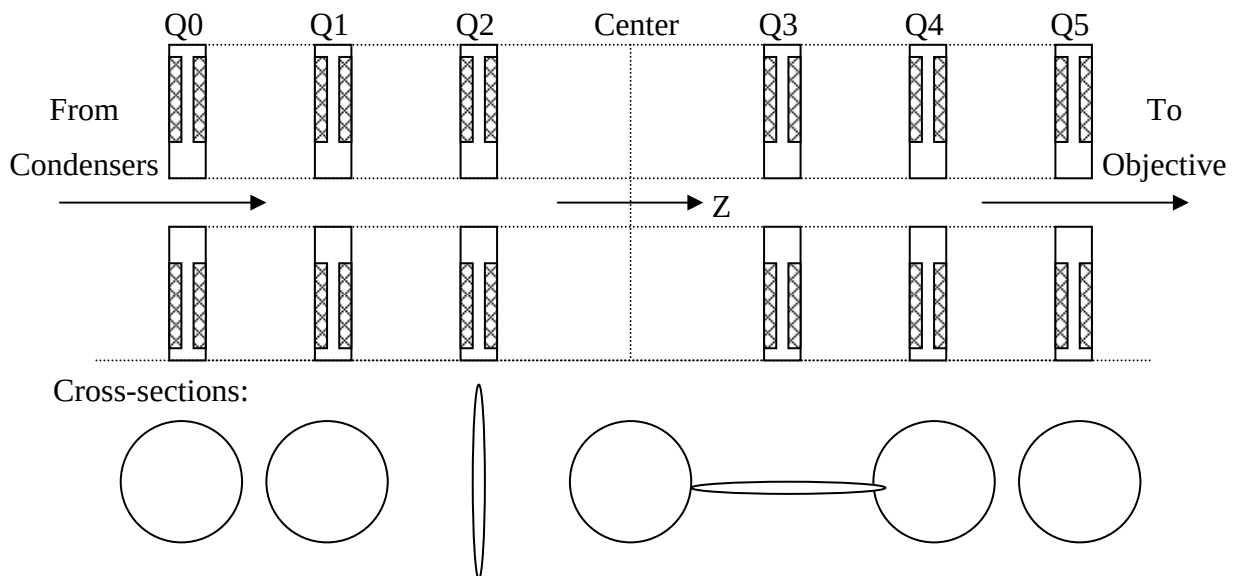


Figure 2.12. Schematic of the quadrupole-octupole corrector in the HB5. This consists of six multipole units (numbered) each capable of producing any multipole field up to an octupole at any angle. Approximate beam cross-sections at the center of each stage are indicated.

2.4.1. HB5 Software

The software used to control the aberration corrected microscopes is divided into two main sections, and on the HB5 this is split across two computers. A Macintosh Power PC running Digital Micrograph software (from Gatan Inc) is used to control the imaging and aberration analysis. This communicates with a PC using custom software (Krivanek et al, 1997a) developed to control the electronics. This arrangement is illustrated in Fig. 2.13.

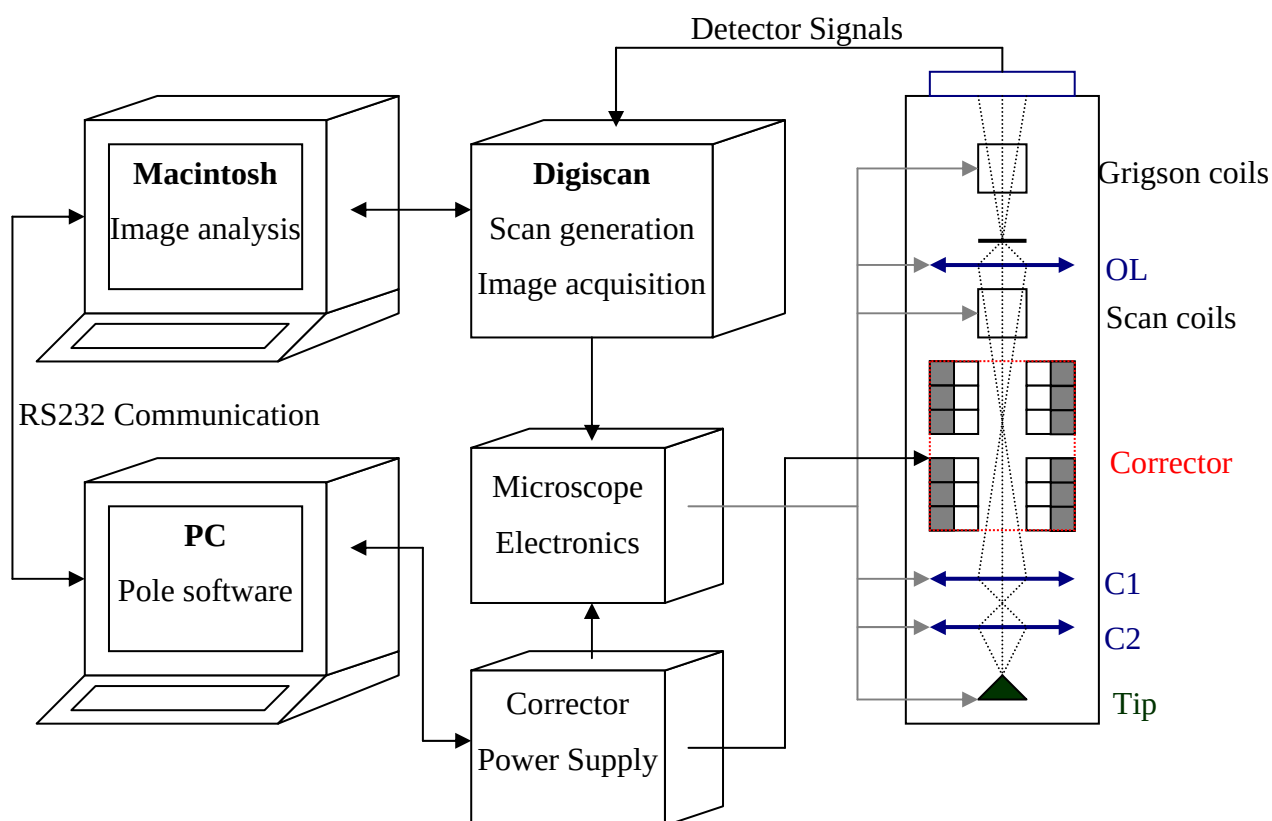


Figure 2.13. Schematic of the computer control of the HB5 microscope. The corrector is directly powered from the corrector power supply unit. The other elements indicated are powered from the original microscope electronics with inputs from the computer control to facilitate automatic adjustment.

One of the reasons the early correctors failed is due to the lack of flexibility in adjustment. It is necessary to be able to introduce any required multipole at arbitrary angles and to correct any lower order imperfections; otherwise the parasitic errors swamp any improvements. This greatly increases the number of controls necessary (with 84 extra controls in the HB5 corrector, and 40 in the HB501 corrector, compared to a normal STEM). The only practical way to obtain a humanly adjustable system is to provide a mechanism for linking basic controls into higher order compound controls. The trim coils are linked together to provide basic multipoles, which are further linked to provide correctly aligned, compound elements, referred to as 'pure'. The hierarchy is schematically illustrated in Fig. 2.14.

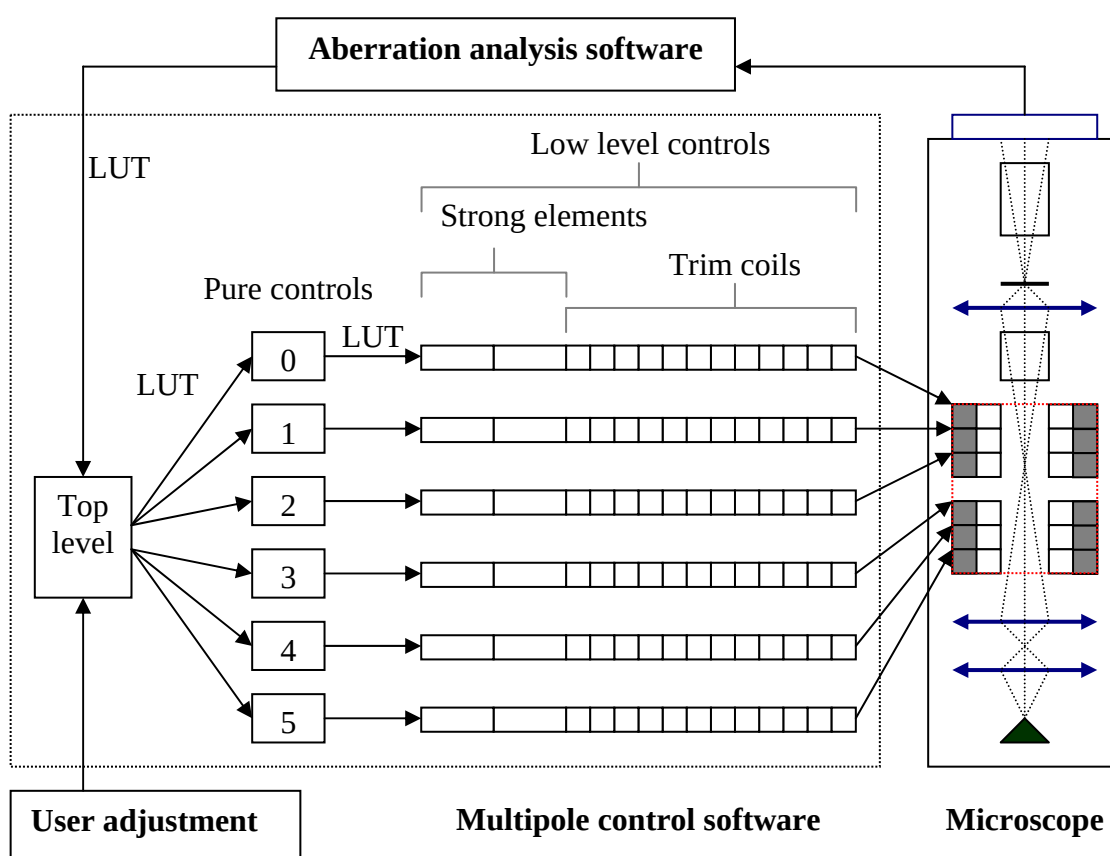


Figure 2.14. Schematic illustration of the hierarchical linking of controls so that one compound and pure controls adjust lower layers of controls, causing the settings of a large number of individual power supplies to be changed. The look-up tables (LUT), which must be calibrated, are indicated.

The PC software can be used with the Ronchigram to provide a convenient way of making pure poles. A higher order multipole is excited and then lower order multipoles are added until the resulting shift of the aperture and unwanted distortions are cancelled, corresponding to the action of an idealised element. For example, a quadrupole will provide a pure quadrupole action only when a suitably aligned dipole, at the same level, is excited by a proportional amount, while an octupole might need both quadrupole and dipole elements to be added. The software can link the adjusted controls into one compound element, which then acts as a pure control. These linkages are then stored in look-up tables, which are calibrated as part of the alignment procedure

Figure 2.14 also indicates where these look-up tables are used to convert the pure controls to the low-level values that are sent to the multipole elements. For completely automatic tuning, it will also be necessary to generate look-up tables for the effects of the corrector elements on the aberration coefficients. This relies on an accurate method of measuring the aberration function. The aberration function should be measured, then a multipole adjusted, and the measurement repeated. Performing this process in a systematic manner allows the change in the aberration function resulting from each auto-adjustment to be calibrated. It is then possible for the computer to automatically adjust the corrector settings. This process has recently been implemented and described in more detail by Dellby et al (2001).

The Macintosh computer is used to control the image acquisition through the Digital Micrograph software (Gatan Inc). This provides control of the scan generation and automatic image acquisition. An analogue output board was used to control the objective lens, condenser lenses, accelerating voltage and Grigson coils in software. The principal benefit of the Digital Micrograph software is that it provides a flexible scripting language and the ability to add user-created code to the central program. This was used extensively for the image analysis that automatically measures the aberration function.

2.4.2. Multipoles

The correctors used in this work are entirely comprised of magnetic multipole elements. These have some advantages over electrostatic elements: They are totally external to the vacuum system, do not require cumbersome insulators inside the drift tube that can charge up, and are more suited to high energy beams. A description of constructing electrostatic multipole elements relevant to aberration correction is given by Archard (1958). Round lenses are normally preferred to multipole elements for the main focusing elements in electron microscopes. This is because of the difficulties in accurately machining and aligning the pole-pieces in multipole elements, the inhomogeneous nature of the materials, and the effects of instabilities. Hence a pure multipole system will normally not perform better than an equivalent round lens system (Koops, 1978). Finally, having a short focal length round lens as the last optical stage provides a useful demagnification of all aberrations and instabilities in all directions, whereas a quadrupole would act very differently in different directions.

The question of which material to use for the pole-pieces has been addressed by Krivanek et al (1998). Permendur is not the ideal choice in the relatively low field regime in which the corrector works (around 0.1T). Instead a low remanence Ni-Fe alloy with magnetic characteristics similar to permalloy was used to obtain a sufficiently linear and repeatable performance, which is essential for tuning the corrector.

To describe the fields inside the multipole elements for numerical calculations, the scalar magnetic potential, Φ , is used, which must obey Laplace's equation (Jackson, 1998):

$$\nabla^2 \Phi = 0 \quad (2.15)$$

In cylindrical polar coordinates (r, θ) , the 2-dimensional solutions to this have the general form:

$$\Phi(r, \theta) = r^n \{ p_n \cos(n\theta) + q_n \sin(n\theta) \} \quad (2.16)$$

Where n is the degree of rotational symmetry, and p_n and q_n represent orthogonal components, as A and B did for aberrations. The 3-dimensional solutions are obtained by the superposition of the 2-dimensional solutions with appropriate weighting functions to describe the axial variation. This is derived fully by Hawkes and Kasper (1994), and using a simplified version of their notation is given up to fifth order by:

$$\begin{aligned}
\Phi = & \phi - \frac{r^2}{4} \phi'' + \frac{r^4}{4} \phi'''' + \dots \\
& - r \left[p_1 \cos(\theta) + q_1 \sin(\theta) \right] + \frac{r^3}{8} \left[p_1'' \cos(\theta) + q_1'' \sin(\theta) \right] + \dots \\
& + \frac{r^2}{2} \left[p_2 \cos(2\theta) + q_2 \sin(2\theta) \right] - \frac{r^4}{24} \left[p_2'' \cos(2\theta) + q_2'' \sin(2\theta) \right] + \dots \\
& - \frac{r^3}{6} \left[p_3 \cos(3\theta) + q_3 \sin(3\theta) \right] + \dots + \frac{r^4}{24} \left[p_4 \cos(4\theta) + q_4 \sin(4\theta) \right] + \dots
\end{aligned} \tag{2.17}$$

Where primes denote derivatives with respect to z , and the strengths of the two orientations of each element are $p_n(z)$ and $q_n(z)$, while the derivatives of ϕ are the round lens terms.

It can be seen that the off-axial terms vary with the axial derivatives of the field strength. It is therefore possible to obtain the off-axial components of the field distribution for idealised elements, if the axial variation is known. This is seen again in the calculation of spherical aberration, where it is only necessary to know the axial field distribution. This property also gives the higher order field components in the numerical simulations (Chapter 4), where a Gaussian distribution is assumed for the primary axial variations and the other components are calculated from the derivatives.

By considering the resulting forces from the field distributions in the multipole elements and the form of the aberration function (equation (2.4)), it can be seen that dipoles act primarily on the beam shift, quadrupoles on astigmatism, and so on. Additionally the axial variation of the distributions, and the extension of the fields beyond the physical extent of the element, will contribute some higher order terms to the aberration function. Similarly the extended lengths of the fields mean that non-linear higher order terms will also be present, for example the spherical aberration of extended hexapoles. These effects are discussed further by Hawkes and Kasper (1994) and in the documentation for the software COSY and TRANSPORT.

The multipoles are centered on the optic axis and classified by their rotational symmetry in a similar manner to the aberrations. Multipoles should always be named by the symmetry of the field they produce and not by the number of pole-pieces that actually contribute to the field. Thus a quadrupole field generated by eight pole-pieces, so that it can be rotated, should never be called an octupole. The magnetic multipoles used in the HB5 corrector are all made up on a 12-pole system. These are then excited in the correct ratios to produce lower order multipoles of suitable alignment. So for example a quadrupole, which has $n=2$, and would ideally be made up of two pairs of carefully shaped north and south poles, is created by exciting the twelve poles in ratios varying as $\cos(2\theta)$. Similarly the strong octupole is generated by exciting the poles in the ratio $\sin(4\theta)$. The change in orientation is necessary for magnetic multipoles, in order to obtain the line cross-over in the right direction for C_s -correction. These excitations are illustrated in Fig. 2.15. The octupole cannot be made up perfectly on the 12-pole system, because of the 8-fold symmetry, but the resulting errors correspond to aberrations of seventh order and can be neglected.

Experimentally it was observed that the higher order effects were quite small for correctly aligned multipoles, because these components due to non-ideal field distributions were much smaller than the lower order effects. In future it would be desirable to model the field distribution inside the multipoles more accurately, but as the major problems were caused by misalignment, this was not a priority. In the second corrector, the chromatic aberration was minimized by reducing the beam diameter in the corrector. This requires that the octupoles be as strong as possible, with consequentially small bores, and so care must be taken to avoid saturation.

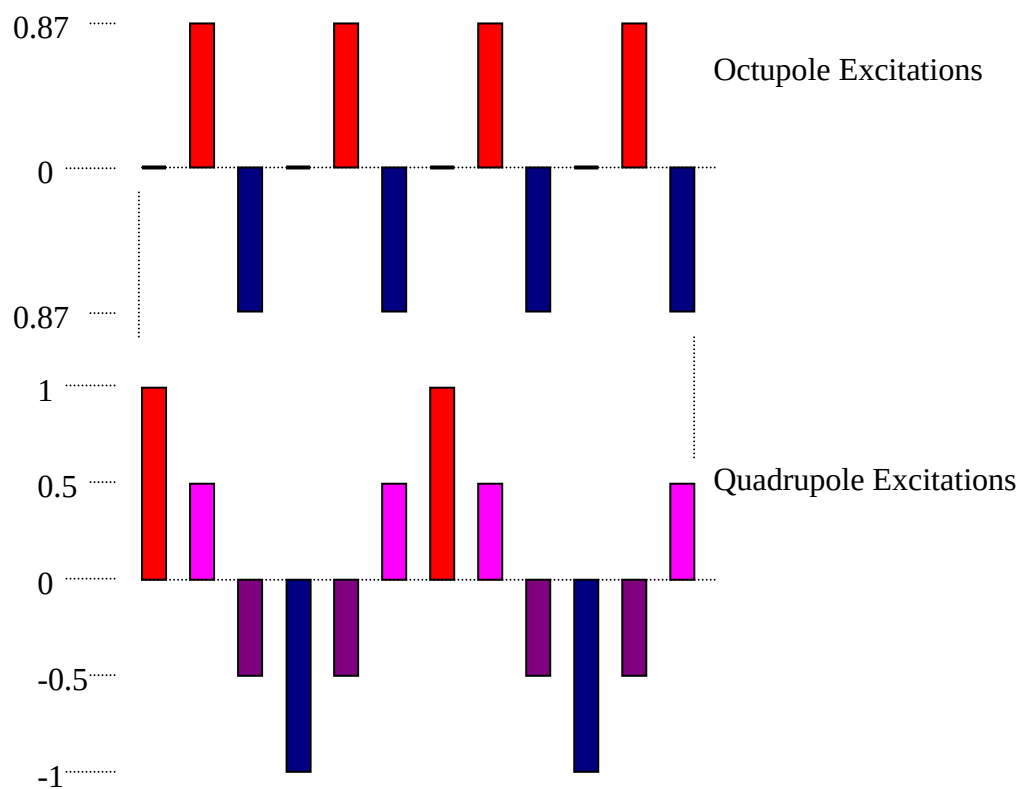


Figure 2.15. Illustration of the relative excitations necessary to produce quadrupole and octupole elements. The octupole does not fit perfectly onto the twelve-pole system, but the resultant errors will be higher order.

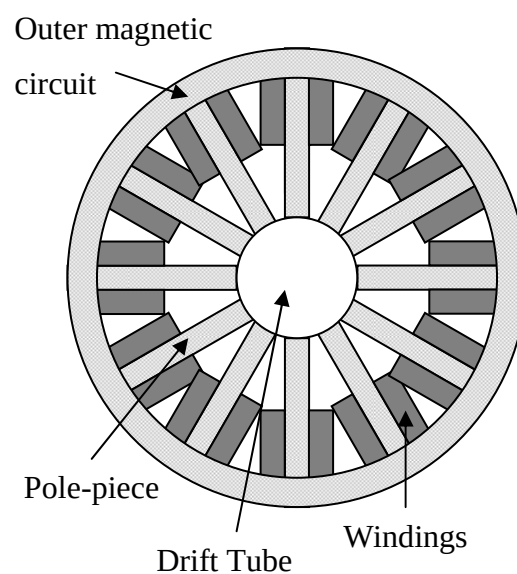


Figure 2.15. Schematics of a multipole element. In practice, the pole-pieces are carefully shaped to allow very accurate mechanical alignment.

2.5. Round lenses

In all modern high-performance transmission electron microscopes, the main optical elements are round magnetic lenses, which have lower aberrations than equivalent electrostatic lenses (for example, Munro, 1997). They also have the advantages of being external to the vacuum system and not suffering from dielectric breakdown. Fig. 2.16 is a schematic diagram of a section through a symmetric magnetic lens. Current flows through the copper wires, and the resulting magnetic field is concentrated by the magnetic core (normally soft iron). The most important aspect of a round lens is the design of the pole-pieces. These will normally be made of a high permeability, high saturation field material such as permendur and require extremely carefully production. Even slight imperfections will have profound consequences for the lens performance. In the VG STEM the pole-pieces are asymmetric with a larger upper section to allow a top entry specimen holder. In symmetric pole-pieces the upper and lower sections are identical, and the specimen is usually inserted from the side. A comprehensive review of the important issues in magnetic lens design is provided by Riecke (1982).

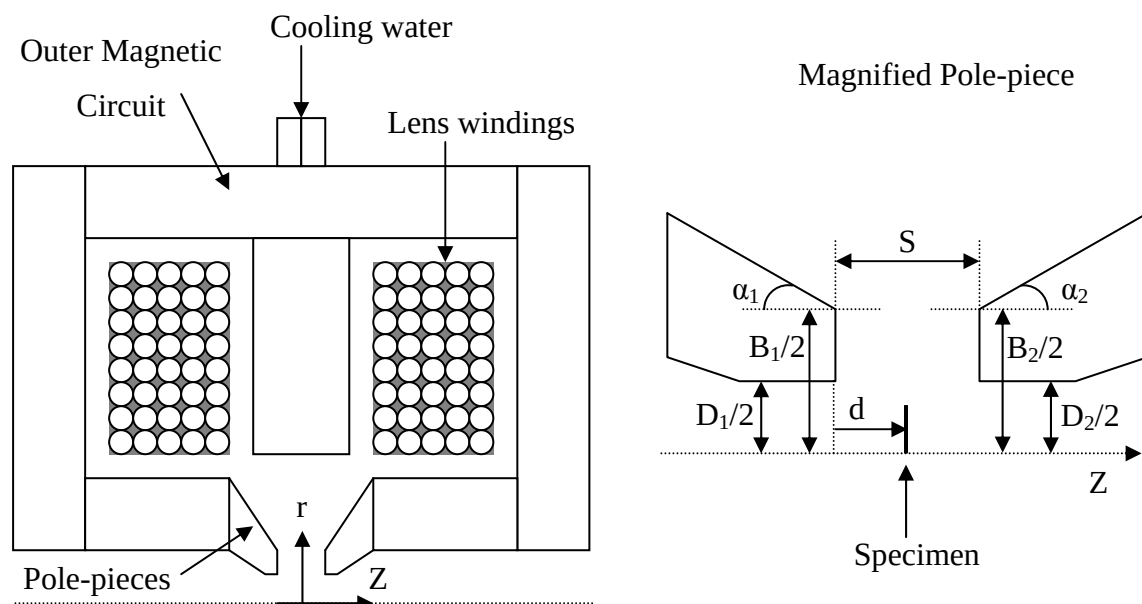


Figure 2.16. Schematic of a cross-section through one half of a round magnetic lens, defining the coordinates used in this section. A magnified view of the pole-piece section is shown, along with the parameters that can be adjusted to optimize the design. In a symmetric pole-piece the upper and lower sections will be identical.

Detailed descriptions of the action of round magnetic lenses are given by Spence (1977) and Hawkes and Kasper (1994). An important result was the proof by Scherzer (1936) that in such a system spherical and chromatic aberrations are unavoidable. The general principle is that the magnetic field is directed principally in the Z-direction with a radial component due to the axial variation. An electron, which enters the field almost parallel to the axis, begins to spiral because of the radial component, resulting in a force towards the axis. This calculation is in principle quite complex, so the **paraxial approximation** is introduced, where the magnetic field is assumed to always be equal to the axial value, $B_z(z)$. The resulting trajectory is a second order differential equation and so has two independent solutions. The values of the spherical and chromatic aberrations are then obtained by integrating along these two solutions. The off-axial field variations are obtained from the axial derivatives, as in the previous section. Using the formulae for spherical and chromatic aberrations, listed by Hawkes and Kasper (1994, Chapter 24), it is thus possible to calculate the properties of round lenses from the axial field distribution. The rotation caused by the magnetic lenses is a little more important in aberration correctors than otherwise in STEM because it changes the apparent orientation of the detectors relative to the non-round elements.

The round lenses used in this work were modeled using the CIELAS2 simulation package. This uses a version of the first order finite element method to calculate the magnetic field (for example, Munro 1997). The principle is that the volume of interest is broken up into a series of small elements in which the magnetic potential varies linearly and the total stored magnetic energy is then minimized. Hence most of the difficulties with this method arise from covering the region of interest with a continuous grid without large changes of scale. The number of points used will increase the calculation time, but enough should be used to describe the field accurately. The issues that affect the mesh layout and the precision of the resulting calculation are considered by Podbrdsky and Krivanek (1988). A typical mesh used is shown in Fig. 2.19, along with the resulting field and flux contours. It is also important to match the saturation properties of the pole-piece material (normally permendur) as closely as possible.

Fig. 2.17 shows the properties of the present HB5 pole-piece calculated using the above methods. The spherical and chromatic aberrations are calculated from the axial field variation determined using the CIELAS2 program. The idealized focal length shown is defined as the distance from the optimal aperture plane to the optimal sample plane. The post-sample compression factor is the ratio by which the angles of rays leaving the sample are compressed. The first of these quantities is important because it shows where the aperture should be positioned. The post-sample compression will determine the range of angles that the detectors subtend, and relates to the magnification in the Ronchigram. These are plotted as a function of cross-over height above the lower pole-piece face, because it was found that the optimum sample position is not always in the center of the gap. Fig. 2.17 agrees well with the measured performance of the high-resolution pole-piece with $C_s=3.5$ mm at a focal length of about 4 mm. This also shows why a similar pole-piece achieves far better performance in the HB501. By lowering the cross-over height, the aberrations are significantly lowered. Also the post-sample compression, and hence field of view in the Ronchigram, are increased. Although it was possible to lower the sample in the present HB5, it was not possible to increase the excitation because of limitations in the objective power supply. New lens windings were wound, but following the construction of the second C_s -corrector, were never installed.

In an uncorrected system the main requirement for the pole-piece design is that the spherical aberration should be minimized. In the C_s -corrected case it is relatively simple to correct large amounts of C_s , so alternative lens designs are then possible. This allows large focal length lenses with room for high-tilt cartridges or in-situ viewing of experiments. Alternatively for the highest resolution images, it will be necessary to minimize the uncorrected aberrations. In a C_s -corrected system, the resolution is principally limited by the chromatic aberration, higher order aberrations and instabilities. The higher-order effects can be minimized by suitable choice of aperture and the instabilities can be reduced by careful design, so the chromatic effects will limit the resolution. Therefore a lens with reduced chromatic aberration and a different pole-piece geometry to C_s -minimized designs is called for. The adjustable parameters in pole-piece design are illustrated in Fig. 2.16. Exploring such geometries numerically was the purpose of two previous papers (Lupini and Krivanek, 1998 and Edgcombe, Lupini and Taylor, 1999).

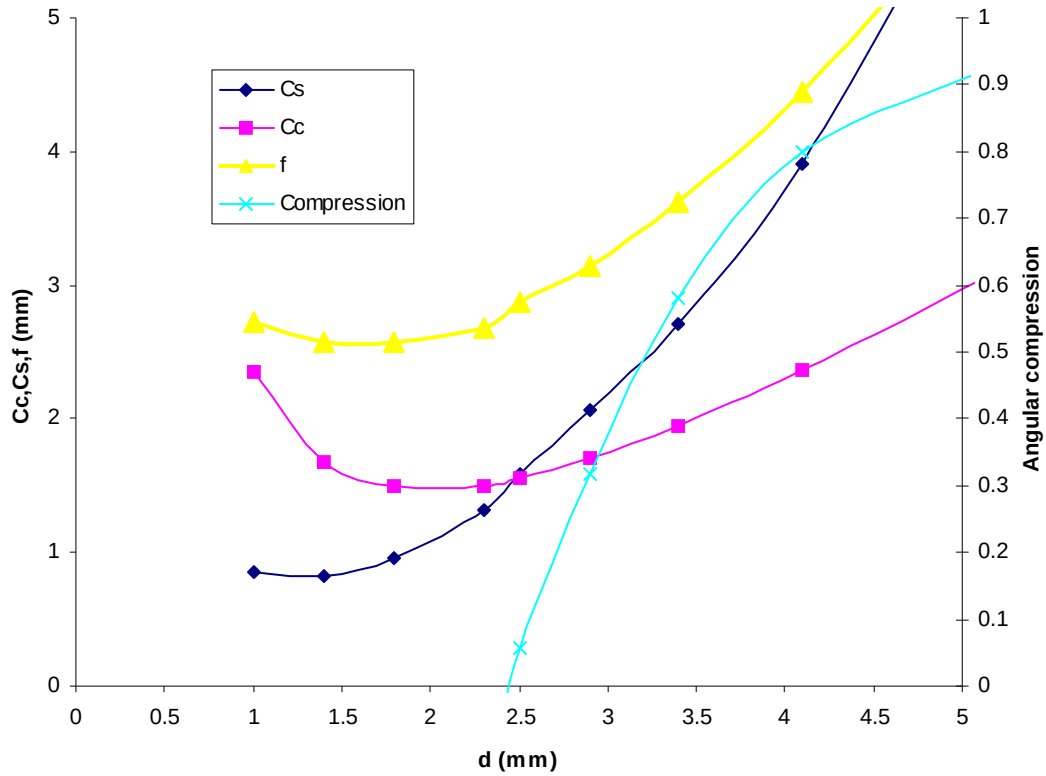


Figure 2.17. Properties of the high-resolution pole-piece for the HB5. Spherical and chromatic aberrations, the (idealized) focal length and the post-sample compression factor are shown as functions of the cross-over height above the lower pole-piece.

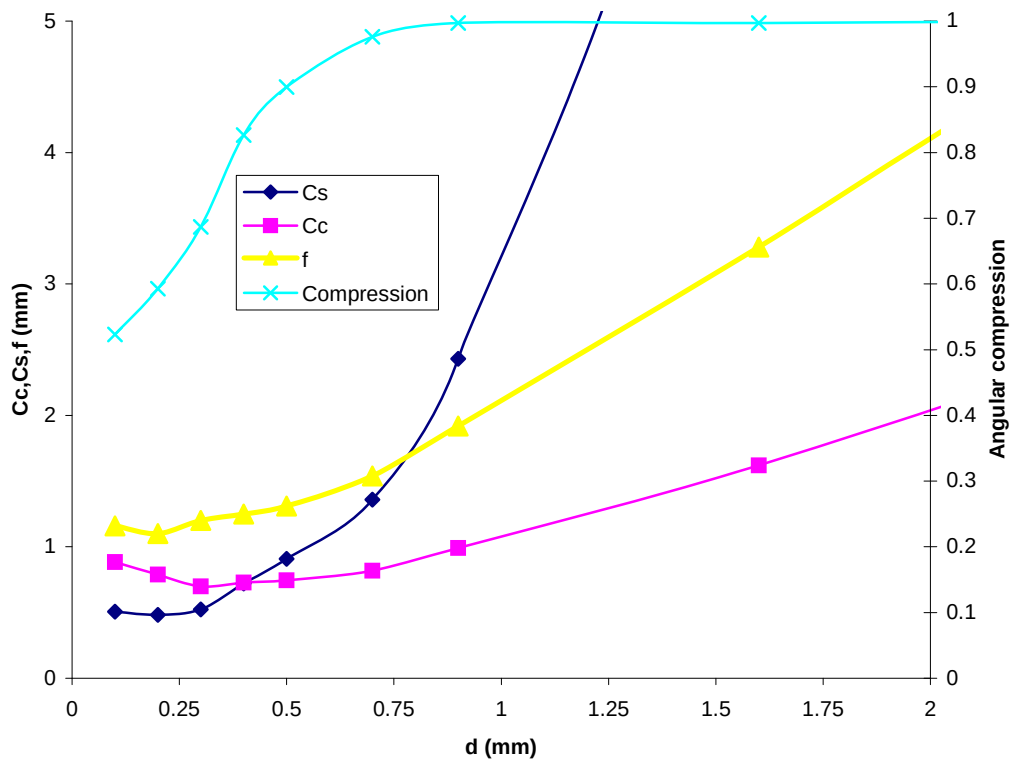


Figure 2.18. Properties of a symmetric pole-piece optimized for lower chromatic aberration. Spherical and chromatic aberrations, the (idealized) focal length and the post-sample compression factor are shown as functions of the cross-over height above the lower pole-piece.

Scaling rules (for example Riecke, 1982) show that it is possible to improve the aberrations of a round lens simply by reducing the pole-piece dimensions. However, constraints on the design of the pole-piece arise because of the need to insert and tilt a specimen, as well as the magnetic saturation. Calculations showed that by changing to a side entry holder and higher excitation windings it would be possible to improve the performance of the objective lens.

The performance of a lens optimized for reduced C_c is shown in Fig. 2.18. This was a symmetric pole-piece with a very narrow bore; the dimensions are given in Table 2.2. Using reasonable constraints on the pole-piece dimensions the optimal C_c value was around 0.7 mm. The C_s of such a lens is also quite low, meaning that this lens would perform well in an uncorrected instrument. The excitation required is rather higher than the existing design and the pole-tips are saturated. This class of lens with a narrow bore relative to the gap size could be experimentally useful, because a suitably designed specimen holder could still allow large tilt angles.

Numerically optimizing lens design was found to be time consuming, with rather too many parameters to manually search. So an auto-optimization routine was developed. This used the Complex method to optimize a figure of merit (in this case the chromatic aberration) as described by Edgcombe, Lupini and Taylor (1999). The output from this calculation is shown in Fig. 2.20, with C_c decreasing as the optimization proceeds. C_s was no longer required to be optimized and showed quite a large variation, so the optimal design would be one with low values of all of the aberrations. The interesting result is that the narrow bore pole-piece designs that this calculation tended to favor begin to display a double peaked field distribution (similar to that found by Tsuno and Jefferson, 1998a,b). This is shown in Fig. 2.19, where the mesh used, the axial field and flux contours are shown. The flux contours indicate that this unusual field distribution is due to the tapering of the pole-piece faces allowing the flux to escape.

Lens	D_1	D_2	B_1	B_2	S	α_1	α_2
HB5 lens (asymmetric)	3	12	9.5	17	4	45	45
C_c optimised (symmetric)	0.6	$=D_1$	0.8	$=B_1$	1	75	75

Table 2.2. Dimensions for the pole-pieces in figures 2.17 and 2.18.

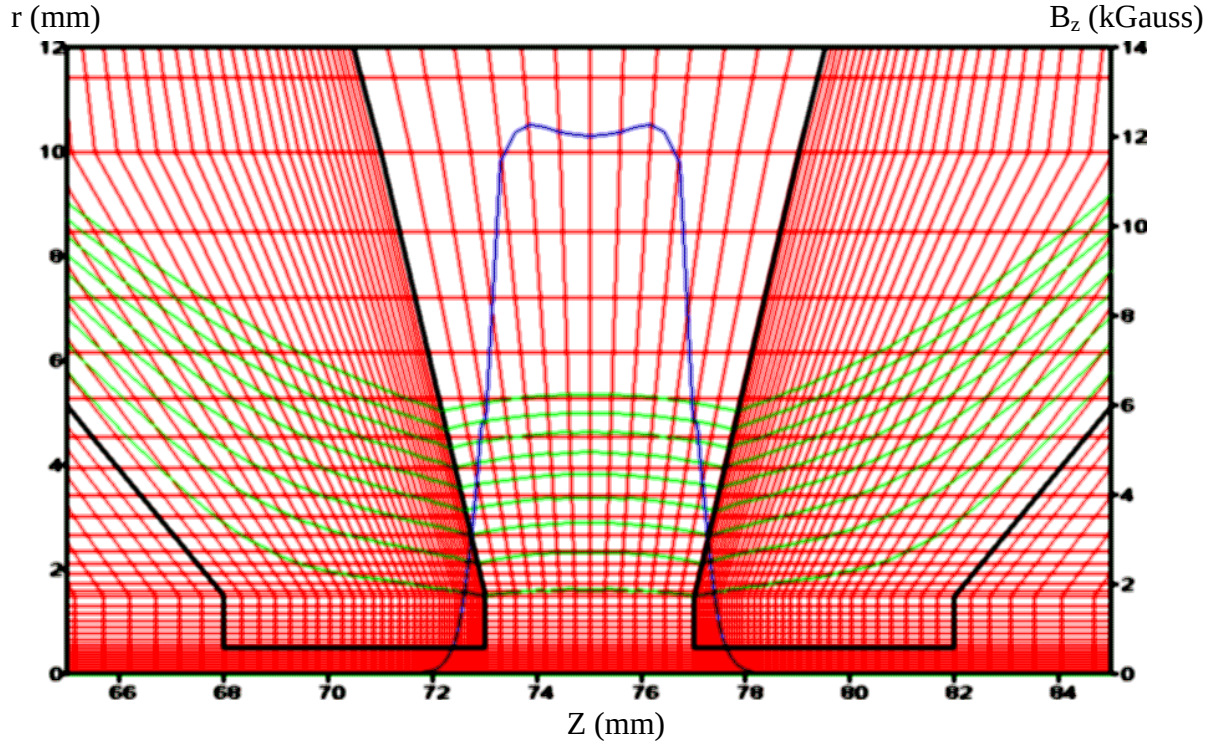


Figure 2.19. Mesh used in the automatic analysis of the round lenses in the finite element analysis. Here a close-up of the sensitive pole-piece region is shown. The finite elements are shown in red, the pole-piece outlines in black, the axial field, B_z is shown in blue and contours of flux density are plotted at intervals of 10^{-5} Wb in green. In this case, the narrow bore and tapered pole-pieces have resulted in a double-peaked field distribution.

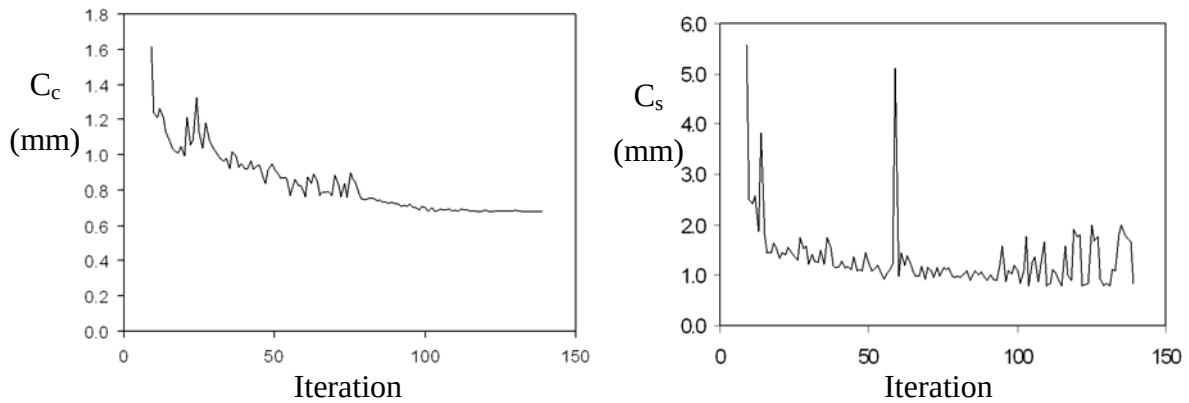


Figure 2.20. Results from auto-optimization of chromatic aberration. As the optimization proceeds the parameters shown in Fig. 2.16 are automatically adjusted to seek the minimum chromatic aberration subject to constraints on the pole-piece dimensions.

2.6. Conclusions

In this chapter the notation necessary to consider the action of an aberration corrector, and the instrumentation on which such a device should be installed, have been introduced. Because of the integral nature of the C_s -corrector to the whole instrument it is essential to consider the complete system that is to be corrected. The principles of STEM operation have been considered and the VG HB5 has been discussed. The general principles behind the hardware and software that comprise the HB5 aberration corrector and how such a device integrates into a complete system have been reviewed. Finally the consequences for round lens design in an era where aberration correctors have become successfully employed have been considered. Such lenses must have low chromatic aberration, and C_s is no longer the major factor. This seems to favor lenses with a very narrow bore relative to the gap. These also have reasonable spherical aberration and seem to be practically useful, and so could have profound consequences for future round lens design.

One conclusion from this review of a C_s -corrected system is that the complexity is significantly increased relative to the uncorrected instrument. The description of the aberration function presented here has shown that up to third order there are twelve aberration coefficients that have to be considered. The aberration corrected HB5 has an additional 84 power supplies over the uncorrected instrument. Thus the computer control developed during this work forms an essential part of the system. The control was divided up into two main parts, the automatic aberration analysis, and the software controlling the corrector electronics. In addition to setting the context, the significant contributions made during the work described in this chapter were: Clarifying the definition of the aberration function, improving the controlling software, and modeling the optical elements.

The essential components of the present system are: A high performance (if old) STEM instrument, a flexible C_s -corrector with electronics of an advanced design, and a sophisticated aberration diagnosis and computer control system. It is this combination of distinct elements that has finally allowed the successful correction of spherical aberration in transmission electron microscopy.

Chapter 3

Wave-optical Imaging

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3.1. Introduction

Due to the wave-like nature of the electron a wave-optical treatment of the imaging in a TEM is essential. In this chapter a summary of the different imaging modes will be presented. This is not intended to be a full description, but to provide sufficient detail to examine those areas of particular relevance to the correction of aberrations. This will then be extended to provide convenient models for the methods used to analyse aberrations. Finally a description of the electron Ronchigram will be presented, which extends the existing models by including aberrations of all orders and symmetries.

3.2. Electron-Specimen Interaction

This report will concentrate mostly on the electron optical stages rather than the interaction between the specimen and the beam. Detailed descriptions of how the wave-function leaving the specimen depends on the sample and the incident wave are given by many authors (for example Spence, 1977 and Cowley, 1975). Reviews are also given by Nellist (1995) and James (1997).

Here it will be assumed that the interaction between the specimen and the incident wave-function can be modelled by a two-dimensional **transmission function**. Thus the exit surface wave-function will be given by the product of the incident wave-function and this transmission function, which can modify both the magnitude and the phase of the wave-function. The physical basis for this approximation is that we have assumed single, perfectly elastic scattering by the specimen. The Ewald sphere (radius= $1/\lambda$) is used in reciprocal space to represent the conservation of energy. In this case the curvature of the Ewald sphere will be sufficiently small, compared to the scale of the reciprocal lattice, that it can be approximated by a plane. (This is because the reciprocal lattice points will be stretched in the z -direction due to the extremely thin TEM specimen, and the scattering angles will be small for high energy electrons). This results in the projection approximation, where it is then possible to regard the crystal potential as being projected along the beam direction onto a plane as shown in Fig. 3.1.

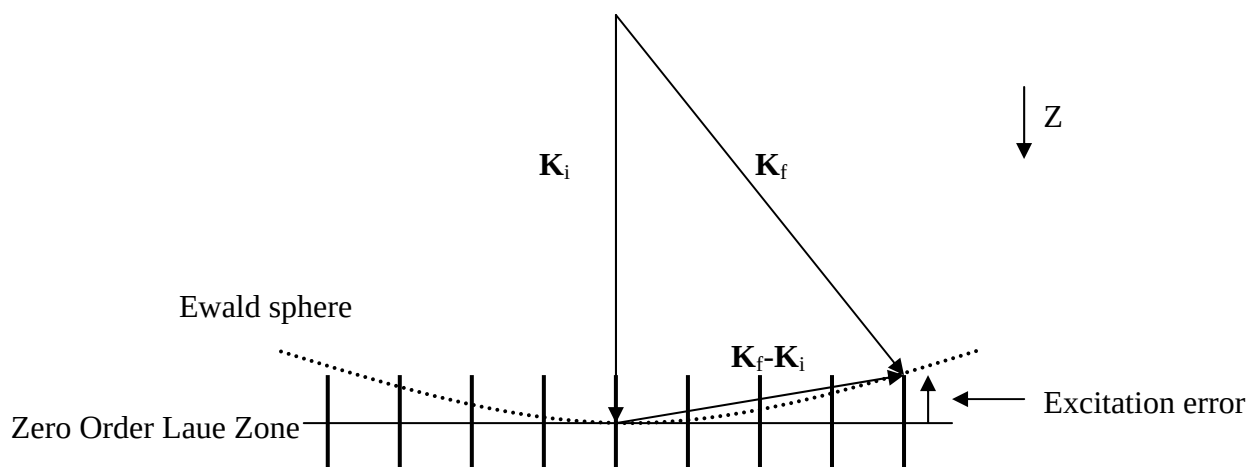


Figure 3.1. Diagram showing the Ewald sphere construction. The truncation of the specimen causes the reciprocal lattice points to be extended in the z -direction.

Therefore, this approach has neglected the dependence of the transmission function on incident angle, $\mathbf{K}_i$. The scattering to a particular final wave-vector, $\mathbf{K}_f$, is then a function only of $\mathbf{K}_f - \mathbf{K}_i$. (Multiple scattering and scattering from higher order Laue zones are also neglected.)

Under the projection approximation, the potential projected into a plane, $V_p(\mathbf{R})$ can be written as the integral of the potential, $V(\mathbf{R}, z)$ through the specimen:

$$V_p(\mathbf{R}) = \int V(\mathbf{R}, z) dz \quad (3.1)$$

An electron passing through this plane will experience a force due to the projected potential. For axial plane wave illumination, if absorption can be neglected, this will result in a pure phase change across the wave-front, depending on the projected potential at each point. This is known as the **phase object approximation** (POA). The specimen transmission function, $\psi(\mathbf{R})$, can then be represented as:

$$\psi(\mathbf{R}) = e^{-i\sigma V_p(\mathbf{R})} \quad (3.2)$$

Where σ is the interaction constant defined as: $\sigma = 2\pi me\lambda / h^2$ (as given by Spence, 1977), h is the Planck constant, and m and e refer to the mass and charge of the electron respectively. For weakly scattering objects the exponential is expanded as a series, neglecting all but the first terms, leaving:

$$\psi(\mathbf{R}) \approx 1 - i\sigma V_p(\mathbf{R}) \quad (3.3)$$

Fourier transforming this gives in reciprocal space:

$$\Psi(\mathbf{K}) = \delta(\mathbf{K}) - i\sigma \tilde{V}_p(\mathbf{K}) \quad (3.4)$$

Where $\tilde{V}_p(\mathbf{K})$ is the Fourier transform of the projected specimen potential:

$$\tilde{V}_p(\mathbf{K}) = \int V_p(\mathbf{R}) e^{2\pi i \mathbf{R} \cdot \mathbf{K}} d\mathbf{R} \quad (3.5)$$

This is known as the **weak phase object approximation** (WPOA). This requires that the axial, un-scattered beam is much stronger than the diffracted terms. If the effects of small amplitude variations are to be included, it is possible to make the projected potential complex (corresponding to weak inelastic scattering, see section 6.1 of Spence, 1977). However, we will neglect these effects and assume that this projected potential is always real.

The weak phase object approximation gives an equivalent exit wave-function to the first Born approximation for single scattering, although without the terms representing propagation through the crystal, because of the projection approximation (as discussed by Nellist, 1995). Higher order terms in this expansion will represent multiple scattering, although again without the propagation terms (Cowley, 1975).

The preceding approximations will only be valid in the limit of very thin samples and high energies. Scattered beams will propagate within the sample and undergo multiple scattering, and so real images will be strongly dependent upon specimen orientation and thickness. This is known as ***dynamical diffraction***.

One method of including the dynamical diffraction effects would be to use Bloch waves, as described in many texts, (for example Hirsh et al, 1977 or Humphreys and Bithell, 1992). In this approach the solution to the Schrödinger equation is written as a series of Bloch waves and their excitations matched to the incident beam. These waves then interfere as they propagate through the crystal and so the dynamical effects are reproduced.

Another equivalent method to include dynamical effects is the multi-slice calculation, developed by Cowley and Moodie (1957). As the name suggests, this involves breaking the specimen down into many small slices normal to the beam. Each slice should be sufficiently small that the phase object and projection approximations are valid, and a Fresnel propagation is then performed to the next slice.

However for the purposes of this report, the WPOA will provide a sufficiently detailed description of the imaging process for thin specimens, and considerably simplifies some of the expressions. Some further reviews of the WPOA are given by Williams and Carter (1996) or Hawkes and Kasper (1994). Where possible the specimen function will be left as $\psi(\mathbf{R})$, to allow for extension, for example to strong phase objects (Spence, 1977, section 3.4), but dynamical effects will be neglected.

3.3. Basic descriptions of imaging modes

Before considering the correction of aberrations in STEM it is important to present a description of the commonly used imaging modes that will be sufficient to illustrate the effects that the lens aberrations have on the recorded images. The approach used here is closely related to that given by Nellist and Pennycook (1997).

We begin by assuming a monochromatically, uniformly and coherently illuminated objective aperture. This can be described by an aperture function, $A(\mathbf{K}_i)$, which gives the phase change of the wave-front across the objective aperture as a function of incident wave-vector, $\mathbf{K}_i$, as:

$$A(\mathbf{K}_i) = H(\mathbf{K}_i) e^{2\pi i \lambda^{-1} \chi(\mathbf{K}_i)} \quad (3.6)$$

Where $H(\mathbf{K}_i)$ is a top-hat function, describing the spatial extent of the objective aperture, which is assumed to be centered on axis. $\chi(\mathbf{K}_i)$ represents the aberration function, and the factor of $2\pi\lambda^{-1}$ is needed because of the way that the aberration function has been defined here. The probe amplitude is then given by the reverse Fourier transform of (3.6):

$$P(\mathbf{R} - \mathbf{R}_0) = \int A(\mathbf{K}_i) e^{-2\pi i \mathbf{K}_i \cdot (\mathbf{R} - \mathbf{R}_0)} d\mathbf{K}_i \quad (3.7)$$

Where an extra phase term ($2\pi i \mathbf{K}_i \cdot \mathbf{R}_0$) has been included to allow for the probe, at position $\mathbf{R}_0$, to be scanned across the sample. It will be assumed that the transmitted wave function is given by the product of this incident wave and the specimen function, $\psi(\mathbf{R})$. The amplitude at the detector plane, $M(\mathbf{K}_f, \mathbf{R}_0)$, will be given by the Fourier transform of this product:

$$M(\mathbf{K}_f, \mathbf{R}_0) = \int P(\mathbf{R} - \mathbf{R}_0) \psi(\mathbf{R}) e^{2\pi i \mathbf{K}_f \cdot \mathbf{R}} d\mathbf{R} \quad (3.8)$$

This can be re-written in reciprocal space as the convolution of the aperture function and the Fourier transform of the specimen function:

$$M(\mathbf{K}_f, \mathbf{R}_0) = \int A(\mathbf{K}_i) \Psi(\mathbf{K}_f - \mathbf{K}_i) e^{2\pi i \mathbf{K}_i \cdot \mathbf{R}_0} d\mathbf{K}_i \quad (3.9)$$

This equation is equivalent to the form given by Rodenburg and Bates (1992). The observable is the intensity, which is given by $|M(\mathbf{K}_f, \mathbf{R}_0)|^2$ as:

$$I(\mathbf{K}_f, \mathbf{R}_0) = \left| \int A(\mathbf{K}_f - \mathbf{K}_i) \Psi(\mathbf{K}_i) e^{2\pi i(\mathbf{K}_f - \mathbf{K}_i) \cdot \mathbf{R}_0} d\mathbf{K}_i \right|^2 \quad (3.10)$$

Where the order of the convolution has been changed to a more convenient form. This equation can be written as:

$$I(\mathbf{K}_f, \mathbf{R}_0) = \iint A(\mathbf{K}_i) \Psi(\mathbf{K}_f - \mathbf{K}_i) A^*(\mathbf{K}_i') \Psi^*(\mathbf{K}_f - \mathbf{K}_i') e^{2\pi i(\mathbf{K}_i - \mathbf{K}_i') \cdot \mathbf{R}_0} d\mathbf{K}_i d\mathbf{K}_i' \quad (3.11)$$

Following Nellist and Pennycook (1997), (3.11) can be Fourier transformed, and becomes:

$$I(\mathbf{K}_f, \boldsymbol{\rho}) = \int A(\mathbf{K}_i) A^*(\mathbf{K}_i + \boldsymbol{\rho}) \Psi(\mathbf{K}_f - \mathbf{K}_i) \Psi^*(\mathbf{K}_f - \mathbf{K}_i - \boldsymbol{\rho}) d\mathbf{K}_i \quad (3.12)$$

Where $\boldsymbol{\rho}$ is the spatial frequency. The intensity recorded at each point in a conventional image can be found by integrating the appropriate form of these equations over the detector function, $D(\mathbf{K}_f)$. Thus (3.12) gives the Fourier transform of the image as:

$$I(\boldsymbol{\rho}) = \iint D(\mathbf{K}_f) A(\mathbf{K}_i) A^*(\mathbf{K}_i + \boldsymbol{\rho}) \Psi(\mathbf{K}_f - \mathbf{K}_i) \Psi^*(\mathbf{K}_f - \mathbf{K}_i - \boldsymbol{\rho}) d\mathbf{K}_i d\mathbf{K}_f \quad (3.13)$$

Where $D(\mathbf{K}_f)$ is the appropriate detector function for the imaging mode being considered.

These equations will form the basis for much of the further consideration of imaging in the STEM.

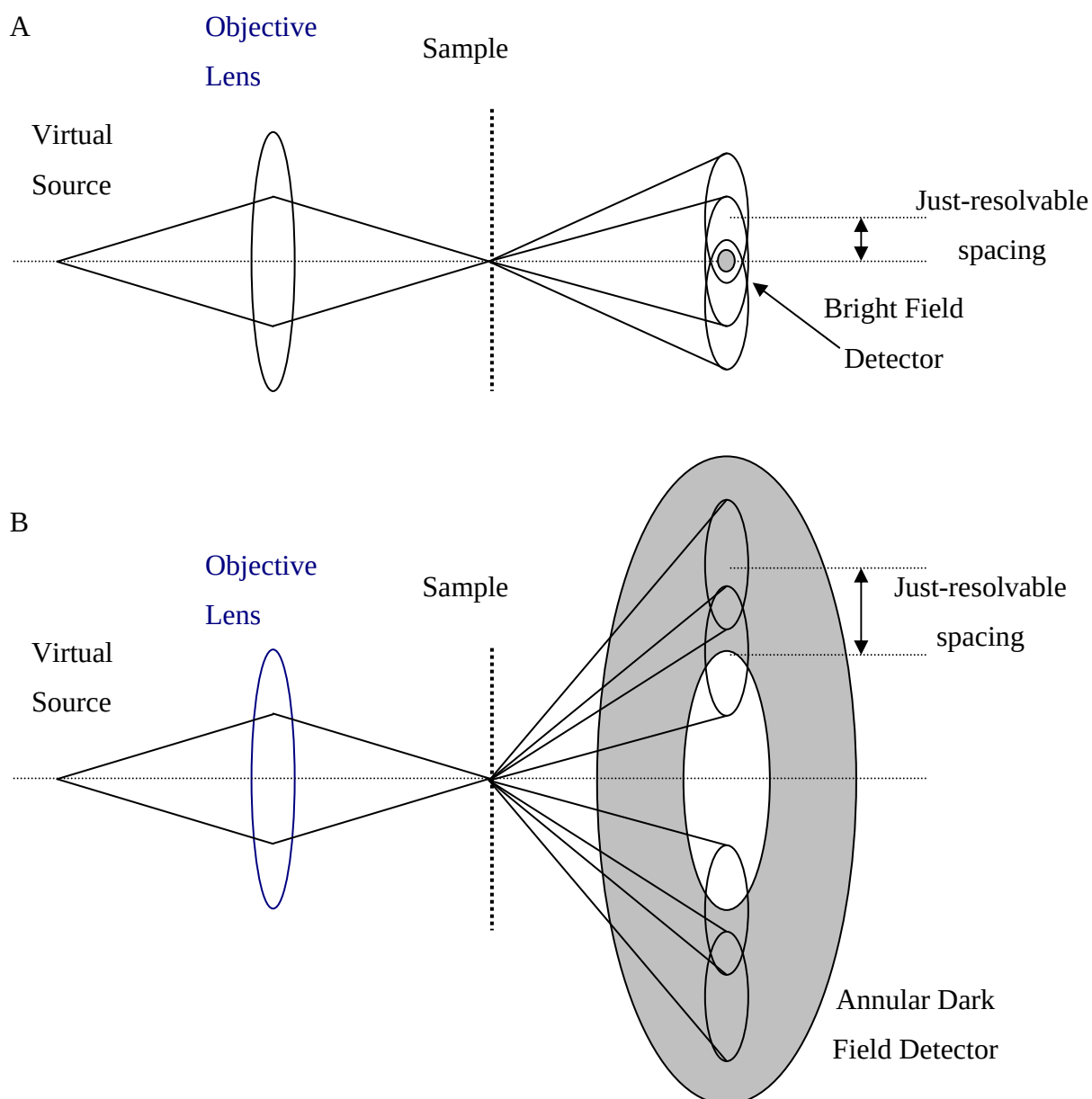


Figure 3.2. Schematic diagram of STEM imaging modes.

(A) Axial Bright Field. Interference between the unscattered and diffracted beams is detected axially as a function of probe position.

(B) Dark Field. Interference between 2 sets of diffracted orders is integrated over the whole detector as a function of probe position.

3.3.1. Bright Field Imaging

In conventional STEM imaging modes, the probe is scanned across the sample and the detector subtends a finite range of angles. We will consider the two extreme limits, of the bright-field (BF) detector being small (and on the axis in the first instance) and the annular dark-field detector covering only very high angles. Therefore, if the bright-field detector is a delta function at $\mathbf{K}_f = 0$ (Nellist and Pennycook, 1997), from (3.10) the intensity in the bright-field image will be given by:

$$I_{BF}(\mathbf{R}_0) = \left| \int A(-\mathbf{K}_i) \Psi(\mathbf{K}_i) e^{-2\pi i \mathbf{K}_i \cdot \mathbf{R}_0} d\mathbf{K}_i \right|^2 \quad (3.14)$$

By the convolution theorem, this can be written as:

$$I_{BF}(\mathbf{R}_0) = \left| \int \psi(\mathbf{R}) P(\mathbf{R} - \mathbf{R}_0) d\mathbf{R} \right|^2 \quad (3.15)$$

Which is just the conventional formulation of bright-field imaging in STEM. By reciprocity, this is exactly the same as bright-field imaging in CTEM. This image is just (the intensity of) the convolution of the specimen function with the probe. The image of the sample is effectively being filtered by the imaging system. This can be illustrated most clearly using the weak phase object approximation in reciprocal space.

If we insert (3.4) into (3.12), then neglecting terms in σ^2 and integrating over $\mathbf{K}_i$, gives the expression for the intensity in the diffractogram of a bright field image as:

$$I_{BF}(\mathbf{K}_f, \boldsymbol{\rho}) = \delta(\boldsymbol{\rho}) + i\sigma \left(A(\mathbf{K}_f) A^*(\mathbf{K}_f + \boldsymbol{\rho}) - A(\mathbf{K}_f - \boldsymbol{\rho}) A^*(\mathbf{K}_f) \right) \tilde{V}(\boldsymbol{\rho}) \quad (3.16)$$

It is assumed that bright field detector is on the axis (at $\mathbf{K}_f=0$), in which case this becomes:

$$I_{BF}(\boldsymbol{\rho}) = \delta(\boldsymbol{\rho}) + i\sigma \left(A^*(\boldsymbol{\rho}) - A(-\boldsymbol{\rho}) \right) \tilde{V}(\boldsymbol{\rho}) \quad (3.17)$$

Which can be written:

$$I_{BF}(\boldsymbol{\rho}) = \delta(\boldsymbol{\rho}) + 2\sigma H(\boldsymbol{\rho}) \sin\left(2\pi\lambda^{-1}\chi_e(\boldsymbol{\rho})\right) e^{-2\pi i\lambda^{-1}\chi_o(\boldsymbol{\rho})} \tilde{V}(\boldsymbol{\rho}) \quad (3.18)$$

Where χ_e and χ_o denote the even and odd parts of the aberration function respectively. (With the parity being determined about the axis, which depended on our choice of $\mathbf{K}_f=0$, and so will change for off-axial imaging. If the specimen was a weak

amplitude object the expression would be similar but with the sine term changed to cosine.)

The phase of a diffractogram is normally changing very rapidly and so is not easily measurable; so the phase variation due to the odd part of the aberration function can be neglected. In that case (3.18) will simplify to the more usual form given in the literature, where often χ is assumed to be purely even (for example Spence, 1977). The $H(\boldsymbol{\rho})$ term is just the frequency limit due to the aperture and $\tilde{V}(\boldsymbol{\rho})$ is the Fourier transform of the specimen potential. The most significant part of this equation is then the $\sin(2\pi\lambda^{-1}\chi_e(\boldsymbol{\rho}))$ term, which is known as the ***phase contrast transfer function*** (PCTF). This causes the intensity of the diffraction pattern to vary with the even part of the aberration function. Some examples are given in Chapter 5.

This expression for bright field phase contrast imaging shows that if there were no aberrations, there would be no contrast in the image at all! For optimum contrast the argument of the sine term would be $\pi/2$ for all spatial frequencies. However in practice the aberrations, especially spherical aberration, cause this argument to vary with spatial frequency. Thus under-focus is usually used to balance the normally unavoidable spherical aberration, in order to obtain a flat region of the PCTF for simple image interpretation (Scherzer, 1949). Examples of PCTFs for different values of defocus and spherical aberration are given by Hawkes and Kapser (1994).

At the spatial frequencies where the PCTF is zero, no information will be transferred to the image. This will happen when $\chi_e(\boldsymbol{\rho}) = n\lambda$, for integer n . This is relevant to the correction of aberrations because locating these zeros thus provides a way of experimentally determining the aberration function, as described by Krivanek (1975, 1977). However only information about the even parts of the aberration surface is contained in a single axial diffractogram.

3.3.2. Annular Dark Field Imaging

The other limiting case considered here is the situation in which it is possible to take the angular size of the detector as being far larger than the angular size of the objective aperture, so $\mathbf{K}_f \gg \mathbf{K}_i$. This corresponds to an ideal high angle annular dark field detector. In this case, it is possible to neglect the dependence of the specimen function on $\mathbf{K}_i$, and the integral in (3.13) will become separable (Nellist and Pennycook, 1997). The Fourier Transform of the recorded intensity will then be given by:

$$I(\boldsymbol{\rho}) = \int A(\mathbf{K}_i) A^*(\mathbf{K}_i + \boldsymbol{\rho}) d\mathbf{K}_i \int \Psi(\mathbf{K}_f) \Psi^*(\mathbf{K}_f - \boldsymbol{\rho}) d\mathbf{K}_f \quad (3.19)$$

Fourier transforming this leads eventually to:

$$I_{ADF}(\mathbf{R}_0) = \int |P(\mathbf{R} - \mathbf{R}_0)|^2 O(\mathbf{R}) d\mathbf{R} \quad (3.20)$$

Where $O(\mathbf{R})$ is a transmission function, which represents the interaction of the beam with the sample, and will be a series of localized peaks at the atomic column positions, as discussed by Jesson and Pennycook (1993).

Equation (3.20) shows that ADF imaging is a convolution of the modulus squared of the probe with the transmission function. This indicates that the ADF image will be free of the contrast reversal effects that complicate the interpretation of the bright field image. The scattering to high angles will resemble Rutherford scattering and show a strong dependence on atomic number, Z , so this mode is often known as Z contrast imaging. Further discussion is given by Jesson and Pennycook (1993) and Pennycook and Nellist (1998).

Fig. 3.2 illustrates some important differences between ADF and axial BF imaging. If we consider two spacings that are just resolvable, then the BF detector is interfering rays that have been through the centre of the bright-field disk with the edge of the diffracted disk. If the objective aperture has a half-angle of $\rho_{\max}$, then the maximum detectable scattering angle for an axial BF detector will correspond to a diffraction angle $\rho_{\max}$. The ADF detector interferes rays that have been through opposite parts of the objective lens. Thus the corresponding maximum detectable scattering angle will be $2\rho_{\max}$ in ADF imaging. Therefore the ADF resolution in this approximation will be twice that of the axial BF mode (Nellist and Pennycook, 1997).

The principle of reciprocity implies that the detector in STEM is equivalent to the source in CTEM, and so ADF STEM with its high-angle detector will correspond to having a large source in CTEM. The van Cittert-Zernike theorem shows that the amount of spatial coherence is inversely related to the source size (Born and Wolf, 1970). Applying this to the source and, by reciprocity, the detectors in STEM, shows that the degree of spatial coherence will depend on both the effective source and detector sizes. Hence the large ADF detector will provide incoherent imaging, while the BF detector is considered to provide coherent imaging. In this work the limitations of finite coherence will not be considered in detail, and it will generally be assumed that the coherence and brightness have been optimised by suitable adjustment of the condenser lenses. Further discussion of the effects of coherence in the electron microscope is given by Spence (1977) and Nellist and Pennycook (1998).

3.3.3. Resolution and Chromatic effects

In bright field imaging, the first zero of the PCTF is often referred to as the ***point resolution***. That is the maximum spatial frequency that is transferred to the image without contrast reversal, and so higher frequency information cannot be simply interpreted. Ways of optimising the point resolution in a C_s -corrected system are discussed by Scherzer (1970) and examined in Chapter 5.

Often the objective aperture will be chosen to prevent beams from beyond the point resolution from contributing to the image. If the aperture is larger than this, then higher spatial frequencies will be present. However, in practice the beam will not be perfectly coherent, because of mechanical and electrical instabilities (particularly the high voltage), the finite extent of the source, space charge effects, and the spread of electron energies from the tip. This ultimately results in a maximum spatial frequency beyond which no information is transferred, which is known as the ***information limit***. The effect of these coherence limitations is modelled as introducing a spread of defocus values, which must be incoherently summed to give the resulting image

(Nellist and Rodenburg, 1994). The incoherent defocus change, Δf , is obtained by assuming that these effects add in quadrature (Spence, 1977):

$$\Delta f = C_c \left[\left(\frac{\Delta E_0}{E_0} \right)^2 + \left(\frac{2\Delta I_0}{I_0} \right)^2 \right]^{\frac{1}{2}} \quad (3.21)$$

Where ΔE_0 and ΔI_0 are the variance of the energy, E_0 , and the variance of objective lens current, I_0 , respectively. These effects are included through the chromatic aberration coefficient, C_c . With non-round optical elements, as in a C_s -corrector, this could vary with direction, but it is generally assumed to be rotationally symmetric.

The effects of C_c on a particular ray are illustrated in Fig. 3.3. The defocus term in the aberration function caused by chromatic aberration will be $\Delta f \rho^2 / 2$ and the corresponding ray deviation $\Delta f \rho$, which is sometimes used to obtain the chromatic broadening of the probe. However this can be a bit misleading, particularly in ADF STEM (Nellist and Pennycook, 1998). For bright-field imaging the effect of C_c is to change the oscillations in the phase contrast transfer function. When this is summed over the range of defocus values present, the rapidly oscillating parts will cancel, reducing the information transfer at higher frequencies, giving rise to the information limit.

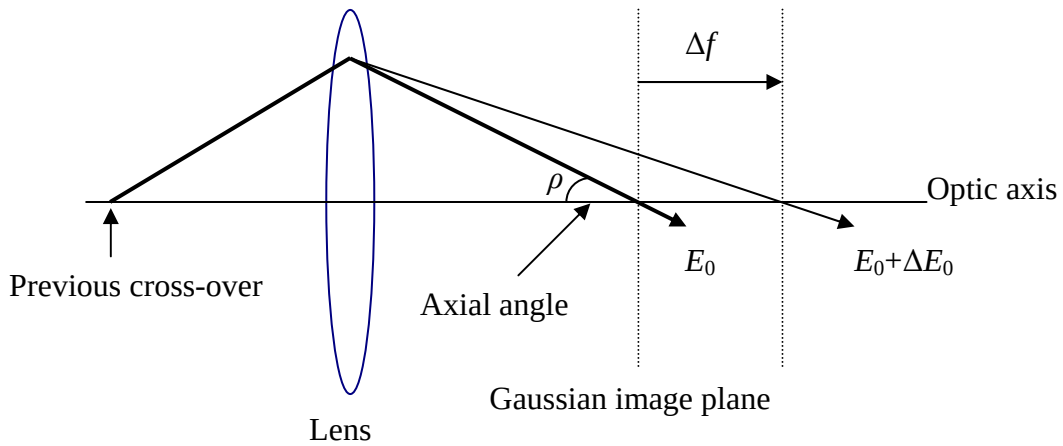


Figure 3.3. Illustration of the effects of chromatic aberration. Higher energy electrons are less strongly focused and so are brought to a focus after the Gaussian image plane.

It is assumed that the distribution of intensity over the range from equation (3.21) is approximately Gaussian. Thus the intensity at the detector plane will just be $|M(\mathbf{K}_f, \mathbf{R}_0)|^2$ multiplied by a Gaussian weighting factor and then integrated over the defocus spread (in the Z direction). This procedure is performed by Nellist and Rodenburg (1994), and assuming only the defocus varies, will result in equation (3.12) being modified to:

$$I(\mathbf{K}_f, \boldsymbol{\rho}) = \int A(\mathbf{K}_i) A^*(\mathbf{K}_i + \boldsymbol{\rho}) \Psi(\mathbf{K}_f - \mathbf{K}_i) \Psi^*(\mathbf{K}_f - \mathbf{K}_i - \boldsymbol{\rho}) e^{-\frac{1}{2}\pi^2 \lambda^2 (\Delta f)^2 (2\mathbf{K}_i \cdot \boldsymbol{\rho} - |\boldsymbol{\rho}|^2)^2} d\mathbf{K}_i \quad (3.22)$$

Where the additional exponential factor represents the damping of the transfer function due to the limited coherence.

Considering which part of the objective lens the interfering rays at a just resolvable spacing have been through also gives some indication of the effects of chromatic aberration (see Fig. 3.2). In the annular dark field case the rays will have been through opposite edges of the objective lens and so will pick up similar phase shifts from the chromatic aberration. (The extra phase shift for rays going through the objective lens at $\mathbf{K}_i$ with a particular energy, $E_0 + \Delta E_0$, will be proportional to $C_c \Delta E_0 |\mathbf{K}_i|^2 / E_0$.) Thus ADF imaging will not be much affected by chromatic effects. In the bright field case for a just resolved spacing, rays that have been through the edge of the lens will interfere with rays that have been through the centre and acquired a different phase shift. The interference fringes will therefore move position slightly with energy. As the images are incoherent sums over a range of energies, which all have slightly different phases in the bright field case, this leads to a reduction of the image contrast at high spatial frequencies.

The damping envelope in axial bright field CTEM at a particular spatial frequency, $\boldsymbol{\rho}$, is normally given (from equation (3.22), or from Spence, 1977) by:

$$E_{chr}(\boldsymbol{\rho}) = e^{-\frac{1}{2}\pi^2 \lambda^2 (\Delta f)^2 |\boldsymbol{\rho}|^4} \quad (3.23)$$

In the more general case, this defocus spread does not lead to a simple envelope in reciprocal space, but rather the more complicated exponential term in

(3.22) should be used. Nellist and Pennycook (1998) show that the damping of the incoherent transfer function for ADF imaging is approximately proportional to $1/|\mathbf{p}|$. This remarkable conclusion shows that chromatic aberration has less effect at higher spatial frequencies in ADF STEM.

So it would be expected that ADF images will have a higher resolution and be less affected by chromatic aberration than BF or by reciprocity CTEM images. Similarly the optics in a STEM are mostly before the sample, so there will be fewer chromatic aberration effects due to the energy losses in the sample compared with a CTEM (Spence, 1977).

This is particularly relevant to the field of aberration correction because both spherical and chromatic aberrations are correctable in principle. Although most aberration correction work has concentrated primarily on C_s -correctors, such systems can increase the point resolution in bright field only up to the information limit. In order to improve the attainable resolution beyond this limit in bright field, some attempt must be made to reduce $C_c \Delta E_0 / E_0$. This aberration correction project uses microscopes with field emission guns because of the lower range of emitted energies, ΔE_0 . Further improvements can be achieved in a variety of ways: by C_c -correction, monochromation, or increasing the voltage. Even in a C_s -corrected system, it appears that one of these measures will be necessary to continue improving microscope resolution, although the relative insensitivity of ADF STEM to chromatic effects will be an advantage.

3.3.4. Diffraction Patterns and Ronchigrams

The final method for obtaining an image in a STEM that will be considered is to have a CCD in the detector plane and record the intensity in the far field. Pixel positions in such an image will correspond to different scattering angles, $\mathbf{K}_f$. This is described in more detail in Chapter 6, and an extensive review is given by Cowley (1978). Thus the image is given by the intensity at the detector, $\left| M(\mathbf{K}_f, \mathbf{R}_0) \right|^2$, given by equation (3.10), as a function of angle for a particular (fixed) probe position, $\mathbf{R}_0$. As shown by this equation, the Ronchigram is the intensity of the convolution of the aperture function with the diffraction pattern of the sample. The effects of limited coherence on this type of image are considered by Cowley (1999).

If we consider the limit of a small aperture, the aperture function will become a delta function. In this case (3.10) will just be the Fourier transform of the specimen, and so the observed intensity will be the diffraction pattern:

$$I_{Diff}(\mathbf{K}_f, \mathbf{R}_0) = \left| \Psi(\mathbf{K}_f) \right|^2 \quad (3.24)$$

As the objective aperture becomes larger, (3.10) can be treated as a convolution:

$$I_{Micro}(\mathbf{K}_f, \mathbf{R}_0) = \left| \Psi(\mathbf{K}_i) e^{2\pi i \mathbf{K}_i \cdot \mathbf{R}_0} \otimes_{\mathbf{K}_i} A(\mathbf{K}_i) \right|^2 \quad (3.25)$$

So, as the aperture is expanded, the aperture function will become a disk (of uniform intensity in this approximation, although with phase depending on the lens aberrations). This disk is then convolved with the diffraction pattern of the specimen, forming a micro-diffraction pattern, which is described by Cowley (1999) or Williams and Carter (1996).

As the size of the of the aperture increases further, these disks will begin to overlap and interfere. Because the illumination is assumed to be coherent, this gives fringes in the overlap region. This approach is further considered in by Nellist and Rodenburg (1994) and used to explain STEM imaging by Spence and Cowley (1978).

A Ronchigram is normally considered to be just a micro-diffraction pattern with a large aperture, so we take the amplitude as unity, $\left| A(\mathbf{K}_f - \mathbf{K}_i) \right| = 1$. From equation (3.10), the intensity in the Ronchigram can be written as:

$$I_{Ronchi}(\mathbf{K}_f, \mathbf{R}_0) = \left| \int \Psi(\mathbf{K}_i) e^{2\pi i (\lambda^{-1} \chi(\mathbf{K}_f - \mathbf{K}_i) + (\mathbf{K}_f - \mathbf{K}_i) \cdot \mathbf{R}_0)} d\mathbf{K}_i \right|^2 \quad (3.26)$$

The Ronchigram will be a mix of real space and diffraction space phenomena and so will be difficult to quantitatively analyse (Krivanek, 1997). It is possible to expand the aberration function in terms of its derivatives as:

$$\begin{aligned} \chi(\mathbf{K}_f - \mathbf{K}_i) &= \sum_{n=0}^{\infty} \frac{1}{n!} (\lambda \mathbf{K}_f \cdot \nabla)^n \chi(-\mathbf{K}_i) \\ &= \chi(-\mathbf{K}_i) + \lambda \mathbf{K}_f \cdot \nabla \chi(-\mathbf{K}_i) + \frac{1}{2} (\lambda \mathbf{K}_f \cdot \nabla)^2 \chi(-\mathbf{K}_i) + \dots \end{aligned} \quad (3.27)$$

Where ∇ is the gradient operator, differentiating with respect to the two directions, (u, v) defined earlier, so the factors of λ are needed to scale these to wave-vectors, $\mathbf{K}$. This expansion can be re-ordered, so that the first few terms are:

$$\chi(\mathbf{K}_f - \mathbf{K}_i) = \chi(-\mathbf{K}_i) - \lambda \mathbf{K}_i \cdot \nabla \chi(\mathbf{K}_f) + \frac{1}{2} (-\lambda \mathbf{K}_i \cdot \nabla)^2 \chi(\mathbf{K}_f) + \dots \quad (3.28)$$

Where we have re-arranged terms so that it is clearly just the normal aberration function for axial bright field imaging, $\chi(-\mathbf{K}_i)$, plus some extra terms in higher orders of $\mathbf{K}_i$ multiplied by derivatives of the aberration function evaluated at $\mathbf{K}_f$. (The negative sign agrees with that in equation (3.14).) This form can be verified by explicitly writing out the aberration function and comparing terms.

In general we cannot neglect the terms in $(-\lambda \mathbf{K}_i \cdot \nabla)^2$ and higher. However if we have an **amorphous** sample in which $\Psi(\mathbf{K}_i) \rightarrow 0$ rapidly as $\mathbf{K}_i$ increases, the relative contribution of these terms to the integral in (3.26) will be less significant than the leading terms in (3.28), and so the higher derivatives can be neglected. This assumption is the essentially the same as that given by Cowley (1986), and really it means that Bragg diffracted terms are being neglected. It is not possible to say that $\mathbf{K}_i$ will necessarily be small (as the aperture will be often be large); we are relying on the sample reducing the relative contribution of these terms. As this assumes Bragg diffraction is not important, this will obviously fail for crystalline samples. It is possible that conditions allowing some of these other terms to become significant accounts for some unusual features occasionally observed in Ronchigrams, such as ghost images at different magnifications (Krivanek, private communication).

The modulus squared will eliminate any terms in the exponential that do not depend on $\mathbf{K}_i$, and so (3.26) gives:

$$I_{Ronchi}(\mathbf{K}_f, \mathbf{R}_0) = \left| \int \Psi(\mathbf{K}_i) e^{2\pi i \lambda^{-1} \chi(-\mathbf{K}_i)} e^{-2\pi i \mathbf{K}_i \cdot (\nabla \chi(\mathbf{K}_f) + \mathbf{R}_0)} d\mathbf{K}_i \right|^2 \quad (3.29)$$

This is very similar to the expression for bright field imaging in (3.14) (although, instead of scanning $\mathbf{R}_0$ we are now observing over a range in $\mathbf{K}_f$). Following a suggestion from Nellist, we can recognize the second exponential as a Fourier transform from reciprocal space to a space over the variable $\nabla \chi(\mathbf{K}_f) + \mathbf{R}_0$ (instead of just $\mathbf{R}_0$ for BF imaging). Thus the intensity in the Ronchigram can be written as:

$$I_{Ronchi}(\mathbf{K}_f, \mathbf{R}_0) = \left| \int \psi(\mathbf{R}) P(\mathbf{R} - (\nabla \chi(\mathbf{K}_f) + \mathbf{R}_0)) d\mathbf{R} \right|^2 \quad (3.30)$$

Where $\mathbf{R}_0$ is just the (constant) probe position, which can be ignored. Comparing this to the expression for bright-field imaging (3.15), it can be seen that (3.30) is just a bright-field image distorted by $\nabla \chi(\mathbf{K}_f)$. It is possible to write $\nabla \chi(\mathbf{K}_f)$ as $M(\mathbf{K}_f)$, where M is a variable magnification, remembering that $\mathbf{K}_f$ will give the position in the Ronchigram. It can therefore be concluded that a Ronchigram is, in this limit, a bright field image of the sample with non-uniform magnification depending on the gradient of the aberration function!

This description is similar to, although more general than, those given by Cowley, (1984) where the effects of finite coherence and the WPOA are also considered. The consequences of this description and the non-uniform magnification can be clarified using a geometrical optics approach, which is discussed in Chapter 6. It might be thought that the higher order terms in (3.28) mean this is not quite a bright field image, but this description is only really valid when those higher order terms do not make any significant contribution. The Ronchigrams of crystalline and amorphous materials will therefore appear significantly different even for exactly the same aberration function, and some comparison is given in Section 6.2.

Although normally a large aperture has been assumed, the size of the aperture was not that significant in this approach (provided it is not so small that the probe incident on the sample is completely dominated by the diffraction from that aperture).

A smaller aperture will cut out the higher spatial frequencies, but this doesn't change the basic features because it was assumed that the specimen function effectively cut out these same frequencies anyway. This is verified experimentally where we see that a smaller aperture just masks off the edges of an otherwise unchanged Ronchigram (see Fig. 3.4). It is the form of the specimen itself that really governs the applicability of this description of the electron Ronchigram.

The approach used here demonstrates one of the most interesting aspects of this imaging mode in a STEM. With a small objective aperture and crystalline sample, the intensity distribution in the CCD plane is the diffraction pattern of the sample. As the aperture size increases, it becomes a pattern of coherently interfering disks, yet in the limit of a large aperture and amorphous specimen it becomes a distorted real-space image of the sample.

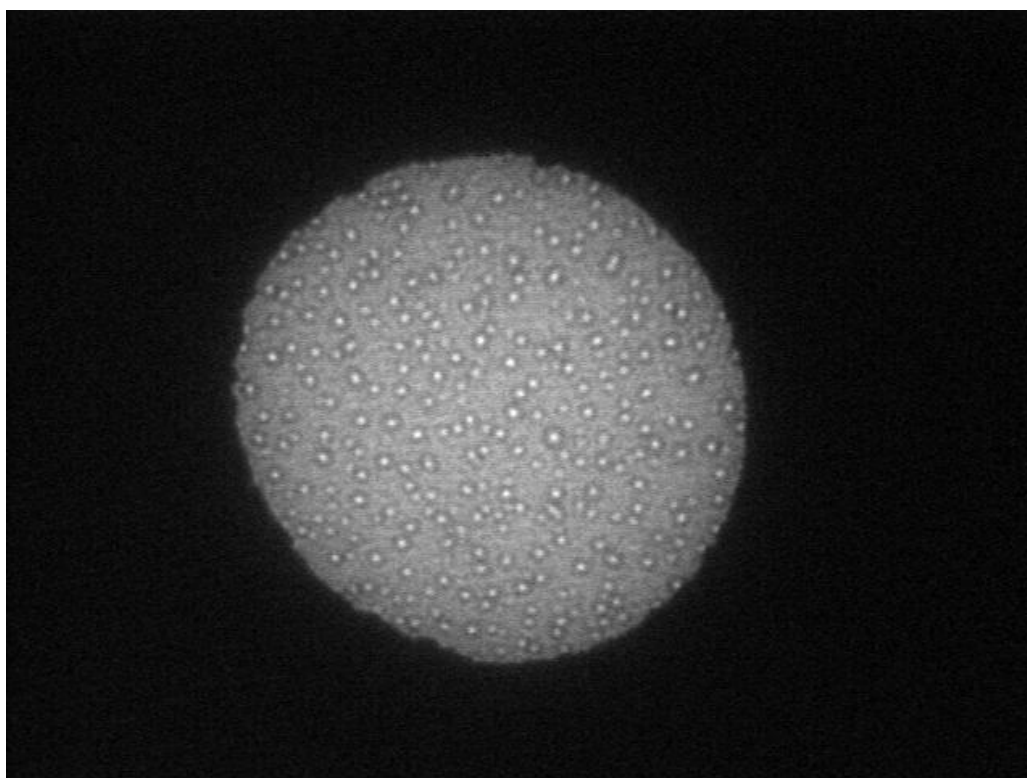


Figure 3.4. An over-focused Ronchigram with a smaller aperture, which just masks off the edges of the image. See also Chapter 6.

3.4. Measuring Aberrations

One variant of the bright field image is obtained if the detector is off-axis. In practice, the Grigson coils are used to tilt the bright-field cone to obtain an off-axis image. The detector is modeled as a delta function at a particular $\mathbf{K}_f$ value, $\mathbf{T}$. This will be familiar as the reciprocity equivalent of tilted illumination in the CTEM. This mode is sometimes used to resolve higher spatial frequencies (with a directional dependence) because it allows the detection of interference between the edges of Bragg diffracted disks. From (3.10), the intensity in an off-axis image can be written as:

$$I_{TBF}(\mathbf{T}, \mathbf{R}_0) = \left| \int A(\mathbf{T} - \mathbf{K}_i) \Psi(\mathbf{K}_i) e^{-2\pi i \mathbf{K}_i \cdot \mathbf{R}_0} d\mathbf{K}_i \right|^2 \quad (3.31)$$

If it is possible to neglect the shift of the top-hat function, $H(\mathbf{K})$, it is only necessary to consider the phase due to the aberration function. We can expand the aberration function in terms of the tilt angle. After repeating the expansion in (3.28), then (3.31) becomes:

$$I_{TBF}(\mathbf{T}, \mathbf{R}_0) = \left| \int A(-\mathbf{K}_i) e^{2\pi i \left(-\mathbf{K}_i \cdot \nabla \chi(\mathbf{T}) + \frac{1}{2} \lambda (\mathbf{K}_i \cdot \nabla)^2 \chi(\mathbf{T}) + \dots \right)} \Psi(\mathbf{K}_i) e^{-2\pi i \mathbf{K}_i \cdot \mathbf{R}_0} d\mathbf{K}_i \right|^2 \quad (3.32)$$

This should be compared to (3.14), in which case we can use the normal properties of Fourier transforms to recognize the terms in $(-\mathbf{K}_i \cdot \nabla \chi(\mathbf{T}))$ as just representing a shift of the image. So the image will be shifted relative to the axial image by an amount depending on the gradient of the aberration function.

The terms $\frac{1}{2} \lambda (\mathbf{K}_i \cdot \nabla)^2 \chi(\mathbf{T})$ will change the **apparent** values of the first order aberrations of the image (defocus and astigmatism), since those are the only terms in K_{iu}^2 , K_{iv}^2 and $K_{iu}K_{iv}$. (Where K_{iu} and K_{iv} are the u and v components of $\mathbf{K}_i$, respectively, again remembering the definition that $|\mathbf{K}| = 1/\lambda$.) The images will therefore appear to have defocus and astigmatism coefficients changed by amounts that depend on the second derivatives of the aberration function (Krivanek, 1975). (In a similar manner, the higher order terms in $\mathbf{K}_i$ will affect the apparent higher order aberrations in the same way, but this is not usually important.)

Thus it can be seen that the bright-field images taken at different tilts will be exactly the same as the bright-field image on axis, just with different aberration coefficients and shifts. Further discussion is given by Krivanek (1976) and McFarlane (1975).

Comparing (3.32) to the equation for the aberration function given earlier, gives the image shifts and the apparent values of the first order aberrations as:

$$\begin{aligned}
 Shift_X &= \left. \frac{\partial \chi(u, v)}{\partial u} \right|_{(u, v)=\mathbf{T}} \\
 Shift_Y &= \left. \frac{\partial \chi(u, v)}{\partial v} \right|_{(u, v)=\mathbf{T}} \\
 C'_{10} &= \frac{1}{2} \left(\left. \frac{\partial^2 \chi(u, v)}{\partial u^2} \right|_{(u, v)=\mathbf{T}} + \left. \frac{\partial^2 \chi(u, v)}{\partial v^2} \right|_{(u, v)=\mathbf{T}} \right) \\
 C'_{12a} &= \frac{1}{2} \left(\left. \frac{\partial^2 \chi(u, v)}{\partial u^2} \right|_{(u, v)=\mathbf{T}} - \left. \frac{\partial^2 \chi(u, v)}{\partial v^2} \right|_{(u, v)=\mathbf{T}} \right) \\
 C'_{12b} &= \left. \frac{\partial^2 \chi(u, v)}{\partial u \partial v} \right|_{(u, v)=\mathbf{T}}
 \end{aligned} \tag{3.33}$$

Where the dashed coefficients are those that would be measured at a particular tilt angle, $\mathbf{T}$.

These expressions are equivalent to those given in by many authors, (for example, Saxton, 1995) where they are derived for bright field CTEM. The form used here is the same as used by Krivanek (1994), which has the advantage that it is easily extendable to arbitrary orders without additional manipulation. These formulae contain the mathematical descriptions of both the methods commonly used to measure lens aberrations in TEM. By collecting a series of images at different tilts and processing them, the derivatives of the aberration function can be extracted.

Cross-correlating the images to find the shifts will give the first derivatives from equations (3.33) (Krivanek and Fan, 1992). This forms the basis of the tilt induced shift (TIS) method. We have seen here that the TIS method applies to any

object in which the exit wave-function can be written as a product of a specimen function and an incident wave-function. The inversion from the gradients to the aberration function is over-determined and so a least squares fit is performed as described in Appendix 2.

Fourier transforming the images and extracting the apparent defocus and astigmatism will give the second derivatives of the aberration function. This is the STEM equivalent of the diffractogram tilt tableau methods described by Zemlin et al (1978).

The diffractogram analysis method can be understood by returning to the WPOA introduced earlier. We have seen that each tilted image is just the same as the axial image, but with different aberration coefficients and so it is possible to use (3.18) to represent the off-axial diffractograms. The important difference is that the aberration coefficients will be the modified forms given in (3.33). Equation (3.18) showed that the PCTF will have zeros where the even part (about that tilt angle) of the aberration function $\chi_e(\mathbf{p}) = n\lambda$, for integer values of n . The diffractograms are divided into angular sections, each of which is cross-correlated to theoretically generated profiles. Analysing the angular variation of the fitted profiles gives the variation of defocus and astigmatism at each tilt. This process is described in more detail by Krivanek and Fan (1992) and Spence (1977, Appendix 1).

This discussion has shown why it is necessary to have more than one diffractogram to measure the odd aberrations, as they will only be present in the off-axial variation. Using several diffractograms of bright-field images taken at several different tilts, it is possible to determine the defocus and astigmatism as a function of angle. These quantities will vary at different tilts as shown by (3.33), which can then be solved to give the second derivatives of the aberration function at those points. It is then possible to fit polynomials to these second derivatives and thus fit the aberration coefficients from this dataset. This is relatively straightforward, but the inversion from the fitted polynomials to the aberration function is not strictly unique because the situation is over-determined. This problem is encountered again in Chapter 6 (and Appendix 2), where the details of this process are considered.

Both of these methods are applicable in the STEM as well as the CTEM. However one problem with applying them in a STEM is that the images are acquired in series over a period of time, during which the sample drift can be sufficient to disrupt the measurement. To overcome the drift, it is possible to acquire the images in parallel. This was implemented by acquiring a single line from each image in the tableau in turn. Unfortunately this lead to long acquisition times on the present system, due to hardware delays in outputting the signals to the microscope. A better method in the STEM would be to use a multi-channel photomultiplier and acquire all of the tableau images simultaneously. This would also require some post-sample optics to couple to the system over a suitable range of angles, and would limit the flexibility.

One of the major problems with these methods in CTEM is that when the illumination is tilted, different parts of the apertures will be illuminated. This will give spurious shift effects. So there is some scope for the application of alternative methods to measure the aberration function.

3.5. Conclusions

In this chapter, convenient descriptions of the imaging modes commonly used experimentally in STEM have been outlined. Bright field, ADF and Ronchigram images have been introduced and described in the presence of aberrations. Some consideration has been given to the definition of resolution and to the effects of limited coherence. The suitability of the two main imaging modes available for use with aberration correctors has been considered. It appears that high angle annular dark field STEM is particularly suitable for C_s -correction, because it is less affected by chromatic aberration than other modes.

The methods that we use experimentally to measure aberrations from tilted bright field have been introduced. Some experimental difficulties with implementing these methods in a STEM (because of the scanned nature of the images) have been pointed out.

The model for the Ronchigrams of amorphous materials has been extended over that found in the literature. For an amorphous specimen, the Ronchigram can be considered to be a bright field image distorted by the gradient of the aberration function (although this does not apply to crystalline materials). It is difficult to determine aberrations in normal images, however the approach used here demonstrates clearly that the appearance of a Ronchigram will directly relate to the symmetry of the aberration function. This model will later be used to provide a convenient basis for more quantitative analysis of the microscope aberration function.

Chapter 4

Aberration Correctors

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4.1. Introduction

In this chapter the operation of aberration correctors is examined with respect to the design of existing and future systems. First the history of aberration correction is briefly reviewed in order to gain an appreciation of the relative importance of the factors involved. Then the different techniques used to calculate the properties of aberration correctors are introduced. Using these methods, the properties of the HB5 corrector design are considered. This corrector will be referred to as corrector 1. Following on from the experimental results (Chapter 7) and the calculations presented in this chapter, a second, improved corrector design was produced. This will be introduced and is subsequently referred to as corrector 2. Finally some simplified models for the behaviour of such systems are presented with regard to future developments in this field.

In the existing correctors, to correct the third order aberrations, significant amounts of lower order aberrations have to be imparted to and then removed from the beam. Because these methods rely on introducing large amounts of new aberrations to the existing lens aberrations and correcting the total, it is obvious that the stability and precision requirements will be extreme. So it can be seen without any detailed consideration that although it is readily possible to build a device for the correction of spherical aberration, it is far more difficult to achieve an overall improvement in resolution. In order to account for the design and properties of the modern aberration corrector, it is first essential to review the previous instruments and to examine the reasons for their lack of practical success.

4.2. A Brief History of aberration correction

In a modern transmission electron microscope, the resolution is ultimately limited by the chromatic and spherical aberrations of the objective lens. Thus Scherzer's demonstration in 1936 that the spherical and chromatic aberrations of conventional round lenses will always be greater than zero appears to set an upper limit on the attainable resolution. It was Scherzer again, in 1947, who first showed some possible methods for overcoming this limit, by breaking one of the four conditions that this proof relies upon. Since that proposal, various attempts to build a C_s -corrector have met with little success until very recently. The history of aberration correction is documented extensively by Hawkes and Kasper (1994), with Archard (1954) and Koops (1978) providing other useful reviews.

The multipole corrector, which relies on breaking the rotational symmetry required for Scherzer's proof is the only aberration corrector that has displayed any real practical success. The other varieties were listed in Chapter 1 and are reviewed by Hawkes and Kasper (1994). There are two main sorts of multipole corrector, which work in slightly different ways. These are referred to as quadrupole-octupole (QO) correctors and hexapole (or sextupole) correctors after the main correcting elements present in each variety.

4.2.1. Quadrupole-Octupole correctors

The use of octupoles to correct the spherical aberration of round lenses was first suggested by Scherzer in 1947. An octupole acting on a round beam will primarily affect the third order aberrations (see for example the TRANSPORT manuals). In 1968, Scherzer and Typke investigated the use of a single octupole to correct the C_s of a round lens, and found that in some circumstances this could increase the resolving power. The problem is that an octupole has the wrong symmetry to correct C_s in all directions, leading to 4-fold astigmatism when acting on a round beam. However if the beam is suitably deformed by first order elements, and rotational symmetry abandoned, it is then possible to obtain only the desired phase change from a series of

octupoles. Using combinations of first order elements to change the beam shape and octupoles to perform the correction, it is thus possible to correct all of the third order aberrations simultaneously (Scherzer, 1947).

Scherzer's proposal for correction of C_s relied on cylinder lenses to break the rotational symmetry of the beam and suitably positioned octupoles to impart the desired third order phase changes. In the years immediately following this suggestion through the early 1950s, such a corrector was constructed by Seeliger (for example, 1949, 1953) and further improved by Möllenstedt (1956). These designs consist of 2 cylinder lenses, 3 octupoles and 1 additional round lens as illustrated in Fig. 4.1(A). Because these designs mix both round and multipole elements, the inherent rotation of any magnetic round lens could potentially cause some problems, so these early designs were purely electrostatic. A detailed description of the action of cylinder lenses is given by Hawkes and Kasper (1994), but the main effect is just equivalent to the superposition of a converging round lens and quadrupole element. These designs are quite complicated, yet not flexible enough to cure all parasitic aberrations.

At a similar time Burfoot investigated the problem of producing theoretical designs with minimized numbers of electrodes (Burfoot, 1953). His proposal relies on producing electrodes of precise shapes to achieve complicated field distributions. This has the effect of producing octupole and quadrupole fields superimposed on an electrostatic round lens field, leading to correction of C_s . These electrodes will have complicated profiles and extreme tolerance requirements (Archard, 1954), so it seems unlikely that such a design would function well in practice. This design would entirely replace the objective lens, and so lose the benefit of the final demagnification this provides. Such systems will never perform as well as designs with round magnetic lenses (for example, Maclachlan, 1974).

The next significant improvement to the design of octupole correctors was suggested by Archard (1954), who pointed out that it was possible to eliminate the cylinder lenses in these designs by using suitable arrangements of quadrupoles. In 1955, Archard then proposed two alternative systems in which the round lenses are also eliminated, and so consist only of **superimposed** quadrupole and octupole fields (1955b). The more flexible design was to form the basis for much of the future work

on quadrupole-octupole correctors. This is the 4-quadrupole version of the design shown in Fig. 4.1(B). The superimposed quadrupole-octupole fields in this design have the desirable property of reducing the total number of multipole elements compared to designs with separate quadrupoles and octupoles. This is obtained at some expense, because it becomes difficult to independently change the field strengths, due to the magnetic saturation of the polepieces. Such a system was investigated experimentally by Deltrap (1964), who found that this corrector was limited by the lack of additional multipoles to cure parasitic aberrations.

Potentially one of the most significant advantages of a quadrupole-octupole corrector is the possibility of also correcting the first order chromatic aberration, as suggested by Scherzer (1947). It was for this reason that the first corrector built by Krivanek and co-workers (1996) was a direct descendent of these designs, with extra multipole elements to increase the flexibility and provide extra strength for the octupole elements. Archard (1955a) went some way towards proposing a practical system for the correction of C_c in 1955, with combinations of ‘two-pole’ electrodes. However a significant result was the finding that superimposed electrostatic and magnetic quadrupole elements can be free of C_c for suitable excitations (Kel’man and Yavor, 1961). Similarly, such a system can be run in a manner as to correct the C_c of the objective lens, because of the different dependence of the effects of the electric and magnetic fields on the electron velocity. Such a system was studied experimentally by Hardy in 1967, who found that the basic quadrupole system made the aperture aberrations worse, and so proposed combining the chromatic aberration correction with the spherical aberration corrector of Deltrap (Hardy, 1967).

Another attempt to build a quadrupole-octupole corrector was that of Crewe and co-workers from the late 1960s onwards (Thomson, 1971, Beck and Crewe, 1974, 1976, Beck, 1977). This system had separate quadrupole and octupole elements, which increases the number of multipoles necessary, but allows them to be adjusted independently. This corrector is shown in Fig. 4.1(C). The most significant problem with this design was the remnant magnetism of the pole-pieces, which prevented suitable alignment (Beck, 1977).

At a similar time, the Darmstadt project showed some success with correction of C_c , but was suspended before demonstrating correction of the geometric aberrations (for example, Rose 1971a,b, Fey 1980, Bernhard 1980). The importance of electrical stability was highlighted, with specifications approaching 1 part in 10^7 . This design consisted of five magnetic octupole elements and three electric elements, allowing correction of both C_s and C_c . This was shown to successfully correct the chromatic aberration (Koops, 1978), although difficulties with aberration measurement and alignment were observed.

Despite all of these efforts, it was not until the 1990s that the quadrupole-octupole corrector actually produced a useful improvement in resolution. The reasons that most of the early aberration correctors did not succeed are varied, with the first problem being the stability of power supplies (as indicated by the Darmstadt project), and the mechanical tolerances of the corrector (for example, Archard, 1954). These obstacles can be surmounted, as demonstrated by Beck and Crewe, who were then defeated by anisotropy in the magnetic circuit (Beck, 1977). The earliest systems, such as those constructed by Deltrap and Hardy suffered from a lack of flexibility in adjustment, with no way to cure the parasitic aberrations. Deltrap (1967) also cited the lack of a suitably accurate method of diagnosing aberrations, which seems also to have plagued many of the other attempts (Koops, 1978). It is only with the advent of computerized diagnostic techniques that accurate alignment of aberration correctors has become routinely possible, and so automatic control should be regarded as essential.

There are also some other possibilities for different designs of quadrupole-octupole correctors (for example Maclachlan, 1974). However, it appears that the anti-symmetric arrangements of quadrupole excitations used here offer considerable advantages in terms of flexibility (for correction of parasitic aberrations and variable first order trajectories) and required field strengths. Obviously other minor variants are possible, such as putting in extra elements at optically equivalent planes, although these do not substantially affect the performance at low orders.

The first quadrupole-octupole corrector to demonstrate a useful improvement in resolution was that of Zach and Haider (1995a,b) in a low-voltage scanning

electron microscope. This was similar to the correctors of Deltrap and Hardy, consisting of superimposed quadrupole and octupole elements. While most of the multipole elements are electrostatic, the presence of magnetic quadrupole elements allowed the correction of first order chromatic effects. This was found to produce an improvement in the device performance, but the resolution of such systems is far below that attainable in transmission electron microscopes.

This was closely followed by the first successful aberration corrector in a STEM, constructed by Krivanek and Dellby (1996, 1997b, 1998). This was an important proof-of-concept because up until this point, no aberration corrector in a transmission electron microscope had displayed a useful improvement in resolution. This corrector is shown in Fig. 4.1(B). It consists of six stages, each composed of a strong quadrupole and octupole, with twelve weaker trim coils. Although only four stages are necessary for the operation of this corrector, the extra elements increase the octupole strength and allow a wider range of focal lengths for the system. The twelve-poles allow any multipole up to and including an octupole to be produced at any angle in each stage, providing enough flexibility to correct all of the parasitic aberrations. This contrasts to the designs of Deltrap and Hardy, where it was not possible to produce extra fields at arbitrary angles. The machining tolerances listed by (for example) Archard (1958), show that it is not possible to mechanically align any such corrector with sufficient accuracy, so this flexibility is a required component of any aberration corrector, as highlighted by Krivanek et al (1997b). The inhomogeneity of magnetic materials and the remnant magnetism were cited as significant problems with the Beck-Crewe corrector (Beck, 1977). The choice of a low remnant Ni-Fe alloy forms an important part of the new correctors (Krivanek et al, 2000a).

Perhaps the most significant difference of the Krivanek-Dellby corrector from any previous device was the sophisticated computer control. This consists of two essentially separate components, operating originally on two distinct computer systems (Krivanek et al, 1998). The first half of the software concerns the automatic measurement of the aberrations. As pointed out by Krivanek et al (1997a), it is not possible to manually decide which multipole control to adjust from conventional images. The Ronchigram was found to be more useful experimentally, because of its relation to the symmetry of the aberration function. The methods used originally are

described in more detail in Chapter 3, and rely on the changes in the off-axis bright-field images. An important development was the invention of an entirely new method of measuring the aberration function by analysis of the Ronchigram as described in Chapter 6 and by Krivanek et al (2000a,b). This allows characterization of the aberration function extremely rapidly, and is closely related to the manual alignment procedures.

The second half of the software concerns the computerized control of the multipole units. An important new development here was to allow the hierarchical linkage of individual coils into compound controls (Krivanek et al, 1997a,b). This allows the construction of ‘pure’ controls, where lower order multipoles are added to higher order multipoles so that they remain optically centered over a range of excitations. This allows the controls to be adjusted without continually having to realign each element. The computers controlling each stage are linked in order to allow these changes to be entirely automated. The development of this feedback process, with completely automated image analysis and communication between the different software components formed a large part of this project (Dellby et al, 2001). Such a mechanism will form an essential part of any device that is intended for more general use.

The first Krivanek-Dellby corrector was the first demonstration that an aberration corrector could usefully improve the resolution in a STEM. The next significant development was the implementation of a second, improved corrector, described by Krivanek et al (1999b), and shown in Fig. 4.1(C). The significant change is that the octupoles and quadrupoles are separated, allowing the octupoles to be saturated without affecting the quadrupoles. Combined with a modified magnetic circuit and smaller pole diameters, this allows far stronger octupole fields. Thus the new corrector is more flexible and able to operate with a smaller beam diameter. This makes adjustments easier and reduces the chromatic aberration. Combined with a better starting instrument (lower C_s , C_c and focal length) a large performance improvement has been produced.

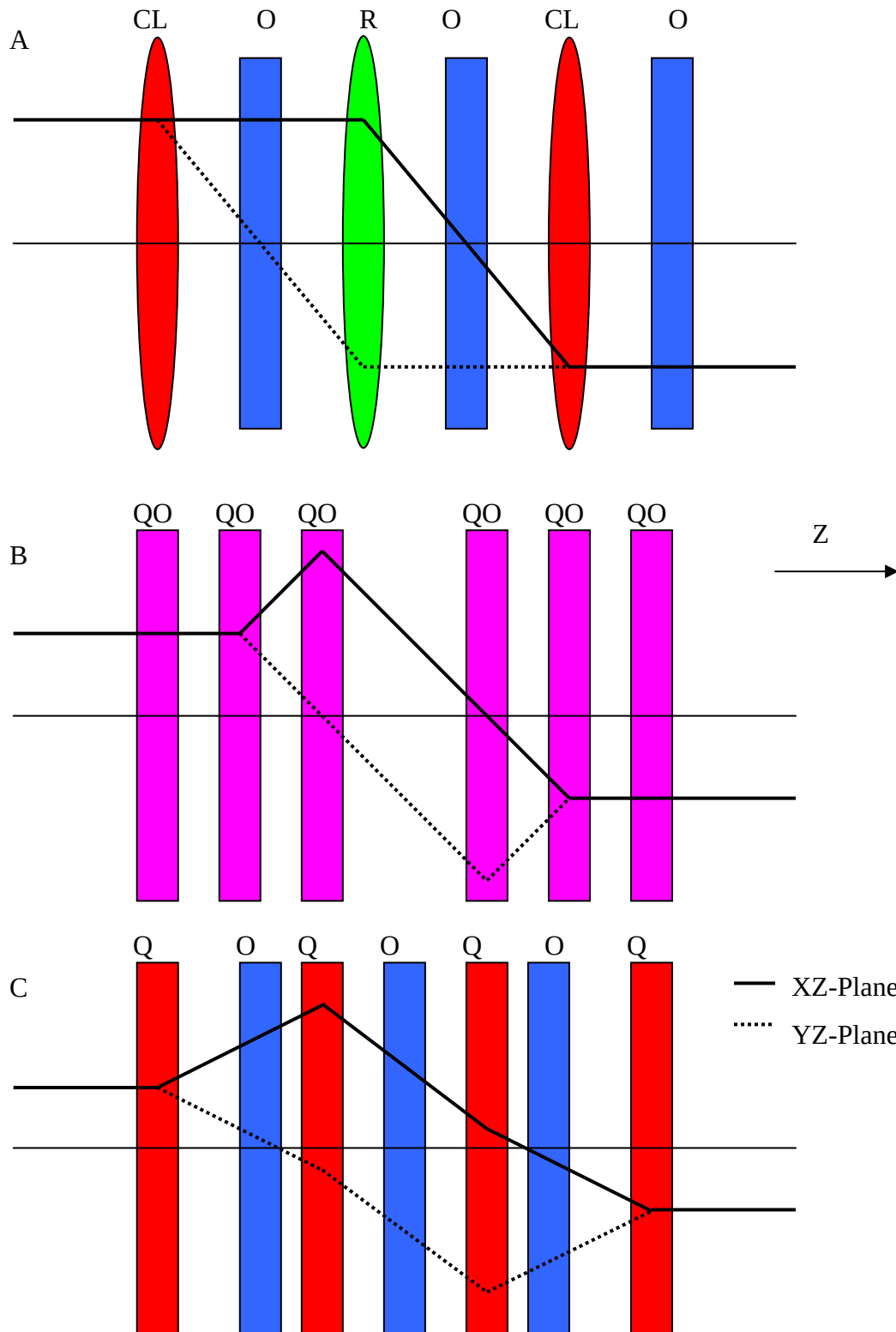


Figure 4.1. Schematics of octupole correctors suitable for a STEM. (Other examples are shown by Archard, 1954.)

- (A) Scherzer's design consisting of cylinder lenses (CL), octupoles (O) and a round lens (R).
- (B) Quadrupole-Octupole corrector (Krivanek-Dellby 1) with combined QO elements. In the Deltrap corrector only the inner 4 elements are present.
- (C) Quadrupole-Octupole corrector (Krivanek-Dellby 2, or Beck and Crewe) consisting of separate quadrupole (Q) and octupole (O) elements.

4.2.2. Hexapole Correctors

The fact that hexapoles can have negative spherical aberration has long been known, although this effect is only secondary to the introduction of 3-fold astigmatism to the beam. Thus the possibility of using hexapoles for correction of spherical aberration was not considered, until the suggestion by Beck in 1979 that the **combination aberration** of two hexapoles could be used for this purpose. The third order aberration of hexapole correctors occurs because the second order trajectory changes cause rays to travel through different parts of the field than those predicted from only the first order trajectories (even in ideal, pure hexapole fields). This means that they experience an ‘extra’ force, causing the third order aberration (as was clearly illustrated by Rose, 1994). It is then necessary to cancel the 3-fold astigmatism while retaining the negative C_s , without causing other aberrations.

Beck’s proposal for eliminating the 3-fold astigmatism inherent in any hexapole system was to have a hexapole doublet, consisting of two **thin** hexapoles of equal and opposite excitations. In this context, thin implies only primary order effects are imparted by the elements (second order for hexapoles). The second hexapole cancels the 3-fold astigmatism of the first, and the combination aberration from such a system is primarily C_s . The strength of this aberration can be calculated geometrically, and varies as LH^2 , where L is the separation of the hexapoles, and H their strength Beck (1979). Thus to achieve negative C_s it is necessary to project the first hexapole into the second, using a doublet of round lenses such that they have a negative effective optical separation. So Beck’s proposal requires two round lenses and two hexapoles as shown in Fig. 4.2(A). Each of the round lenses has a focal length of f , the distance between the lenses is $2f$ and the image is inverted, giving an optical separation of $-2f$, assuming the lenses are very close to the hexapoles. The problem with this corrector is pointed out in the original paper, where it is shown that the fourth and higher order aberrations of such a device will be significant. Similarly the off-axial aberrations mean that this would not be appropriate for a CTEM.

The remaining hexapole correctors work in a slightly different manner to this original suggestion. In these systems, the hexapoles are taken to be **extended**, and so each one imparts both 3-fold astigmatism and C_s to the beam.

Perhaps the most obvious way of correcting C_s using a hexapole without leaving significant 3-fold astigmatism is to have a converging beam cross the axis in the center of the hexapole element (Fig. 4.2(B)). Such a corrector was investigated by Crewe and Kopf (1980), and was indeed found to correct the third order aberration but still at the expense of introducing significant fourth order aberrations (C_{43}). Rose (1981) also examined a very similar system, where the extended hexapole is in fact 2 different hexapoles with the cross-over between them. In all of these systems, the combination aberrations will cause residual higher order and off-axial aberrations (Rose, 1981 and Shao, 1988b).

In 1980, Crewe and Kopf suggested the possibility of having two hexapoles separated by a round lens. This is illustrated in Fig. 4.2(C), which has coupling lenses either side of the central hexapole-lens-hexapole unit. The important difference between this and Beck's original suggestion is that the effective optical distance between the two hexapoles is zero, so there is no combination aberration in this system. The C_s and rotation of the field lens will degrade this idealized model, but such a system would be expected to improve the performance of a STEM where axial aberrations can be neglected. The main problem with this modified system is that the off-axial aberrations would be significant (Crewe and Kopf, 1980). (A simple explanation is that a single, perfect lens will project one plane to another, with axial rays having their angles preserved. However, off-axial rays will have their angles changed with distance off axis, giving rise to additional aberrations.) The properties and higher order effects of such systems are further discussed by Shao (1988a,b).

While off-axial aberrations are not usually significant for STEM aberration correctors, they must be considered for any useful CTEM corrector (Rose, 1981). The aplanatic corrector system proposed by Rose (1990) addresses this problem, and is shown in Fig. 4.2(D). This uses a round lens **doublet** to obtain zero effective optical separation, while also preserving the angles of off-axial rays between the two planes. (That is to say it obeys the Abbe sine condition, hence 'aplanatic'.) Thus the off-axial rays are transferred correctly, and there is no combination aberration. This means that the spherical aberration correction can be retained without significantly degrading the off-axial aberrations. Such a system has recently been constructed by Haider et al (1998, 1999, 2000) and has been shown to improve resolution in a CTEM.

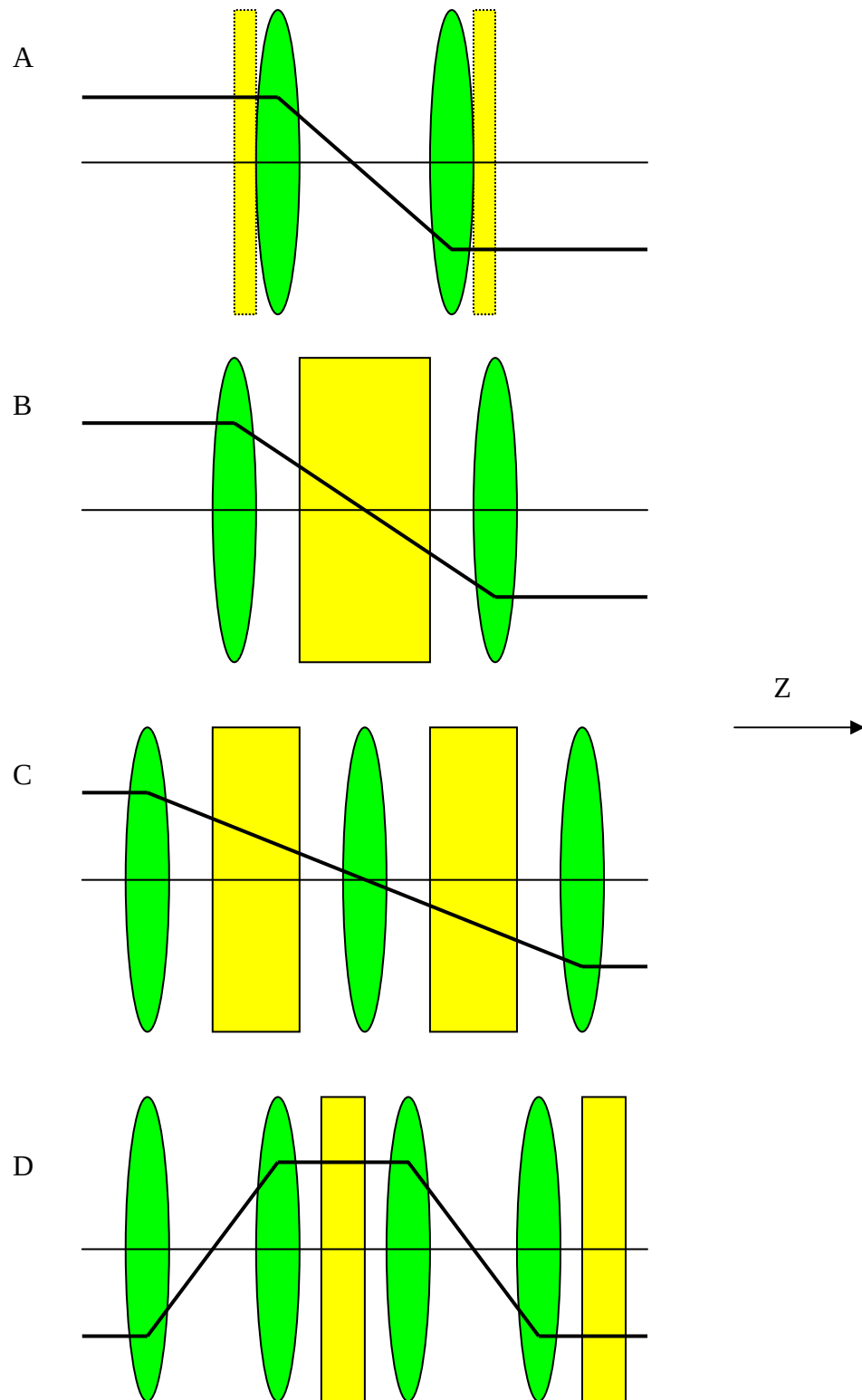


Figure 4.2. Schematics of hexapole correctors.

- (A) 2 *Short* hexapoles (yellow) separated by a pair of round lenses (in green) (Beck, 1979).
- (B) Single (or double) hexapole (yellow) with cross-over in the center. (Crewe and Kopf, 1980).
- (C) Twin hexapole system with intermediate lens (Crewe and Kopf, 1980). Surrounded by two coupling lenses, further lenses can be added to reduce high order effects (Shao, 1988).
- (D) Aplanatic system with hexapoles separated by round lens doublets (Rose, 1990).

4.3. Trajectory Simulations

Having introduced the different varieties of aberration correctors and given a brief summary of their properties, it is now necessary to examine the approaches that have been used in this work in more detail. Two classes of methods were used to calculate the electron trajectories through the corrector system, which are based on direct numerical integration or transfer matrix methods. These have different advantages, but it was considered essential to produce calculations by independent methods to gain confidence in the predictions. The other practical consideration of course was the relative availability of software. TRANSPORT calculates only up to third order, so initially it was necessary to construct an alternative method to examine the important fourth and fifth order aberrations.

4.3.1. Numerical Integration

The numerical integration method consists simply of treating each electron as a ballistic particle and solving the full relativistic equations of motion, using a Runge-Kutta integrator to calculate the trajectories. The axial aberration coefficients are determined by shooting a series of rays, at different starting angles from the source, through the system. The intersections of these rays with the sample plane will depend on the gradient of the aberration function. So for each ray intersecting this plane it is possible to perform a least-squares-fit to obtain a polynomial relating the positions and angles of the rays. This is used to obtain the aberration coefficients, as described in Appendix 2. Since it was found to be tedious to adjust the multipole values by hand, a simplex optimizer was implemented to allow multidimensional adjustments of the corrector elements. This is like the Complex method used in the automatic round lens optimization (Edgcombe et al, 1999) and would potentially be a useful method for the automatic adjustment of the corrector. All numerical routines necessary were taken from the book *Numerical Recipes in C* by Press et al (1988).

The problems with this method were that it was rather easy to accumulate rounding errors, in the fitting as well as the trajectory tracing, and that it is inherently inefficient. The advantages of this method are that it can calculate to any order and operate on any calculated field distribution. This includes round lenses as well as

unusual elements such as those proposed by Burfoot (1953). A good check on the consistency of results was to increase the step size and number of rays used. This approach could have been developed further, but appears to have been largely superseded by the COSY software.

The most serious disadvantage of this method was that it is necessary to know the field distribution in some way. Using the constraint that the fields must obey Laplace's equation, it is possible to obtain the higher order components of field distributions for a known variation along the axis, as shown in Chapter 2. One physically realistic approximation for the field distribution in a single multipole element is a Gaussian field profile along the axis. However this is still somewhat arbitrary, because the magnetic fields leak out beyond the physical extent of the elements. An experimental confirmation of this was found with the quadrupoles running in the anti-symmetric mode normally used. In corrector 1, quadrupole 1 appeared to be weaker than the surrounding quadrupoles 0 and 2. This was probably due to the outer multipoles reducing the extent of the fringing fields from the inner element, and also applied to quadrupole 4 (see Fig. 4.3). In practice, the method used was to apply the sharp-cut-off approximation and use an "effective length." Here the length is extended by an amount equal to 1.1 times the multipole radius, in order to include the field leakage beyond the element. Quadrupole fringing fields are discussed by Lee-Whiting (1970) and Grivet and Septier (1960), and depend on the multipole design and environment.

Similarly, no investigation was carried out as to how a multipole made on our twelve-pole construction differs from an ideal multipole. This could be done by a finite element method, but no suitable 3-dimensional modeler was easily accessible. This criticism applies equally to the transfer matrix methods, although in the literature for these, considerable effort has been devoted to matching experimental results by including the fringing fields. Experimentally factors such as other multipoles being excited and hysteresis were also significant, so better models for the field distributions were not investigated, although this could provide some scope for future work.

4.3.2. Transfer Matrices

The use of optical transfer matrices is described in the manuals for TRANSPORT (Carey et al, 1995) and COSY (Berz, 2000). The basic principle is that the beam at any point in the system is described by a vector, $\mathbf{X}$, such that:

$$\mathbf{X} = (x, x', y, y', l, \delta) \quad (4.1)$$

Where the components, denoted by X_i , are given by:

x = The horizontal displacement of a beam with respect to the central trajectory.

x' = The inclination of this beam to the central trajectory in the xz -plane.

y = The vertical displacement of this beam with respect to the central trajectory.

y' = The inclination of this beam to the central trajectory in the yz -plane.

l = The path length difference between this ray and the central trajectory.

δ = The fractional momentum deviation of this ray from the central trajectory.

The two components x' and y' are not strictly angles, although they are normally referred to as such. In TRANSPORT, they are the tangents of the angles, and in COSY they are the sines of the angles. The differences become significant only at third order and above, but for high-order calculations there is the possibility of quite significant differences being introduced. Here, the tangent is used throughout (because this is more accessible in the experimental situation), other than in the COSY calculations.

The transfer matrices then simply relate the description of the beam at a later point in the system, $\mathbf{X}(1)$, to the earlier conditions, $\mathbf{X}(0)$, by:

$$X_i(1) = \underline{\mathbf{R}}_{ij} X_j(0) + \underline{\mathbf{T}}_{ijk} X_j(0) X_k(0) + \underline{\mathbf{U}}_{ijkl} X_j(0) X_k(0) X_l(0) \quad (4.2)$$

Where the summation convention has been used. Thus $\underline{\mathbf{R}}$ is the 6x6 first order matrix giving the linear dependence of the beam on the earlier conditions. $\underline{\mathbf{T}}$ is the 6x6x6 second order matrix, $\underline{\mathbf{U}}$ the 6x6x6x6 third order matrix and so on. This notation is very general, including chromatic, axial and off-axial aberrations, and is extendible to any order. However, it is apparent that the number of terms makes this slightly unwieldy. Details of how to convert this to other notations are given in Appendix 2.

Through matrix manipulation it is thus possible to calculate the total transfer matrices for a system composed of elements for which the individual transfer matrices

can be determined. A significant disadvantage of the TRANSPORT software is that it is limited to third order calculations, and so cannot be used for the higher order calculations necessary for the newest corrector designs.

For analysis of particular designs of aberration correctors, the flexibility of COSY seems to make it a very useful tool. This software uses differential algebra techniques, which are far beyond the scope of this thesis, to allow calculations of arbitrary order. This approach is more advanced than that of TRANSPORT, because of the ability to perform higher order calculations and it is less likely to run into rounding errors than simple ray tracing. Finally it is possible to include a converter from the transfer matrix notation to the aberration coefficient notation into the COSY code itself, meaning that it is unnecessary to process the output in additional software. The main inaccuracies for the calculations will then arise because the fields present in the corrector will not correspond to the idealized models in the software.

One useful possibility with the transfer matrix method is to perform first order calculations in a spreadsheet such as Excel. This provides the ability to have a real-time graphical display, and automatic optimization of trajectories. It is also possible to include higher-order elements, although this eventually becomes rather complicated. Similarly, simple models for all of the electron optical elements can be included in such a way that the focal properties are calculated automatically from the measured currents. This was found to be an essential tool during the alignment of the corrector system. The formulae used to convert currents to field strengths for the multipole elements are those given by Breese et al (1996). For the round lenses, it was found that fitting the measured focal lengths to the currents and using the simple models outlined in Chapter 2, was better than calculating these in more detail. Typical output from such calculations are shown in Fig. 4.3 and 4.4.

4.4. Properties of C_s -correctors

The first order trajectories for the complete C_s -corrected HB5 in a typical operating mode are shown in Fig. 4.3. It is essential to consider the whole column because the effects of the corrector change significantly with the position of the cross-overs (this is examined in Section 4.4.2). The VOA is used to define the beam before it enters the corrector. The condenser lenses control the demagnification and the apparent position of the virtual source. The trajectories shown are for a convergence angle of 20 mrad in the objective lens. This mode is capable of correcting large amounts of spherical aberration, but is very sensitive to instabilities, because of the large beam diameter in the corrector.

Externally, the corrector behaves just like a round lens with approximately one to one imaging and negative spherical aberration. The position of the apparent source and the intermediate cross-over are indicated in Fig. 4.3. In this case, to ensure anti-symmetric trajectories within the corrector, they are equal distances from the central plane. This anti-symmetry is important because it will cancel out some of the parasitic aberrations caused by not passing through the centre of the optic elements. In a similar way to the normal operating mode of the HB5, the intermediate cross-over was initially situated at the SAD plane.

The first order trajectories within the corrector are solely influenced by the quadrupoles in an ideal system. The purpose of these elements is to obtain line cross-overs at the centre of the octupoles and then to re-assemble the beam to be round and stigmatic. An important point for C_s -correction is that, as the location of the apparent source moves along the axis, the positions of the line cross-overs and the intermediate cross-over shift. This has the main effect of changing the magnification of the corrector as seen from the objective lens, as shown in Fig. 4.4. An additional effect is that the size of the line cross-overs in the octupoles are no longer equal. The corrector should not be run in that mode, as the trajectories are no longer anti-symmetric. So it is apparent that the first order trajectories through the corrector are set-up for one particular position of the apparent source. It is therefore necessary to adjust the quadrupole settings in the corrector for different source locations, to make the trajectories within the corrector anti-symmetric again. It is clearly also necessary to

have two condenser lenses to allow the apparent source to be positioned correctly and still allow variable demagnification of the tip.

It is obvious from the field distributions given in Chapter 2 that the significant aberrations in a particular system will depend on the symmetry. The details of the relation between system symmetry and the permitted aberrations are summarised by Hawkes and Kasper (1994). In general, the largest aberrations will match the symmetry of the elements used, but misaligned elements will also have lower order effects of different symmetries. For example, quadrupole aberrations are primarily 2-fold and first order (C_{12}). A correctly aligned 'ideal' QO corrector will therefore have no second order aberrations. Higher order effects, such as the third order aberrations of quadrupoles or hexapoles, have some dependence on the symmetry. However, these also depend on the coupling between field and position, and so there may be extra terms of different symmetry. Thus for a correctly aligned QO corrector the significant aberrations will be 2 and 4 fold, first and third order. For a hexapole corrector the significant aberrations will be primarily 3 and 6 fold, second and fourth order. The second order effects in the experimental systems investigated here (all QO correctors) are therefore mainly due to mis-alignments.

In QO correctors, the third order effects are due primarily to the octupoles, but there is also a relatively small contribution from the quadrupoles, mostly as C_5 . Ideally aligned octupoles are assumed to act only on the third order aberrations, although in practice there were significant amounts of other aberrations caused by the inhomogeneity of the magnetic pole-pieces. The reason for the locations of the octupoles is apparent from these simulations. The octupoles at the line cross-overs act on the spherical aberration, but also give significant amounts of the other third order aberrations. The ratio of the coefficients $C_3:C_{32}:C_{34}$ generated by these octupoles acting on perfect line cross-overs would be 3:4:1 at O2 and 3:-4:1 at O3. (In the TRANSPORT notation, these would act only on the U_{1222} and U_{3444} coefficients and so these ratios come from the conversion tables listed in Appendix 2). In the simulations, the magnitudes of the ratios were found to be approximately 2.7:3.7:1. This is because the octupoles have a finite length and so the beam is really a narrow ellipse rather than a perfect line cross-over. Thus by exciting these two octupoles with

equal strengths it is possible to eliminate the C_{32} . The remaining octupoles act on a round beam to cancel the C_{34} , resulting in adjustable C_3 as required.

Table 4.1 compares the magnetic fields calculated from the currents output to the corrector to the values produced in the simulations for the HB5. The formula used for the magnetic field strength at the pole-tips was (Breese et al, 1996):

$$B_0 = \frac{\mu_0 n N I}{a} \quad (4.3)$$

Where μ_0 is the permeability of free space, and N is the number of turns used in the winding, carrying current I . The multipole rotational symmetry is given by n , in the same way as for the aberration terms (so, 2 for quadrupoles), and the pole-tip radius is a . This formula is based on an ideal multipole, and does not include saturation. The magnetic field inside the element at a radius, r , will also have a radial dependence such that the radial component, B_r , is given by (Breese et al, 1996):

$$B_r(r) = B_0 \left(\frac{r}{a} \right)^{n-1} \quad (4.4)$$

Although these formulae do not include saturation, one important result is that for a particular maximum flux density at the pole-tips, the field at the beam radius can be increased using a smaller pole-tip radius.

Experimentally the first stage was to set up the quadrupoles to produce a focal length for the corrector such that the cross-over is at the SAD for a virtual cross-over an equal distance (approximately 170 mm) before the corrector. The pre-corrector cross-over is located from the calibration of the condenser lenses in Chapter 2, and the current measured across the sense resistors. The amount of C_s -correction is altered by changing the excitation of the condenser lenses, so these cross-overs will subsequently move. In this mode the corrector would contribute about 8 mm to the C_c of the system and be capable of correcting large amounts of C_s . So in operating modes the cross-over was moved further down the column to approximately the same position as the scan coils to further demagnify the corrector effects. This will require subsequent small adjustments to the quadrupole strengths to minimise the parasitic aberrations.

Experimentally the quadrupole settings were found to be slightly different from those predicted by the models. This is because the cross-over positions, hysteresis and the other elements excited on the same pole-pieces were found to alter the strengths required. A more accurate agreement would depend on a detailed calculation of the fields. The comparatively large disagreement in the settings of the weakly excited end quadrupoles (Q0 and Q5) is probably due to having strong octupoles excited on the same pole-pieces. These are also adjusted to stigmatize the beam and to couple to different cross-over positions.

The octupole settings required depend strongly on the demagnification in the objective lens, which is adjusted by moving the locations of the cross-overs. One requirement is that the octupole excitations should be similar in order to obtain similar magnetic properties in all stages. To achieve the required octupole strength with this constraint will then require extra octupoles acting on the round beam (stages 0,1,4, and 5), because the diameter is lower there than at the line cross-overs (stages 2 and 3 in Fig. 4.3). The octupole settings are consistent with the correction of 3.5 mm of C_s for the objective lens providing an angular magnification of about 40. In this mode the corrector would contribute almost 2 mm to the chromatic aberration, and the instabilities are further demagnified (relative to the original mode). This was thus found to be a better mode to operate the corrector in, although it could be further improved by more demagnification in the objective lens, which would require stronger octupole fields.

This corrector was found to work well and provided a resolution improvement in the VG HB5 (Krivanek et al, 1998). However it was apparent that this suffered from some problems, mostly because of instabilities of the original microscope and of the electrically noisy room the microscope was located in. One further problem was that having all of the multipoles generated by the same elements meant that it was difficult to align, because a small change in any one control would also alter all of the other field components. Similarly when the pole-pieces begin to saturate, the optical centres of the elements shift. This is one reason why the octupole excitations in Table 4.1 are lower than the quadrupole settings. As the fields increase, the relationship between the current and the fields becomes less linear, making it far harder to accurately adjust the first order trajectories.

The chromatic aberration of this corrector was also quite large, contributing several millimetres at typical operating settings. It would be possible to reduce the intrinsic C_c of the corrector slightly by reducing the overall length, as it depends only on the first order trajectories. Unfortunately, shortening the octupoles would further reduce their effectiveness. Therefore the overall length should be kept short, but this is constrained by the need to generate sufficiently strong octupole fields to correct the spherical aberration.

A final problem was the objective lens of the HB5 microscope. Even with the high-resolution pole-piece, this had spherical and chromatic aberration values of around 3.5 mm and a focal length of approximately 4 mm. The uncorrected system is clearly not optimised, and with the corrector on a better instrument, improved results would be expected. One important consequence of this first corrector was that it enabled the factors that would affect the design of an improved corrector to be identified.

Element	Field calculated from corrector settings	Values predicted from simulations	Element	Field calculated from corrector settings	Values predicted from simulations
Q0:	0.0324	0.0145	O0:	-0.1463	-0.1547
Q1:	-0.2681	-0.2461	O1:	-0.1448	-0.1547
Q2:	0.2355	0.2197	O2:	0.1072	0.1022
Q3:	-0.2421	-0.2197	O3:	0.1019	0.1022
Q4:	0.2676	0.2461	O4:	-0.1448	-0.1547
Q5:	-0.0330	-0.0145	O5:	-0.1420	-0.1547

Table 4.1. Calculated multipole fields in kilo-Gauss for corrector 1 to obtain suitable trajectories in the VG HB5. The fields are defined at the tips of the pole-pieces as for the TRANSPORT software. The HB5 values are calculated from the currents going to the multipole elements and the number of turns.

4.4.1. A Second Corrector

The second quadrupole-octupole corrector designed by Krivanek and Dellby is more closely related to the corrector shown in Fig. 4.1(C). The general principle of the corrector operation is unchanged, with octupoles acting on line cross-overs in the same way. In this design the quadrupole and octupole elements are physically separated, allowing the octupoles to be run such that the pole-pieces are just saturated. This means that the other elements can be more independently adjusted and have larger field strengths, without encountering non-linear magnetic properties. In this design the line cross-overs are no longer quite at the center of the octupoles. As was apparent with the first corrector, this will change the ratios in which the coefficients are adjusted, but because of the symmetry of the first order trajectories, this does not involve a significant change in the action of the corrector.

Another important improvement in this new design is that the distances to the round lenses are reduced. It will be seen that the combination aberrations depend on the distance between the corrector and the objective lens, and so this change reduces these parasitic effects. The first order trajectories are shown in Fig 4.5. There is no cross-over between the corrector and the objective lens; there are now virtual cross-overs on either side of the corrector.

Finally the design of the multipoles has been improved. The significant changes are that the outer magnetic circuit is improved and the diameter of the multipoles has been reduced relative to the first design. Together with the separation of the quadrupole and octupole elements, this allows stronger fields to be generated. This is an important change because the beam in the corrector can have a smaller diameter relative to that in the objective lens, but still generate enough phase change to correct the spherical aberration. In the example shown, the beam diameter in the corrector is reduced by almost a factor of 10 relative to corrector 1. This decreases the contribution of the corrector to the chromatic aberration and reduces the sensitivity to stray fields and instabilities in the corrector.

The quadrupole fields required in this corrector are lower than the first because they are defined at the pole-piece tips, which have a smaller radius. This is

apparent from equations (4.3) and (4.4). A similar result applies to the octupoles, where the required field at the pole-tips decreases as the cube of the radius. Thus for a fixed maximum flux density at the pole-tips, the effective octupole strengths will increase. This allowed all of the C_s -corrector effects to be demagnified, reducing the chromatic aberration and instabilities by a large factor. Again the exact octupole excitations depend on the ratio of beam diameters in the corrector to the objective lens. In this configuration the calculated contribution of the corrector to the chromatic aberration is below 0.1 mm (disregarding the effect of the condenser lenses).

The VG HB501 on which the new corrector is installed is also significantly improved from the HB5 on which the original was developed. This machine has a lower starting C_s and C_c (of 1.3 mm) and a shorter focal length of approximately 1.7 mm. This reduces the amount of negative C_s needed and allows a greater demagnification in the final stage. In this case the beam in the corrector is about the same diameter as in the objective aperture, and the objective lens provides an angular magnification factor of about 150. These improvements meant that this corrector thus performed rather more effectively and produced a significant improvement in the instrument resolution.

Element	Values predicted from simulations	Element	Values predicted from simulations
Q1:	-0.0678	O1:	-0.712
Q2:	0.0903	O2:	0.816
Q3:	-0.0903	O3:	-0.712
Q4:	0.0678		

Table 4.2. Calculated multipole excitations in kilo-Gauss for corrector 2. The fields are defined at the tips of the pole-pieces as for the TRANSPORT software. The octupole strengths listed are compatible with an objective lens having $C_s=1.3$ mm, $f=1.7$ mm.

4.4.2. Scaling aberration coefficients

It has been stated that the effects of the corrector depend on the position of the intermediate cross-over, and that the corrector properties have been determined for one particular apparent source position. This is because the ratio of the beam diameter in the corrector to that in the objective lens governs the ratio in which these two sources of C_s contribute to the final aberration function. The reason for this can be used to predict the scaling of aberrations with this factor. Consider a system consisting of a planar aberration corrector and a planar objective lens as shown in Fig. 4.6. The aberration corrector imparts a phase change across the wave-front as a function of radial angle, ρ_1 . Thus when the angles are magnified in the objective lens, the same phase change will be associated with a different angle, ρ_2 , such that:

$$\rho_2 = M \rho_1 \quad (4.5)$$

Where M is the angular magnification of the objective lens, which is equivalent to this ratio of diameters, and is the reciprocal of the spatial demagnification. Thus scaling rules could also be derived by considering the demagnification of the effects of aberrations between two cross-overs.

Now, we must ensure that the phase across the wave-front is independent of the coordinate system used to describe it. Thus because the phase change associated with each of the aberration coefficients (C_n) of order, n , is associated with a particular term in radial angle, ρ^{n+1} , we can equate terms to give:

$$C'_n \rho_1^{n+1} = C''_n \rho_2^{n+1} \quad (4.6)$$

Where C'_n is the aberration coefficient at the corrector plane and C''_n , the associated coefficient referred to the objective lens plane. Using equation (4.5), the coefficients are thus related by:

$$C'_n = C''_n M^{n+1} \quad (4.7)$$

Hence the spherical aberration, C_3 , scales as M^4 . The chromatic aberration, defocus, and sensitivity to instabilities all scale as M^2 . It is therefore possible to change the amount of C_s corrected without altering the corrector settings, merely by altering the ratio of diameters (for instance by moving the position of the intermediate cross-over). Hence the demagnification of all of the corrector effects depends on the settings of the condenser and objective lenses.

Such a method also shows how to add aberrations at different positions: They must be added in such a way as to take into account the magnification at each point. The limitation of this approach is that it does not allow for the variation of trajectory due to the aberrations, or (equivalently) that M is a function of angle in an aberrated system. If such an approach were generalised to include these effects, it would ultimately become equivalent to the matrix notation introduced earlier.

Hence by adjusting the ratio of the beam diameter in the corrector to the objective lens it is possible to magnify or demagnify the aberrations imparted by the multipole system. Which of these two options is desired depends on the relative strength of the C_s imparted by the octupoles and the objective lens (at the same beam diameter). The other factor that might affect this choice is the magnitude of the other aberrations that are imparted to the beam by the corrector. From Chapter 5 it is apparent that the chromatic aberration of the beam is a significant factor, and so it is desirable for the additional C_c imparted by the corrector to be demagnified as much as possible.

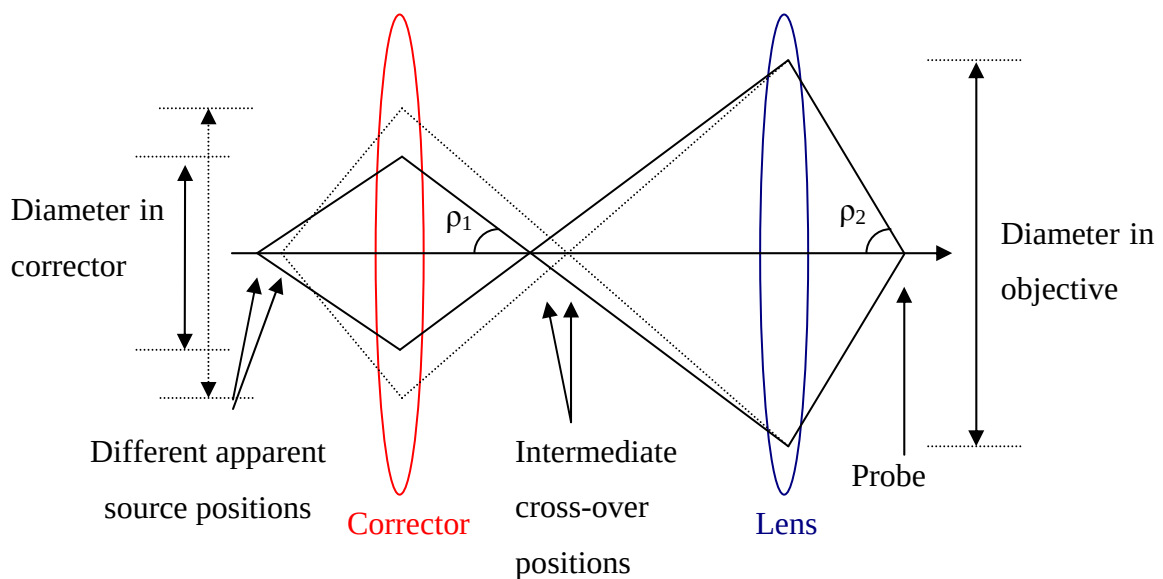


Figure 4.6. Ray diagram for two different rays showing how the position of the cross-overs affects the beam diameters in the corrector and the back focal plane of the objective lens. By shifting the crossovers using the condenser lenses it is possible to change the diameters and the ratio of the diameters.

4.5. Stabilities

The importance of stability in the corrector, as well as the rest of the microscope is frequently mentioned, (for example Krivanek et al 2000a, Beck and Crewe 1976, Fey 1980). In this section the effects of small errors in the position or excitation in the elements are considered. These can be used to estimate the alignment and stability requirements. From the previous section it is apparent that these will be intimately connected to the amount of demagnification provided by the objective lens.

The presence of third order elements shows the importance of alignment, as any errors in the first order trajectories mean that the beam experiences a significantly different force due to the large variation of the octupole field with position. Thus changes in the lower order aberrations are typically associated with non-linear higher order effects due to the change in trajectories. One result affecting stabilities is that the primary aberrations imparted by the multipoles are proportional to the currents (without saturations), and we know that the sensitivities to different aberrations decreases rapidly with order. Thus small variations in the power supply will be more critical for the lower order elements that change the trajectories. Misaligned higher order elements will give strong lower order effects, so for these elements it is the alignment that will be critical. This means that stability will still be important to avoid the spurious lower order effects changing the first order trajectories.

The stabilities given here are intended as estimates because they are rather sensitive to rounding errors due to the limited precision of the TRANSPORT output. Obviously there is some mixing between coefficients of the same order, because these are expressed in the same matrix elements. Also COSY and TRANSPORT output in different coordinate systems and the secondary effects are non-linear. (This is why alternative calculation methods were investigated.) Most importantly they can change by orders of magnitude by altering the demagnification, or by using different first order trajectories through the corrector. The calculations here are for the second corrector using the objective lens with a focal length of 1.7 mm and C_s of 1.3 mm. Using less demagnification would increase the sensitivity to these fluctuations, as described above. The effect of these aberrations on the resolution is discussed in Chapter 5.

Aberration	Octupole 3		
	B field 10 ppm	Alignment 1 μm	Rotation 1 mrad
C1	-	1.5 nm	-
C12	-	1.7 nm	-
C21	-	3 μm	-
C23	-	1 μm	-
Third order	10 nm	600 nm	30 nm

Aberration	Quadrupole 1			Quadrupole 4		
	B field 10 ppm	Alignment 1 μm	Rotation 1 mrad	B field 10 ppm	Alignment 1 μm	Rotation 1 mrad
C1	-	8 nm	-	-	10 nm	-
C12	0.3 nm	-	1 nm	0.3 nm	5 nm	1 nm
C21	-	10 μm	-	-	8 μm	-
C23	-	2 μm	-	-	-	-
Third order	50 nm	2 μm	160 nm	80 nm	800 nm	260 nm

Table 4.3. Approximate magnitudes of changes in aberration coefficients relating to small changes in the alignments or excitations of aberration correctors. Changes in rotation affect the *B*-type of each aberration most strongly. Other changes act primarily on the *A*-type. Very small changes are neglected, and the signs will depend on the direction of the change and position of the element.

Table 4.3 was produced by initialising the simulations with the first order trajectories calculated for the correctors, using models to represent the lens actions as accurately as possible. The octupole strengths were then adjusted to provide enough negative C_s to correct the objective lens aberrations. The sensitivity to variations in power supplies were calculated by adjusting the strengths of individual elements by amounts equal to ten parts per million (ppm). The effects of rotation were obtained by rotating individual elements by 1 mrad about the optic axis. This was found to primarily affect the *B*-type of each aberration as would be expected. All of the other changes only really affected the *A*-type because of the orientation. The effects of displacement were obtained by shifting the axis, in the *x*-direction, before and after single elements. The magnitude of each of these changes was selected to be

appropriate to the experimental conditions. It is apparent that the displacement was the most critical factor, producing more and larger changes than the other sorts of variations.

It is also apparent that because of the anti-symmetry used in these correctors to impart and then remove the lower order aberrations, many of the elements have opposite effects on the beam. Thus linking power supplies such that any variations are linked would lessen the degradation of resolution but limit the flexibility of adjustment.

Such simulations also show how to use an aberration corrector to eliminate parasitic aberrations. Many aberrations are in principle straight-forward to correct, such as quadrupoles acting on astigmatism, hexapoles on three-fold and octupoles on four-fold effects. This comes from considering the action of the poles on a round beam. Exceptions include the total C_s , which is adjusted by changing the demagnification of the corrector, and the C_{32} aberration, which is affected by the ratio of the two octupoles acting on the line-cross-overs. The B -types of each aberration are generally affected by using the B -type of the appropriate multipole, other than for C_{32} . In the simulations the easiest way to adjust this is to use the B -type quadrupoles to rotate the line cross-overs in the octupoles slightly. This indicates some of the difficulties that will be encountered later in the alignment of the corrector, where it will be necessary to adjust many controls and perform many subsequent corrections in order to change some of the aberration coefficients. Finally it should be noted that shifting the alignment of the corrector relative to the objective lens acts primarily on coma, as would be expected from two misaligned sources of opposite spherical aberration. Thus a dipole in the final stage of the corrector will provide a convenient method for adjusting the coma.

Fifth order aberrations in the mark 2 corrector are approximately 100 nm, with no fourth order effects in a correctly aligned system. Because some of the fifth order effects are due to the combination aberrations, these cannot be trivially demagnified as for the rest of the corrector effects. Thus it can be concluded that demagnification reduces the contribution of the chromatic effects and instabilities and

misalignments, but the fifth order aberrations are potentially a problem in this sort of corrector.

One advantage of the quadrupole-octupole corrector is that it is also possible to correct the chromatic aberration. The stability requirements for a chromatic aberration corrector were calculated by Rose (1999). These were found to be in excess of 1 part in 10^7 . This is because a C_c -corrector balances electric and magnetic fields. To convert the present quadrupole-octupole corrector to also correct C_c would involve additional electrostatic quadrupole elements inside the present quadrupoles. Simulations show that the magnetic quadrupoles would require approximately twice their present strength in order to eliminate just the C_c from the corrector (not even the objective lens). The required voltages would be approximately 1 kV. The problem arises because the corrector effects (including any C_c) are being demagnified in the objective lens, which was essential to reduce the stability requirements. This means that the necessary field strengths to correct the objective lens C_c will be impractically high. The alternative is to reduce the demagnification, which increases the susceptibility to instabilities and stray fields. Thus the stability requirements will be increased relative to the present correctors, and so such a device is probably not practical yet.

4.5.1. Hexapole correctors

The only hexapole corrector considered was of a similar design to that proposed by Rose (1990) and constructed by Haider et al (1998). This uses projecting doublets to eliminate the off-axial aberrations. Although these are less important in STEM, both doublets should be retained as this provides the ability to project aberrations into different planes. The difference relative to a CTEM corrector is that the corrector is now placed before the objective lens (as for the other STEM correctors). This uses the same objective lens as for the QO corrector and the excitations have been adjusted to values suitable for a 100 keV design.

An aligned corrector of this variety with perfect projecting lenses has no significant axial aberrations to third order, other than the desired negative C_5 . Because of the projecting doublets the fourth order axial aberrations are very low, typically a few nanometres. Non-ideal properties of the lens doublets will also significantly affect the performance of these correctors. Using the projecting lenses it is possible to project the correction into the back focal plane of the objective lens and thus remove the round fifth order aberration. By shifting this plane fractionally it is also possible to ensure that this eliminates the intrinsic C_5 of the objective lens. However the hexapoles also impart significant amounts of six-fold aberration to the beam, which it is not possible to eliminate in this way. This was typically about a millimetre of C_{56} , but varied with the hexapole strength. The other problems with this sort of corrector seemed to be the contribution of the round lenses to the chromatic aberration (particularly relevant for CTEM) and the extreme sensitivity to the alignment of the hexapoles. The chromatic aberration is a particular problem for CTEM imaging and has been calculated by Rose (1981) as approximately 0.4 mm. Also a small misalignment produces a lot of astigmatism. Both of these problems could be improved by demagnifying the corrector more, as for the quadrupole-octupole correctors. This would require stronger hexapoles, which appears to degrade the higher order aberrations (particularly C_{56}).

Effects	Changing second Hexapole			Round lens
	B field 10 ppm	Alignment 1 μm	Rotation 1 mrad	Focus 10 ppm
C1	-	-	-	-
C12	-	500 nm	-	-
C21	-	1 μm	-	-
C23	4 nm	1 nm	20 nm	-
Third order	10 nm	3 μm	-	40 nm

Table 4.4. Approximate magnitudes of changes in aberration coefficients relating to small changes in the magnetic fields, alignment or rotation of a hexapole aberration corrector. Changes of less than 1 nm have been set equal to zero.

4.5.2. Higher-order combination Aberrations

When spherical aberration and all the lower order aberrations have been corrected, it is not totally obvious which factors will dominate the resolution of an electron microscope. The main possibilities are the higher order, chromatic and off-axial aberrations. In general these are all made worse by the action of an aberration corrector. The case of the single hexapole corrector provides one pessimistic scenario; that the higher order aberrations will be made so much worse that no significant improvement in resolution is possible. To avoid this possibility, the higher order aberrations must be considered. Shao (1988a,b) provides an extensive investigation of the higher order effects of multipole systems.

To construct a simple view of the higher order effects, the aberrations are considered to be imparted to the beam in specific planes. In an idealized corrector the aberrations would be corrected in the same plane that they originate. This is clearly not achievable in practice; even if the elements could all be physically situated in the correct planes, the overlapping fields would prevent correct operation. The fact that the positive and negative aberrations originate in different planes means that although it is still possible to correct the C_s , there will be errors introduced as higher order terms, which are known as **combination aberrations**. The off-axial terms will also vary in a similar way, although these will not be considered here (see Rose, 1981 for some discussion).

The fifth order aberrations are normally masked by the third order aberrations in an uncorrected microscope. The approach given by Shao (1988b) treats both the aberration corrector and the objective lens as perfectly thin. This is summarized in Fig. 4.7, which shows the simplest situation. A C_s -corrector acts on a parallel round beam of radius r_0 . This introduces a third order aberration, S , which will be proportional to the octupole strengths, and deflects rays by $-Sr_0^3$. This propagates a distance L , to an aberrated lens with focal length f , where the rays arrive at slightly different positions due to the corrector deflection. The position and slope at each

plane are denoted as r_N and ρ_N , respectively in Fig. 4.7. The ray arrives at the sample with a third order aberration that can be cancelled by adjusting the corrector strength and produces a fifth order remainder term.

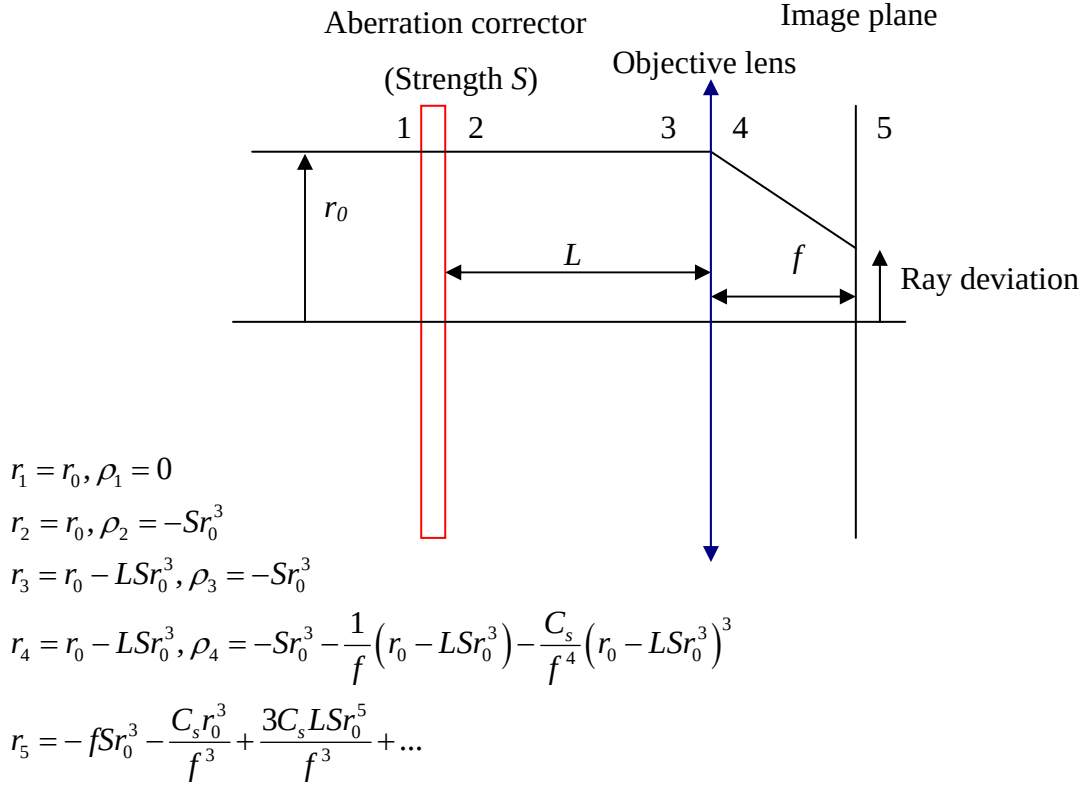


Figure 4.7. An idealized correction system consisting of a perfectly thin corrector and objective lens, both acting on a round beam. The positions r_N , and deflections ρ_N , at each plane are shown.

This approach shows that the strength of the third order aberration required will be:

$$S = \frac{-C_s}{f^4} \quad (4.8)$$

The field strength required is proportional to the amount of C_s to be corrected, but scales with the inverse of the fourth power of the focal length. So reducing the objective lens focal length dramatically increases the octupole fields required, because it demagnifies the corrector effects.

Similarly the fifth order aberration due to the corrector is:

$$C_5 = \frac{3C_s^2L}{f^2} \quad (4.9)$$

As observed by Shao (1988b), the combination aberrations get worse with increasing separation. However the magnitude and even the sign of the combination aberration can be changed by changing the sign of the third order aberration, or by using suitable projector lenses (as for Beck's corrector, Fig 4.2(A)). It is thus possible to project the apparent sources of the third order aberrations into different planes. A similar technique is used in the Rose hexapole corrector where the corrector is imaged with two round lenses into the objective lens to eliminate combination aberrations.

The quadrupole-octupole corrector has interesting combination effects because the third order aberrations are imparted in three different planes. Thus a slight modification of this argument should be used in the case of non-rotational symmetry. Treating the three components of the correction (X,Y and C_{34a} type) as each happening in separate planes, it is apparent that for a QO corrector, the fifth order aberrations will not be rotationally symmetric (as there are different distances for each component). This means that the C_5 , C_{52} and C_{54} aberrations will all be made worse in a QO corrector.

However, it is now possible to deliberately shift the optical positions of the planes in which the third order aberrations are imparted, in order to adjust the fifth order aberrations. This could be done with extra lenses (round or quadrupole) between the corrector and objective to alter the apparent position of the corrector. This will slightly degrade the off-axial aberrations, but this is less significant in a STEM. Thus by adding several quadrupoles to a QO corrector, it is possible to use the combination aberrations of such systems to correct the fifth order aberrations! A similar result also applies to changing the excitation of the projecting lenses in a hexapole corrector.

As will be investigated in the next Chapter, the resolution in a C_s -corrected system is limited by the fifth order and chromatic aberrations. C_5 is normally cited as the resolution limit and used to control the choice of aperture in a C_s -corrected microscope (Krivanek et al 2000a). In a system with C_5 also corrected, the sixth and seventh order aberrations will have to be considered, although the chromatic aberration will still provide the ultimate information limit. The ability to correct the fifth order spherical aberration is therefore expected to provide an important

improvement to the next generation of aberration corrector, but will probably not be as significant as the correction of the third order aberrations.

4.6. Conclusions

In this chapter the history of aberration correction has been reviewed. This has suggested that the reasons that previous aberration correctors did not provide a useful increase in resolution were mostly related to alignment. They either did not possess sufficient degrees of freedom to allow all of the parasitic aberrations to be corrected, or there were insufficient methods for diagnosing the necessary adjustments. The choice of magnetic material for the pole-pieces was also seen to be a critical factor, with magnetic remnance making alignment impossible for the Beck and Crewe corrector.

The first C_s -corrector designed by Krivanek and Dellby has been described. One conclusion was that it is desirable to demagnify the corrector effects by keeping the beam diameter about the same width in the corrector as in the objective lens, and to have a short focal length. This reduced the chromatic aberration and the sensitivity to misalignments and instabilities. However, such a change requires very strong octupole fields to produce a large effect that can be demagnified sufficiently. The beam size in the objective aperture, and thus the corrector, will then depend on the angles necessary to form the probe and the focal length of the objective lens. This suggests that the best performance from aberration correctors will be seen in already optimised systems, although some improvement was seen in the HB5 instrument.

An understanding of the behaviour of this device has lead to the design of a second corrector. This second device is optimised, according to the work described in this chapter, to be able to correct the spherical aberration, but also to have minimised contributions to the instabilities and chromatic aberration. This is the first C_s -corrector to achieve such a useful performance increase in a STEM and is presently being installed in several instruments worldwide. Intriguingly it appears that modifications of the existing aberration correctors will be able to act on the fifth order aberrations.

The main problem with the present design has been shown to be the accurate alignment of the corrector elements, although with sufficient flexibility and suitable computer control such problems can be overcome. Thus the present aberration correctors are indeed capable of improving modern STEM instruments.

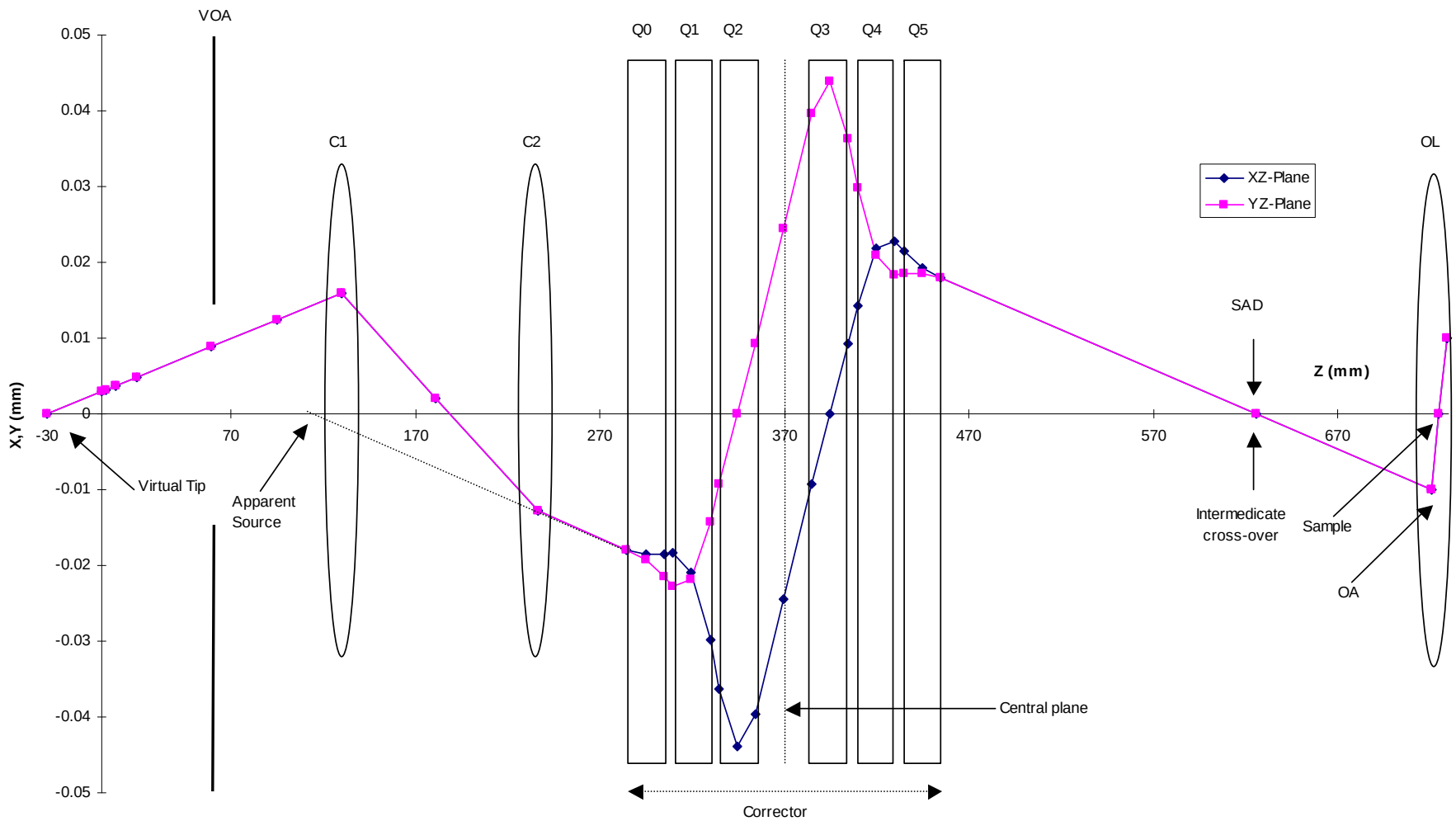


Figure 4.3. The first order trajectories in the C_s -corrected HB5. This is the initial mode, able to correct large amounts of C_s , but very sensitive to instabilities. The condenser lens (C1 and C2), objective lens (OL) and combined quadrupole-octupole elements (Q0-5) are indicated.

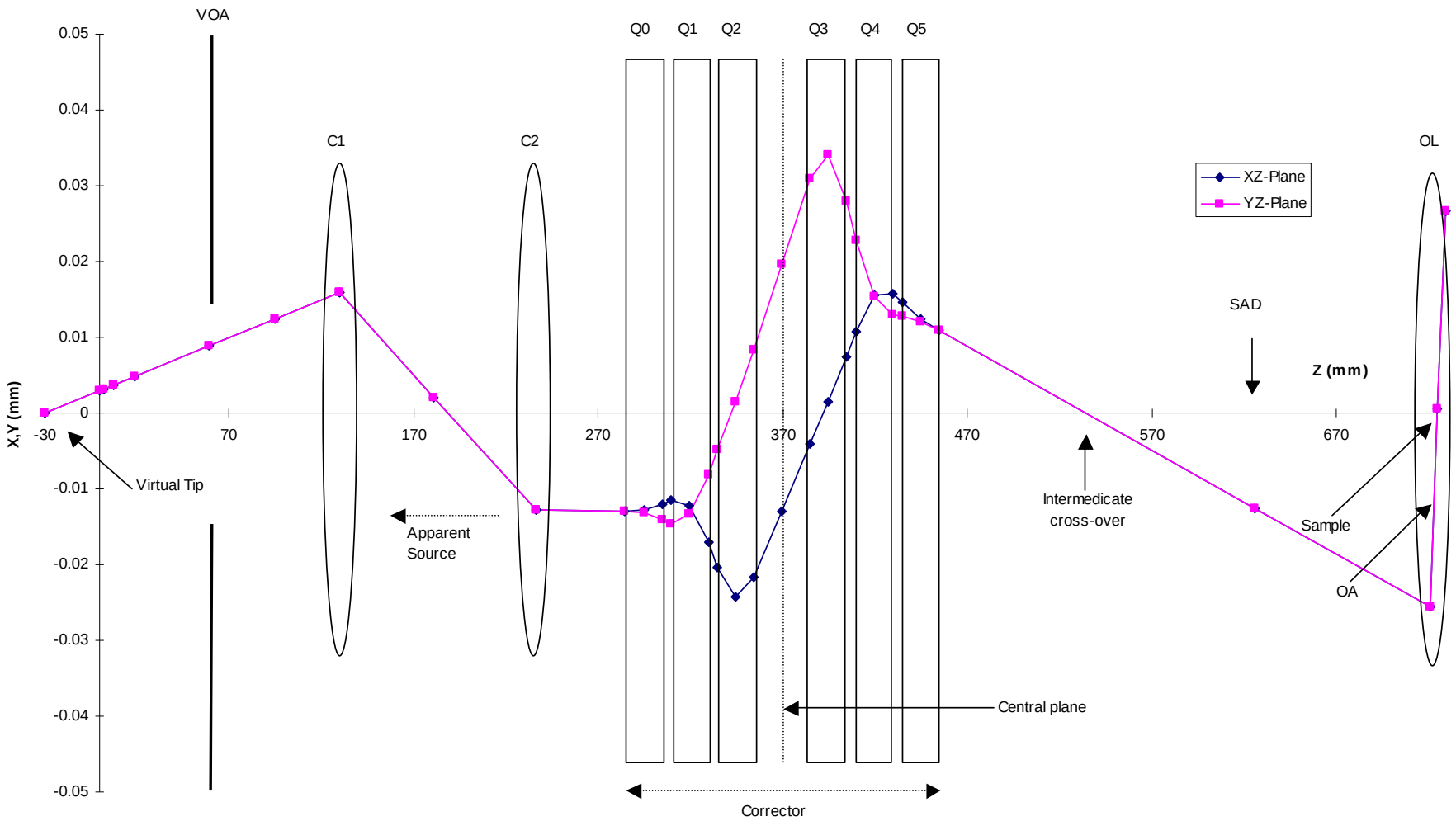


Figure 4.4. First order trajectories in the C_s -corrected HB5 where the condenser lenses have been adjusted to shift the apparent source. The line cross-overs are no longer in the centre of the octupole elements, and the intermediate cross-over has shifted. It is therefore necessary to now adjust the quadrupole settings so that the trajectories in the corrector are anti-symmetric again.

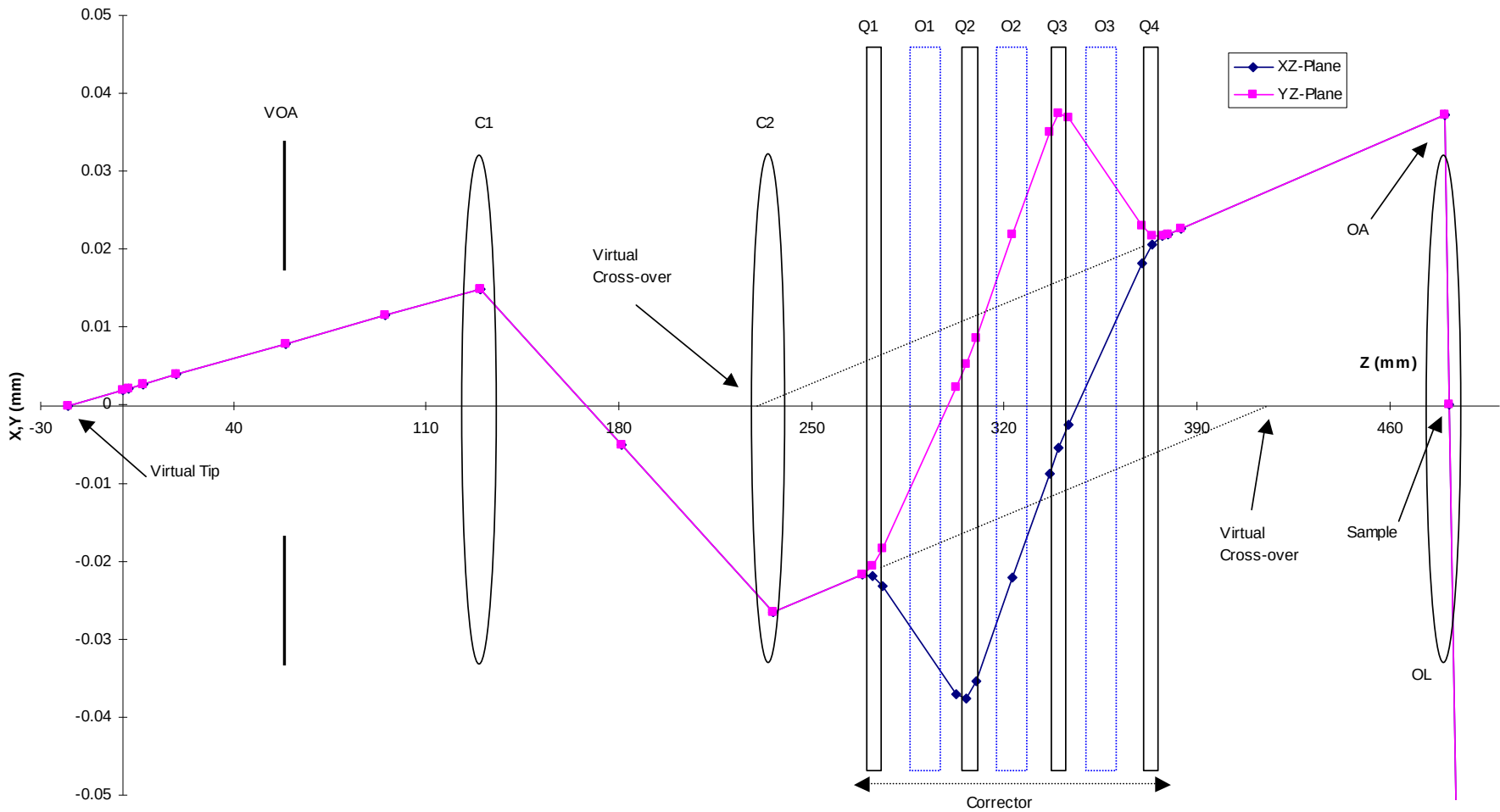


Figure 4.5. First order trajectories in the second C_s -corrector on the HB501. The condenser lens (C1 and C2), objective lens (OL), quadrupole and octupole elements (Q0-4, O1-3) are indicated. The beam diameter is significantly smaller than in corrector 1.

Chapter 5

Resolution Limits in C_s -corrected Systems

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5.1. Introduction

In this chapter the effects of different aberrations on the probe size will be considered, along with their implications for high-resolution imaging and future corrector design.

Before the present aberration correctors, it was adequate to consider the resolution limit as being due to the spherical aberration of the objective lens. In a C_s -corrected system it has not thus far been made clear which of the uncorrected factors will limit the attainable resolution, although there have been several suggestions. The most significant remaining factors will be the chromatic aberration of the system (Lupini and Krivanek, 1998), the higher order aberrations (Scherzer, 1970) and any uncorrected (sometimes known as parasitic or residual) aberrations (Haider et al, 2000). This last category also governs the precision to which the aberration coefficients must be measured, adjusted and stabilized for C_s -correction to offer real improvements in resolution.

In ADF STEM imaging the resolution will depend on the current distribution in the probe. This contrasts with the conventional bright field image where the PCTF must be considered. This discussion will concentrate on resolution in ADF images, because in this mode the attainable resolution is higher and the sensitivity to C_c is reduced relative to the bright field image. In order to investigate the resolution limits, a method for calculating the probe intensity distribution for a given aberration function will be described. Using the intensity distribution as a measure of the probe diameter, methods for defining and optimizing the resolution will be examined. The resolution limiting effects are then numerically determined and tolerances for the various aberration coefficients are derived. It is thus possible to determine the accuracy to which it is necessary to set the aberration coefficients, and to give stability requirements for the design and construction of aberration correctors.

5.2. Probe Size Calculations

The propagation of the electron wave through the microscope was accomplished by means of Fourier transforms as described earlier. The probe can thus be calculated as the Fourier transform of the electron wave in the objective aperture plane. Using the notation introduced in Chapter 3, the probe intensity for monochromatic illumination, I_p , is simply given by the modulus squared of equation (3.7):

$$I_p(\mathbf{R}) = \left| \int A(\mathbf{K}_i) e^{-2\pi i \mathbf{K}_i \cdot \mathbf{R}} d\mathbf{K}_i \right|^2 \quad (5.1)$$

The effects of chromatic aberration described earlier imply that for an electron at energy, E , rather than the mean, E_0 , the defocus, C_1 , is modified to:

$$C_1' = C_1 + C_c \frac{E - E_0}{E_0} \quad (5.2)$$

While this is adequate for the case of round lenses, for an aberration corrected system, the other aberration coefficients may also vary in similar ways. In the present design, the corrector contributes less than one quarter of the total chromatic aberration, and so any non-round chromatic effects will be small and can be ignored. The distribution of energies emitted by the tip is assumed to be Gaussian. The fraction of electrons with energy in the range E to $E+dE$ is given by:

$$dP(E) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(E-E_0)^2}{2\sigma^2}} dE \quad (5.3)$$

Where the standard deviation, σ , is given by the spread of energies emitted by the tip. The current distribution, J_p , is given by the incoherent superposition of the probe intensities at each energy, by:

$$J_p(\mathbf{R}) = \int I_p(\mathbf{R}, E) \frac{dP(E)}{dE} dE \quad (5.4)$$

In practice a discrete sum over a suitable range of energies is used to replace the integral in (5.4). The chromatic aberration of the corrected Nion STEM was measured to be 1.7 mm, and the energy spread used for all calculations was chosen to be $\sigma=0.25$ eV. This value is given by James (1997), and the choice of values for the energy spread is reviewed by Spence (1977). Hence it is possible to calculate the intensity distribution in the probe under the influence of any combination of aberrations and diffraction. This approach is essentially the same as that outlined by Haider et al (2000).

5.2.1. Evaluation of Resolution

The question of what resolution can be achieved in an aberrated imaging system has been examined by several authors, for example Crewe (1987), Michael and Williams (1987) or Hawkes and Kasper (1994), and dates back to the work on optical systems by Rayleigh (1896).

In an uncorrected microscope, underfocus is used to balance the spherical aberration (Scherzer, 1949). An objective aperture is used to cut out the high angle aberrated rays, and provides a frequency cut-off. (All aperture sizes given refer to the aperture half-angle.) The exact choice of parameters to optimize the resolution of the microscope will vary with imaging mode and the precise definition of resolution. The formulae normally given for the parameters are very similar, often changing only in the precise values of the constants involved. Some example formulae are given in Table 5.1, along with the actual values that will be used to model the probe, although others can be found (see Hawkes and Kasper, 1994, or Williams and Carter, 1996). Scherzer obtains values for these constants by optimizing the PCTF over a particular range (Scherzer, 1949). It is also possible to analyze the current distribution (Hawkes and Kasper, 1994), or to assume that the contributions from diffraction and spherical aberration add in quadrature (Crewe, 1987). The different resolution limiting effects are summarized in Table 5.2 for the case where each aberration is acting in isolation from the others. This allows a qualitative prediction of which of these round aberrations is most significant for a given imaging system.

The difference in a C_s -corrected system is that the spherical aberration is adjustable. The C_s is then used with defocus to balance the uncorrected aberrations (particularly C_5) in the same way as defocus was used previously to balance the uncorrected C_s . For optimal bright field imaging, the PCTF should be close to unity over a wide range of spatial frequencies. Scherzer (1970) gives values for C_s and defocus that achieve this in a corrected system by restricting the phase shift to $\pi/2 \pm \pi/4$. These are also listed in Table 5.1.

As most of the calculations in this chapter were performed before the C_5 of the present system was measured or calculated reliably, a value of 10 mm was used for C_5

(from Krivanek et al, 1999a, and also used by Haider et al, 2000). Subsequent calculations revealed that the value for C_5 is a little higher than this in the present corrector design (calculated as around 100 mm, and measured initially as 60 mm, depending on the demagnification in the objective lens). In hexapole correctors this could be eliminated, but there was typically about 3 mm of C_{56} . In uncorrected lenses, the C_5 is either much smaller or masked by the C_s . In future correctors, this aberration will be minimized or corrected, so the value of 10 mm for C_5 is a reasonable choice for calculations.

To illustrate the effects of different values of C_5 on the calculations, the argument of Brown (1981) showing the resolution in an uncorrected STEM can be repeated for the C_s -corrected STEM. The various possible resolution-limiting factors, which are listed in Table 5.2, are plotted out in Fig. 5.1, where the resolution improvement due to C_s -correction can be seen. At very large angles, the resolution again becomes limited, this time by C_5 . In practice this is partly balanced by suitable choice of C_s and defocus, as described, to give a better resolution than indicated by this graph. The simple argument used to generate this figure is suitable for illustration purposes, but the more detailed calculations described in section 5.2 will now be used to obtain more accurate values for the expected resolution.

For ADF imaging and EELS the parameter that determines the resolution is the probe diameter. The two measures that will be examined here are the full width half maximum (FWHM) and the diameter containing a given fraction of the intensity distribution. To calculate these, equation (5.1) is numerically evaluated by setting up a suitable aperture function, Fourier transforming and taking the intensity, with a final discrete sum used in place of the integral in (5.4). Performing this in 2-dimensions enables the effect of non-round aberrations to be included. The FWHM is calculated in two perpendicular directions aligned with the axes about which the aberration function is defined. The diameter containing a proportion of the current is obtained by integrating the intensity distribution, and does not distinguish between different orientations.

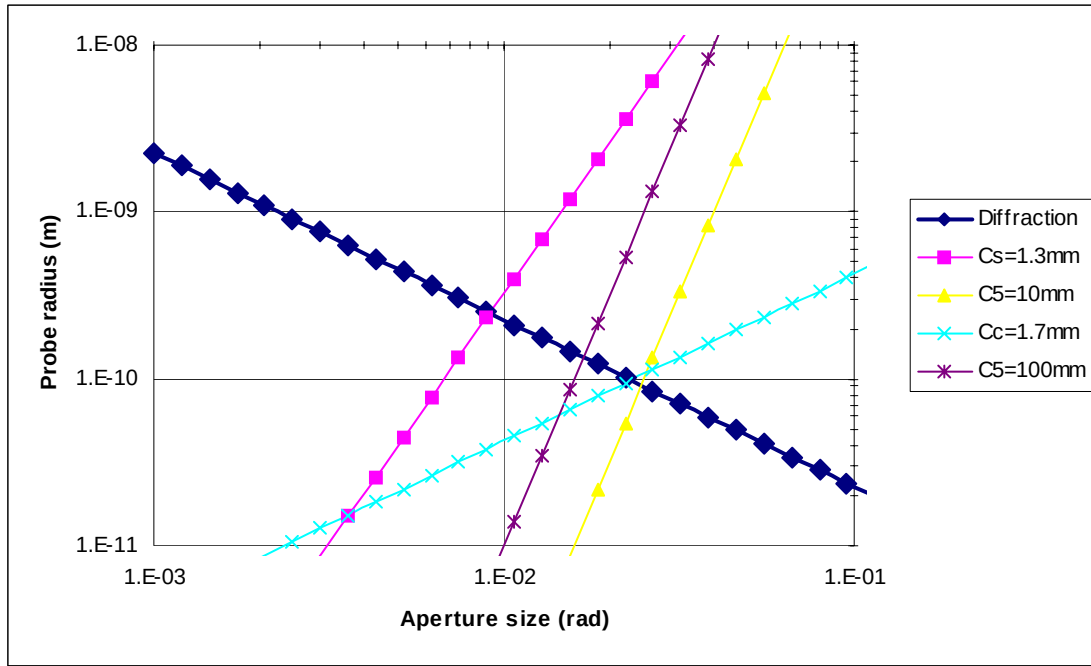


Figure 5.1. Probe size contributions from the entries in Table 5.2, calculated at 100 keV, on a logarithmic scale. The optimal aperture size and resolution are given by the intersection of the diffraction curve with the uncorrected aberrations. The C_s curve is taken at the disk of least confusion. No attempt has been made to balance C_5 with lower order aberrations in the same way, so that line represents a significant overestimation.

No.	Source	Aperture	Defocus	C_3	Resolution
(1)	Uncorrected (BF)	$1.51(\lambda/C_3)^{1/4}$	$-1.16(\lambda C_3)^{1/2}$	-	$0.66(C_3\lambda^3)^{1/4}$
(2)	Uncorrected (ADF)	$1.41(\lambda/C_3)^{1/4}$	$-(\lambda C_3)^{1/2}$	-	$0.43(C_3\lambda^3)^{1/4}$
(3)	Scherzer (BF)	$1.75(\lambda/C_5)^{1/6}$	$2.0(\lambda^2 C_5)^{1/3}$	$-3.2(\lambda C_5^2)^{1/3}$	$0.33(C_5\lambda^5)^{1/6}$
(4)	Krivanek (ADF)	$1.65(\lambda/C_5)^{1/6}$	$1.05(\lambda^2 C_5)^{1/3}$	$-2.4(\lambda C_5^2)^{1/3}$	$0.37(C_5\lambda^5)^{1/6}$
(5)	Haider et al (ADF)	ρ	$0.144 \rho^4$	$-0.92 \rho^2$	-
(6)	Numerical (ADF)	ρ	$0.165 \rho^4$	$-0.92 \rho^2$	-

Table 5.1. The different parameters given for optimal imaging given by several authors. (1) Scherzer (1949), (2) Hawkes and Kasper (1994), (3) Scherzer (1970), (4) Krivanek et al (2000a), (5) Haider et al (2000), (6) Values from the numerical calculations here. Where a particular resolution is claimed, this is shown in the resolution column. (1) and (2) correspond to uncorrected systems, while the remaining entries correspond to systems limited by C_5 only. In (5) and (6), ρ is used to represent the size of the aperture, which is optimized by some other means.

For EELS, the signal to noise ratio will depend on what fraction of the probe intensity is centered on the region of interest, and so for some applications it is likely that the 90% diameter might define the resolution (Michael and Williams, 1987). Rose (1981) describes the 59% diameter as a useful measure of resolution, which is the diameter containing 59% of the electron current. This coincides with the first zero of the Bessel function if a diffraction limited probe is used, and is the measure used by Haider et al (2000).

In practice, such a limit does not always correspond to the maximum achievable resolution. The complications arise because in the general case the probe consists of a central maximum surrounded by smaller peaks (the so-called probe tails). The optimal probe in the uncorrected case is obtained by using underfocus to balance the effects of the spherical aberration, as described by Scherzer (1949). However, by using extra underfocus it is possible to obtain a narrower central peak at the expense of putting more current in the probe tails (Nellist and Pennycook, 1998). This means that higher spatial frequencies are transferred to the image but the signal to noise ratios will be degraded. Thus the variation in the constants in Table 5.1 for the different modes will depend on the aspect of the probe that is being optimized. To calculate the experimentally achievable resolution fully would really involve a far more detailed calculation, including details of the specimen, beam damage, scattering process and numerous other effects. However it is satisfactory to disregard these in order to just obtain limits for the resolution (Crewe, 1995).

Effect	Radius	Comments
Diffraction	$0.61\lambda/\rho$	First zero of Airy disk
Spherical aberration	$0.25C_3\rho^3$	At disk of least confusion
Chromatic aberration	$C_c\rho\Delta E/E$	Only the energy spread is included
Source size	Mr_s	Demagnification (M) of source size (r_s)
5 th order spherical aberration	$C_5\rho^5$	For no compensating aberrations

Table 5.2. Effects of different factors on the probe size in isolation. Other effects, such as space charge, are discussed by Crewe (1987). In the final entry, no attempt has been made to balance the effects of C_5 with any lower order aberrations, so this is a significant over-estimation.

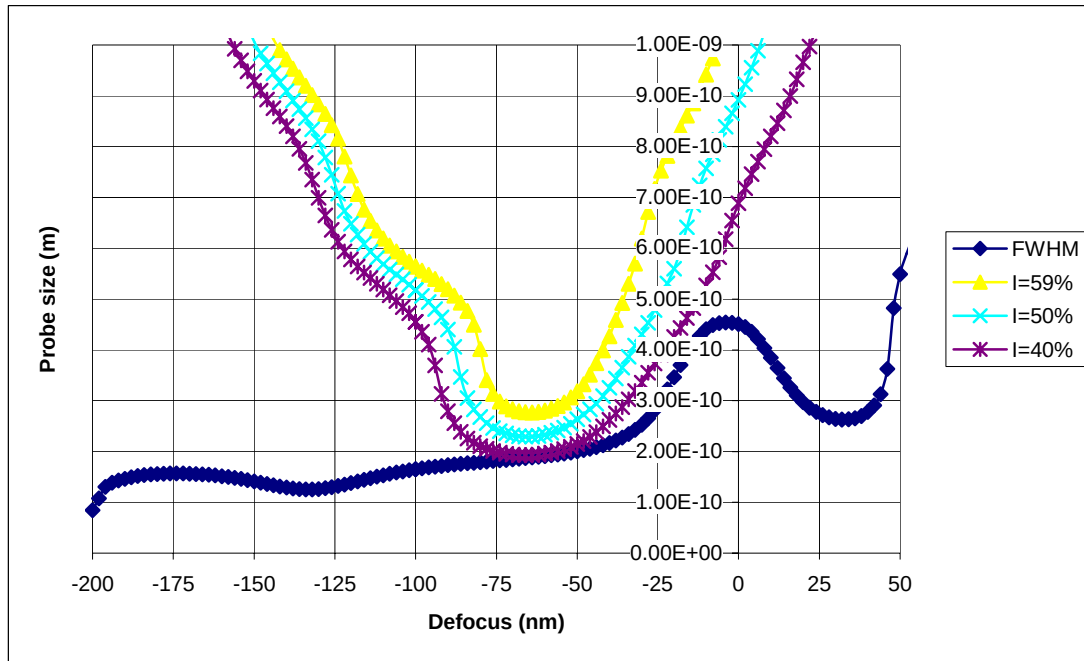


Figure 5.2. Different measures of probe diameter for an uncorrected system as a function of defocus. Typical parameters for the VG HB501 are used, with $C_s=C_c=1.3$ mm, aperture size=10.3 mrad. The probe diameters containing different proportions of the current and the FWHM are indicated.

Figure 5.2 shows the FWHM and current distribution for a probe for an uncorrected 100 kV VG STEM, limited by spherical aberration of 1.3 mm. The formulae given in table 5.1(2) predict that the optimal probe would be obtained for an aperture of 10.3 mrad half-angle and an underfocus of 69 nm. Using this aperture it can be seen that the 59% diameter is a minimum at approximately 65 nm underfocus. All of the current distributions have minima close to this value, with more underfocus favored for the lower current boundaries. The FWHM continues decreasing with underfocus, but the fraction of intensity contained in this central peak becomes progressively less. The changes in slope of the intensity measures are because of the uneven distribution of intensity between the central maximum and subsidiary peaks. One observation is that including the chromatic aberration in this calculation makes almost no difference to these curves because of the small aperture. This is consistent with the insensitivity of ADF imaging to chromatic effects demonstrated by Nellist and Pennycook (1998).

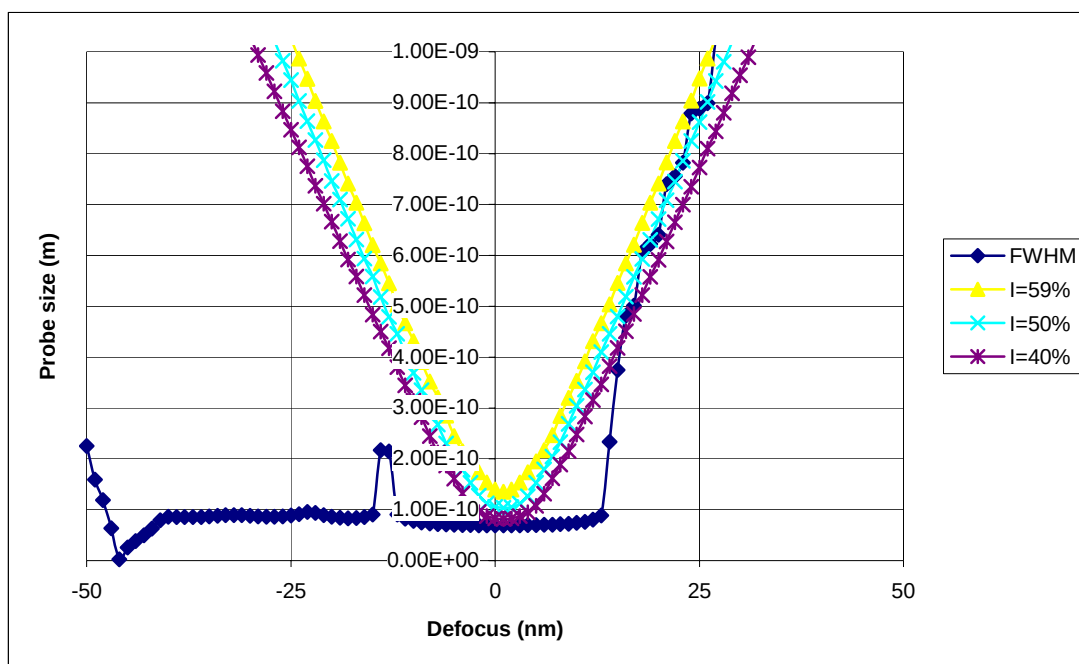


Figure 5.3. Different measures of probe diameter for a C_s -corrected system as a function of defocus. Parameters used are $C_c=1.7$ mm, $C_5=10$ mm, C_s optimised from Table 5.1(6), and aperture size=28 mrad. The probe diameters containing different proportions of the current and the FWHM are indicated. The irregularities in the FWHM are due to the discrete calculation method.

Figure 5.3 shows the FWHM and current distribution in a C_s -corrected STEM at 100 keV. The aperture used is 28 mrad, which closely matches some of the later experiments, but is significantly smaller than the optimal for the 10 mm of C_5 used. This smaller aperture is actually preferable because of the chromatic aberration of the system. Because it is now possible to adjust the spherical aberration to balance C_5 , the optimal defocus is almost zero. The most obvious improvement is that at this optimal defocus all of the resolution measures are improved by approximately a factor of 2, relative to the uncorrected case. One disadvantage is apparent from the change of scale necessary on the defocus axis: The corrected probe is far more sensitive to defocus. Because the aperture is larger, all aberrations (which scale as a function of aperture size) will have more effect. This leads to the important general conclusion that the tolerances to which each aberration must be set will scale with aperture size.

Figures 5.2 and 5.3 show several possible measures of resolution. From these, it is clear that the FWHM is not always going to be a useful measure, because it will favor the case of a very narrow central maximum containing very little current. It was also found that the FWHM measured automatically sometimes behaved rather irregularly (Fig. 5.3). This was because the probe intensity has oscillations, just like the normal Airy disk diffraction pattern. If the sampling is not fine enough to match these, then the exact point where the half maximum falls could shift quite substantially. This meant that the FWHM was not a good measure in the cases with a small central maximum and large probe tails. The intensity distribution measures were much less affected by this problem. (As with all calculations involving a discrete finite FFT, care must be taken to ensure that the phase does not change by too much between pixels and that the implicit periodicity does not affect the calculation. A useful check was to change the sampling and verify that the probe size did not change significantly). Thus the current distribution measures were found to be more useful as practical limits.

In the uncorrected case, the resolution predicted from Table 5.1(2) is 2.1 Å, which is just between the 40 and 50% diameters. Thus the 50% diameter is certainly an achievable resolution and it seems that 40% may just be possible in an optimised system. In the C_s -corrected Nion STEM, images showing 1.23 Å spacings have been recorded in a system with $C_c=1.7$ mm at 100 keV, using an aperture of approximately 27 mrad. This is between the 50 and 59% diameters, and is largely limited by mechanical instabilities, so it appears that the 59% diameter is too conservative a choice. When the aperture size and defocus have been optimised, the 40% diameter gives results that correspond closely to the FWHM used by Krivanek et al (2000a), so this represents a useful (and slightly optimistic) resolution limit (see Fig. 5.5). Conveniently, the similarities between the three intensity distribution curves show that it is possible to scale the results for slightly different current thresholds without incurring serious errors.

Figure 5.4 shows calculated probes from a corrected and an uncorrected 100 keV STEM. These have been normalized to have the same total beam current. The equation defining brightness, B , was introduced in Section 2.3: For surfaces normal to the beam, the current through the solid angle element, $d\Omega$, from a geometrically

demagnified source area, dS , was given by $dF = BdSd\Omega$. Although the aperture, and hence $d\Omega$, is larger in the C_s -corrected system, the geometrical demagnification of the source will also have to be increased by the same factor to avoid the source size limiting the probe diameter (Krivanek et al, 1999a). So at the highest resolution attainable, there is no net increase in the total probe current. However, it is apparent that the current density is greatly increased. Conversely, in any given probe diameter there will therefore be a lot more current, and the probe tails visible in Fig 5.4 are greatly reduced, making a C_s -corrected probe better suited to EELS analysis.

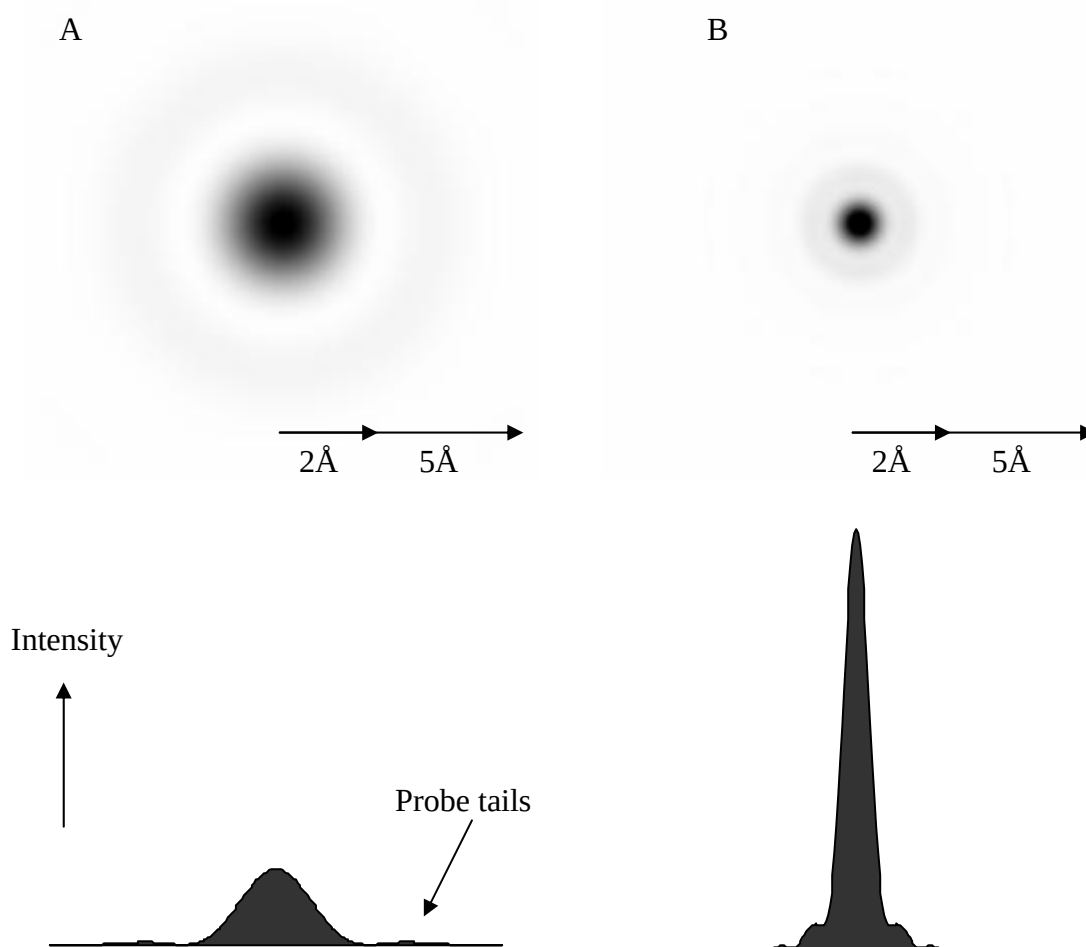


Figure 5.4. Cross-sections through calculated STEM probes at 100 keV. The probes are normalized such that the same total intensity is in both probes, and are plotted on the same scale.

(A) Uncorrected. $C_s=C_c=1.3$ mm, defocus=-69 nm, aperture=10.3 mrad. FWHM=1.85 Å, 59% diameter=2.8 Å, 50% diameter=2.3 Å, 40% diameter=1.94 Å.

(B) C_s -corrected. $C_s=10$ mm, $C_c=-7.2$ μm, defocus=1 nm, $C_c=1.7$ mm, aperture=28 mrad. FWHM=0.7 Å, 59% diameter=1.4 Å, 50% diameter=1.0 Å, 40% diameter=0.8 Å.

5.2.2. Optimizing Imaging Parameters

Using the formulae in Table 5.1, it is now possible to examine the effects of the different aberration coefficients on the expected resolution in a C_s -corrected system. This can be used to provide estimates for how precisely the residual aberrations need to be corrected, and for how accurate the measurement process must be to allow effective computer control. The first step is to determine a way to optimise the remaining parameters. The formulae given in Table 5.1(5) are particularly convenient because they allow optimisation of C_s and defocus at variable aperture sizes. Evaluating the probe size measures numerically indicates that adjusting these values to those in Table 5.1(6) by increasing underfocus produced a slightly smaller probe.

The optimal probe size as a function of aperture half-angle is shown in Fig. 5.5. As noted by Krivanek et al (2000a), the chromatic aberration has almost no effect on the FWHM. However, the current distribution measures (which were shown to be more useful estimates) were significantly altered. Thus, because of the chromatic aberration, the calculated probe sizes do not always agree with the predictions from Table 5.1. This contrasts with the C_s -limited probes where C_c had very little effect on the intensity distribution. This is because the effect of C_c increases with aperture size, and the apertures optimised for C_s will be very large, meaning that the chromatic effects become increasingly important. At the optimal aperture size, it is the chromatic aberration that limits the resolution. Therefore to optimise the current distribution it is necessary to reduce the aperture size compared to the very large C_s optimised aperture.

It can be assumed that the effects in Table 5.2 add in quadrature. Williams and Carter (1996) demonstrate this by balancing the contributions from diffraction and C_s to choose the optimal aperture size. Similarly it is possible to balance C_c and diffraction. Equating these two factors gives the optimal aperture as:

$$\rho^2 = 0.61\lambda E / C_c \Delta E \quad (5.5)$$

As a rough rule of thumb, if this is lower than that predicted from Table 5.1, then it is the chromatic aberration rather than the higher order aberrations that limits the resolution. To confirm this numerically, the aperture size was varied to find the optimum aperture for different values of chromatic aberration (using the 59%

diameter, which is normally very close to the minima of the other current measures). In practice the aperture size from equation (5.5) was slightly too small, and larger apertures were favoured. A better match to the simulations was produced by choosing an aperture optimised for C_c , ρ_{Cc} , as:

$$\rho_{Cc}^2 = 0.68\lambda E / C_c \Delta E \quad (5.6)$$

This is shown in Fig. 5.6, where the aperture size that gives the smallest probe for different values of C_c is plotted. This suggests that in a STEM with $C_s=10$ mm the resolution is limited by the chromatic aberration rather than the higher order aberrations.

For the C_c -limited case, the resolution measures approximately scale with the diffraction limit for the aperture in equation (5.6). When all other parameters are optimised, the FWHM diameter, limited by C_c , d_{Cc} , is given by:

$$d_{Cc} = 0.61\lambda / \rho_{Cc} \quad (5.7)$$

The 40% diameter behaves similarly, with the constant in equation (5.7) adjusted to 0.67, the 50% diameter with the constant as 0.80 and the 59% diameter as 0.97 (these constants scale closely with the fraction of current enclosed). So this equation is a rather optimistic estimate (see Fig. 5.7).

The attainable resolution as a function of the chromatic aberration is illustrated in Fig. 5.7. For values of C_c above 1 mm the chromatic aberration limits the resolution, and it is only at small values that the fifth order effects become significant, and all of the resolution measures behave in a similar way. This also shows that the 59% diameter does not predict sub-Angstrom resolution at 100 keV until chromatic effects are dealt with. Therefore monochromation or C_c -correction would be likely to benefit a C_s -corrected, particularly in EELS. Hence a pole-piece that reduces the chromatic aberration even at the expense of other aberrations could be a significant improvement (some consideration of this is given in Chapter 2). It is thus proposed that the chromatic aberration should be used to determine the choice of aperture in a C_s -corrected system operating at moderate voltages.

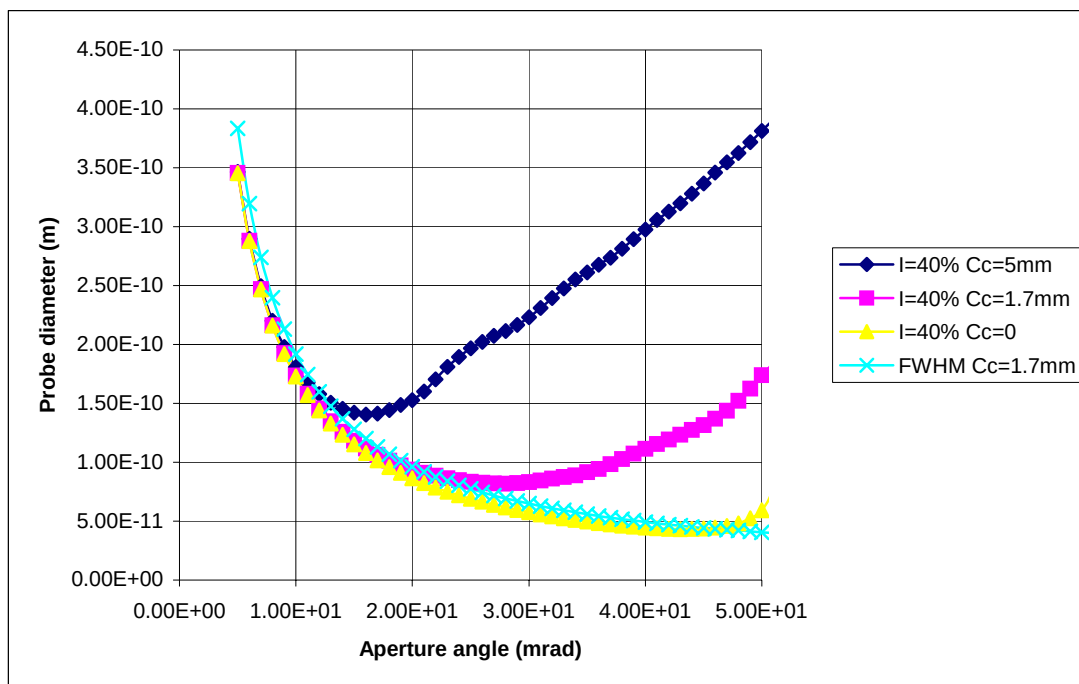


Figure 5.5. Probe size from calculations as a function of aperture angle, for several values of chromatic aberration. The optimal aperture for the value of $C_5=10$ mm is about 44 mrad, but when the chromatic aberration becomes significant this is no longer the best choice. The other imaging parameters are still optimised according to Table 5.1. Chromatic aberration has far less effect on the FWHM than on the current distribution measures, so only one example is shown.

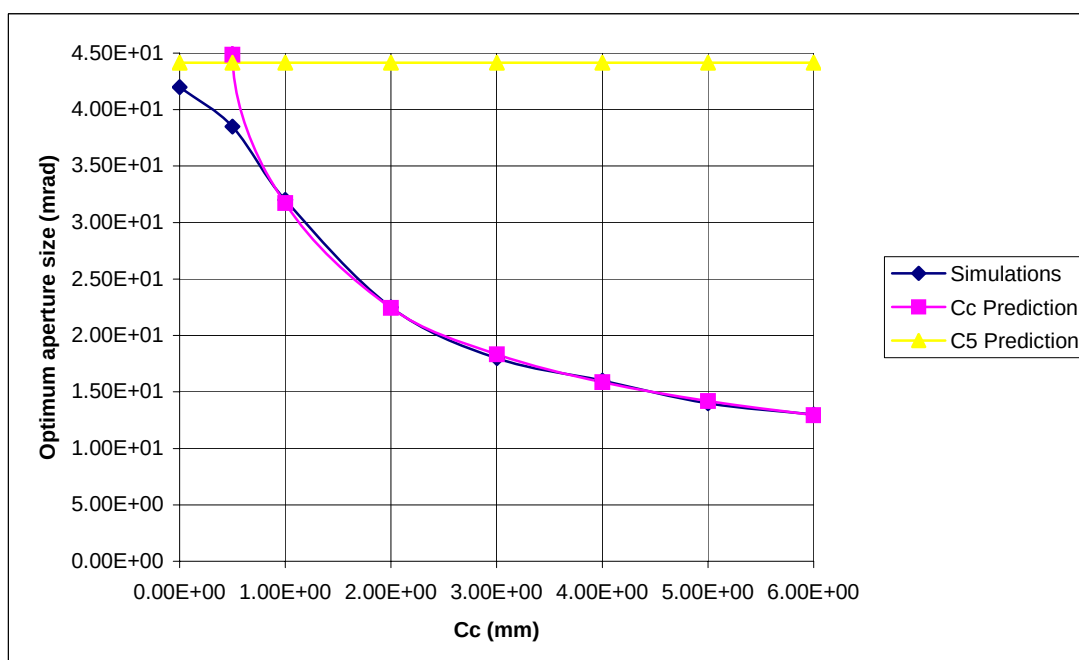


Figure 5.6. Optimal aperture sizes at 100 keV, for $C_c=1.7$ mm, $C_5=10$ mm. The simulations curve is generated by adjusting the aperture until the probe size is minimised, using the 59% diameter. The C_c prediction is from equation (5.6), and the C_5 prediction from Table 5.1(4).

5.2.3. Changing Accelerating Voltage

One method of overcoming the limit imposed by equation (5.7) is to increase the accelerating voltage. This has the advantages of reducing the wavelength, and assuming that the stabilities are kept in proportion the range of focal lengths due to the energy spread, $\Delta f = C_c \Delta E / E_0$, will be decreased. Fig. 5.8 shows the resolution limits calculated for a range of C_c and C_5 values as a function of energy.

Using the value of 10 mm for C_5 , Fig 5.8 shows that resolutions below 1Å are possible at moderate voltages, and it is the chromatic aberration that limits the probe size. This value for C_5 is a better than the value measured from the mark 2 corrector (around 100 mm), but significantly worse than expected in the hexapole or next generation of QO correctors (better than 1 mm). Even large changes in C_5 , up to 1 m, do not have much effect, because both of the C_5 resolution limits in table 5.1 scale as $C_5^{1/6}$. However, the chromatic aberration has a far more dramatic effect, scaling as $C_c^{1/2}$, so even small increases in C_c will degrade the performance. At higher energy values both of the limits on the probe size decrease with a sharper improvement seen in the C_c curve.

One suggested compromise between the problems associated with higher energy and the resolution improvements is to use a 200 keV accelerating voltage (Krivanek et al, 2000a). At this point, even the pessimistic resolution measures are below 1Å. A review of the problems and benefits associated with increasing accelerating voltage is given by Spence (1977). While displacement damage is worse at higher voltages, ionization damage actually decreases (Glaeser, 1975). Therefore increasing the voltage will not necessarily increase all sample damage, and so this is a very promising future direction.

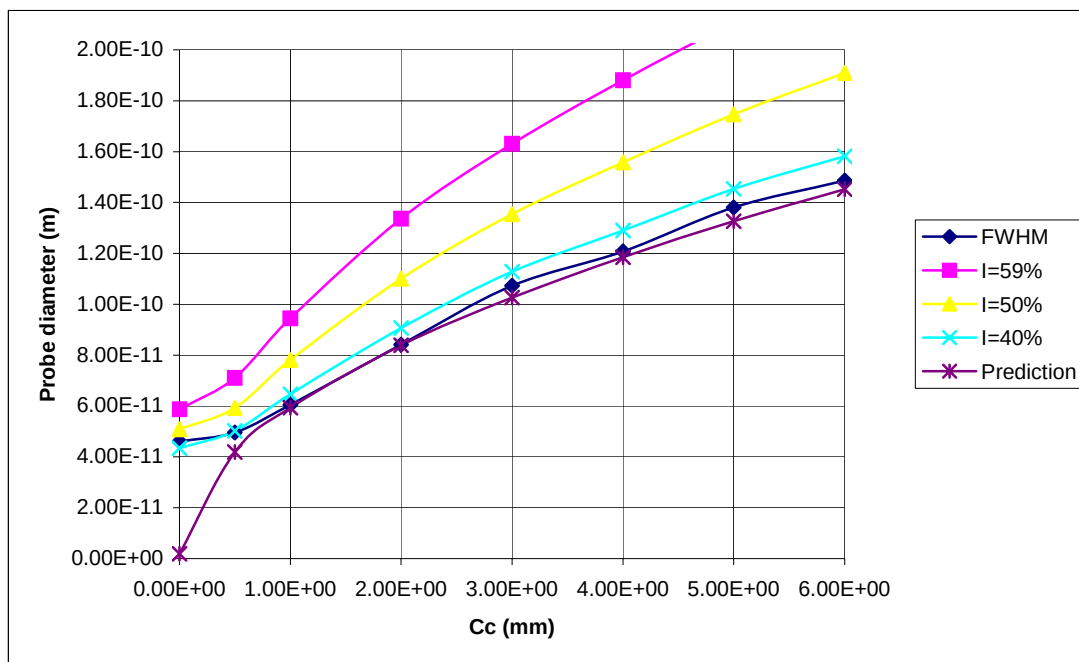


Figure 5.7. Optimal resolution from the numerical calculations that can be obtained at 100 keV for different values of C_c . $C_s=10$ mm, with defocus and C_s optimised according to Table 5.1(6) and aperture optimised as for Fig. 5.6. The predicted curve is based on equation (5.7). At the measured value of $C_c=1.7$ mm the 3 resolution measures from the current distribution are: 1.2 Å, 1.0 Å and 0.8 Å, respectively. Only below $C_c=1$ mm will the C_s limit the resolution.

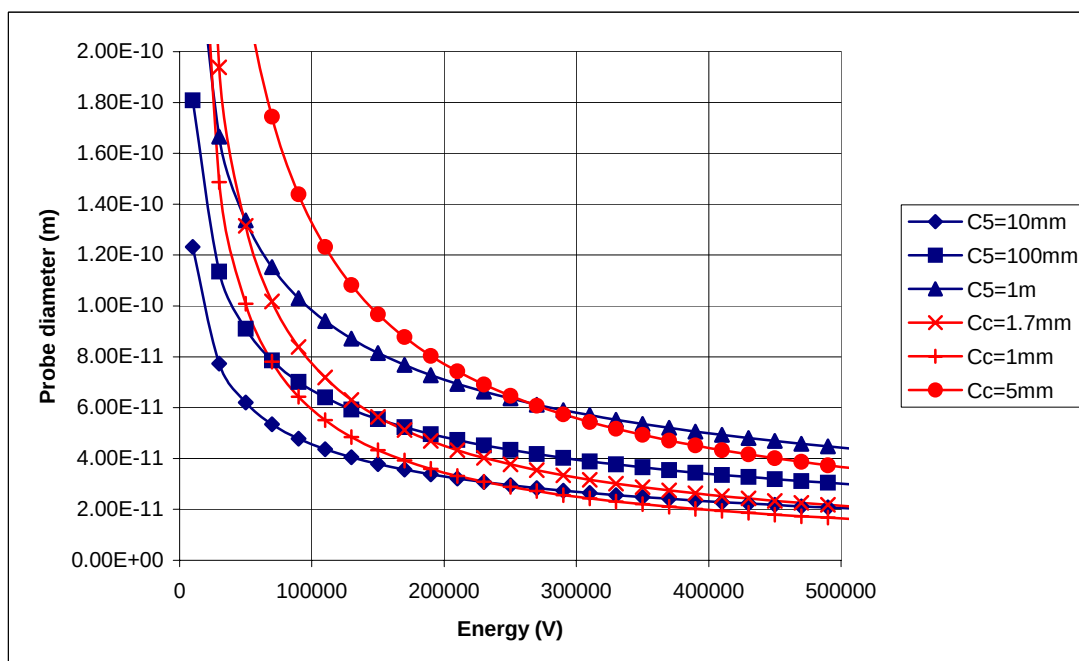


Figure 5.8. Optimal resolution that can be obtained as a function of energy for different values of C_s and C_c . The resolution limit for C_s is calculated from Table 5.2(4), while the C_c limit is based on equation (5.7), which is slightly optimistic.

5.3. Stability criteria

In an unaligned C_s -corrected system, the resolution will firstly be limited by the parasitic aberrations. If these are not minimized then there will be no improvement in resolution. Having defined some measures of the expected resolution it is now possible to determine the precision to which these must be reduced in a C_s -corrected system. It is likely that being unable to measure and adjust these is the main single reason that the earlier aberration correctors were not practically successful. Within this context, the definition of a *tolerance* is how much of a given aberration can be present before it prevents the achievement of a target resolution.

It is clear that the effects of higher order aberrations in particular will depend on aperture size. Hence one benefit of the earlier need to use smaller apertures than the optimal choice for a given C_s is that the stabilities to which the parasitic aberrations must be set will not be so precise. This is because the effects of all aberrations increase with aperture size, and so the effect of undesired aberration will decrease with a smaller aperture. This is shown in Fig. 5.9, where the effect of defocus is shown for several different apertures in a 100 keV system with $C_c=1.0$ mm. A few nanometers change in defocus corresponds to a few parts per million change in the focal length for a typical objective lens. This is a similar scale to the range of focal lengths due to chromatic spread, and will lead to extreme stability requirements. At 10 mrad the probe is almost totally unaffected by a 5 nm change in defocus, but at apertures above 40 mrad, this change yields a worse resolution than the uncorrected case. Thus the choice of aperture will depend heavily on the magnitude of the parasitic aberrations, highlighting the importance of measuring and adjusting these effects.

Simple scaling rules for tolerances can be derived by assuming that the critical factor is the phase change at the aperture edge. So if the aperture is scaled by a factor, S , the effects of aberrations of order, n , will be scaled by S^n . This also applies to the tolerances. Thus if the aperture size is doubled, the precision to which first order aberrations must be stabilized will be twice as strict, and eight times as strict for the third order aberrations. This simple argument is in excellent agreement with the numerical scaling of tolerances noticed by Haider et al (2000).

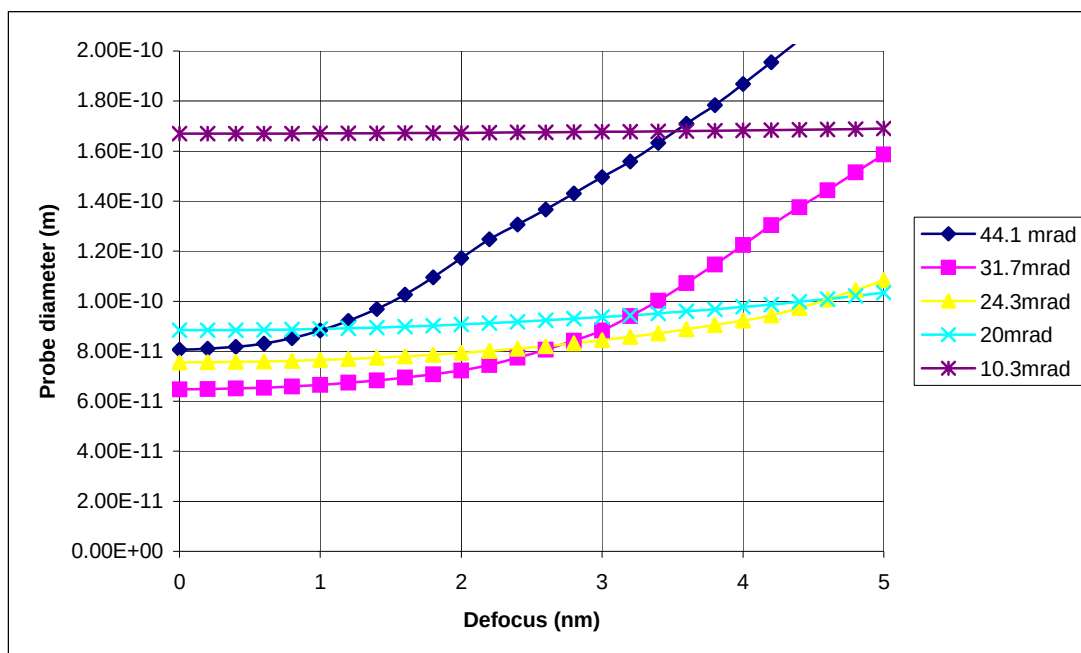


Figure 5.9. Effect of defocus (given as a displacement from the optimised value) on probe size (40% diameter) for different aperture sizes. $C_c=1$ mm, $C_s=10$ mm, at 100 keV. The optimal aperture size for C_s is 44 mrad, for C_c it is 24 mrad and for the uncorrected case would be 10 mrad. The sensitivity to defocus increases with aperture size.

However, some care is necessary when applying this rule in general. The intersects of the curves in Fig. 5.9 depend on the diffraction limit (and C_c), while the shapes of the curves depend on the aberration and aperture size. Thus, the scaling only really applies in the region where the degradation in resolution due to the aberration is sufficiently large compared to the diffraction limit. Table 5.3 shows this quite clearly. The scaling rule applies between the aperture sizes 31.7 and 24.3 mrad. At an aperture size of 20 mrad, the target resolution of $1\text{\AA}$ is close to the diffraction limit, so the scaling is not so accurate.

Fig. 5.10 shows the effects of residual aberrations of different orders. The effects of the aberrations decrease with order because of the aperture sizes (less than 1 radian). It is clear that all of the parasitic aberrations have less effect at the smaller aperture size. There is also some variation with the symmetry of the aberrations.

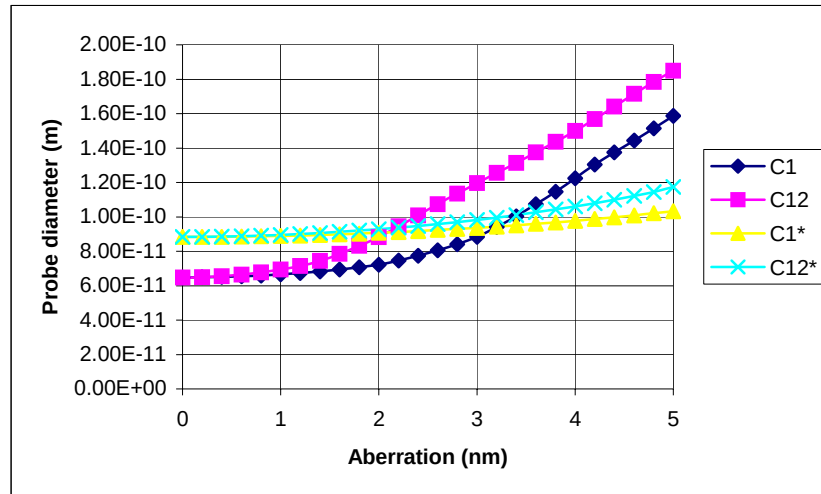
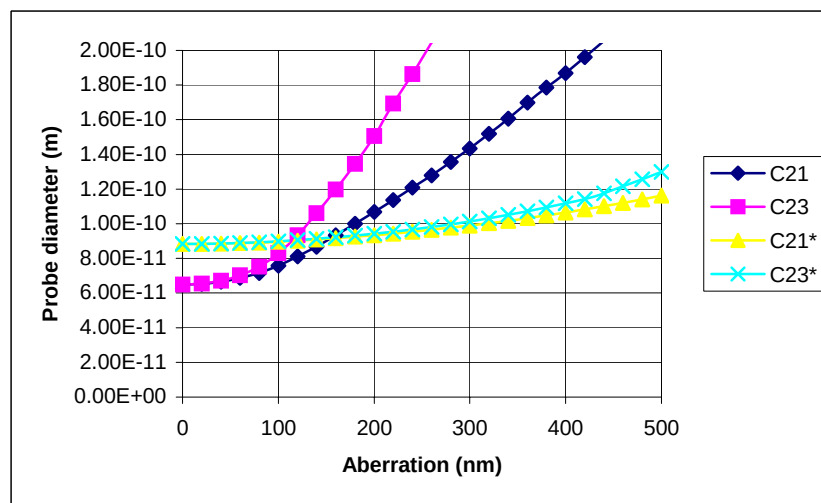
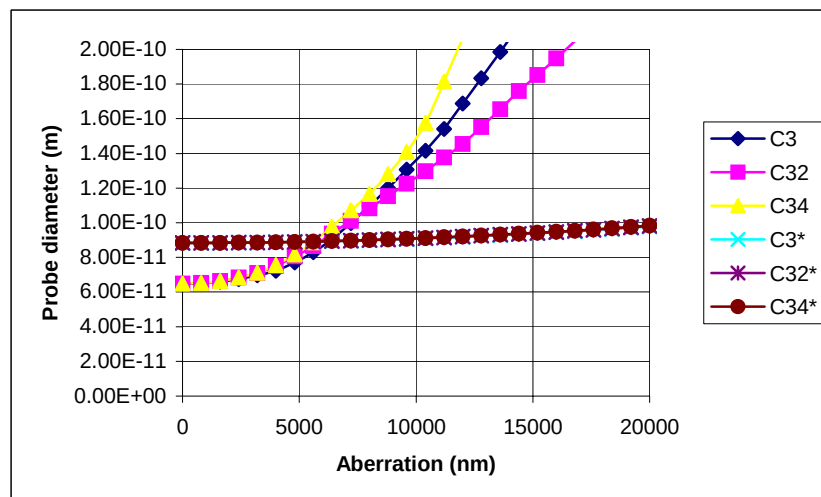
1st Order2nd Order3rd Order

Figure 5.10. (A) The effect of different aberrations by order using the 40% diameter. First to third order aberrations are shown (on different horizontal scales). Two aperture sizes are examined in each case, 31.7 mrad, (the optimal value in this system) and 20 mrad denoted by asterisks.

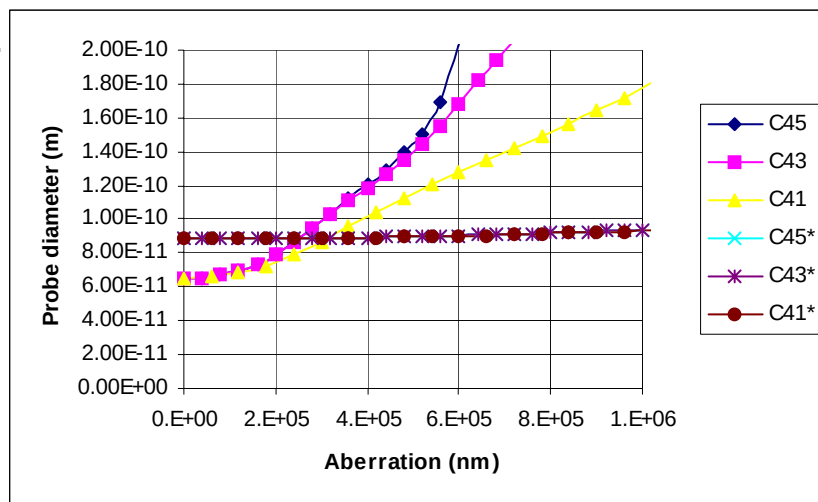
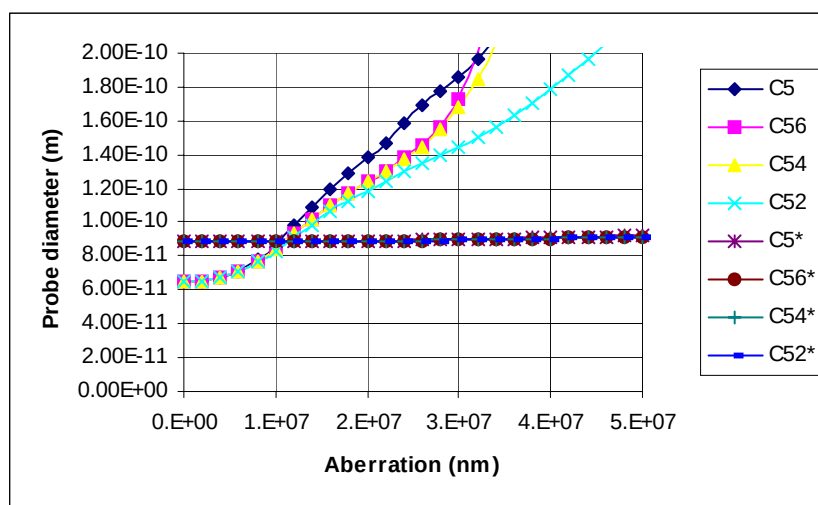
4th Order5th Order

Figure 5.10. (B) The effect of different aberrations by order, using the 40% diameter criteria. Fourth and fifth order aberrations. Two aperture sizes are examined in each case, 31.7 mrad, (corresponding to the optimal value in this system) and 20 mrad denoted by asterisks. Other parameters used were: $C_c=1.0$ mm, $C_5=10$ mm, $E_0=100$ keV, $dE=0.25$ eV, with all other values optimised according to Table 5.1.

Table 5.3 shows the accuracy to which each aberration coefficient must be set in order to obtain a value for the 40% diameter measure of resolution of better than $1\text{\AA}$. These figures are based on considering each aberration in isolation. As is shown by Haider et al (2000), when all of the aberrations are added the resulting resolution can be significantly worse. However in the same way that defocus is used to balance spherical aberration, non-round aberrations can be adjusted to balance different order aberrations of the same symmetry, so this is not always detrimental!

Aberration	Symbol	Maximum value for 1 Å resolution		
Aperture size: (mrad)		31.7	24.3	20
Defocus	C_1	3.4 nm	4.6 nm	4.4 nm
Astigmatism	C_{12}	2.4 nm	3.3 nm	3.3 nm
Axial coma	C_{21}	180 nm	290 nm	310 nm
3-fold Astig	C_{23}	130 nm	240 nm	280 nm
Spherical	C_3	7.2 μm	16 μm	22 μm
2-fold C_s	C_{32}	7.1 μm	16 μm	22 μm
4-fold Astig	C_{34}	6.6 μm	15 μm	22 μm
5-fold Astig	C_{45}	0.3 mm	0.9 mm	1.5 mm
3-fold C_4	C_{43}	0.3 mm	0.9 mm	1.5 mm
4 th order coma	C_{41}	0.4 mm	1.0 mm	1.6 mm
5 th order C_s	C_5	12 mm	44 mm	89 mm
6-fold Astig	C_{56}	14 mm	50 mm	99 mm
4-fold C_5	C_{54}	14 mm	50 mm	99 mm
2-fold C_5	C_{52}	14 mm	50 mm	100 mm

Table 5.3. Precision with which the aberration coefficients need to be set if a resolution of 1 Å is to be attained in a 100 keV STEM. $C_5 = 10$ mm. $C_c = 1$ mm. The criterion used is the 40% diameter, so these values represent minimum stability requirements.

The figures given in Table 5.3 are very close to those given by Krivanek et al (2000a) and Haider et al (2000). The reasons for the differences can be summarized as: The tolerances change slightly (but as will be seen, not significantly) with accelerating voltage and some of the published figures were calculated at 200 keV. Different values are assumed for the source energy spread and value of C_c . The tolerances scale significantly with aperture size and the choice of target resolution. The tolerances will also change with how the resolution is defined, with the published values each using different criteria (the FWHM or different fractions of the current considered resolvable).

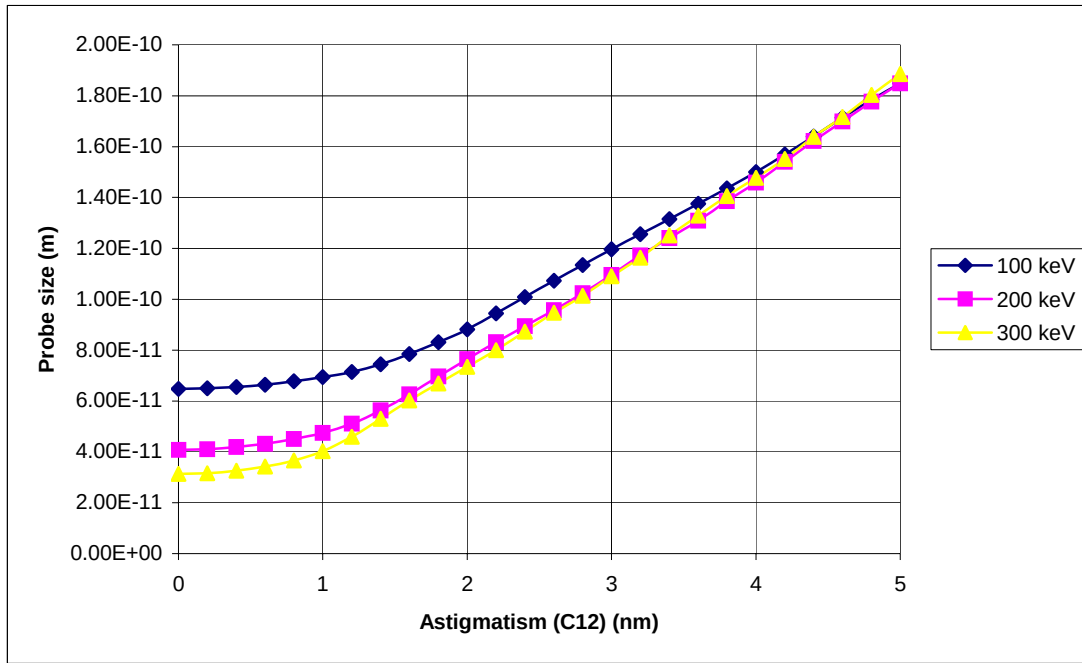


Figure 5.11. Effect of astigmatism on probe size at different accelerating voltages. The angular size of the aperture is kept constant (at 31.7 mrad, the optimal value in this system). $E_0=100$ keV, $C_c=1.0$ mm and all other parameters are optimized according to Table 5.1.

The sensitivity to parasitic aberrations was not normally found to change significantly with accelerating voltage at constant angular aperture size. Some variation was sometimes observed, but in general the stability tolerances were relatively unchanged over the 100 to 300 keV range considered. An example of this is shown in Fig. 5.11. Provided the parasitic aberrations are kept small enough, the resolution is improved due to the reduced wavelength at higher energies. However once the parasitic aberrations become too large they, not the wavelength, limit the resolution. (Like the earlier scaling rules, this argument depends on the significant contribution to the probe size being caused by the aberration itself, not some other effect, such as C_c .) This result is consistent with a geometrical optics viewpoint where the ray deviation does not depend on the energy. Increasing the accelerating voltage does have the advantage that a smaller aperture will give the same resolution, and so in that case the stability requirements will be less severe. The optimal aperture would also increase with accelerating voltage, corresponding of course to an improved resolution (although that can lead to higher stability requirements).

5.4. Bright Field Resolution

This chapter has concentrated on the resolution achievable in ADF STEM because of the higher possible resolution and relative insensitivity to chromatic aberration outlined in Chapter 3. It is now possible to use the formulae in Table 5.1 to examine the phase contrast transfer function expected in an aberration corrected system and verify this numerically. Once again it is the chromatic aberration and the stabilities that will determine the achievable resolution. These are included using the damping envelope from Section 3.3.3. Plotting the PCTF without this damping, but with small errors in defocus, shows the cause of this damping and why C_c is so important for bright-field imaging: At larger apertures even a single nanometer error in defocus gives a significant change in the PCTF, and so the chromatic focus spread causes high spatial frequencies to be attenuated sharply.

Figure 5.12 shows the PCTF for corrected and uncorrected systems at 100 keV. In the uncorrected case the chromatic aberration has almost no effect. The point resolution can be taken as the point where the PCTF has its first zero. This gives a resolution of 3.3 Å, agreeing accurately with the prediction in Table 5.1(1). This doesn't allow for techniques such as pass-band imaging that can transfer higher spatial frequencies (see Spence, 1977, for discussion), but these will be less relevant in C_s -corrected systems because of chromatic effects.

In the C_s -corrected case the higher spatial frequencies are significantly reduced because of the chromatic aberration. A practical measure of the information limit is the point where the chromatic damping factors on the PCTF fall below $1/e$, (Hawkes and Kasper, 1994). For no chromatic spread, the resolution would be better than 0.8 Å, although not quite as high as that predicted in Table 5.1(3). However, when the chromatic damping is included, the expected resolution will be about 1.8 Å. This compares poorly to even the conservative 59% diameter resolution for ADF STEM and is much worse than the more optimistic estimates (see Fig. 5.7). Hence, because the effects of C_c increase with aperture size and the apertures are larger, the insensitivity of ADF imaging to chromatic effects will be a significant advantage. It thus appears that the most dramatic improvements in resolution due to C_s -correction will be seen in ADF images.

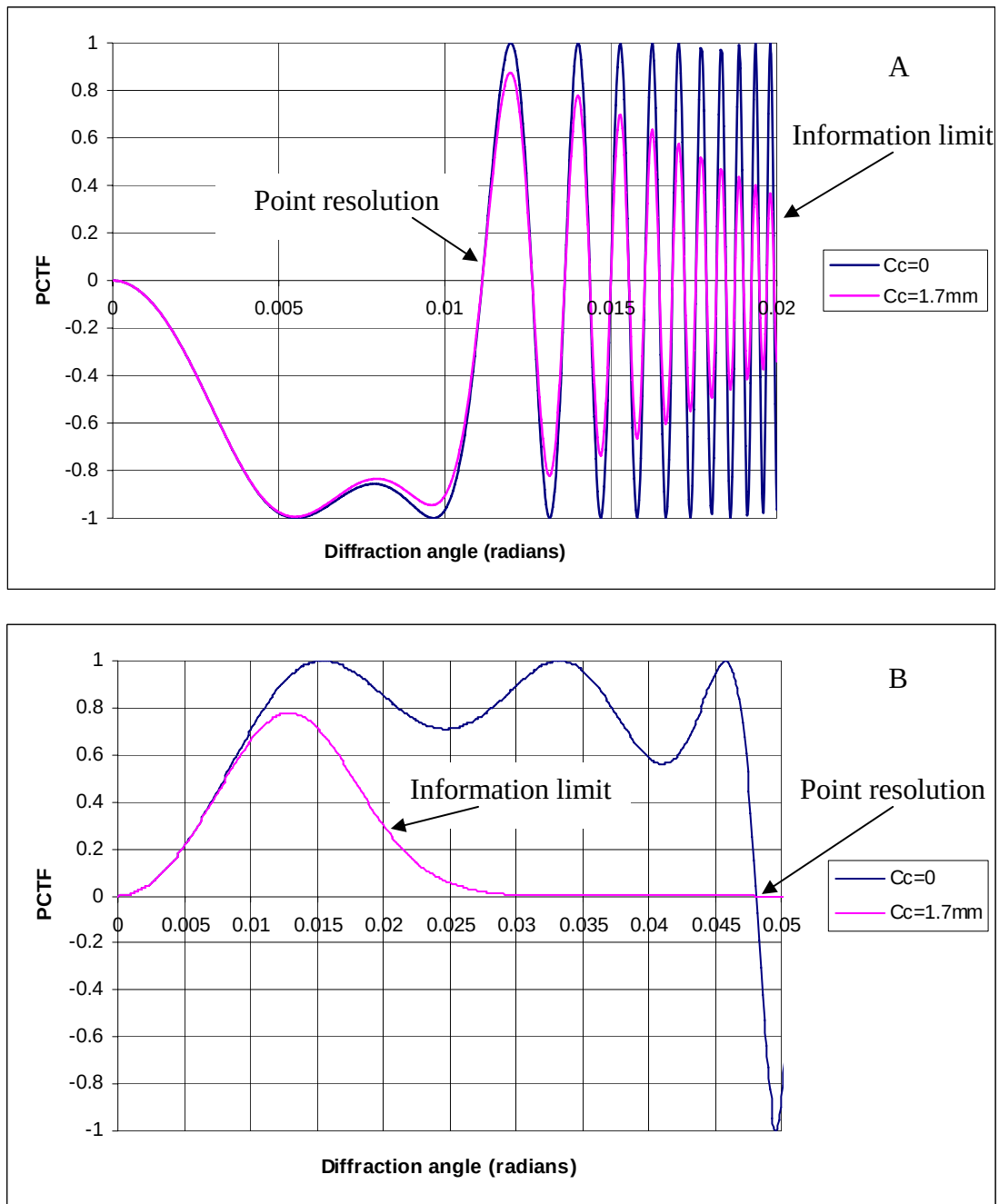


Figure 5.12. The calculated bright field phase contrast transfer functions for (A) uncorrected ($C_s = 1.3 \text{ mm}$) and (B) corrected ($C_s = 10 \text{ mm}$) systems (on different scales). All adjustable parameters are optimized according to Table 5.1, where the apertures just cut out the high frequency oscillations. $E_0 = 100 \text{ keV}$, energy spread $= 0.25 \text{ eV}$. The change of sign is because in (A) underfocus balances positive C_s , but in (B) overfocus and negative C_s balance positive C_s . The uncorrected system is almost totally unaffected by chromatic aberration below the point resolution, but in the corrected case, the information limit is below the point resolution.

5.5. Conclusions

The resolution limits in ADF STEM under the influence of aberrations up to fifth order have been calculated numerically. The current distribution was found to form a better measure of the achievable resolution for ADF STEM than the FWHM. As would be expected, the resolution in the C_s -corrected case is significantly improved. An expression has been given for the resolution limit due to chromatic aberration. If this limit is less than the limit from C_5 alone, then C_c and not the fifth order aberration will limit the resolution. All future aberration correctors should thus be designed in order to rectify or not significantly increase the chromatic aberration.

The tolerances necessary for the achievement of $1\text{\AA}$ resolution have been presented. These also show the precision necessary in the measurement and adjustment of the aberration function. Simplified scaling rules for the effect of aberrations with aperture size have been presented, from which it is apparent that the effects of parasitic aberrations will also be reduced at smaller aperture sizes. At larger aperture sizes, the precision with which all aberrations must be set and power supplies stabilized becomes extreme, so it is these factors rather than the high order aberrations that will limit the resolution. It will probably be necessary to increase the accelerating voltage beyond 100 keV to obtain the highest resolution images. This is because of the decreased sensitivity to chromatic aberration and the reduced wavelength. It will then be possible to obtain a smaller probe at the same aperture sizes, or to use smaller apertures to achieve a given resolution, reducing the stability requirements.

It has been demonstrated that the probe current density will be greatly increased in a C_s -corrected system, even in the case that the total current remains constant. (The current may be constant, despite the larger apertures used after C_s -correction, because the increased source demagnification is needed at higher resolutions). The probe tails are greatly reduced and the current is concentrated in the central peak. This means that a C_s -corrector will give higher resolution EELS analysis as well as higher resolution imaging. The resolution limits given here correspond well to the experimental ADF results, and compare very favorably to the calculated bright field resolution limits, because of the relative sensitivity to C_c . Therefore it is in ADF STEM that the most benefit will be observed from C_s -correction.

Chapter 6

The Electron Ronchigram

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6.1. Introduction

This Chapter will consider some of the general features encountered experimentally in the shadow image that are of particular relevance to aberration correction. This discussion will be concentrated on the shadow images of amorphous materials, for which we will introduce and demonstrate a new method for measuring the aberrations of a STEM.

The shadow image in the scanning transmission electron microscope has been examined in some detail by several authors, most notably by Cowley (1978, 1986). The electron shadow image is directly related to the optical shadow image, which is used to test optical lenses and mirrors. These images are known as Ronchigrams, after Vasco Ronchi (1964) who developed this technique in the 1920's, and so the equivalent images in the STEM are referred to as electron Ronchigrams. The configuration for obtaining a Ronchigram of a weakly scattering object is essentially the same as that required to obtain an inline hologram, originally proposed by Gabor as a method of overcoming spherical aberration (Gabor, 1948).

The typical configuration for obtaining a Ronchigram is illustrated in Fig. 6.1. It is closely related to the more conventional imaging modes of a STEM. As in these other modes, the beam is focused to a sharp probe at the sample. If a conventional image was being obtained, then the probe would be scanned across the sample and the integrated signal at the detector (which spans a finite angular range in the far-field) recorded as a function of probe position. It is however also possible to record the entire 2 dimensional transmitted image in the far field. If only one probe position is used then the resulting image is known as a shadow image or electron Ronchigram (Cowley, 1979a,b).

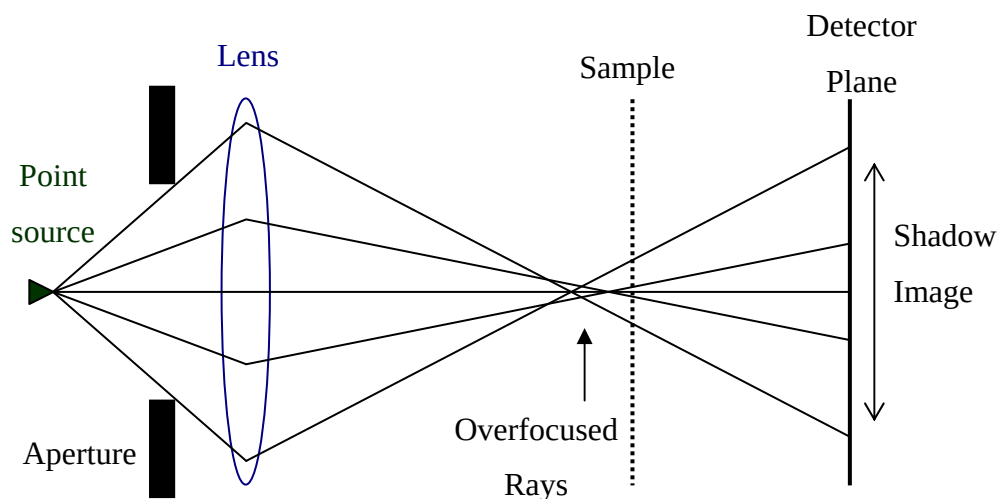


Figure 6.1. Schematic of a typical configuration for obtaining a Ronchigram in either light or electron optical systems. Spherical aberration has caused rays at different angles to be focused in different planes.

The optical Ronchigram is frequently used to analyse the quality of optical elements. A point source is used to provide illumination, which is focused by the element to be tested onto a diffraction grating positioned before a screen. Interference patterns are produced and analysed to give the properties of the element (Ronchi, 1964). The effects can be seen qualitatively because the position of the cross-over will move due to the aberrations and so the magnification in the shadow image will vary as a function of angle. Thus we might expect the appearance of the Ronchigram to be very sensitive to the aberration function. Exactly the same argument applies to the electron Ronchigram. In a relevant example, Deltrap (1964) manually analysed the magnification of a large grid to assess the performance of an aberration corrector.

In the electron microscope, the Ronchigram is employed to provide information on the imaging conditions, but is not widely used for the analysis of materials. Techniques based on analysing the Ronchigram seem particularly well suited for determining the aberration function in the STEM, where regular crystalline structures could be used to provide near perfect diffraction gratings. Indeed, using the Ronchigram as a method of measuring lens aberrations in STEM was proposed and demonstrated by Lin and Cowley (1986), and has since been used by other authors (James et al, 1998). This method relies on the observation and analysis of the interference fringes from a known (crystalline) sample.

By identifying the rings of infinite magnification in the Ronchigram of an amorphous sample (shown in Fig. 6.2 and 6.3), it is possible to measure the *ratios* of defocus and spherical aberration (Cowley, 1979a,b). In a previous paper we proposed that this might be extended to all aberrations if the rings of infinite magnification could be found, by a method such as finding loci of minimum variance (Rodenburg and Lupini, 1999). However it is difficult to locate the rings of infinite magnification automatically and accurately. It will also be shown that this location doesn't provide quite enough information to solve for general super-positions of aberrations. So this is not the best method for measuring the aberration function.

In addition, the Ronchigram can be used to manually align and stigmatize the microscope (Cowley 1979a, Nellist 1995). The Ronchigram is a mix of real and diffraction space information with a magnification that depends on the derivatives of the aberration function (Krivanek, 1997). Hence quantitative interpretation is difficult. As a qualitative manual method of alignment however, Ronchigram analysis is extremely useful, because the symmetry in the Ronchigram is closely related to the symmetry of the aberration function.

It should also be apparent that a more complete dataset than any of the aforementioned imaging modes could be recorded. It is possible to obtain a Ronchigram for each probe position in a conventional image. This would correspond to a 4-dimensional dataset containing information over a range of scattered angles for each probe position. This approach is the basis for super-resolution image reconstruction procedures (Rodenburg and Bates, 1992). The results presented in this chapter are closely related to such work, although here attention will be restricted to single Ronchigrams or sets of a small number of different probe positions only.

The effects of limited coherence and chromatic aberration on Ronchigrams in general are discussed by Nellist and Rodenburg (1994) and by James (1997). These effects result in some blurring and a loss of contrast, in almost exactly the same way as for bright field images, but they will be ignored here.

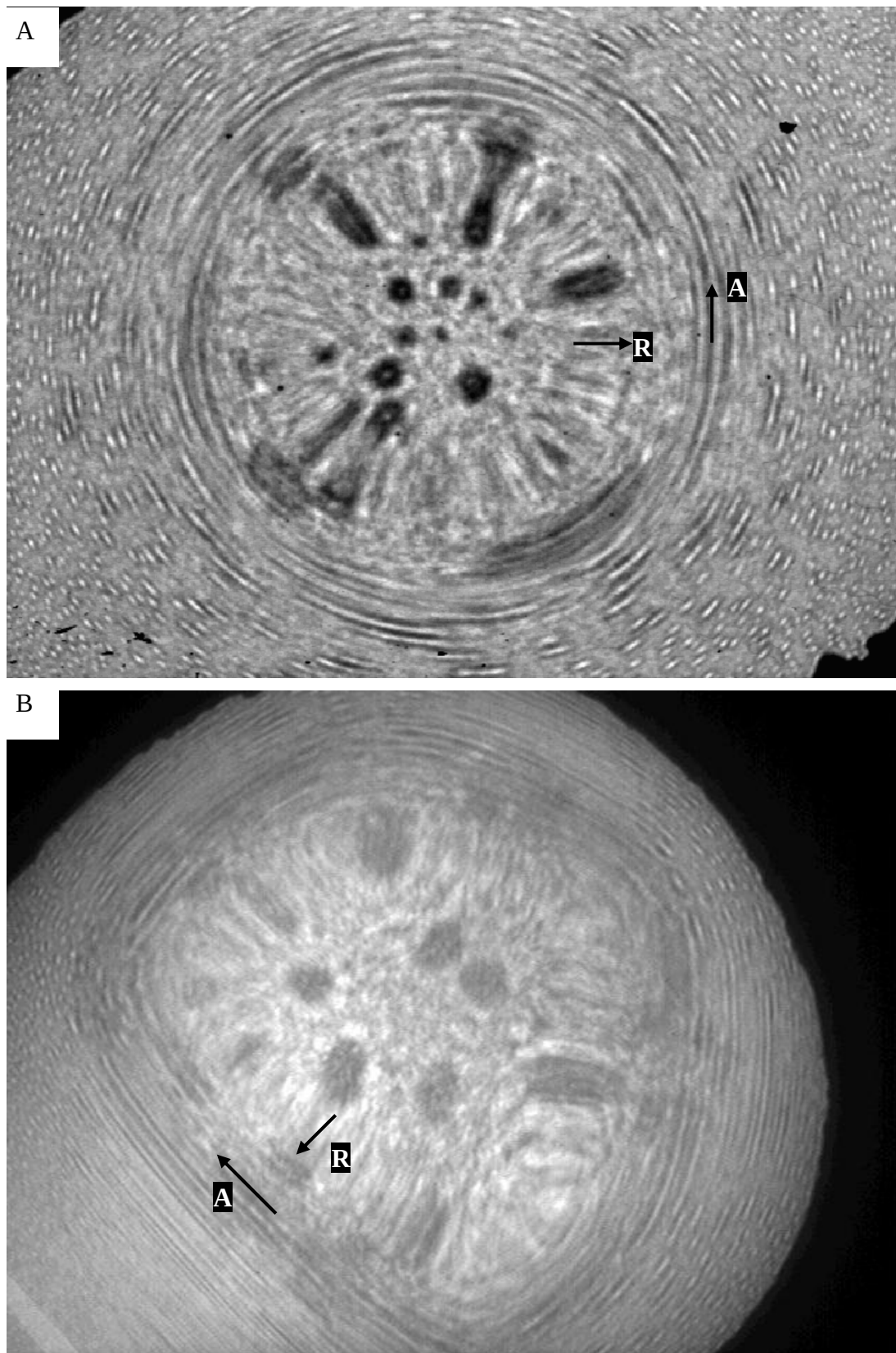


Figure 6.2. Ronchigrams taken in underfocus. Example points of infinite radial (R) and azimuthal (A) magnification are indicated.

(A) In a system dominated by underfocus and C_s , these points form obvious circles.

(B) In a partially corrected system with non-round aberrations, these points form distorted loci.

6.2. Wave Optical Considerations

The applicability of a wave optical approach to the electron Ronchigram was outlined in Chapter 3, and will now be considered further. It is worth mentioning two of the limitations of this approach. Firstly, it was assumed that the specimen could be represented by a 2 dimensional transmission function, this neglects the dependence of the electron-specimen interaction on different incident angles and all dynamical effects. Secondly, it was assumed that the diffraction is purely in the Fraunhofer regime. Experimentally, there are also some Fresnel effects seen at holes in the sample and the small gold particles on the standard test specimen. The appearance of these changes depending on whether the Ronchigram is over or under focused (Fig. 6.18). This is just like at a knife-edge (Cowley and Disko, 1980), but for small round particles. These effects are caused by the Fresnel propagation from the probe cross-over to the specimen. By convention, the rotation from the magnetic lens is also ignored. Despite these limitations, this model provides a good description of the main features in the Ronchigram. The relevance of the Ronchigram to aberration correction rests in the possibilities it offers of quickly and conveniently assessing the aberrations, so the existing techniques to measure these will be briefly reviewed.

Consider the micro-diffraction pattern, as discussed in Chapter 3, from a hypothetical specimen structure that scatters into 2 diffraction spots only. Let these be at reciprocal points $\mathbf{g}$ and $\mathbf{h}$ with complex amplitudes $\Psi_{\mathbf{g}}$ and $\Psi_{\mathbf{h}}$ respectively. In the ideal case, each disk will have uniform intensity and varying phase. Where the disks overlap, the interference will give fringes, which are often encountered experimentally. Equation (3.10) gives the intensity in the Ronchigram for this case as:

$$\left| M(\mathbf{K}_f, \mathbf{R}_0) \right|^2 = \left| A(\mathbf{K}_f - \mathbf{g}) \Psi_{\mathbf{g}} e^{2\pi i (\mathbf{K}_f - \mathbf{g}) \cdot \mathbf{R}_0} + A(\mathbf{K}_f - \mathbf{h}) \Psi_{\mathbf{h}} e^{2\pi i (\mathbf{K}_f - \mathbf{h}) \cdot \mathbf{R}_0} \right|^2 \quad (6.1)$$

Within the overlap region we need consider only the phase part of the aperture function and can thus write the intensity as:

$$\begin{aligned} \left| M(\mathbf{K}_f, \mathbf{R}_0) \right|^2 &= |\Psi_{\mathbf{g}}|^2 + |\Psi_{\mathbf{h}}|^2 \\ &+ 2|\Psi_{\mathbf{g}}||\Psi_{\mathbf{h}}| \cos \left[\mathcal{G}_{\mathbf{g}} - \mathcal{G}_{\mathbf{h}} + 2\pi \left(\lambda^{-1} \chi(\mathbf{K}_f - \mathbf{g}) - \lambda^{-1} \chi(\mathbf{K}_f - \mathbf{h}) + \mathbf{R}_0 \cdot (\mathbf{h} - \mathbf{g}) \right) \right] \end{aligned} \quad (6.2)$$

Where $\mathcal{G}$ denotes the phase of the electron wave associated with each spot. Thus we see the incoherent sum of the intensities and a coherent interference term. This is

discussed in more detail by Nellist (1995). The exact form of the fringes that are generated by the cosine interference term will vary depending on the aberration function. An example of this is shown in Fig. 6.4, and a summary of these 2-beam fringes for different aberrations is given by Mansuripur (1997). This accounts for the fringes seen in experimental Ronchigrams, and the variation of these interfering fringes as a function of probe position provides a useful model for conventional STEM imaging modes (see Spence and Cowley, 1978).

Considering the 3-beam case, the resulting expression is somewhat lengthier, but can be simplified by taking the central unscattered beam (on the axis) to be much stronger than the two scattered beams. This corresponds to the weak phase object approximation for an idealised planar sample. The scattered beams are set to have equal intensities and opposite scattering vectors. In the region where all 3 beams interfere, this gives:

$$\begin{aligned} \left| M(\mathbf{K}_f, \mathbf{R}_0) \right|^2 &= |\Psi_0|^2 + O(|\Psi_g|^2) \\ &+ 2|\Psi_0||\Psi_g| \left[\cos\left(\pi\lambda^{-1}\left(\chi(\mathbf{K}_f - \mathbf{g}) + \chi(\mathbf{K}_f + \mathbf{g}) - 2\chi(\mathbf{K}_f)\right)\right) \right] \times \\ &\left[\cos\left(\pi\left(\lambda^{-1}\chi(\mathbf{K}_f - \mathbf{g}) - \lambda^{-1}\chi(\mathbf{K}_f + \mathbf{g}) + 2\mathbf{g}\cdot\mathbf{R}_0\right) + \mathcal{G}_g\right) \right] \end{aligned} \quad (6.3)$$

This is equivalent to the equation given by Wang and Cowley (1995), but in a slightly more general form. The second cosine term is essentially the 2-beam case, and is modulated by the first cosine term. This results in contrast reversal, the loci of which, for defocus and C_s only, will be in ellipses, as shown in Fig. 6.4. Thus by fitting series of ellipses to these points, it is possible to determine both C_s and defocus. This is covered in more detail by Lin and Cowley (1986), where the effects of other aberrations are also included.

The main problem with the above methods for measuring aberrations is that they only work for samples with suitable structures and spacings. Secondly, it can be difficult to accurately locate the points of contrast reversal, giving a relatively large uncertainty in the measurements. It is also not clear how successfully they could be automated.

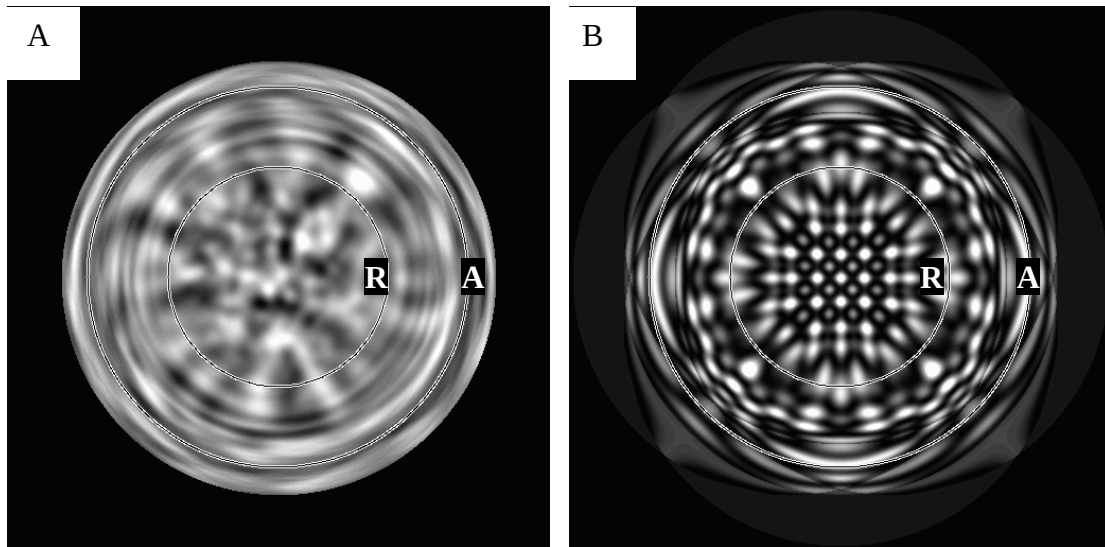


Figure 6.3. Simulated Ronchigrams with underfocus and C_s only for:

- (A) An amorphous sample – the rings of infinite radial (R) and azimuthal (T) magnification and the objective aperture (OA) are marked.
- (B) A crystalline sample – the rings of infinite magnification (marked) are still present but are less visible.

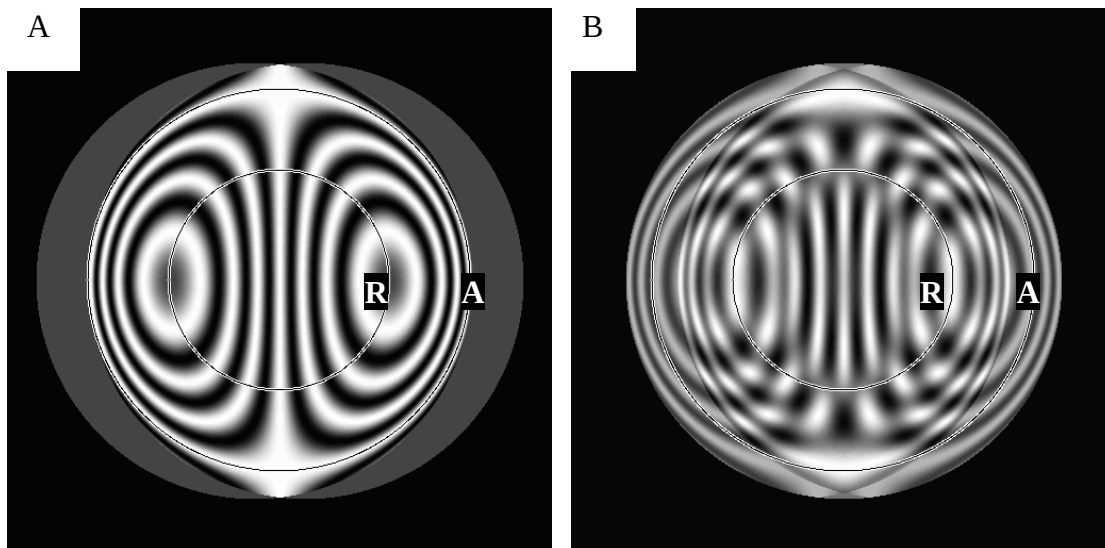


Figure 6.4. Simulated Ronchigrams with underfocus and C_s only for:

- (A) 2 beam case where the interference fringes are distorted by the spherical aberration.
- (B) 3 beam case where the fringes undergo a contrast reversal in ellipses. The locations of the circles of infinite magnification for a corresponding amorphous sample are indicated. The loci of contrast reversal depend on the defocus and C_s as discussed by Cowley (1979b).

The values used in these simulations were: $C_1 = -400$ nm, $C_3 = 1.3$ mm, Aperture = 20 mrad, all spacings = 0.8 nm. Details of the simulations are given in Appendix 3.

6.3. A Geometrical Optics Approach to the Electron Ronchigram

It was shown in Chapter 3 that with the Fourier optics approach, for an amorphous sample, we can take the intensity in the Ronchigram as being equivalent to a distorted bright-field image. Because the sample is amorphous, there is no Bragg diffraction, and so the significant features in the Ronchigram will be dominated by the lens properties. This model is consistent with the geometrical optics approach and thus we can now consider that as a convenient description of the Ronchigram, for a suitable specimen.

A detailed treatment of geometrical optics is given by Born and Wolf (1970). The relationship between the geometrical and wave optical aberrations was introduced in Chapter 2. A ray travelling at a given angle (u, v) to the optic axis will intersect the Gaussian focal plane at a point with coordinates given by the gradient of the aberration function, $\nabla\chi(u, v)$. The central assumption we make is that the intensity at a pixel in the Ronchigram will depend on where the ray that strikes that pixel intersected the sample. For a known sample (in which Bragg diffraction can be ignored) it is then possible to construct the geometrical Ronchigram by deforming an “image” of the sample with the gradient of the aberration function. Following Chapter 3, this “image” is just the bright-field image. The approach used here resembles the “transport of intensity” considerations, which are used to measure the wave-front aberrations in astronomy (reviewed by Ichikawa et al, 1988 and Rousset, 1994). This comparison reveals that we are effectively considering the geometric propagation of the intensity of the exit wave-function from the sample to the far field, which is the approach used by Bajit et al (2000). The overlap between these two models could form the basis for further work. (Reassuringly, if the notation introduced here is used to consider moving the detector along the axis, it ultimately provides the same final expression for the changes of intensity in the resulting image).

It is assumed that the part of the sample illuminated by the beam is on the optic axis and that the post specimen field has no effect. We further assume that the detector, in this case a CCD, is much bigger than the area of the sample that is

illuminated by the probe. This is illustrated in Fig 6.5. For a sufficiently large sample to detector spacing, L , the position at which a ray strikes the detector will be proportional to the angle that that ray makes with the optic axis. Hence the intensity at a pixel in the Ronchigram, at position $\mathbf{Y}$, will be due to the ray travelling at an angle $(u, v) = \mathbf{Y} / L$ to the optic axis. This ray therefore intersected the sample at $\mathbf{X}$, given by:

$$\mathbf{X} = \nabla \chi \left(\frac{\mathbf{Y}}{L} \right) \quad (6.4)$$

The intensity in the Ronchigram will then be given by:

$$I_{\text{Ronchi}}(\mathbf{Y}) = S(\mathbf{X}) \quad (6.5)$$

Where $S(\mathbf{X})$ is a function describing the intensity of each ray leaving the sample at point $\mathbf{X}$. This will be identical to the wave-optical description, equation (3.30), if we recognise that the wave-vector, $\mathbf{K}_f$ is just the position at the detector suitably scaled $(\mathbf{Y}/L)$ and take $S(\mathbf{X})$ to be the intensity of the bright-field image.

It is reassuring to discover that this description of the Ronchigram is essentially the same limiting expression as from the wave-optical expression for an amorphous sample. The approximation made there was that Bragg diffraction was not important, so we would expect agreement. Thus (again) we may view the Ronchigram as being a bright-field image of the sample with a variable magnification depending on the gradient of the aberration function.

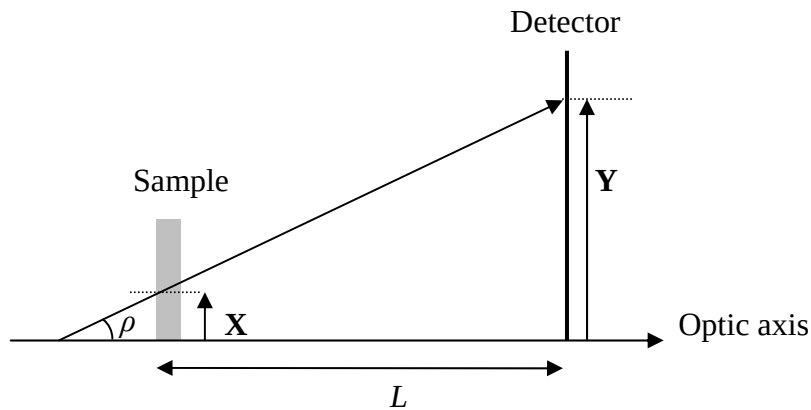


Figure 6.5. Schematic illustrating magnification in a shadow image. $\mathbf{X}$ and $\mathbf{Y}$ are both vectors. Normally L is large and $\mathbf{Y} \gg \mathbf{X}$.

6.3.1. Magnification in the Shadow Image

When considering magnification it is easiest to first consider only the 1-dimensional case. This corresponds to the situation where all of the aberrations are cylindrically symmetrical, which will be true for a correctly aligned and stigmatized round lens system.

From figures 6.5 and 6.6 it is possible to define two sorts of “magnification”:

The first is essentially the same as that given by Cowley (1986). This relates the distance off axis that a given point is in the sample plane to how far off axis the same point will be imaged to at the detector plane. Here this will be referred to as the **absolute** magnification.

The second will be referred to as the **differential** magnification. This relates the size of a feature in the sample plane to how large it will appear to be in the detector plane. This is defined in the limit of an infinitesimally small feature, because both magnifications vary as a function of position. As the name suggests, it is essentially just the derivative of the absolute magnification.

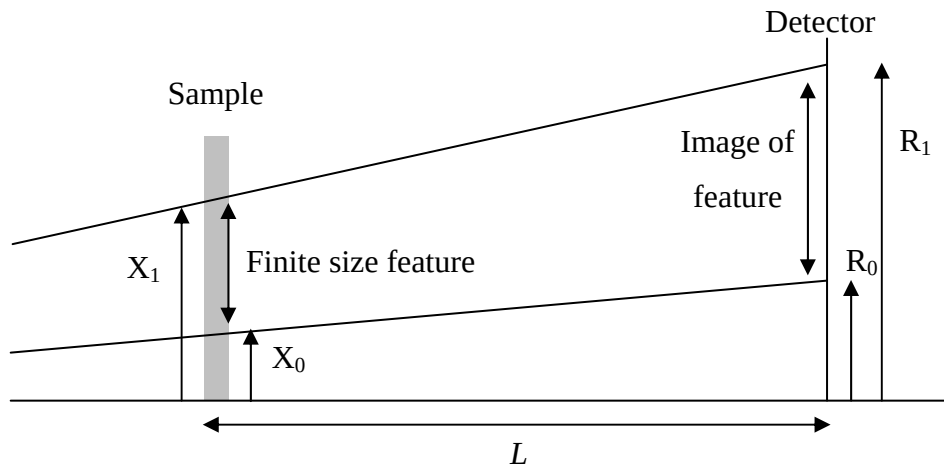


Figure 6.6. Diagram illustrating geometrical magnification in a 1-dimensional shadow image. The points X_0 and X_1 at the sample are imaged to the points R_0 and R_1 in the detector plane.

Figure 6.6 illustrates both sorts of magnification that it is useful to examine in 1-dimension. Consider the limit of two points at the sample (X_0 and X_1) separated by an infinitesimal amount: These will be imaged to the points R_0 and R_1 at the detector, by rays traveling inclined at angles ρ and $\rho+d\rho$ to the optic axis, respectively. In the

rotationally symmetric case, the aberration surface will be a function of the axial angle, ρ , only. So the aberration function can be written as $\chi(\rho)$. All of the derivatives with respect to the azimuthal angle will be zero and the ray deviations will be in the radial direction only. Hence the radius at which a ray intersects the sample will be given by:

$$X_0 = M \left(\frac{\partial \chi}{\partial \rho} \right) \quad (6.6)$$

Where the constant M has been introduced to allow for any extra scaling factors, such as refractive index in optics, but we take as unity. Then if the next ray, travelling at angle $\rho + d\rho$ is considered, for a sufficiently small $d\rho$, this will intersect the sample at:

$$X_1 = M \left(\frac{\partial \chi}{\partial \rho} \right)_{\rho=\rho+d\rho} = M \left(\frac{\partial \chi}{\partial \rho} + \frac{\partial^2 \chi}{\partial \rho^2} d\rho \right) \quad (6.7)$$

Respectively, these points will be imaged at:

$$R_0 = X_0 + \rho L = M \frac{\partial \chi}{\partial \rho} + \rho L \quad (6.8)$$

And:

$$R_1 = X_1 + (\rho + d\rho)L = M \left(\frac{\partial \chi}{\partial \rho} + \frac{\partial^2 \chi}{\partial \rho^2} d\rho \right) + (\rho + d\rho)L \quad (6.9)$$

So, the absolute magnification will be given by the ratio of the 2 distances R_0 and X_0 :

$$M_A = \frac{R_0}{X_0} = 1 + \frac{\rho L}{M \frac{\partial \chi}{\partial \rho}} \approx \frac{\rho L}{M \frac{\partial \chi}{\partial \rho}} \quad (6.10)$$

Similarly, the differential magnification will be given by the ratio of the distance between X_0 and X_1 , and the distance between R_0 and R_1 :

$$M_D = \frac{R_1 - R_0}{X_1 - X_0} = 1 + \frac{L}{M \frac{\partial^2 \chi}{\partial \rho^2}} \approx \frac{L}{M \frac{\partial^2 \chi}{\partial \rho^2}} \quad (6.11)$$

Where it has been assumed that the length, L , is sufficiently large that constants of order unity can be neglected in these expressions.

This can be summarised as showing that the absolute magnification is inversely proportional to the first derivative of the aberration function, and the

differential magnification is inversely proportional to the second derivative (curvature) of the aberration function.

The two circles of infinite magnification in a Ronchigram from a microscope, aligned such that the only aberrations present are under-focus and spherical aberration have been discussed extensively by Cowley (1978, 1979a) and are shown in Fig 6.2. The geometrical model provides a convenient explanation for both circles of infinite magnification in this rotationally symmetric case. These will be where our two different definitions of magnification each tend to infinity. Thus the two radii of interest will be where:

- (1) (Radially) adjacent rays cross each other at the sample, corresponding to the case of infinite differential magnification. This is the point at which the caustic crosses the sample plane (Cowley and Disko, 1980). Each ray at this radius will go through the same point of the sample as the next ray out in the radial direction. The magnification will thus tend to infinity in the radial direction, characterised by radial streaks of uniform intensity. Hence this is known as the circle of infinite **radial** magnification. This can therefore be identified as the point where the second derivative of the aberration function is zero:

$$\frac{\partial^2 \chi(\rho)}{\partial \rho^2} = 0. \quad (6.12)$$

- (2) All rays at a particular radius cross the optic axis at the sample plane, corresponding to the case of infinite absolute magnification. Because of the circular symmetry, all the rays at that radius have been through the same point on the sample. Thus the magnification will tend to infinity in the azimuthal direction and there will be a ring of approximately uniform intensity. This is known as the circle of infinite **azimuthal** (or circumferential, or tangential) magnification. Either side of this radius there is a reversal of the sign of the absolute magnification. For the combined test specimen this will also be associated with a reversal of the contrast of the gold blobs (due to Fresnel diffraction), as the ray cross-overs switch from above to below the sample. This angle can therefore be

identified as the point where the first radial derivative of the aberration function is zero:

$$\frac{\partial \chi(\rho)}{\partial \rho} = 0. \quad (6.13)$$

Thus for spherical aberration (C_3) and defocus (C_1) only, the aberration function will be:

$$\chi(\rho) = \frac{1}{2} C_1 \rho^2 + \frac{1}{4} C_3 \rho^4 \quad (6.14)$$

Evaluating the first and second derivatives of this gives the circle of infinite radial magnification at:

$$\rho_{radial} = \sqrt{\frac{-C_1}{3C_3}} \quad (6.15)$$

And the circle of infinite azimuthal magnification at:

$$\rho_{azim} = \sqrt{\frac{-C_1}{C_3}} \quad (6.16)$$

These results are consistent with those given by Cowley (1986), which are derived from wave optical considerations. The location of these two circles is found experimentally to agree with these predictions, and they also correctly locate the rings in purely wave-optical simulations. This is illustrated geometrically in Fig. 6.7.

The geometrical optics approach illustrates why there are two different styles of circles of magnification. However, it should be observed that this simple argument (like that in Cowley, 1986) is only really valid for the circularly symmetric aberrations. Since these round aberrations dominate most of the images in any conventional electron microscope, this condition gives approximately correct values for cases with small amounts of non-round aberrations. Clearly if neither of these conditions is satisfied, we will not see any rings of infinite magnification, which is why they are normally seen only in under-focus for positive spherical aberration. However, in the situation of an aberration corrected microscope, it is possible to obtain negative C_5 . In that case, these rings can be present in over-focus. This argument also demonstrates that higher order round aberrations will also give circles of infinite magnification under appropriate circumstances.

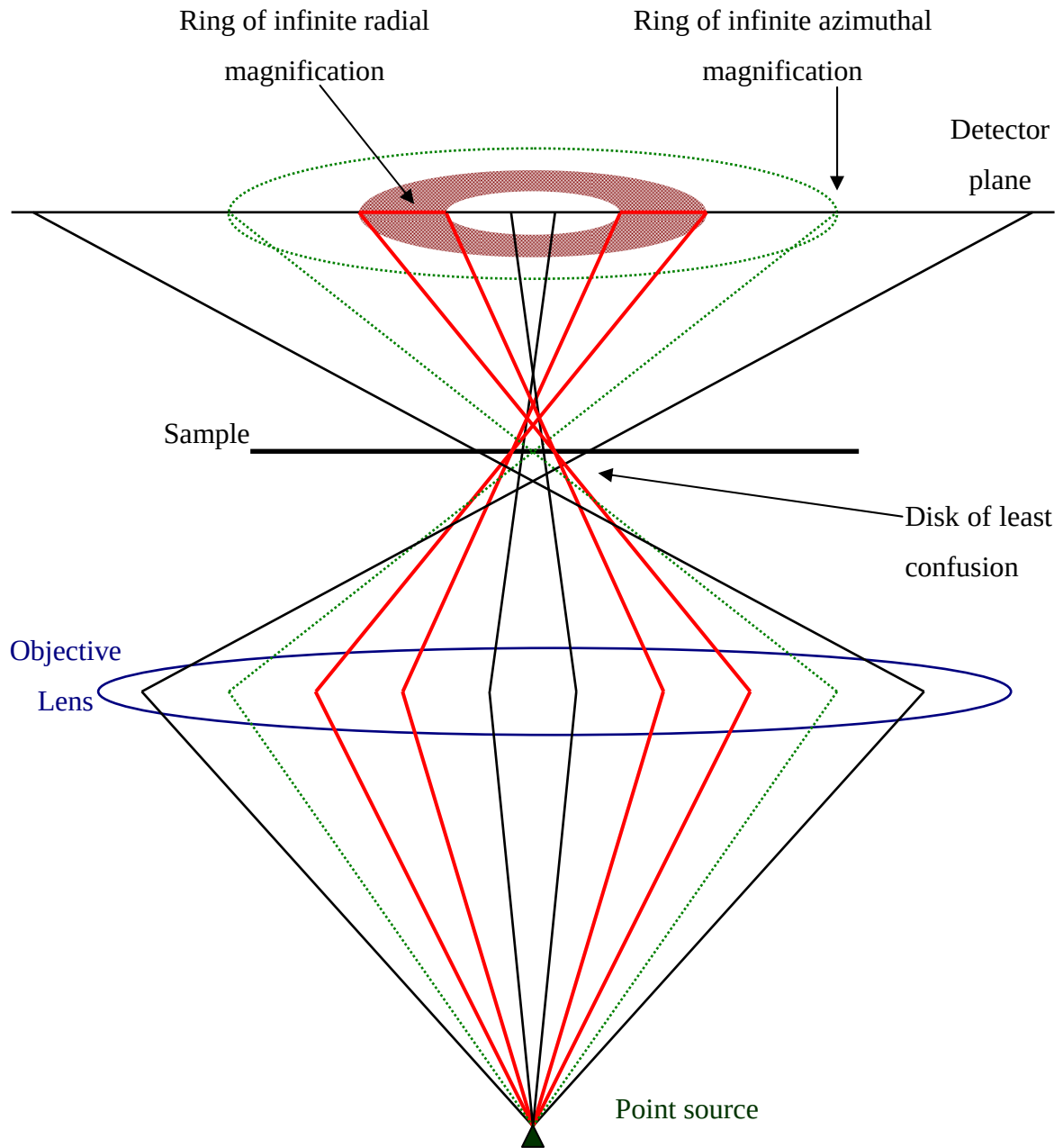


Figure 6.7 Schematic of the formation of a Ronchigram using a geometrical optics approach, for the circularly symmetric case of defocus and spherical aberration only. The green dotted lines indicate rays that cross the optic axis at the sample, and so all of the rays on the dotted circle have been through the same point on the sample. The re weighted lines indicate radially adjacent rays that cross each other at the sample, so those rays have been through points at the same radius. These 2 circles of infinite magnification are seen experimentally in Fig. 6.2, with simulations in Fig. 6.3. The disk of least confusion has a diameter of $0.5C\rho^3$ at an underfocus of $0.75C\rho^2$ from the paraxial focus, for an aperture of half-angle ρ .

6.3.2. Two Dimensional Magnification

In the 2-dimensional case, things are a little more complicated. The rings of infinite magnification will not always be circles. The simple 1-dimensional argument is then no longer the correct way to proceed. The question of what form the points of infinite magnification take in the general case has been tackled in an earlier paper, but no satisfactory answer has yet appeared (Rodenburg and Lupini, 1999). Examples of non-round loci of infinite magnification are given in Fig. 6.2 and 6.9.

Simulations suggest that the equations derived for the rotationally symmetric case are no longer valid for non-round aberrations. So further consideration is necessary to locate these points. Comparison to the transport of intensity equation used in adaptive optics (Ichikawa et al, 1988) suggests that they might be located where the Laplacian of the aberration function is zero, but again this was not consistent with simulations. However, equation (6.5) suggests that we should just consider the Ronchigram imaging process as a mapping from the sample to the image plane. Suppose that we are transforming between the coordinate sets at the detector (u,v) and the sample (ξ,η) , as defined in Fig. 6.8. In the limit of an infinitesimal area element, dA , the ratio of the areas in the 2 coordinate systems is given by the Jacobian, J , as (see Arfken, 1985 or Riley, 1974, for derivations):

$$dA_{\xi\eta} = J dA_{uv} \quad (6.17)$$

Where:

$$J = \frac{\partial \xi}{\partial u} \frac{\partial \eta}{\partial v} - \frac{\partial \xi}{\partial v} \frac{\partial \eta}{\partial u} \quad (6.18)$$

Here, equation (6.4) gives:

$$\xi = \frac{\partial \chi}{\partial u}, \eta = \frac{\partial \chi}{\partial v} \quad (6.19)$$

This mapping is not conformal because it does not obey the Cauchy-Riemann equations and so angles will not always be preserved. From equations (6.18) and (6.19) the appropriate Jacobian is therefore given by:

$$J = \frac{\partial^2 \chi}{\partial u^2} \frac{\partial^2 \chi}{\partial v^2} - \frac{\partial^2 \chi}{\partial u \partial v} \frac{\partial^2 \chi}{\partial v \partial u} \quad (6.20)$$

Where as usual: $\frac{\partial^2 \chi}{\partial u \partial v} = \frac{\partial^2 \chi}{\partial v \partial u}$.

Hence the Jacobian gives the ratio of the area of a small element in the sample to the area of the element at the detector that it is imaged into. This is just the 2-dimensional analogue of the earlier differential magnification, as shown in Fig. 6.8. The 2-dimensional differential magnification can therefore be defined as:

$$M_{2D} = \frac{dA_{uv}}{dA_{\xi\eta}} = \frac{1}{J} \quad (6.21)$$

So we will expect infinite magnification of 2-dimensional area elements where a finite patch at the detector, dA_{uv} , is the image of a patch at the sample of zero area, $dA_{\xi\eta} = 0$. This will clearly be satisfied where $J=0$.

For aberration functions in which the Jacobian has zeros, the magnification will locally tend to infinity at those points. Exactly what will be visible there in the image is not as obvious as in the 1-dimensional case. The appearance will vary with the direction that the magnification is tending to infinity and how contiguous any such regions are. The rotationally symmetric case is once again a useful example. The difference between the two circles of infinite magnification is just the direction that the magnification tends to infinity. Here:

$$\chi(u, v) = \frac{1}{2}C_1(u^2 + v^2) + \frac{1}{4}C_3(u^2 + v^2)^2 \quad (6.22)$$

Then J is given by:

$$J = (C_1 + C_3(3u^2 + v^2))(C_1 + C_3(u^2 + 3v^2)) - (2C_3uv)^2 \quad (6.23)$$

Using $\rho^2 = u^2 + v^2$, the condition $J=0$ is therefore satisfied when:

$$0 = (C_1 + 3C_3\rho^2)(C_1 + C_3\rho^2) \quad (6.24)$$

Which gives the same 2 critical radii as before, (6.15) and (6.16), and hence **both** circles of infinite magnification are solutions of $J=0$.

It is thus adequately justified to conclude that the loci of infinite magnification are given by the points that satisfy:

$$\frac{\partial^2 \chi}{\partial u^2} \frac{\partial^2 \chi}{\partial v^2} - \frac{\partial^2 \chi}{\partial u \partial v} \frac{\partial^2 \chi}{\partial v \partial u} = 0 \quad (6.25)$$

This approach has successfully predicted the locations of both of the rings of infinite magnification in the rotationally symmetric case. These are both solutions of equation (6.25), although this approach did not give any useful information about the appearance of these points, unlike the simpler 1-dimensional argument. However, the two loci are circles and have such different appearances only in the special case of a rotationally symmetric system, so the fact that we do not know which sort of infinite magnification is present is not particularly significant. Simulations indicate that this approach is indeed successful for all symmetries and orders of aberration that are considered. (An example simulation is shown in Fig. 6.9.) Exactly the same condition will also be found in the probe shifting analysis, which serves to confirm this result.

The circles of infinite magnification are sometimes visible for crystalline samples, but not always as the 3-beam case indicates. It appears that the magnification does tend to infinity at those points, although the discrete diffraction pattern has sampled the aberration function only at particular points. Thus the simple description based on amorphous materials is not valid for crystalline samples and these circles cannot always be located in such cases, as illustrated in Fig. 6.3 and 6.4.

In summary, we have clarified the appearance and location of rings of infinite magnification that are seen experimentally in Ronchigrams of amorphous materials, and related the magnification in these images to the aberration function. This will be shown to be relevant to the issue of aberration correction by providing a convenient new method of automatically analysing the aberration function of the microscope.

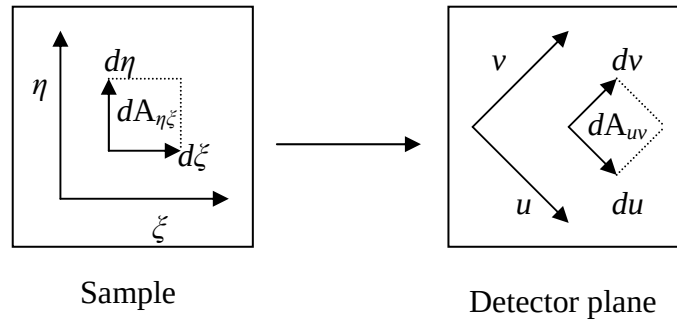


Figure 6.8. Illustration of 2-dimensional differential magnification showing the coordinates used. The small patch at the sample, $dA_{\eta\zeta}$, is mapped onto a small patch, dA_{uv} , at the detector plane. We expect infinite magnification where a zero area patch at the sample is mapped to a finite size patch at the detector.

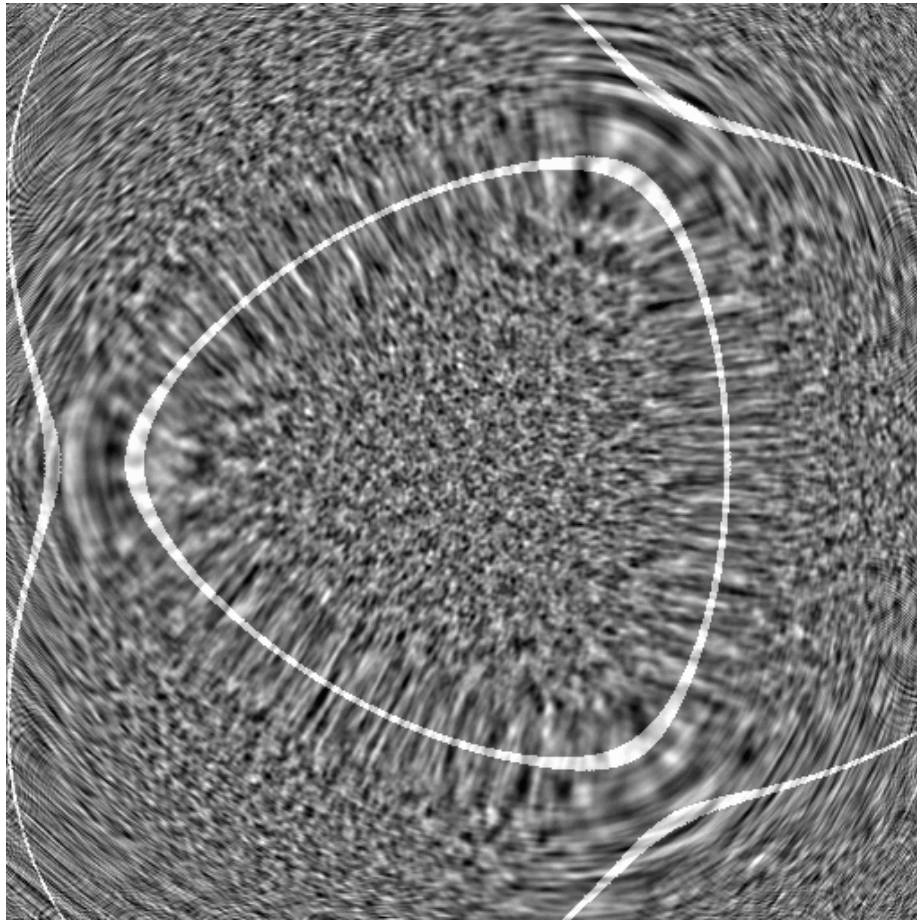


Figure 6.9. A (wave-optical) simulation of a Ronchigram with defocus=-1.5 μm , $C_s=1$ mm, and $C_{23a}=8$ μm . In this case the loci of infinite magnification are no longer circles. The points that satisfy equation (6.25) are highlighted, and can be seen to accurately locate the loci of infinite magnification.

6.4. Quantitative Ronchigram Analysis

Having given a relation between the magnification in the Ronchigram and the lens aberrations, it is possible to quantitatively determine the aberration surface. Exactly how this should be done will clearly depend on how the magnification is extracted. The first way of doing this that will be considered is to move the probe.

6.4.1. Effects of Probe Movement

If the STEM probe is shifted by a small amount then we would expect the apparent motion in a sufficiently small patch in the Ronchigram to depend on the magnification at the angle that patch corresponds to. As for equation (6.5), we write the intensity in the Ronchigram, I_0 , as:

$$I_0(\mathbf{Y}_0) = S(\mathbf{X}_0) \quad (6.26)$$

Where $\mathbf{X}$ is position at the sample and $\mathbf{Y}$ is position at the detector, which are related by:

$$\mathbf{X}_0 = \nabla \chi \left(\frac{\mathbf{Y}_0}{L} \right) \quad (6.27)$$

If the probe is shifted by a small amount $d\mathbf{X}$ to a position $\mathbf{X}_1$ such that $\mathbf{X}_1 = \mathbf{X}_0 + d\mathbf{X}$, then each part of the image, which starts at $\mathbf{Y}_0$, will shift by $d\mathbf{Y}$, to the point $\mathbf{Y}_1 = \mathbf{Y}_0 + d\mathbf{Y}$. For sufficiently small shifts so that we need only retain the leading order terms, we can write:

$$\mathbf{X}_1 = \nabla \chi \left(\frac{\mathbf{Y}_0 + d\mathbf{Y}}{L} \right) \approx \nabla \chi \left(\frac{\mathbf{Y}_0}{L} \right) + \nabla \left(\frac{d\mathbf{Y}}{L} \cdot \nabla \chi \left(\frac{\mathbf{Y}_0}{L} \right) \right) \quad (6.28)$$

For this to be equivalent to the definition of $\mathbf{X}_1$ requires:

$$d\mathbf{X} = \nabla \left(\frac{d\mathbf{Y}}{L} \cdot \nabla \chi \left(\frac{\mathbf{Y}_0}{L} \right) \right) \quad (6.29)$$

Thus we have related the probe shift, $d\mathbf{X}$, to the **apparent** shifts of infinitesimal parts of the image, $d\mathbf{Y}$. Terms on the right-hand-side of this equation can be recognised as the second derivatives of the aberration function. To clarify the notation, this can be written in terms of the ray inclinations to the optic axis (u and v), which were given by:

$$(u, v) = \frac{\mathbf{Y}}{L} \quad (6.30)$$

Then (6.29) becomes:

$$d\mathbf{X} = \left(\frac{\partial^2 \chi}{\partial u^2} du + \frac{\partial^2 \chi}{\partial u \partial v} dv, \frac{\partial^2 \chi}{\partial u \partial v} du + \frac{\partial^2 \chi}{\partial v^2} dv \right) \quad (6.31)$$

Where du and dv are the components of the image shifts defined such that $(du, dv) = d\mathbf{Y} / L$. For compactness, this is expressed as:

$$d\mathbf{X} = \underline{\mathbf{H}} \frac{d\mathbf{Y}}{L} \quad (6.32)$$

Where $\underline{\mathbf{H}}$ is the so called Hessian matrix of second derivatives of the aberration function evaluated at (u, v) :

$$\underline{\mathbf{H}} = \begin{pmatrix} \frac{\partial^2 \chi}{\partial u^2} & \frac{\partial^2 \chi}{\partial u \partial v} \\ \frac{\partial^2 \chi}{\partial v \partial u} & \frac{\partial^2 \chi}{\partial v^2} \end{pmatrix} \quad (6.33)$$

The determinant of this matrix is the Jacobian, which was introduced earlier as the ratio of the differential areas in the two coordinate frames; the zeros of which we claimed gave the loci of infinite magnification. It is then possible to predict the expected image shifts in terms of a known aberration function for a given probe shift, as:

$$d\mathbf{Y} = L \underline{\mathbf{H}}^{-1} d\mathbf{X} \quad (6.34)$$

Where $\underline{\mathbf{H}}^{-1}$ is the inverse of $\underline{\mathbf{H}}$:

$$\underline{\mathbf{H}}^{-1} = \frac{1}{\frac{\partial^2 \chi}{\partial v^2} \frac{\partial^2 \chi}{\partial u^2} - \frac{\partial^2 \chi}{\partial u \partial v} \frac{\partial^2 \chi}{\partial v \partial u}} \begin{pmatrix} \frac{\partial^2 \chi}{\partial v^2} & -\frac{\partial^2 \chi}{\partial u \partial v} \\ -\frac{\partial^2 \chi}{\partial v \partial u} & \frac{\partial^2 \chi}{\partial u^2} \end{pmatrix} \quad (6.35)$$

Which will tend towards infinity where the determinant of $\underline{\mathbf{H}}$ tends to zero. This confirms that the points where (6.25) was satisfied can indeed be identified with the loci of infinite magnification in the Ronchigrams, as the expected shifts will also tend to infinity at those points.

Having related apparent shifts in the image to the probe movement, it is now possible to determine the aberration function at each point in the image. Consider shifting the probe by two different amounts $d\mathbf{X}_1$ and $d\mathbf{X}_2$, giving apparent shifts $d\mathbf{Y}_1$ and $d\mathbf{Y}_2$. Then if these both started from the same point, $\mathbf{X}_0$, it is possible to find the Hessian matrix by the following method: Construct a matrix $d\underline{\mathbf{X}}$, such that the

columns are the two column vectors $d\mathbf{X}_1$ and $d\mathbf{X}_2$, and similarly construct $d\mathbf{Y}$, such that the columns are the two vectors $d\mathbf{Y}_1$ and $d\mathbf{Y}_2$. Then equation (6.32) gives:

$$d\mathbf{X} = \frac{1}{L} \mathbf{H} d\mathbf{Y} \quad (6.36)$$

We can thence solve for $\mathbf{H}$ at each point using:

$$\mathbf{H} = L d\mathbf{X} d\mathbf{Y}^{-1} \quad (6.37)$$

This notation makes it clear that to find the second derivatives of the aberration function, we now just have to invert the matrix $d\mathbf{Y}$. This will fail if the determinant of $d\mathbf{Y}$ is zero (or experimentally, just too close to zero). The best way of avoiding this will be to choose $d\mathbf{X}$ carefully, for instance by using two orthogonal shift directions. It is also desirable to sample both directions evenly in order to obtain unbiased values for the aberration coefficients, so although orthogonality is not required, it will reduce the errors.

In order to use this technique to measure the aberration function of the system, we must first measure the apparent shifts as a function of position within the Ronchigram image. Then the local aberration surface can be reconstructed from the second derivatives thus found. Knowing all four second derivatives provides more than enough information to reconstruct the aberration function, $\chi(u, v)$, uniquely at each point (apart from constant or linear terms). From these discrete values, it is possible to fit a suitable aberration function for the imaging system. The redundancy in this method will also provide information about the rotation of the objective lens, as a function of angle. In the current software, this is presently treated as a constant and used to check the accuracy of the analysis. A least squares fitting method is then used to find the aberration coefficients. The method used for reconstructing the aberration function from its second derivatives is described in Appendix 2.

This technique, which will be referred to as “Ronchigram Analysis,” was used successfully to measure the aberration function, both in simulations and experimentally.

6.4.2. Patterns, Patches and Correlations

Having presented the theoretical basis of the Ronchigram Analysis technique, some of the details of the implementation can now be considered. The first stage is to accurately measure the apparent shifts in the Ronchigram image with changes in probe position.

The apparent shifts are extracted by means of cross-correlations between Ronchigrams taken at different probe positions. As this is being done as a function of angle, the Ronchigram must be divided into small subsections and the correlations done between corresponding patches in different Ronchigrams. Each patch will therefore correspond to a particular scattering angle, although the finite extent of each patch means that it will really cover a range of angles. This procedure is illustrated in Fig. 6.10. Patches are cut out of each Ronchigram and masked in both real space and the Fourier domain. A cross-correlation is simply a convolution without the associated change of sign for the second function, so it is most simply accomplished using the Fast Fourier Transform.

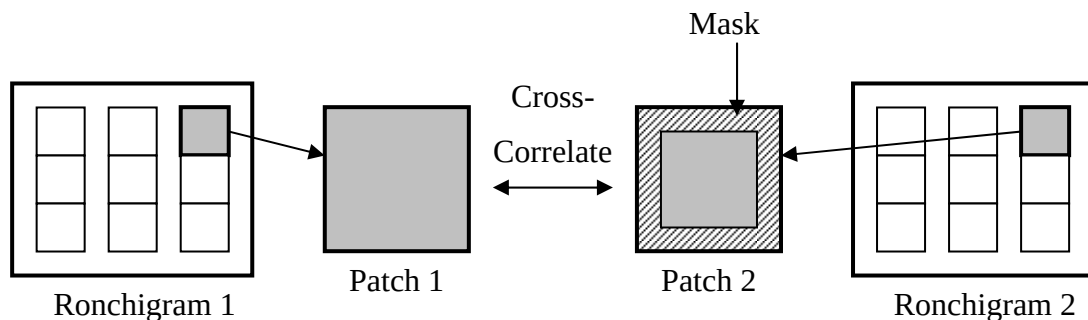


Figure 6.10. Ronchigram 1 is taken at one probe position and divided up into square patches. This is repeated for Ronchigram 2 at a different probe position. Each patch from Ronchigram 2 is masked at the edges and cross-correlated with the patch at the same position in Ronchigram 1. The cross-correlation will show how much the features in the patch have shifted. Patches can overlap, but sizes should be a power of 2. The number of patches used is really limited only by processing speed.

This process is repeated for a number of different patches at different angles, which generates a table of shifts. Equation (6.37) is used to convert these to a table of second derivatives as a function of angle. These are then used to calculate the aberration function, as described in Appendix 2.

The patches are filtered in the Fourier domain to remove spurious effects from the acquisition process. It is also possible to boost certain spatial frequencies to help the cross-correlation, which improves the accuracy in the best case. However, this was also found to enhance the noise, leading to an increased chance of a totally spurious correlation.

Because the discrete finite Fourier transform implicitly assumes periodicity, the borders of one of the patches must be masked in real space, (with the average intensity) to avoid wrap-around errors. The pixel shifts should be smaller than the width of the masked area to obtain a symmetric cross-correlation peak. Distortion of this peak is a problem because it is necessary to interpolate to find the maximum to an accuracy of better than 1 pixel. This is only easily possible if the peak is symmetric. The interpolation method used here was to fit a quadratic to the values in the cross-correlation using the pixels either side of the highest valued pixel. For undistorted, simulated images, the errors in the measured shifts were better than 0.01 pixels at the typical patch sizes used. With suitably small image distortions and shifts, repeated measurements suggest that the accuracy achieved with real Ronchigrams was around 0.1 pixels in practice. Experimentally the probe movement and angular ranges used were adjusted until the pixel shifts were around 10-20 percent of the patch size. This was small enough to produce **reliable** cross-correlations and also large enough to reduce the percentage error due to the limited accuracy. The places in the Ronchigram that the patches are taken from govern how the angular space is sampled. Square patterns were normally used, but circular patch layouts were also implemented in order to obtain less biased estimates for the coefficients (shown in Fig. 6.11).

The difficulties with this method arise because the magnification varies within each patch, so the cross-correlations are not always good. In order to obtain a patch with uniform magnification, we require each patch to subtend a small angle. However, cross-correlations in the presence of noise will need as many pixels as possible to give

accurate results, requiring each patch to be as large as possible. These conflicting requirements do not present serious difficulties, but will ultimately determine the achievable accuracy. One further possibility is to perform a second cross-correlation to obtain corrections based on the first measured value. However, our earlier description also means that the image will be distorted, which will still affect the cross-correlations. Using the expected shifts to undo the distortion forms the basis of the “unwarping” method. Some other techniques exist to find shifts accurately (Saxton, 1994) but these didn’t seem to work well here, probably because of the small size of the patches used.

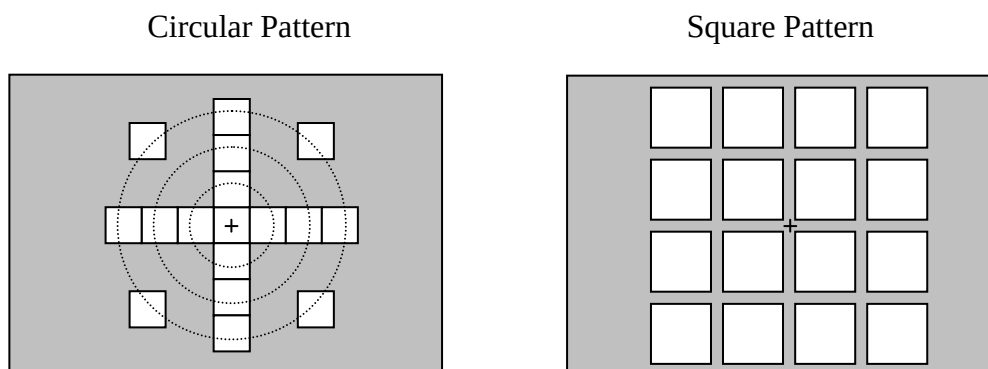


Figure 6.11. Example patch patterns used in the Ronchigram analysis. The shaded area represents a single Ronchigram. Rectangular and circular grids are shown. The location of the optic axis is indicated with a cross. The patches are not drawn to scale and can overlap. The location of the patches can be adjusted to sample angular space appropriately, and the size varied such that each patch subtends a suitable angle. Any number of patches can be used, and should depend on the order of the analysis.

6.4.3. Drift Subtraction

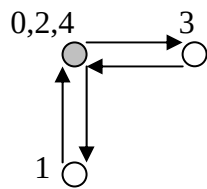
As with any technique that relies on taking sequential images and finding the relative shifts, this method is susceptible to specimen drift, which would give spurious results in the cross-correlations. Experimentally this method should be less susceptible to this problem than other aberration measuring procedures in STEM, because of the relative speed of image acquisition. The Ronchigrams are recorded on a CCD and this can obtain suitable quality images at TV rates. In practice, to boost the signal to noise ratio, the signals were often averaged over several frames, but this does not significantly increase the acquisition time. Total acquisition times were typically below 1 second for a complete set of Ronchigrams. Similarly the scan coils can move the probe extremely rapidly and accurately in a STEM. This avoids the aperture charging and spurious shifts that can cause problems with beam tilting in CTEM.

To improve the accuracy of this technique, two different methods were employed. Rather than just recording Ronchigrams at two probe positions, we can then record a third Ronchigram at the first position. The sets of shifts between the first and second Ronchigrams, and between the third and second Ronchigrams are then measured. It is then possible to average these two measurements, giving the shifts with the linear drift cancelled. A less satisfactory way of achieving the same effect would just be to average the two Ronchigrams from the first point, and find the shift between this compound image and the second one. For drifts much smaller than the image feature size, this will perform satisfactorily, but it might give somewhat spurious results for large drifts. In general this is a slightly less desirable method, but it does have the advantage of being very fast. In practice, both of these methods seemed to perform well experimentally.

It is also essential to use two orthogonal probe directions, and so the 2-dimensional analogues of this method are illustrated in Fig. 6.12. This will require a series of five Ronchigrams, taken at the three different positions indicated. After some experimentation, more probe positions were used on the opposite sides of the original position. This effectively uses twice the probe movement that was possible between just two positions. Again, by averaging the measured shifts, the linear drift can be

cancelled. Similarly this could be extended to any number of positions limited only by how long it takes to acquire and process the Ronchigrams.

(A) 3 Probe positions
5 Ronchigrams



(B) 5 Probe positions
10 Ronchigrams

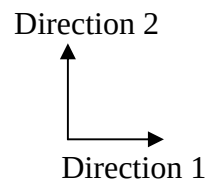
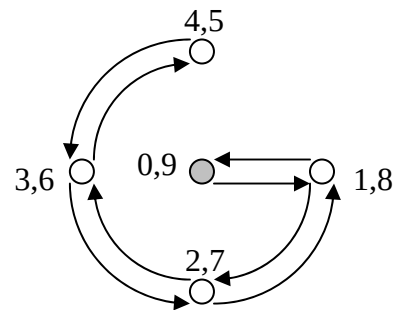


Figure 6.12. Probe positions used to test this technique. Starting at the central position, a series of Ronchigrams is recorded at each marked position in the sequence shown. It is necessary to use 2 orthogonal directions as shown. Pattern (A) uses 5 images, and (B) uses a total of 10 images, on *either side* of the central position. This improves the accuracy with more measurements and larger shifts (between opposite pairs) at the expense of longer processing. Directions 1 and 2 are defined.

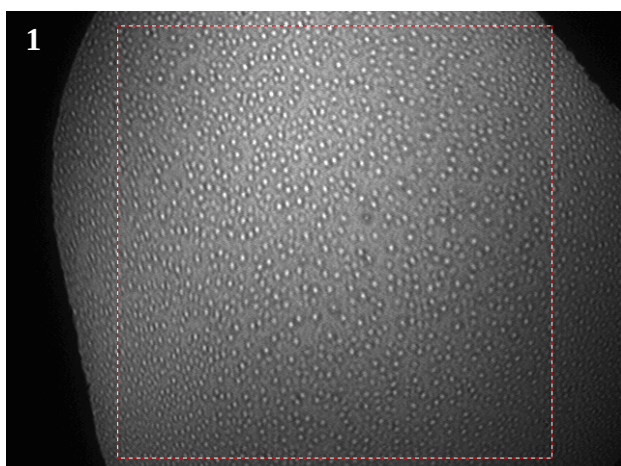
6.4.4. An Implementation of the Ronchigram Analysis

As an extension of the theoretical analysis of the Ronchigram that has been given, it was possible to develop software to automatically measure the aberration function based on the Ronchigram analysis method. Details of the implementation are further described in the appendices.

The graphical output from the first stage of the analysis is shown in figures 6.13 to 6.15. In this example, a series of Ronchigrams is used from the aberration corrected HB501. These correspond to a C_s -corrected system that has not been very accurately aligned, and so has significant parasitic aberrations remaining. First a series of Ronchigrams is recorded, shifting the probe to each of the 10 positions shown in Fig. 6.12(B). The two Ronchigrams at each point are simply averaged, as described, to cancel the linear drift. This results in a tableau of five Ronchigrams, as shown in Fig. 6.13. These are then divided up into small patches, and cross-correlations are performed to find the shift as a function of position in the Ronchigram. This generates two sets of shifts for each direction, which are averaged to give only one set of shifts, with a reduced error. These are now referred to as the image shifts. The average shift for the whole image and the shifts at each point, are indicated by the arrows in Fig. 6.13.

The components of the shift at each point are displayed again graphically, as shown in Fig. 6.14. The brightness in these images is used to indicate the magnitude of the shifts. This requires a series of four images, for the two components of the shifts (x and y), for the each of the two probe movement directions (1 and 2). These are then converted to the four second derivatives of the aberration function, using equation (6.37), which are displayed in Fig. 6.15. Finally those are converted to the aberration coefficients and displayed, as shown in Table 6.1.

For reference, the probe shifts and second derivatives for the example case of C_s and underfocus only are shown in figures 6.16 and 6.17, respectively. These correspond closely to the forms that are predicted by the formulae for the second derivatives given in Chapter 2.



Schematic of the probe position on the sample for each of these images:

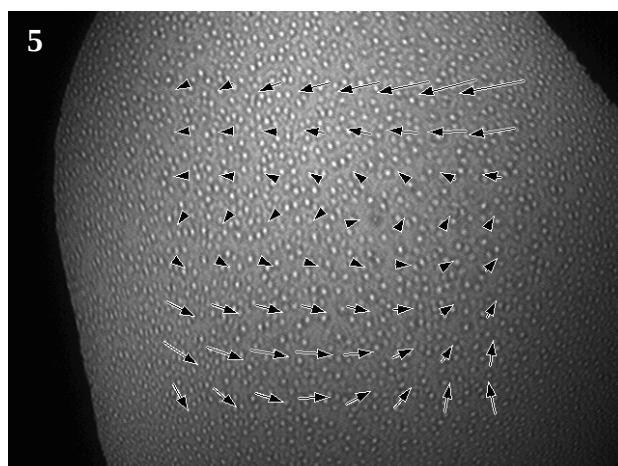
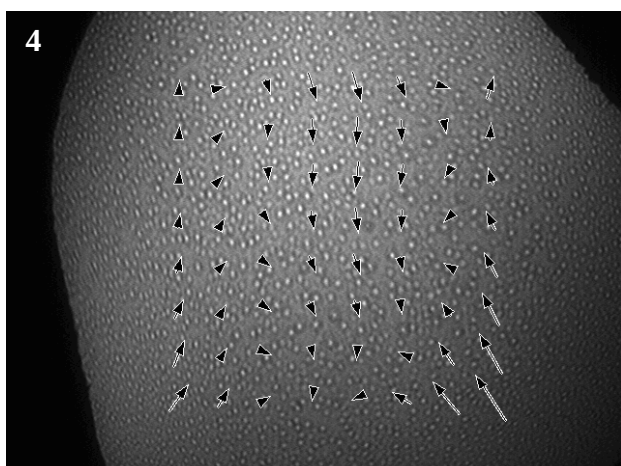
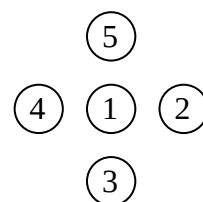


Figure 6.13. A tableau of 5 Ronchigrams from the corrected Nion HB501 in over-focus for small movements of the probe. The schematic indicates the direction of the probe shifts. The box in image 1 indicates the area used for the cross-correlations. The arrows in images 2 and 3 indicate the direction of the average shift of the images (magnified by a factor of 12). The arrows in images 4 and 5 indicate the shifts as a function of position (with the average subtracted) and then magnified by a further factor of 4.

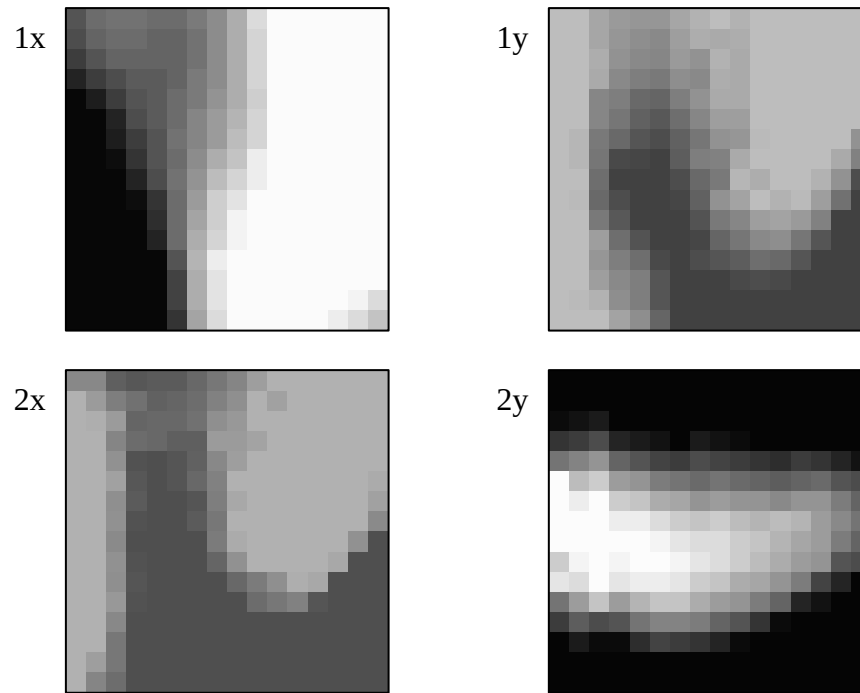


Figure 6.14. Apparent shifts in the Ronchigrams shown in Fig. 6.13. The intensity of the pixels represents the magnitude of the shift in the patch that each pixel corresponds to, on a rather arbitrary scale. These are separated into the x and y components of the shifts for each of the two probe movement directions 1 and 2.

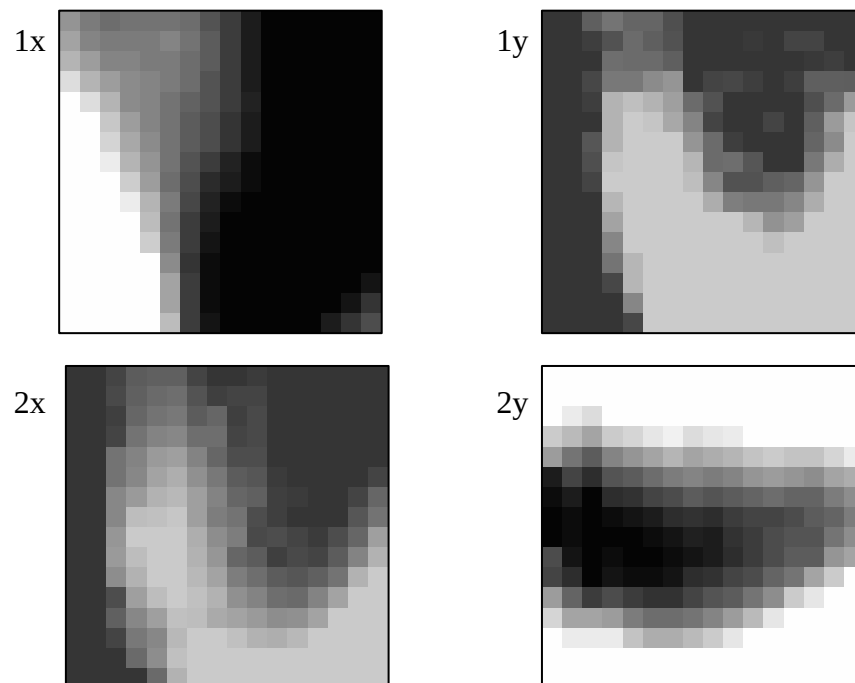


Figure 6.15. The shifts from Fig. 6.14 are translated into the second derivatives of the aberration function. These appear similar because the axes of the shifts were chosen to correspond approximately to the camera orientation. These are then converted to aberration coefficients, given in table 6.1, which can then be displayed graphically as in Fig. 6.19.

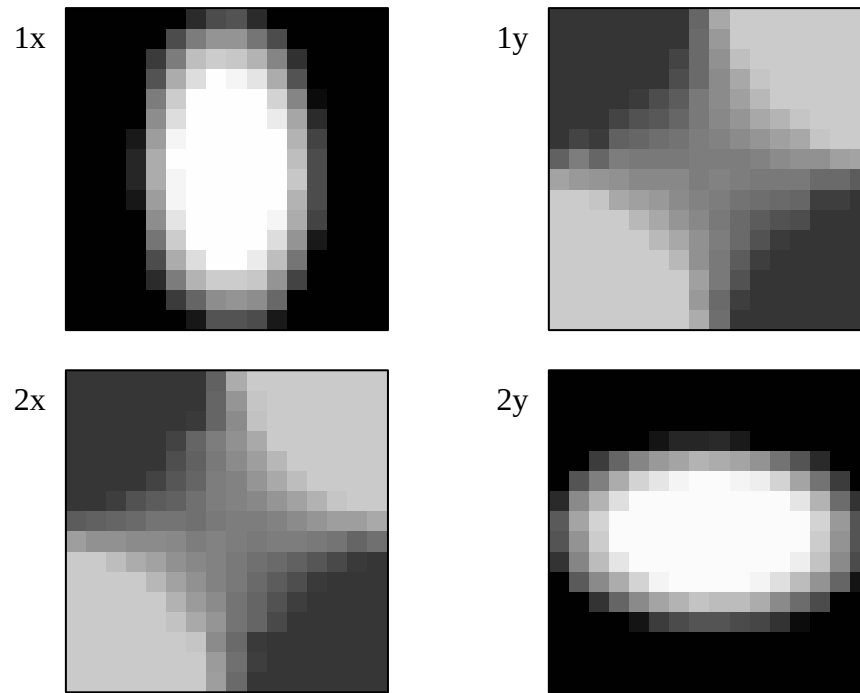


Figure 6.16. Apparent shifts of the Ronchigrams for the example case of C_s and defocus only, measured from a simulated tableau of Ronchigrams. Here the values used were $C_s = 0.1$ mm defocus $= -1.2$ μm .

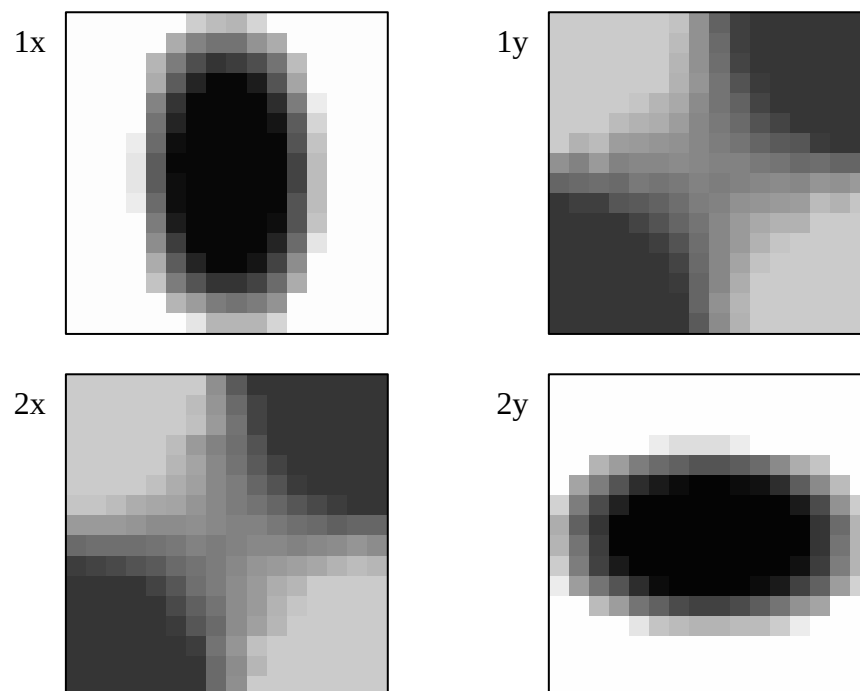


Figure 6.17. The shifts from Fig. 6.16 are translated into second derivatives of the aberration function. Again the similarities are due to the fact that the shift axes were aligned with the camera orientation. These correspond closely to the formulae for the second derivatives given in Chapter 2.

Coefficient	First Pass	Second Pass
C_1	977.557	969.307
C_{12a}	98.1279	104.498
C_{12b}	-38.821	-34.6081
C_{21a}	-4728.05	-5475.62
C_{21b}	-4302.94	-5278.04
C_{23a}	-2599.97	-2670.46
C_{23b}	-1380.06	-1601.48
C_3	116727	75371.3
C_{32a}	-67585.1	-76558.5
C_{32b}	-11191.5	-215.422
C_{34a}	41473.6	44779.8
C_{34b}	-19260.5	-13751.9
C_{41a}	-1.92E+06	-805433
C_{41b}	4.78E+06	4.94E+06
C_{43a}	-1.36E+06	-1.73E+06
C_{43b}	3.09E+06	3.22E+06
C_{45a}	21023.9	-119931
C_{45b}	852314	853231
C_5	3.91E+07	6.30E+07
C_{52a}	2.91E+07	2.26E+07
C_{52b}	-1.52E+07	-1.74E+07
C_{54a}	-2.48E+07	-2.65E+07
C_{54b}	-1.47E+07	-1.79E+07
C_{56a}	678640	4.23E+06
C_{56b}	-1.65E+06	-975371

Table 6.1. Measured aberration coefficients (in nm) from the Ronchigram analysis for the Ronchigrams in Fig. 6.13. The first pass is used to obtain an estimate for the aberration function, which is then used as the basis for the unwarping procedure allow a more accurate second pass. In this case, although the spherical aberration has been greatly reduced, the parasitic aberrations have not yet been fully adjusted. (These figures are a little worse than the more carefully tuned case presented by Dellby et al, 2001).

Table 6.1 shows the aberration coefficients that were measured from this tableau of Ronchigrams. These show that although the spherical aberration has been reduced to below 0.1 mm, all of the parasitic aberrations have not been eliminated. These are similar to the table of values that has recently been published for the same microscope (Dellby et al, 2001). The main difference to the values in that publication is in this example the corrector had not been as accurately aligned, and so the parasitic aberrations were significantly larger. Both sets of measurements show that the C_s has been almost totally cancelled and the residual C_s is about 6 cm. The 3-fold and 6-fold aberrations are small as would be expected from a correctly aligned quadrupole-octupole corrector.

At this point it is essential to verify that the measured aberration function is approximately correct. Further experimental and simulation tests for accuracy and repeatability are given in Chapter 7. However, one qualitative method used was to generate a simulated Ronchigram with the coefficients that have just been measured. This is illustrated in Fig. 6.18. The Ronchigram in overfocus does not particularly suit this purpose. However, in underfocus, circles of infinite magnification are visible in the experimental image. Re-measuring the aberration function and then simulating a Ronchigram in underfocus results in images (B) and (D) shown in Fig. 6.18. The loci of infinite radial magnification have been manually traced in the experimental images. It should be remembered that these points are just solutions of equation (6.25). This shape has then been indicated on both the experimental and simulated Ronchigrams. The similarities between the real and simulated Ronchigrams show that the measured aberration function is a reasonable one. The differences between the real and simulated images indicate that the simulation is not quite perfect (possibly caused by some sort of rotation). Details of the simulations are given in Appendix 3. The contrast reversal above and below focus is not observed in the simulations because Fresnel effects are not included. The aperture is also distorted by the aberration function, but has not been included in the simulations. The measured aberration functions are plotted out in Fig. 6.19. Also shown is a cross-section through each plot, which indicates where the loci of infinite magnification occur.

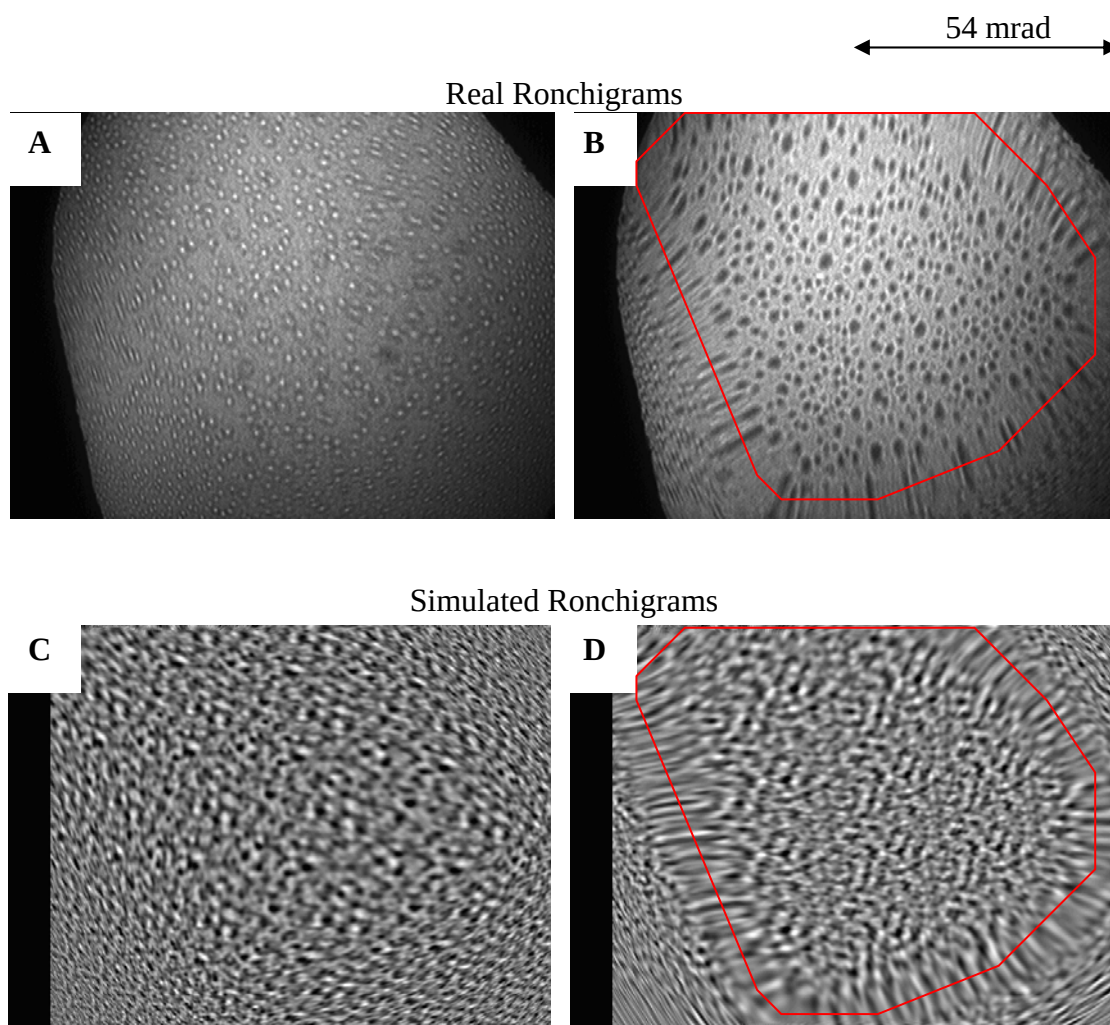


Figure 6.18. Real and simulated Ronchigrams:

- (A) A real Ronchigram taken at an overfocus of approximately 1000 nm showing no infinite magnification. This corresponds to the coefficients measured in Table 6.1.
- (B) A real Ronchigram taken at an underfocus of approximately 1300 nm showing infinite magnification in the radial direction (marked), with gold blobs appearing dark.
- (C) Simulated Ronchigram from the coefficients in Table 6.1, equivalent to image (A).
- (D) Simulated Ronchigram equivalent to image (B). The same loci of infinite magnification as for (B) has been marked.

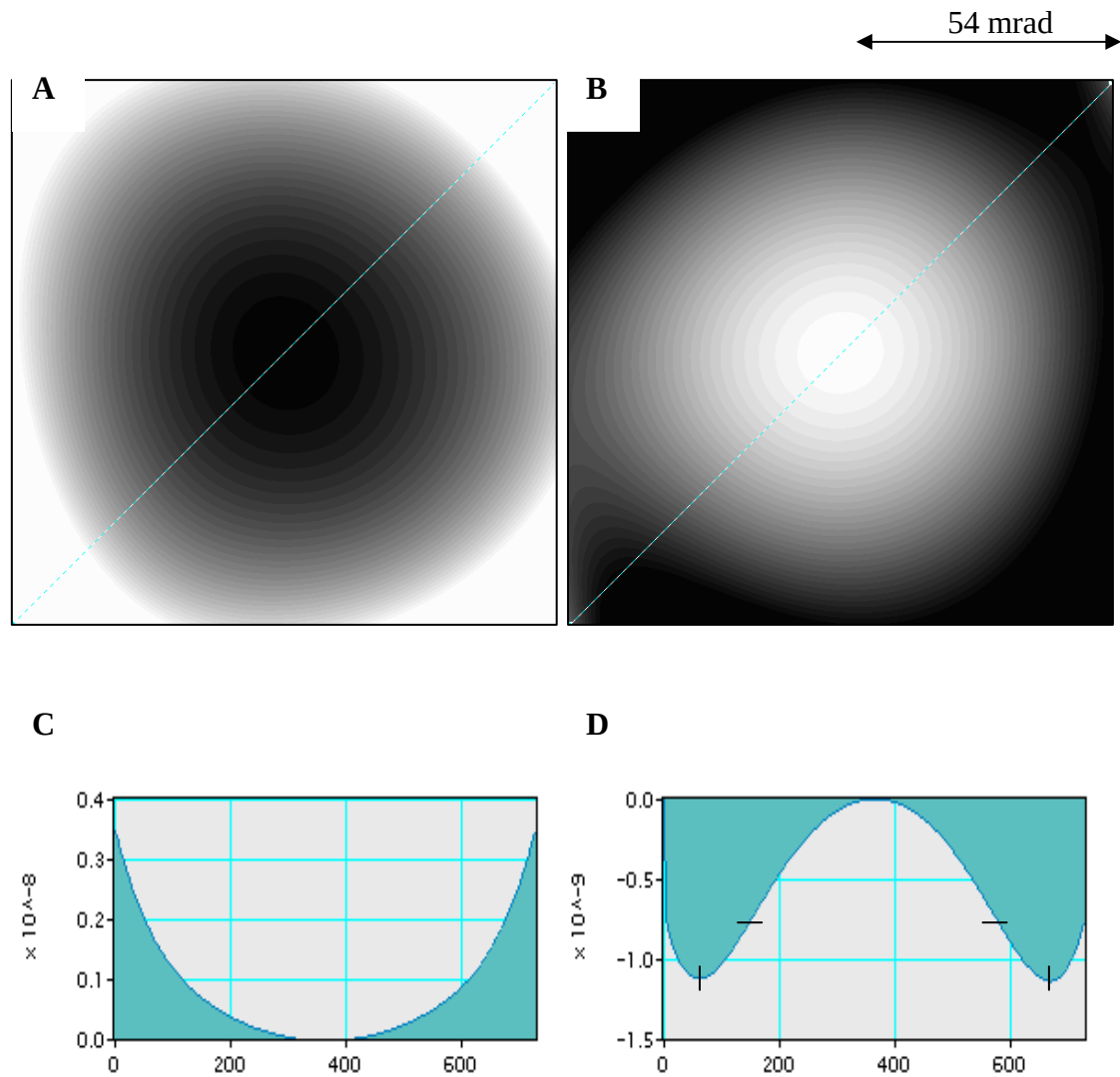


Figure 6.19 Measured aberration functions corresponding to Ronchigrams (A) (underfocus) and (B) (overfocus) in Fig. 6.18. Here (C) and (D) show the value of the aberration function along the cross-sections indicated. The difference between underfocus and overfocus can be seen in the 2 different cross-sections. The approximate locations of the turning points (corresponding to infinite azimuthal magnification) and where the second derivatives are zero (corresponding to infinite radial magnification) are shown with vertical and horizontal bars respectively. These can be identified in Fig. 6.18, although they are partially obscured by the aperture and the edge of the CCD.

6.4.5. Unwarping

Unfortunately the cross-correlations that have been discussed so far will not always work. The reason is that the changes in the Ronchigram image will be both a shift and a deformation. This is demonstrated particularly at the circles of infinite magnification.

The method used to improve the cross-correlation accuracy is to introduce a second pass to the procedure and undo the distortion effects. This applies in the situation that the aberration function is approximately known from a first pass. One solution would be to attempt to re-map all of the Ronchigrams onto a flat aberration surface. This suffers from two main drawbacks. The first problem is that this mapping involves inverting the aberration function, which is not a unique transformation. The second is that we have sampled our Ronchigram at discrete pixels, so when we re-map, the coordinate space is not sampled evenly. A better solution to this problem is therefore to re-map only to a nearby part of the aberration surface. Then any distortions or sampling problems are likely to be less severe. Also, the re-mapping onto a nearby part of the aberration surface is ideally suited to a second pass, as this can be done in such a way as to provide corrections for the aberration function.

Initially the aberration function must be measured in a first pass. Equation (6.34) is then used with the measured aberration function to calculate the expected pixel shifts between our shifted and central Ronchigrams at each point in each of the Ronchigrams for the known probe movements. This is used to map each pixel in the shifted images back by the expected amount to where it appeared in the central (unshifted) image, using interpolation between pixels. The results of this process are illustrated in Fig. 6.20, showing five (in principle) identical images.

The cross-correlations between patches then proceed as in the first pass. If the aberration coefficients are correct, then all of the measured shifts will be zero. Any non-zero values will reflect small corrections to the originally measured aberration function. The new shifts are now more accurate, because this process has compensated for the changes in magnification. So the measured values are added to the shifts from the first pass and a revised aberration function is calculated.

This procedure can be repeated iteratively, obtaining improved estimates for the aberration function until it converges. In practice the acquisition time was much less than the processing time, and so a new tableau of images was acquired each iteration. This is not essential, but should minimize any problems that would be caused by one spurious Ronchigram. The errors from a series of iterations acting on a simulated tableau of Ronchigrams are shown in Fig. 6.23. It can be seen that the accuracy of all of the coefficients improves with iteration. On the fourth iteration in this example, there was some confusion between astigmatism and C_{32a} , which is possibly due to the square shape of pixels. In practice, two passes seemed a useful compromise between speed and accuracy. The two-pass procedure is summarised in Table 6.2.

The graphical output from the second pass is illustrated in Fig. 6.21. This shows the differences between the images 1 and 3 in Fig. 6.20 before and after the unwarping process, by simply subtracting the intensities. Before the unwarping process, there is clearly a difference between the two images. Afterwards they are almost identical, indicating that the unwarping has been successful. Again the pixel shifts are output graphically, as shown in Fig. 6.22. These corrections are all below one pixel on original shifts of approximately 10 pixels.

The main theoretical error with all of this analysis is that the shifts are a finite size, and so it is not totally valid to neglect the higher order terms (e.g. third derivatives) in the expansions used to derive (6.34). Clearly the re-mapping will fail at the circles of infinite magnification. However the accuracy achieved by this method seems to at least equal other comparable methods for finding shifts in distorted images (for example Sjö Dahl, 1995). Experimentally it was found that this process increased the repeatability of the measured coefficients relative to the first pass, as seen in Chapter 7. Fig. 6.23 clearly shows that the unwarping procedure improves the measurement accuracy of all of the coefficients.

Pass 1:

1. Acquire and display a tableau of 5 Ronchigrams while shifting the probe.
2. Extract patches and perform cross-correlations between those patches to find the **apparent** shifts at points within the images.
3. Display the shift vectors as arrows on the images.
4. Convert the shifts to second derivatives at each point and display.
5. Fit polynomials to the second derivatives, and convert those to an aberration function.
6. Display coefficients and error checking information.

Pass 2:

1. (Optional) Acquire a new tableau of Ronchigrams.
 - 1a. Perform the unwarping, based on the previous aberration function.
 2. Perform the cross-correlations using the unwarped images.
 - 2a. Any remaining shifts are due to mis-measurements in the first pass, so add these corrections to the shifts from pass 1.
- Rejoin pass 1 at step 3.*

Table 6.2 A summary of the 2 stage Ronchigram analysis procedure. Pass 2 can be repeated without repeating pass 1 for small or no change to the aberration function.

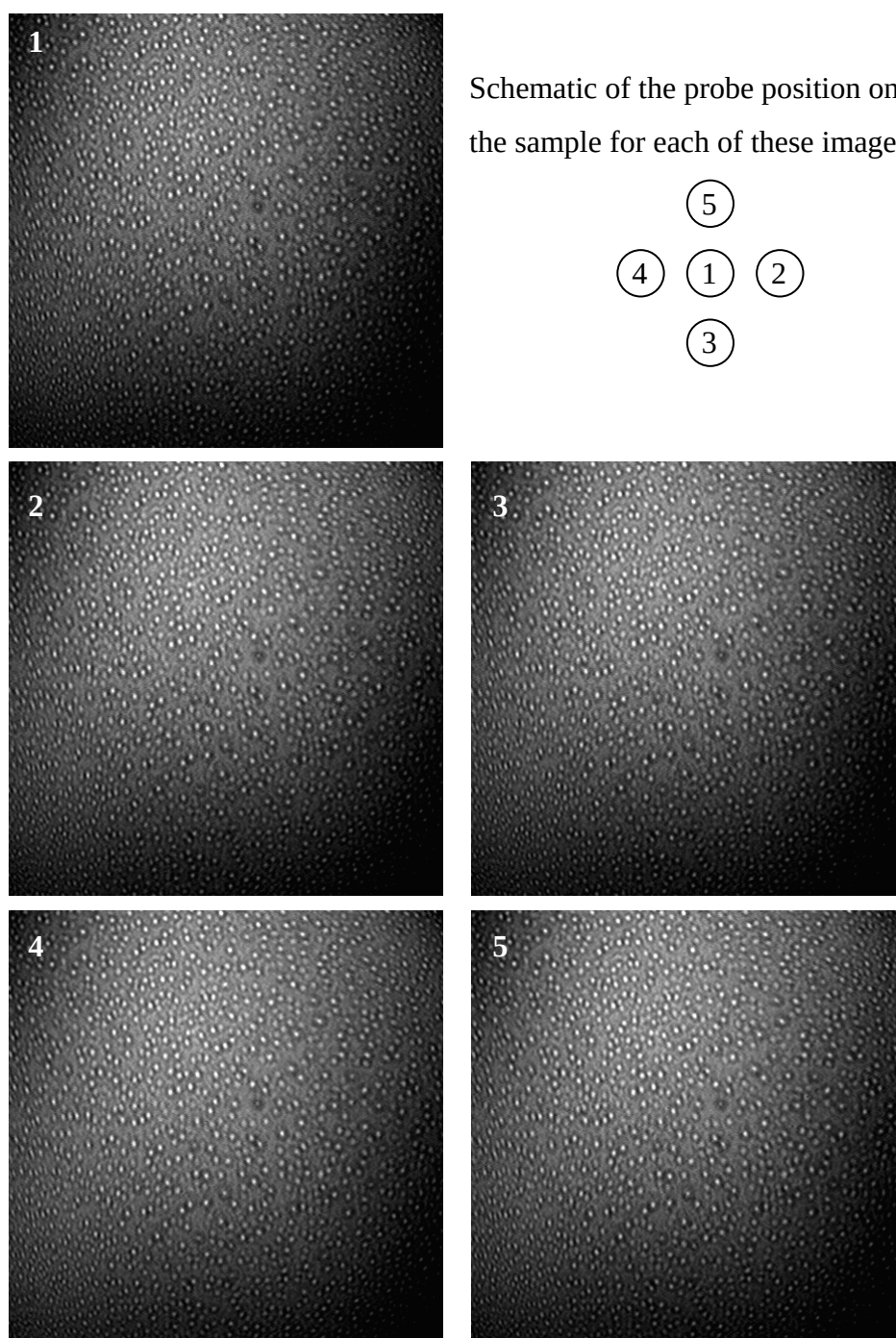


Figure 6.20. Output from the “unwarping” procedure. The sections of the images that are used for the cross-correlations are cut out from each of Ronchigrams shown in Fig. 6.13. Each section is then mapped onto the same position as the unshifted image (1), using the aberration function measured in the first pass to allow for any changes in magnification. All of these images should be identical if the unwarping procedure is perfect.

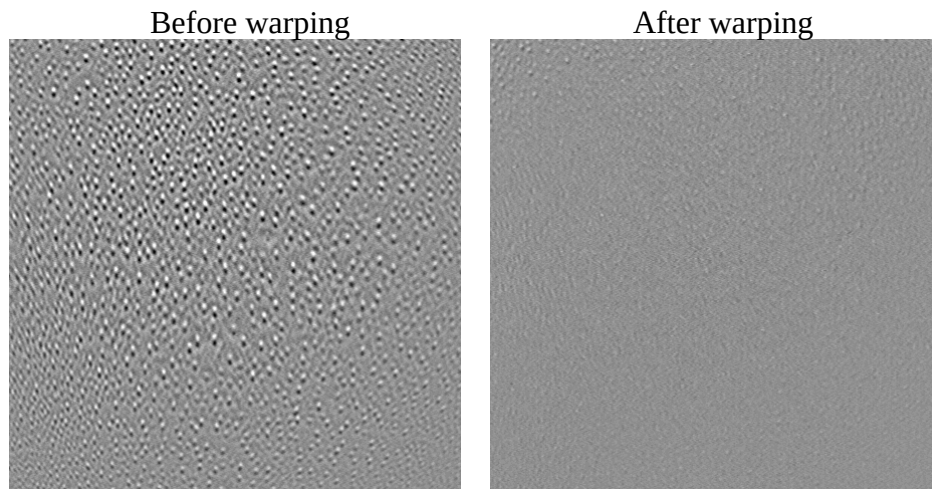


Figure 6.21. Subtracting the intensities in the image provides a visual clue to the extent of the shifts. The before warping image shows the intensity differences between the unshifted image 1 and the shifted image 3 of Fig. 6.13. The warped image shows the intensity differences between the same two images after the warping process. The same intensity scale is used, and the lack of contrast shows that the two images are now almost identical.

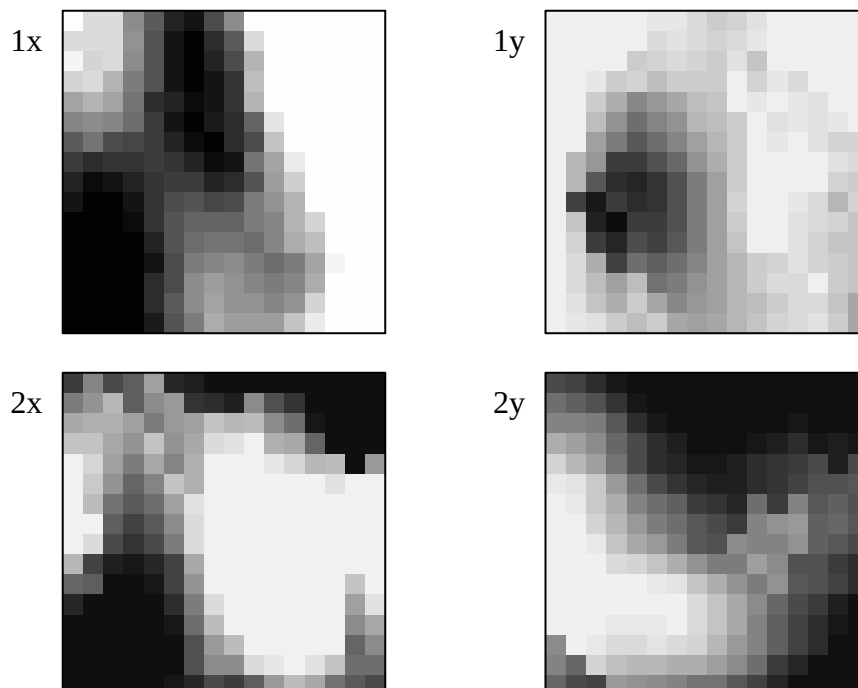


Figure 6.22. Output from the second pass from Fig. 6.20, showing the correlation errors, which give the corrections to the aberration function. All correlation errors are below 1 pixel in this case. These are the errors of the shifts in the directions 1 and 2, separated into x and y

components. Although there are strictly two sets of these images for each direction, only the averages (with the drifts cancelled) are used.

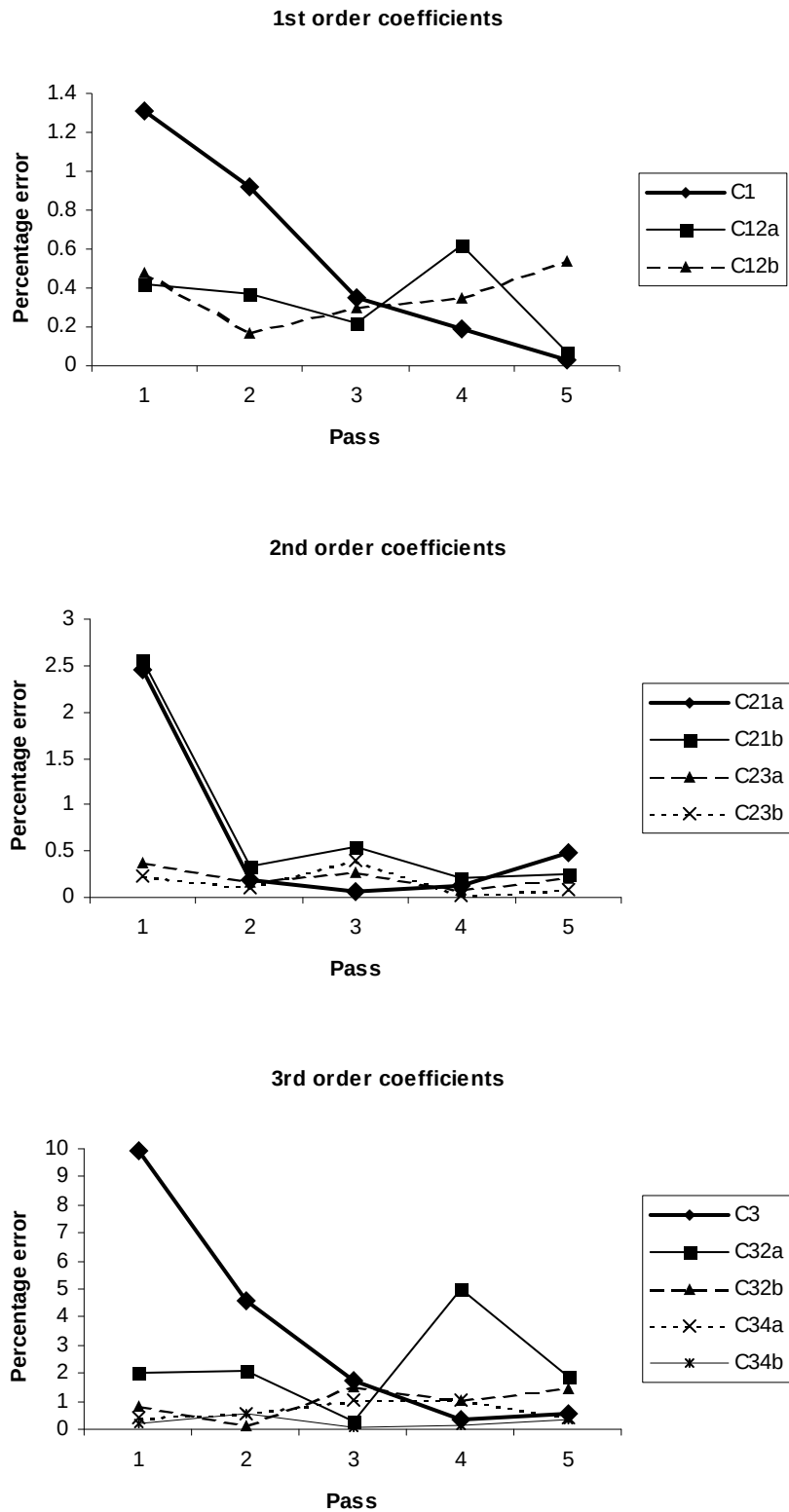


Figure 6.23. Percentage errors in the aberration coefficients over several iterations of the unwarping procedure, measured from a simulated tableau of Ronchigrams. The aberrations used were $C_1 = -1500$ nm, $C_3 = 1$ mm, with all other coefficients zero. The percentage errors are scaled to

the largest aberration value of each order, with 40 μm used for the second order. The remaining parameters were exactly the same values used when measuring aberrations on the HB501.

6.4.6. Effect of Changing the Accelerating Voltage

Alternative suggestions to shifting the probe to determine the aberration function were to alter either the accelerating voltage of the microscope or the current in the objective lens. However these do not provide enough information to solve for the aberration surface, but (respectively) allow the voltage centre and the current centre of the optical system to be determined automatically. This is a useful calibration for the probe shift method, where it is necessary to somehow locate the optic axis.

To a first order, it is possible to approximate changing the accelerating voltage of the microscope by a change in the defocus, assuming that the other aberrations are unchanged. Similarly, the effect of changing the objective lens can be modelled in exactly the same way. We can repeat the analysis in section 6.4.1 for the situation where the accelerating voltage is changed by dV . Considering a ray at $\mathbf{Y}_0$ being imaged to $\mathbf{Y}_0 + d\mathbf{Y}$ as before, we eventually arrive at:

$$0 = \nabla \left(\frac{\partial \chi}{\partial V} \left(\frac{\mathbf{Y}}{L} \right) dV \right) + \underline{\mathbf{R}} \underline{\mathbf{H}} \frac{d\mathbf{Y}}{L} \quad (6.38)$$

Where we have neglected any probe shift with voltage change and all of the rotation effects have been included in the rotation matrix $\underline{\mathbf{R}}$. (This is somewhat arbitrarily because there is rotation from both the pre- and post- sample lens fields, which have different effects). Thus, this does not provide enough information to determine all of the second derivatives by changing the voltage, as there are no longer orthogonal directions available (as was needed for the probe movement). Assuming that defocus is the largest term in the aberration function and that our other coefficients do not change, using: $\Delta f = C_c \Delta V / V_0$ (from Chapter 2) gives:

$$C_c \frac{dV}{V_0}(u, v) = \underline{\mathbf{R}}(C_1 + \dots) \frac{(du, dv)}{L} \quad (6.39)$$

Hence for the case where defocus is the most significant aberration the shifts will all be linear away from the optic axis. (The movement as the voltage is changed thus appears to be in a spiral from the centre because of the rotation of the magnetic lens.) In practice, this provided an automatic method for finding the voltage or current centres. These are simply defined as the point where the shifts were zero, although the

rotation and the change in magnification with voltage were found to reduce the accuracy of the cross-correlations.

6.4.7. Fourier Transforms of the Shadow Image

The key to the previous Ronchigram analysis was that it provided a method for obtaining the magnification as a function of angle. In Chapter 3 it was shown that Ronchigrams of amorphous materials are essentially bright field images with a distortion varying with the gradient of the aberration function. Taking patches from the Ronchigram, we would expect the Fourier transforms of each patch to resemble the diffractograms of BF images. As for the diffractogram analysis, the distortion will change off axis with the second derivatives of the aberration function. This is illustrated in Fig. 6.24. The symmetry present in these diffractograms suggests that interpolating the maximum as a quadratic to determine the shifts from the cross-correlations to sub-pixel accuracy (Section 6.4.2) was a reasonable assumption.

It is then possible to determine the variation in the diffractograms of the patches, in exactly the same way as for the diffractogram analysis method. Thus for a Ronchigram recorded over a suitable angular range and with sufficient pixels it is in principle possible to extract the aberration function from a single image, as was previously suggested. This appears to be a significant improvement over any of the present methods which all need several images. The problem with this proposal is that the gradient of the aberration function varies over each patch. Each Fourier transform will therefore be the incoherent superposition of the transforms from a range of different angles (destroying the PCTF oscillations that we are looking for). This proposed method would therefore only work well in regions small enough that ***the gradient of the aberration function is approximately constant***. However it will also be necessary for each patch to have enough pixels to provide good diffraction patterns.

The advantage of this method is that diffractograms are less affected by drift and a single Ronchigram can be acquired extremely rapidly. In a corrected microscope the aberration function is almost flat and it will be possible to use quite large patches successfully. Thus this method will probably be useful for the fine tuning of the corrector settings. With some development it could be extremely

powerful, but has not yet been successfully implemented. It is likely that the wavelet transform could be used for this purpose (Krivanek et al, 2000b).

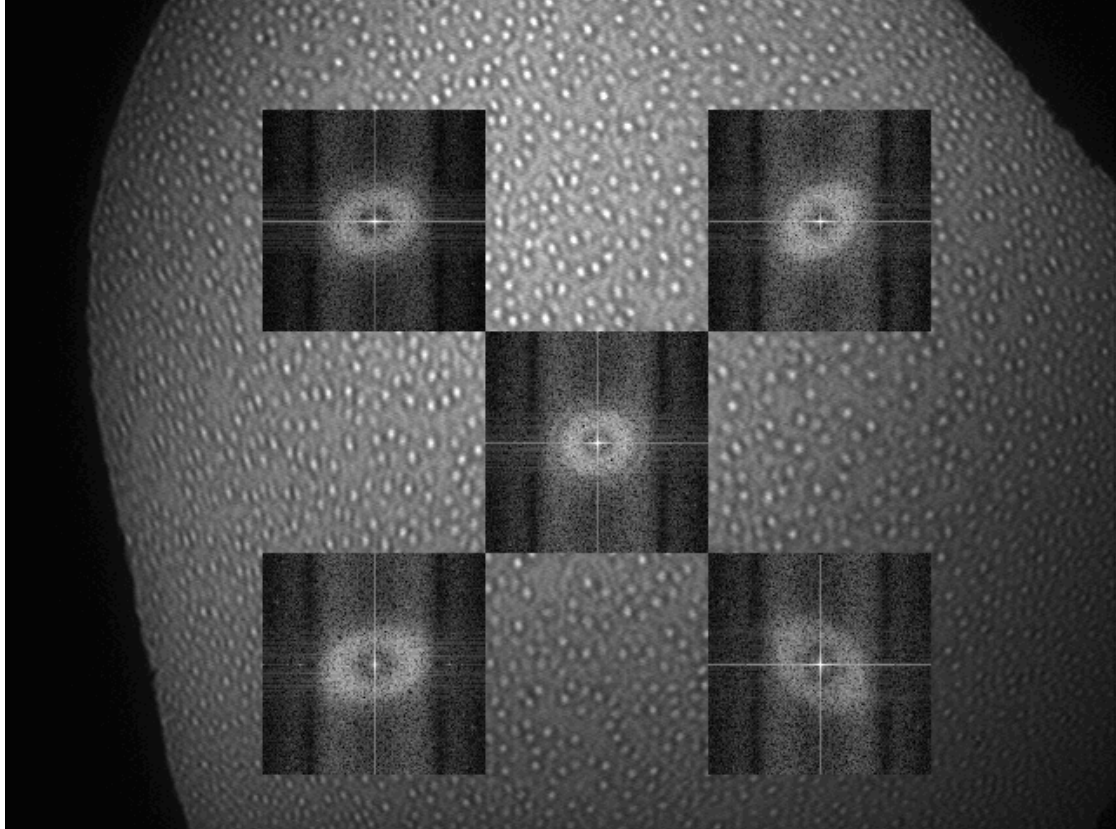


Figure 6.24. An over-focused Ronchigram with the Fourier transforms of small patches taken at different angles shown. The apparent astigmatism changes in a similar manner to that in the diffractogram method of measuring the aberration function.

6.5. Conclusions

A geometrical optics approach to the Ronchigrams of amorphous materials has been presented. It has been shown that this accurately locates the points of infinite magnification. For samples in which this description is applicable, it leads directly to a new method of measuring the aberration function of a STEM. This has been successfully implemented on both the HB5 and the HB501 microscopes, and tested up to fifth order.

Combining the description of the Ronchigram in Chapter 3 with the WPOA, it appears likely that an additional method based on Fourier transforming patches of the Ronchigram could be used to measure all aberrations. This shows that it is in principle possible to determine the aberration function from a single Ronchigram (as proposed by Rodenburg and Lupini, 1999) although there are some practical difficulties. Both methods described here are extendable to lens testing in light optics with a sample such as a ground glass slide. Existing methods of determining aberrations have been described as “time-consuming” (Lin and Cowley, 1986), so these techniques could well have uses outside the area of electron microscopy.

The Ronchigram analysis method is well suited to an aberration corrector because as the correction progresses the aberration surface will get flatter, allowing higher angle patches of the Ronchigram to be used. It is important to take patches from appropriate angles in the Ronchigram, because it is necessary to sample quite large angles in order to find the high-order coefficients reliably. Thus as the leading aberrations are corrected, the accuracy of measuring the remaining defects improves. Ronchigram analysis would also benefit from having post sample optics to allow a more flexible range of angles to be incident on the CCD.

The disadvantages of the other methods for measuring the aberration function were that they do not make efficient use of the available data in the STEM. In comparison to these, the Ronchigram analysis method seems better suited to the scanning transmission microscope, although the other techniques are well suited to a CTEM. The new method requires a maximum of 10 Ronchigrams, acquired in less

than one second, irrespective of order. For a comparable fifth order measurement, the tilt induced shift method would require a tableau of six by six images. This becomes increasingly time-consuming, and any specimen drift will degrade the accuracy of the measurement. On the existing instrument, switching to the new method produced a speed increase from several minutes to around 20 seconds for an equivalent third order measurement (and under one minute for fifth order, which was not implemented in the original tilt induced shift software). The important distinction is that the older methods were bandwidth limited by the performance of the microscope, while the image acquisition for the new method is inherently faster, and most of the time is spent in post-processing. This is only limited by computer speed and on a modern PC is currently a few tens of seconds (depending on order) for code almost totally written in the Digital Micrograph script language. As computing power and CCD resolution increase it will be possible to improve the accuracy of this method by using more patches, higher angles and larger numbers of images. With some optimization and newer computers, this method will become faster and more accurate.

One reason for the lack of success in early aberration correctors was the difficulty in measuring the aberration function (Deltrap, 1964). The Ronchigram analysis method proposed and implemented here seems to be extremely useful in practice and forms an essential part of the autotuning process (Krivanek et al, 2000b and Delby et al, 2001).

Chapter 7

C_s -corrected Imaging

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7.1. Introduction

In this chapter some of the experimental results that have been achieved will be presented. These will demonstrate that, for the first time, an aberration corrector has successfully improved the resolution of a working STEM instrument.

The alignment of the C_s -corrector forms an essential part of the experimental procedure. Indeed, the corrector constructed by Crewe failed because it could not be aligned successfully (Beck, 1977). This procedure is one of the most important experimental aspects in the developmental system. In a system intended for more general users this would have to be significantly automated.

The Ronchigram analysis method that has been described in Chapter 6 will be investigated by simulation and experimentally on the aberration corrected HB5. This technique forms an essential part of the automatic correction process that is presently being developed (Dellby et al 2001, Krivanek et al 2001).

Finally some of the first experimental results from the two aberration correctors that have been described will be presented. The subsequent resolution limitations now seem to be due to instrumental instabilities unrelated to the aberration corrector. These images demonstrate that the aberration corrector has provided a real increase in the usable resolution of the STEM.

7.2. Ronchigram Analysis

A theoretical outline of the Ronchigram analysis technique was given in the previous chapter. For the Ronchigram analysis to be useful in the aberration corrected STEM, it is necessary to know how accurately the aberration coefficients are measured. To investigate this, the software developed to implement the Ronchigram analysis was used to analyze a tableau of simulated Ronchigrams. These are generated with known aberration coefficients, using the simulation method described in Appendix 3. The important aspect is that the simulation is based on the full wave-optical theory in Chapter 3, not the geometrical simplification on which the Ronchigram analysis method is based. This thus provides further evidence as to the validity of the analysis that a geometrical simulation would not.

The results of the analysis of simulated Ronchigrams were illustrated in Fig. 6.18, and the details are given in Table 7.1. These are based on the analysis of a tableau calculated with aberration coefficients that were chosen to be experimentally realistic (defocus of -1500 nm and C_s of 1 mm). The field-of-view, probe shifts and magnification were simulated using values that were also used experimentally (given in Appendix 1). The first row is the error in the coefficients in a first order pass of the Ronchigram analysis. The second row gives the error after two runs of the unwarping procedure, which gave results significantly closer to the real values. The row giving the required accuracy is the precision to which the aberration coefficients must be set to reduce the probe size below 1 Å, from Chapter 5.

The estimated errors in Table 7.1 are an estimate of the uncertainty in the measurement of a single aberration coefficient, dC_n , due to the limited accuracy of the shift detection. Using the results of Chapter 6, this is calculated as the error (in nm) on a single measurement at the largest angle used, $\theta_{\max}$, given by:

$$dC_n \propto 10 / \theta_{\max}^{n-1} \quad (7.1)$$

Where the constant of proportionality was chosen such that 1000 nm of defocus gave a shift of 10 pixels in the cross-correlation, and the cross-correlation error was assumed to be 0.1 pixels. The maximum angle was 20 mrad, which is the value at the center of the outer correlation patches, and so is less than the total field of view. This could also be used to estimate the accuracy for different usable angular ranges.

The values given in Table 7.1 are the errors on a single set of measurements. In practice, taking repeated measurements would reduce the random errors. This is offset by the expectation that images from a real system will be less ideal. Similarly, the number of data points should be adjusted appropriately for the order of the analysis. The precision required for the coefficients will also vary with the aperture size, so that if a larger aperture is used the precision needed will increase.

Table 7.1 shows that the Ronchigram analysis first pass did not quite achieve the accuracy required to obtain 1 Å resolution. The main inaccuracies seemed to be caused by using insufficient numbers of patches, distortions within each patch changing the average shift, or by not taking patches from high enough angles. The last two of these problems are partly solved by the unwarping process. It is apparent from this table that the unwarping procedure reduces the errors sufficiently for this process to form a useful tool for the achievement of 1 Å resolution.

Order	1 st	2 nd	3 rd
First Pass	20 nm	1 µm	100 µm
Second Pass	5 nm	0.2 µm	20 µm
Estimated Error	10 nm	0.25 µm	8 µm
<i>Required for 1 Å</i>	<i>5 nm</i>	<i>0.3 µm</i>	<i>20 µm</i>

Table 7.1. The precision with which the aberration coefficients of simulated Ronchigrams were determined in a third order measurement, using exactly the same values that were used to measure these coefficients experimentally. The first row is the error from pass 1 of the measurement process, the second is the error after the unwarping stage. The estimate is the predicted error in the coefficients. The final row shows the precisions that are necessary to achieve 1 Å resolution in the same system (from Chapter 5).

Testing the Ronchigram analysis on the simulations showed that it could be used to measure aberrations of all symmetries, and worked up to fifth order, provided the field of view was large enough. In general, the errors in any particular aberration are proportional to the largest aberration of the same order. This is because all aberrations of the same order are based on different combinations of the same matrix elements. With real images recorded on the HB5, it was also found that aberrations that match the symmetry of the pixels in the images, and the probe shifts, were occasionally susceptible to higher errors. These problems were all minimized by using more patches in the correlations.

With spherical aberration uncorrected, overfocused Ronchigrams often gave better results because the circles of infinite magnification present in underfocus could give unreliable cross-correlations. The ideal defocus for taking Ronchigrams was different from the ideal defocus for ADF imaging (where ideally all rays would cross the sample at the same point). The ideal defocus for the Ronchigram analysis will depend on the other aberrations present. Obviously there should not be any patches of infinite magnification in the region used for the cross-correlations. However, the defocus should not be so large that the shifts from the other aberrations are completely swamped. As the correction process continues, the area usable in the Ronchigrams will increase. This is because the aberration surface becomes flatter, which allows higher angles to be used, with a subsequent increase in accuracy as the correction progresses. Evidence for this was also found in the simulations. When the aberration function was very flat, the errors in the second pass could be reduced below 1 nm for the first order aberrations, and 100 nm for the third order aberrations. From the simulations, the most important factors in getting the analysis to work well were to have enough cross-correlation patches and to sample the aberration function at high enough angles.

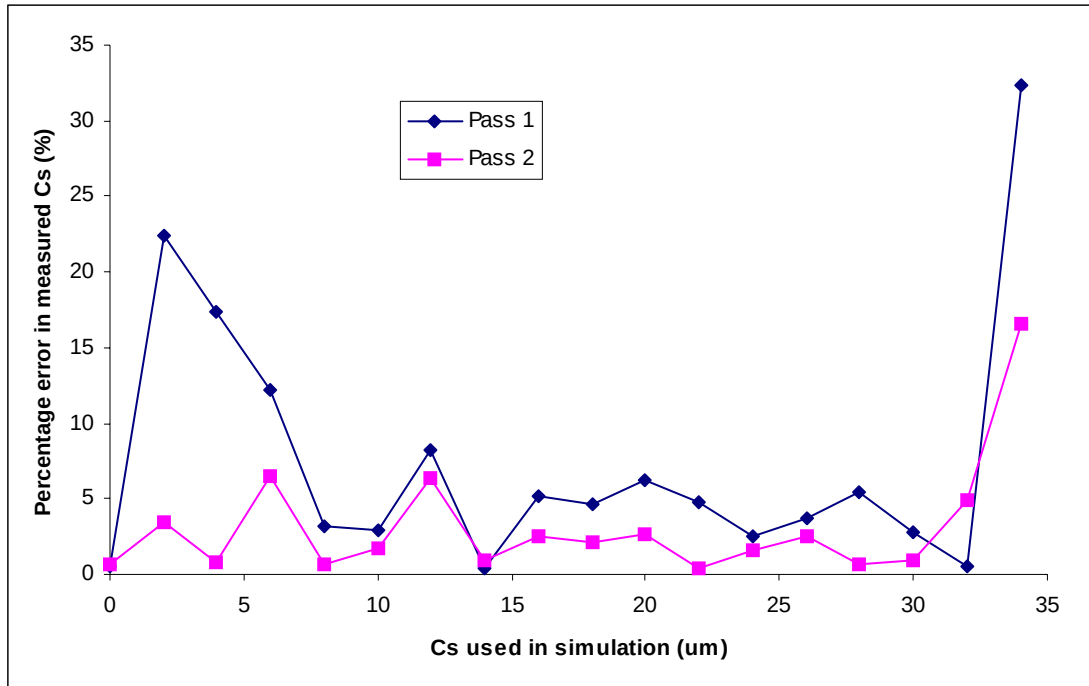


Figure 7.1. Absolute value of the percentage error in the spherical aberration measured from the first and second pass of the Ronchigram analysis routine against the value used to simulate the tableau. The simulations used -400 nm defocus, giving circles of infinite magnification at high angles. (At $C_s = 0$, the value $1 \mu\text{m}$ is used to generate a percentage error, which is small because there was no change of magnification across the Ronchigram.)

Two further theoretical tests on the Ronchigram analysis are illustrated in Figures 7.1 and 7.2. These are based on the analysis of a nearly corrected system with low values of C_s , because it is in this case that the accuracy of measurement is most important. Fig. 7.1 shows the percentage error in the measurement of the C_s , as the amount of spherical aberration in the simulation is increased, keeping everything else constant. The coefficients were found most accurately at low values of C_s , although a small error in the first pass could then give a large percentage error. The second pass provides an improvement in the accuracy in almost every case. At larger values, the circle of infinite magnification encroaches upon the regions used for the correlations, giving unreliable shifts. (Here, the total field of view was up to 40 mrad, and so circles of infinite magnification became visible and the cross-correlations started to fail at C_s values of above $30 \mu\text{m}$.) For larger values of C_s , either the defocus would have to be increased, or the angles used in the correlations reduced. This would remove the effects of the circle of infinite magnification, but reducing the angles would degrade the accuracy.

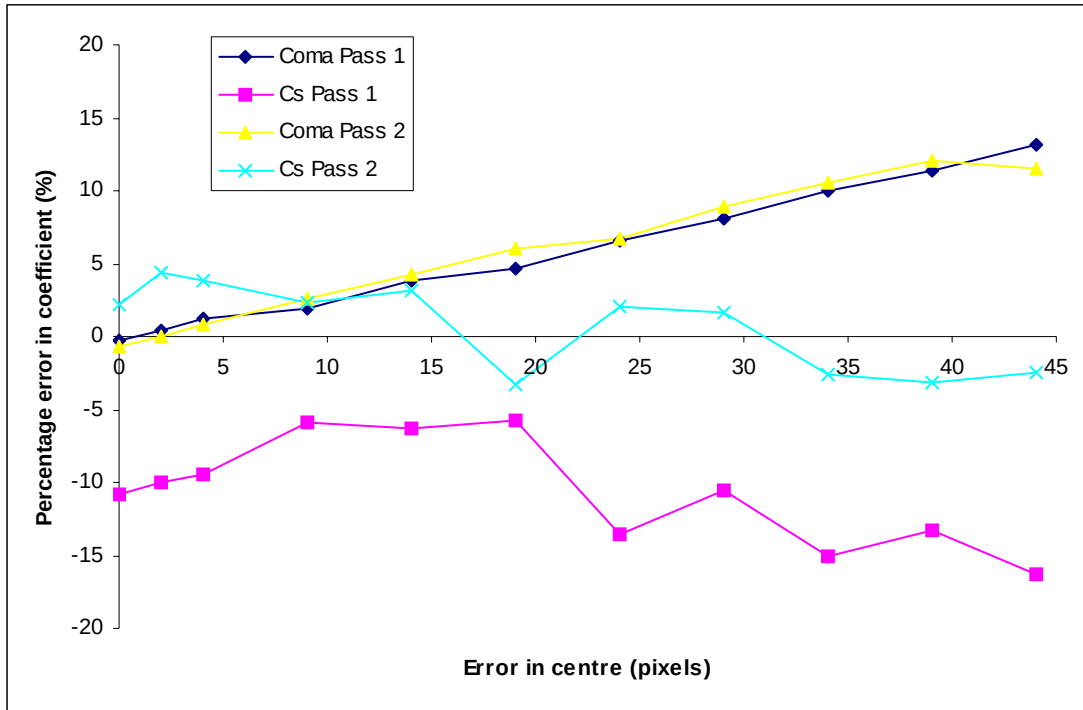


Figure. 7.2. Errors in the aberration coefficients from a simulated tableau of Ronchigrams, plotted against the error in determining the optic axis. 1 Pixel corresponds to approximately 0.1 mrad in this case. The errors in the first order aberrations were approximately constant and around 1 percent in this simulation.

Fig. 7.2 shows the effects of not obtaining the correct position of the optic axis on the aberration coefficients. The first order coefficients were not affected, because they do not depend on the location of the centre. The accuracy of the third order coefficients did not change dramatically. The measured (second order) coma did show a linear dependence on the displacement of the assumed optic axis from that used in the simulations. This would be expected because spherical aberration generated about a displaced axis has exactly the same form as C_s about the optic axis, plus an additional coma term. This is just the same principle that allows the dipole in the final stage of the corrector to adjust the coma experimentally, by shifting the axis of the corrector relative to the objective lens. These errors were calculated for a corrected system with a residual C_s of 10 μm , and the errors from an uncorrected microscope would be larger.

7.2.1. Experimental Test of Ronchigram Analysis

To implement the Ronchigram analysis on the STEM, the main calibrations that are necessary are the angular range in the Ronchigram and the size of the probe shifts. The angular range was found by using a diffraction pattern and measuring the angles that correspond to particular lattice spacings. Graphitised carbon regions on the test specimen gave a ring from the 3.4 Å spacing at a radius of 267 pixels on the HB5. The total field of view across the detector in the horizontal direction was thus approximately 26 mrad. On the HB501, the total field of view in the Ronchigram found in the same way was 108 mrad. This difference is due to the greatly increased post-sample compression from the objective lens. The calibration changes with the sample height and objective lens excitation. This needs to be repeated if the specimen position or optical conditions change significantly. The restricted range of angles available in the Ronchigrams meant that it was not possible to reliably measure aberrations any higher than third order on the HB5.

The scan coil strengths were found in exactly the same way that the magnification is normally obtained. The specific method used was to obtain graphitised carbon fringes in the bright field image, and measure the spacing between scan lines from this known spacing. This varies with the magnification setting being used. A small adjustment to the probe shifts was included to correct for the scan coil effects not being equal in all directions. On the HB5, this correction was about 6 percent.

When implementing the Ronchigram analysis on the HB5, it was apparent that it was necessary to adjust the probe shift used with the defocus settings. Ignoring other aberrations, the change in the Ronchigram will depend on this reciprocal of the defocus. Thus at low defocus values it was necessary to use smaller probe shifts. At defocus values around 10 µm, used for testing, shifts of about 5 nm worked well. This also provides an estimate of the sensitivity to drift, which should ideally be much smaller than these shifts over the acquisition time. The acquisition time on this instrument was less than 2 seconds, and the post-processing of the images took a further 50 seconds for a reliable measurement of the third order aberrations. As for the

simulations, the repeatability was found to depend on suitable adjustment of the angular range and the number of cross-correlation patches used.

An experimental test of the Ronchigram analysis was to measure the defocus as the objective lens current was changed, which is shown in Fig. 7.3. The horizontal scale is given by the current through the objective lens windings. Two measurements were taken at each defocus setting, which are plotted, along with the final line of best fit to the results. The slight non-linearity could be caused by the changing post-sample lens field, magnetic saturation of the pole-pieces, or the non-linear dependence of the focal length on the lens current. Flashing the tip would also change the measured defocus value at the same lens settings. With this pole-piece, one click of the medium knob of the objective lens power supply corresponded to 2.2 μm of defocus.

Taking repeated measurements at around 5 μm defocus gave a standard deviation of around 90 nm for the first order aberrations from the first pass. The second pass reduced the standard deviation to approximately 60 nm for the first order aberrations. This is better than 1 percent, but not as precise as needed for 1 Å resolution. The standard deviations of the coefficients from a further series of measurements on the HB5 are shown in Table 7.2. These are very similar to the accuracies of the bright field measurements given in Table 7.3. The first order coefficients seemed more repeatable from the Ronchigram analysis, although the higher order coefficients were slightly more uncertain. This is because the limited post sample compression on the HB5 objective lens meant that high angles (more than 13 mrad) could not be recorded by the present CCD. Initial results from the Nion HB501 suggest that the higher quality Ronchigrams and larger angular range (up to 53 mrad) gave far more accurate and repeatable measurements at higher orders.

Another test used was to measure the spherical aberration of the objective lens, using the analytical pole-piece. In the HB501, this pole-piece has an optimal C_s value of 3.1 mm, but in the HB5, values as high as 7.4 mm have been measured (Krivanek et al, 1997a). In Chapter 2, it was shown that this was caused by the change of the sample position and the lower power of the HB5 objective lens (there will also be a contribution from the C_s of the condenser lenses). In practice, the HB5 power supply was modified to increase its current output and the sample position was adjusted to

reduce the C_s value. In underfocus, the Ronchigram analysis gave a C_s value of 3.9 mm with a standard deviation of 0.3 mm. This value is in the expected range, and appears to be consistent with a slightly degraded performance relative to the improved HB501 lens design.

The results of both passes over a range of objective lens settings are shown in Fig. 7.4. There also appeared to be a slight variation of the C_s value recorded with the defocus setting, as indicated by the linear fit line in Fig. 7.4. This becomes larger with increasing lens current, and so is not likely to reflect a real change in the coefficient, which would be expected to decrease (Chapter 2). The change in the measured value is in fact caused by the post-sample compression increasing with objective lens current. Therefore the calibration of the angles in the Ronchigram changes, so that higher angle rays become visible. Thus the measured C_s value would be expected to increase. The reliability of the final measurement depends on the accuracy of the calibrations. In the future, these could be measured more accurately and the post-sample compression determined at a variety of field strengths.

The variations of the coefficients also depended on the quality of the Ronchigrams recorded. So, experimentally the errors could be reduced by optimising the section of the Ronchigram used to cross-correlate within for each defocus setting. Ultimately this was limited by the range of angles recorded, and the relatively poor quality of the Ronchigrams that could be recorded from the HB5.

Order	1 st	2 nd	3 rd
Average Coefficient	-5100 nm	9.2 μ m	3.9 mm
First Pass	96 nm	3.7 μ m	0.66 mm
Second Pass	66 nm	3.1 μ m	0.31 mm

Table 7.2. Standard deviation from a series of 10 measurements of the aberration coefficients from the Ronchigram analysis method. This shows the accuracy with which the coefficients were measured. The first row gives the values of the coefficients at which the errors were measured.

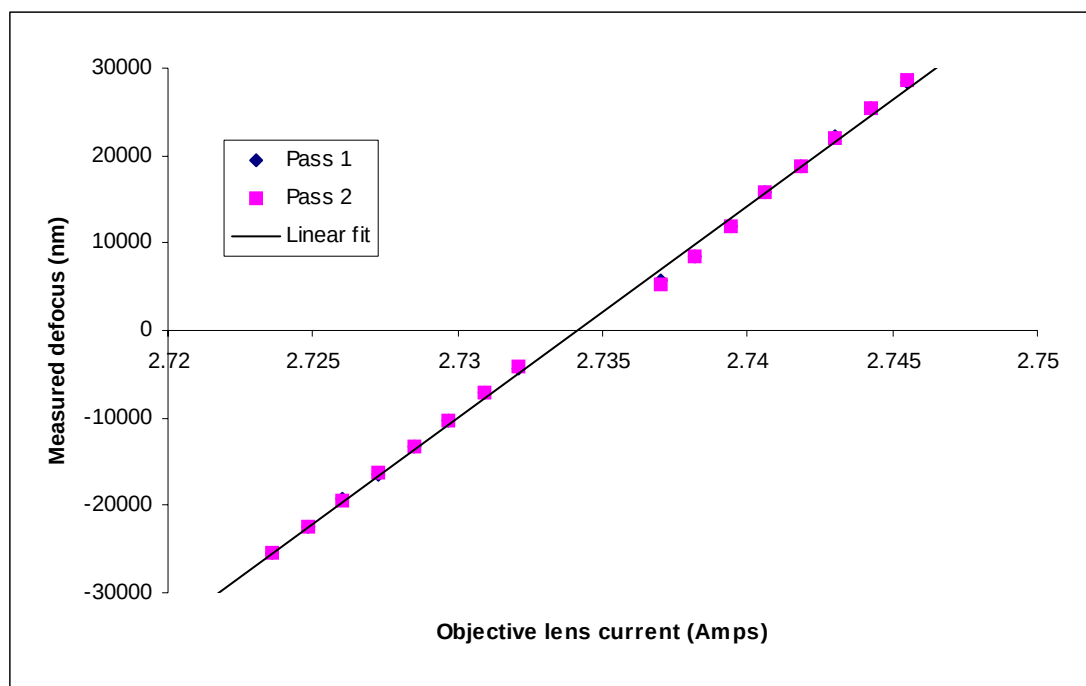


Figure 7.3. Measurement of defocus on the HB5 plotted as a function of objective lens setting, measured from the voltage across the sense resistor. The “medium” knob on the objective lens power supply is adjusted either side of defocus and the defocus measured by the Ronchigram analysis (first pass only). The points are the results of the measurements (two passes at every setting) and the solid line is the line of best fit to these results.

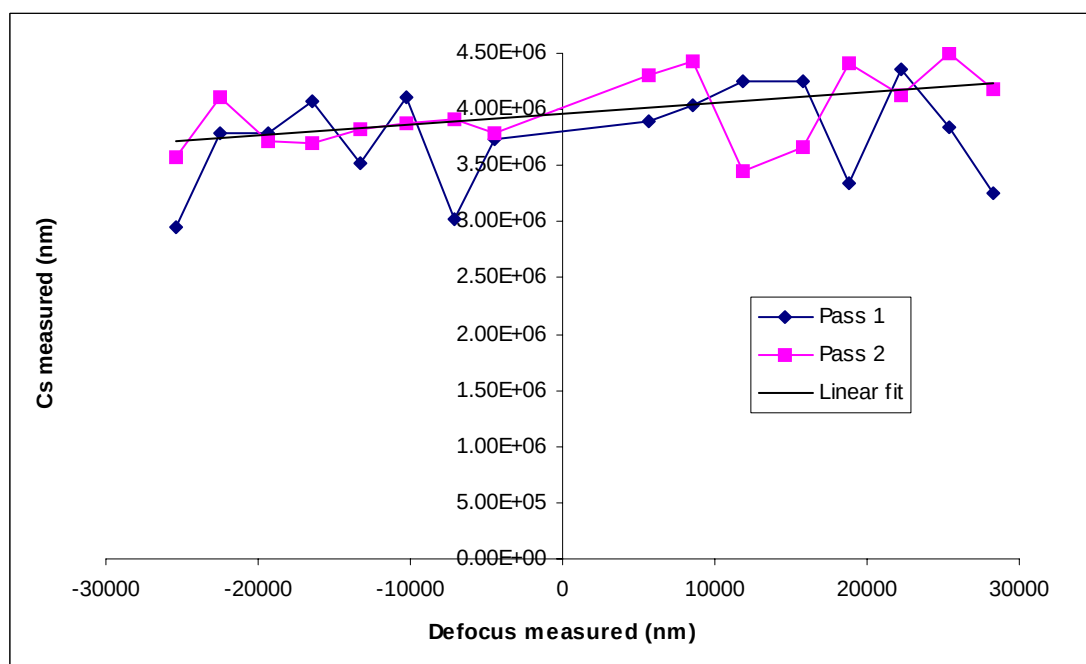


Figure 7.4. Variation of the spherical aberration measured with objective lens setting. The measured C_s value is plotted against the defocus (which increases with lens current). This shows both random errors from the cross correlations and a systematic increase with lens current.

7.2.2. Aligning a C_s -corrected Microscope

The alignment of the C_s -corrector forms one of the most important experimental aspects of using such a system. A practical description of how to align an uncorrected VG HB501 STEM is given by Nellist (1995). The usual technique is based on aligning the elements in a systematic way, wobbling the strengths to locate the ideal aperture positions. For example, the location of the objective aperture is adjusted so that the shift of the ADF image is minimised as the strength of the objective lens is wobbled. In the HB5, the Ronchigram provided a convenient method of aligning the microscope.

In the C_s -corrected HB5, the VOA is used as the beam-defining aperture to prevent the beam encountering the edge of the column. Experimentally it was found that an insulating surface between the SAD and OA would charge up if this were not done. In the absence of a strongly diffracting sample, the Ronchigram shows the beam shape at the detector. Thus with all of the optical elements turned off, the VOA appears as a disk (Fig. 7.5).

When using the Ronchigram to align an aperture to a round lens, the principle is to shift the aperture until altering the lens excitation causes the disk to expand and contract centred on the same point with no lateral motion. If the corrector is not being used, this is performed for the VOA and the objective lens. The first condenser lens is then aligned (using the shift and tilt screws) with the objective turned off, so that it causes no shift of the aperture image with changing excitation. The shift and tilt of the condenser lens are then both iteratively adjusted until this is true for all excitations of both lenses. This is repeated for both condensers and iterated until they are also mutually aligned. In practice it is necessary to use the alignment coils to perform the fine adjustment and to compensate for any shifts in the VOA position.

The most important optical element in an electron microscope is usually the objective lens and so that is used to define the optic axis. However in a C_s -corrected system the corrector, as the most complex element, is used to define the optic axis. Fortunately the combination of the presence of the strong quadrupoles in the corrector and the Ronchigram provides a convenient mechanism for aligning the corrector.

Exciting the quadrupole elements distorts the circular VOA image to an ellipse, if they are correctly aligned. However, for an unaligned element there will also be a parasitic shift proportional to the amount of misalignment. Thus to centre the VOA with respect to the first quadrupole, Q_0 , the procedure is to wobble the strength of Q_0 and shift the VOA until only a pure distortion (and no shift) is seen. Higher order multipoles produce similar figures of the appropriate symmetry, although only the octupoles were sufficiently strong for this to be encountered regularly in practice. The effects of changing the excitations of some of the optic elements in the first stage of the corrector are shown in Fig. 7.5. It can be seen that there are slight irregularities in the images, which are caused because the pole-pieces are saturating unevenly and the centres of the elements shift.

Having used Q_0 to position the VOA, the condenser lenses are aligned to this axis, as for the uncorrected case. Then the last quadrupole, Q_5 is used to define the axis in the corrector. Q_5 is wobbled and a dipole added at the same level as Q_0 , until Q_5 also produces no image shift when wobbled. A dipole is then added at the level of Q_5 to align the objective lens so that it too causes no shift of the image. The alignment of the condensers proceeds as before. Although now Q_0 is also wobbled, and so the twin criteria of no image shift with condenser excitation and maintaining the pure action of the quadrupole allows both shift and tilt to be set correctly. Again it is normally necessary to iterate over much of this procedure until all elements are mutually aligned.

The alternative alignment strategy would be to bring all of the corrector elements onto an axis defined by the gun and the objective lens. However, this was not successful, because it was found to require very high excitations of the trim coils in the HB5 corrector.

The remaining quadrupole elements in the corrector are brought onto the axis defined by Q_0 and Q_5 by adding lower order poles and trim coils to each high order multipole. The ability to do this forms an essential part of the software control of the aberration corrector, described in Chapter 2. The pure quadrupoles were produced by adding dipoles to the quadrupole elements, until changing the excitation produced no

shift in the shadow image (see Fig. 7.5). The pure octupoles were produced in a similar way by adding all lower order poles, including quadrupoles.

Having created the pure quadrupole controls, the condenser lenses and the quadrupoles were set to the strengths predicted from the simulations in Chapter 4. This is done for each stage, in order, up the corrector. Even with the pure controls, after each quadrupole was excited, it was necessary to make further adjustments using dipoles so that the image of the aperture remained centred. This showed that the rapid first order simulations, described in Chapter 4, were essential, because there is no simple method for locating the beam inside the corrector. After the quadrupoles were correctly excited, the octupoles were activated in the same order. Again this requires small additional multipoles and would be extremely difficult without pure controls.

The Ronchigram also provides a convenient method for positioning any other apertures used, such as the collector aperture. The location of any aperture is visible in exactly the same way as the VOA. It can then be moved to the optic axis, which is identified as the centre about which the objective lens expands the Ronchigram image.

This procedure is rather more complicated than the normal alignment process. However, like the coarse mechanical alignment of the uncorrected microscope, it does not need to be repeated every time the STEM is used. In order to ensure that the settings could be re-used it was necessary to regularly degauss during the alignment procedure. The corrector settings are then saved and reloaded each time the corrector is used. It was only the fine adjustments to the aberration function that needed to be performed on a regular basis, and these were also performed using the Ronchigram.

Aberrations are identified by a non-symmetric appearance of the Ronchigram. An example of this is given in Fig. 7.6, where an over-excited octupole gives the Ronchigram a 4-fold appearance. It is easiest to distinguish between different aberrations by wobbling the objective lens and looking for non-symmetric motion. The correction was relatively easy for the low-order aberrations, although occasionally it was necessary to realign elements with the dipoles. Experimentally it was found quite difficult to see the difference between coma and 3-fold astigmatism, because of the similar forms of these aberrations.

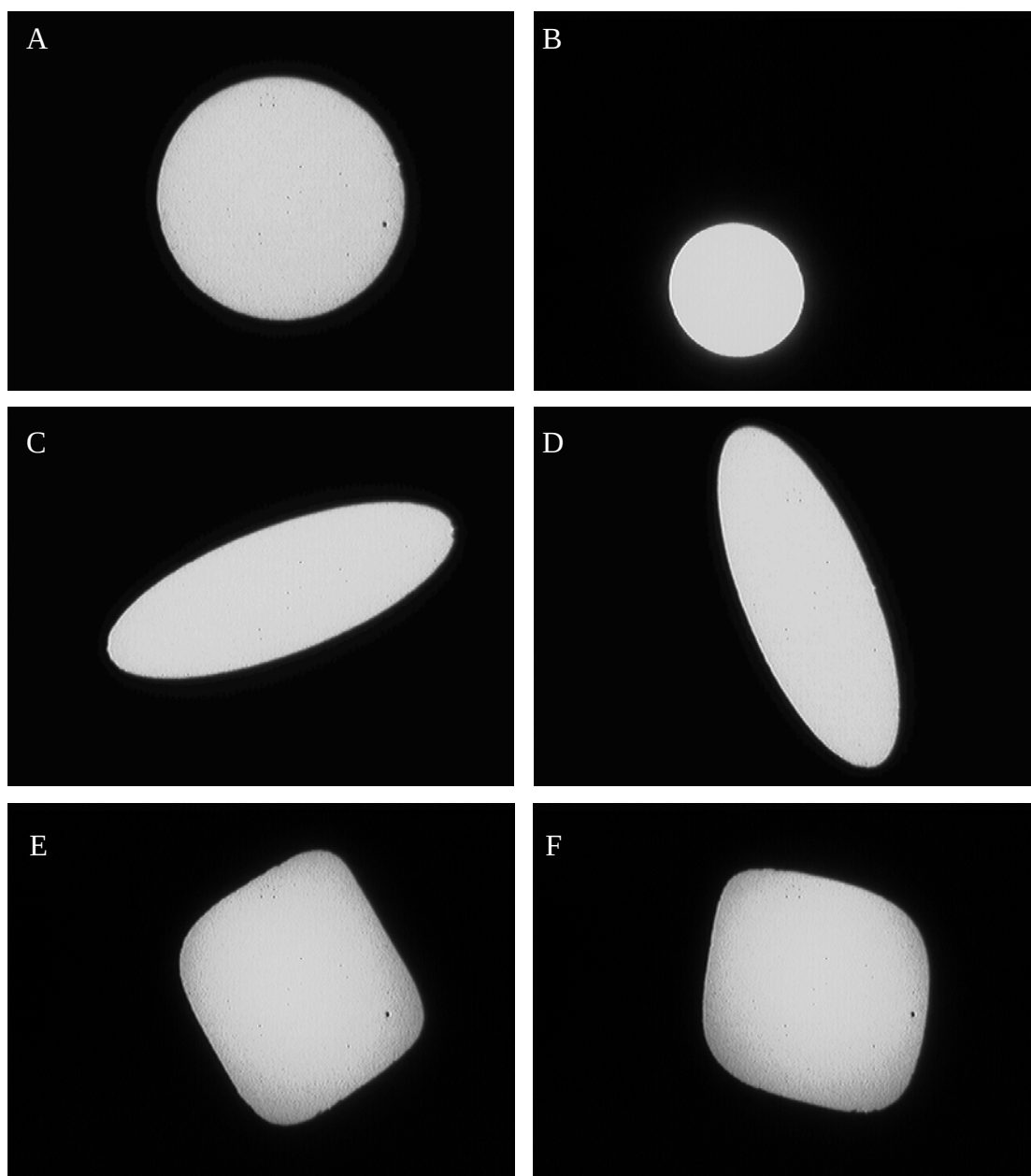


Figure 7.5 Shadow image of the VOA on the Ronchigram detector without a sample and with the objective lens weakly excited as a projecting lens.

- (A) The VOA only.
- (B) Exciting the objective lens causes the image of the VOA to change in size. Here the aperture has been misaligned and so the position also shifts.
- (C) Exciting Quadrupole 0 causes the beam to be distorted into an oval
- (D) Changing the sign of Quadrupole 0 changes the direction of the distortion.
- (E) Exciting Octupole 0 causes the beam to be distorted with 4-fold symmetry.
- (F) Changing the sign of Octupole 0 changes the direction of the distortion. Misalignments mean that this has a small shift and is not perfectly 4-fold.

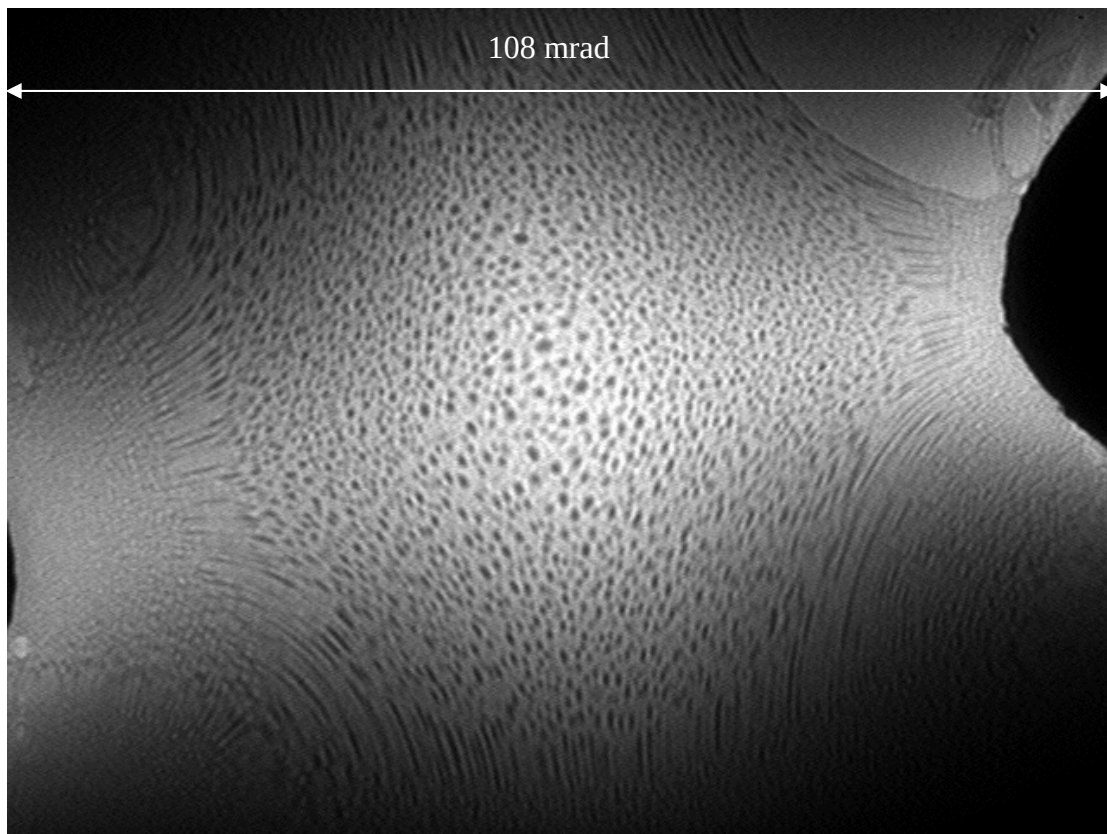


Figure 7.6. The symmetry of the aberration function is reflected in the appearance of the Ronchigram. An octupole excited too strongly has given this Ronchigram a very 4-fold appearance. Loci of infinite magnification are visible. In agreement with Chapter 6, these are not circles because of the 4-fold aberration.

The controls used to adjust the different aberration terms were given in Chapter 4. The manual fine adjustment of the high-order aberrations was more difficult than for the lower order aberrations. This was ultimately one of the limiting factors in the performance of the first corrector. In order to also correct the higher-order aberrations, it was essential to develop the Ronchigram analysis routine. This is just starting to be applied to the automatic alignment, and was used during the acquisition of all of the images from the mark 2 corrector.

7.3. Ronchigrams with Adjustable C_s

The effects of spherical aberration are most clearly seen in the electron Ronchigram. In Chapter 6 it was shown that spherical aberration leads to circles of infinite magnification in underfocus, at the points where the first and second radial derivatives of the aberration function are zero. Fig. 7.7 shows Ronchigrams taken at different defocus values from the HB5. At large values of defocus (either overfocus or underfocus), this dominates the aberration function over the angles detected by the CCD and so the magnification is approximately uniform. At zero defocus, the spherical aberration is the only term in the aberration function, and so the magnification in the Ronchigram is infinite at the center and decreases at higher angles. The above and below focus images in this figure also demonstrate the contrast reversal of the gold blobs and of the sample edges, due to Fresnel effects.

Fig. 7.8 shows the effect of C_s in underfocus. These are taken with the analytic pole-piece and so there is very little post-sample compression. At low defocus values the image becomes rather blurred, and so the circles of infinite magnification are not obvious. At moderate defocus values the radial circle of infinite magnification is visible. Using the Grigson coils to shift the image shows that both of the circles of magnification are visible at higher angles.

The amount of spherical aberration imparted to the beam by the corrector is adjusted by using the condenser lenses to change the beam diameter and cross-over positions. During the set-up of the corrector, one interesting stage was encountered when the beam diameter in the corrector was too large. This meant that the aberration of the objective lens was over-compensated, resulting in negative spherical aberration (about -4 mm). This is most clearly seen in the electron Ronchigram, as shown in Fig. 7.9. The Fresnel effects from the gold particles and the hole in the carbon film show that the centre of the Ronchigrams are over-focused. At larger angles the circles of infinite magnification are visible because the spherical aberration is negative.

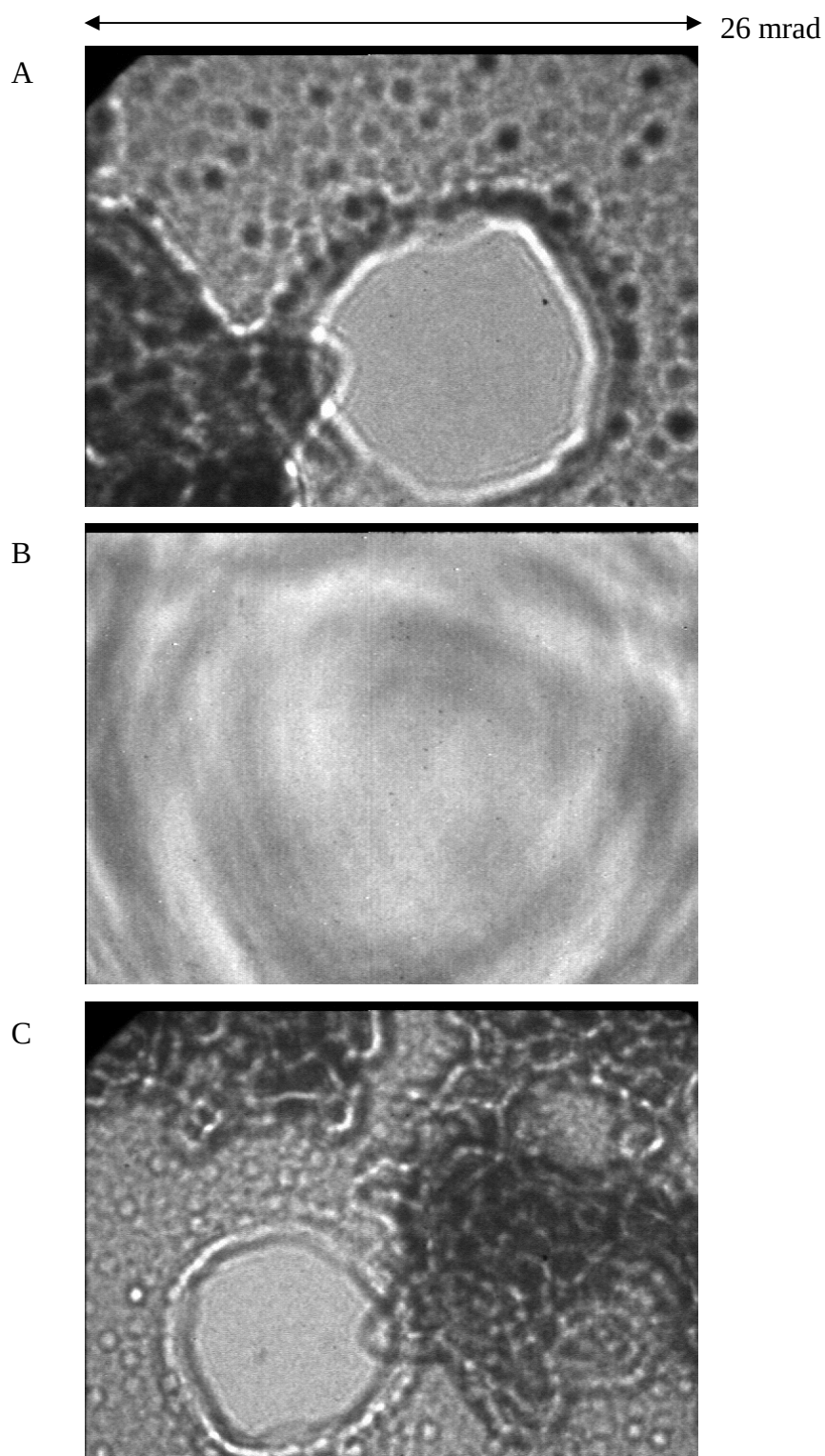


Figure 7.7. Ronchigrams taken with spherical aberration (about 3.5 mm) at different defocus values from the HB5, for the combined test specimen.

- (A) Moderate underfocus.** The magnification is approximately uniform with increased magnification at the edges due to spherical aberration.
- (B) In-focus (amorphous carbon film).** The magnification at the center tends to infinity.
- (C) Moderate overfocus.** Again the magnification is uniform, but the contrast on the gold blobs has changed, due to Fresnel effects.

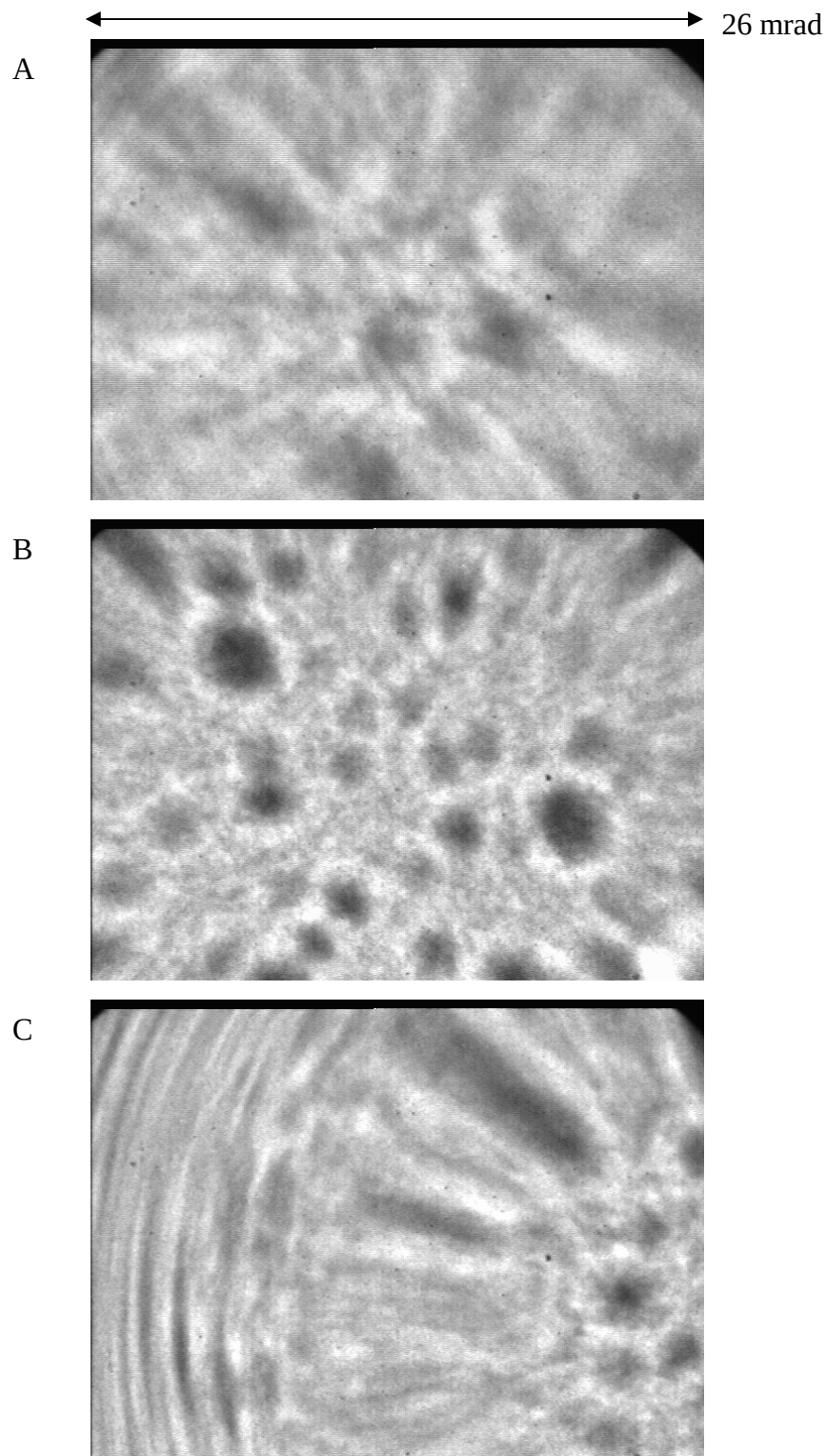


Figure 7.8. Ronchigrams showing the effect of spherical aberration in underfocus from the HB5.

- (A) Small underfocus. The circles of infinite magnification are just visible.**
- (B) Increased underfocus. The circle of infinite radial magnification is visible, but the radius has increased.**
- (C) At the same underfocus as for (B), using the Grigson coils to shift the center of the Ronchigram to one side so that both circles of infinite magnification (radial and azimuthal) are visible.**

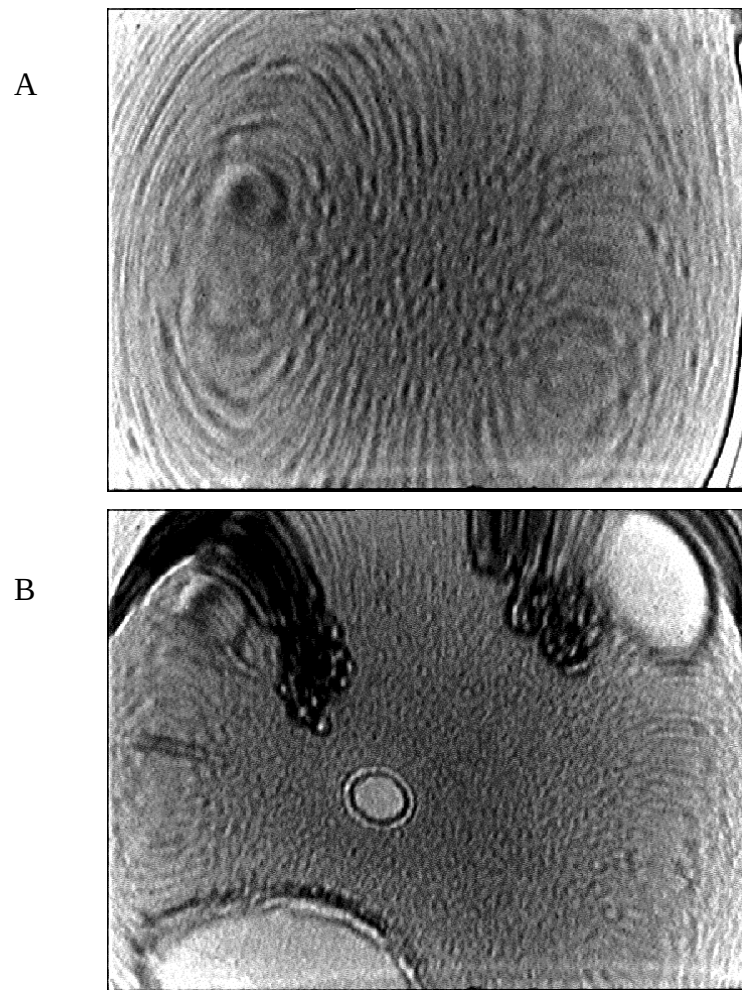


Figure 7.9. Ronchigrams taken with negative spherical aberration.

- (A) The circles of infinite magnification are visible in overfocus.**
- (B) At increased overfocus, only the circle of radial magnification is now on the CCD, but the Fresnel fringes around the hole in this section of the amorphous carbon film show that the centre of this image is in overfocus.**

Following the full alignment of the C_s -corrector, it was possible to record electron Ronchigrams with the spherical aberration eliminated. Figure 7.10 shows Ronchigrams taken without spherical aberration a small distance either side of the Gaussian focus. If the only aberration present is defocus then the magnification will be uniform and the features in the images above and below focus are similar except for the Fresnel effects. From Chapter 6, in the rotationally symmetric case, the magnification was given by the reciprocal of the second radial derivative of the aberration function. For defocus only, therefore the magnification will just change sign either side of the Gaussian focus. This means that the positions of the features in the Ronchigrams are reflected through the position of the optic axis.

Figure 7.11(A) shows a Ronchigram taken in underfocus from a region of graphitised carbon. The lattice fringes corresponding to the 3.4 Å spacing are visible with no pronounced directional dependence. This shows that the instabilities have been reduced below this level over the time necessary to record the Ronchigram. The second derivatives of the aberration function are constant over the range of angles in the image and so the fringes appear straight. Figure 7.11(B) shows a Ronchigram that is taken with no defocus. With no aberrations present the rays go through the same part of the sample (or equivalently, the entire Ronchigram is the infinitely magnified image of a single part of the sample), and so there is no contrast in the Ronchigram.

The Ronchigrams demonstrate that the parasitic aberrations have been reduced to such a level that they have no significant effect over the range of angles recorded in these images. The next test of the aberration corrector was then to attempt conventional STEM imaging with zero C_s .

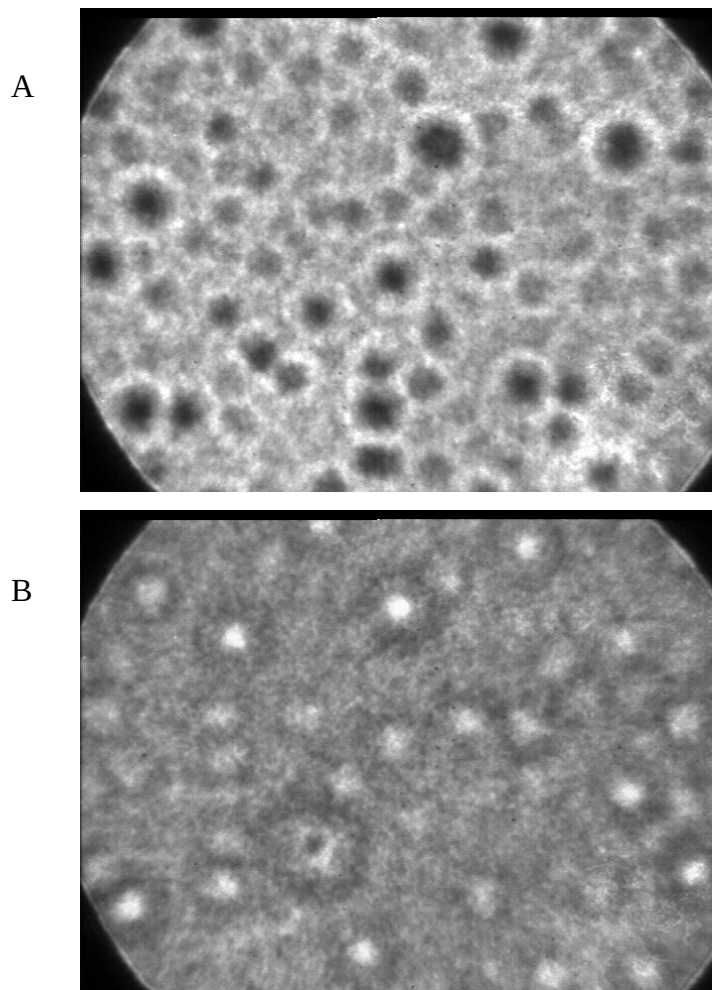


Figure 7.10. Ronchigrams taken with spherical aberration set to zero.

- (A) Underfocus. The circles of infinite magnification are no longer visible because the spherical aberration is zero.**
- (B) Overfocus. No circle of infinite magnification is present in over-focus, indicating that the spherical aberration has not been over-corrected. Other than the Fresnel effects, the image is essentially an inverted version of (A).**

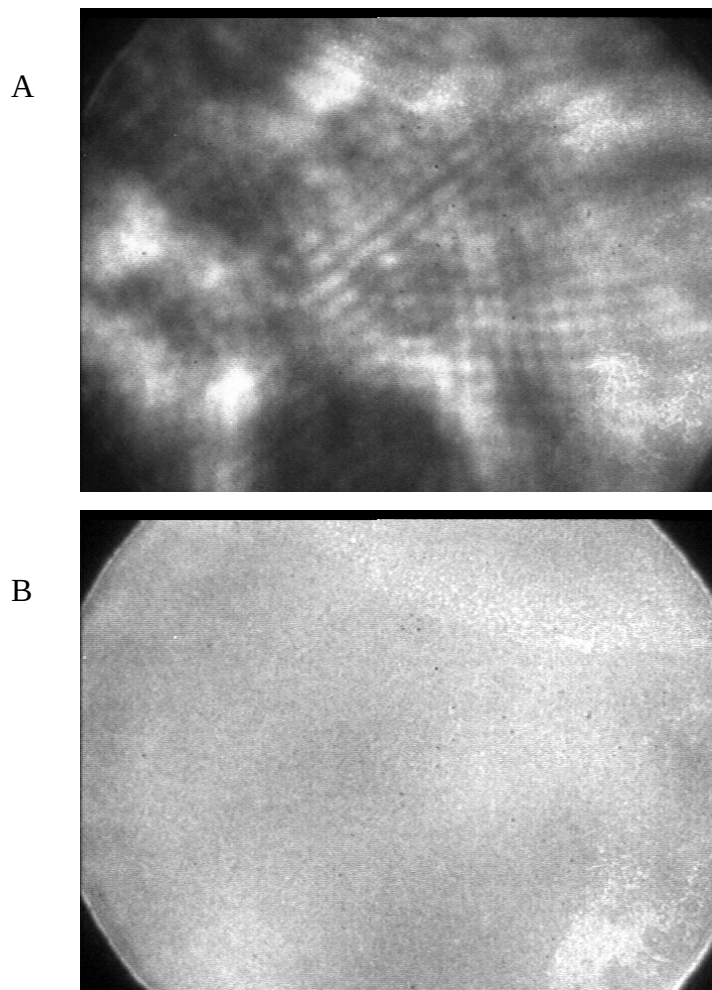


Figure 7.11. Ronchigrams taken with zero spherical aberration.

- (A) Graphitised carbon fringes taken in underfocus. The only significant term in the aberration function is defocus, and so the magnification is uniform and the fringes are straight across the whole image.**
- (B) In focus. The aberration function is flat and the Ronchigram is an image of a single point on the sample and so it shows very little contrast.**

7.4. Imaging with Adjustable C_s

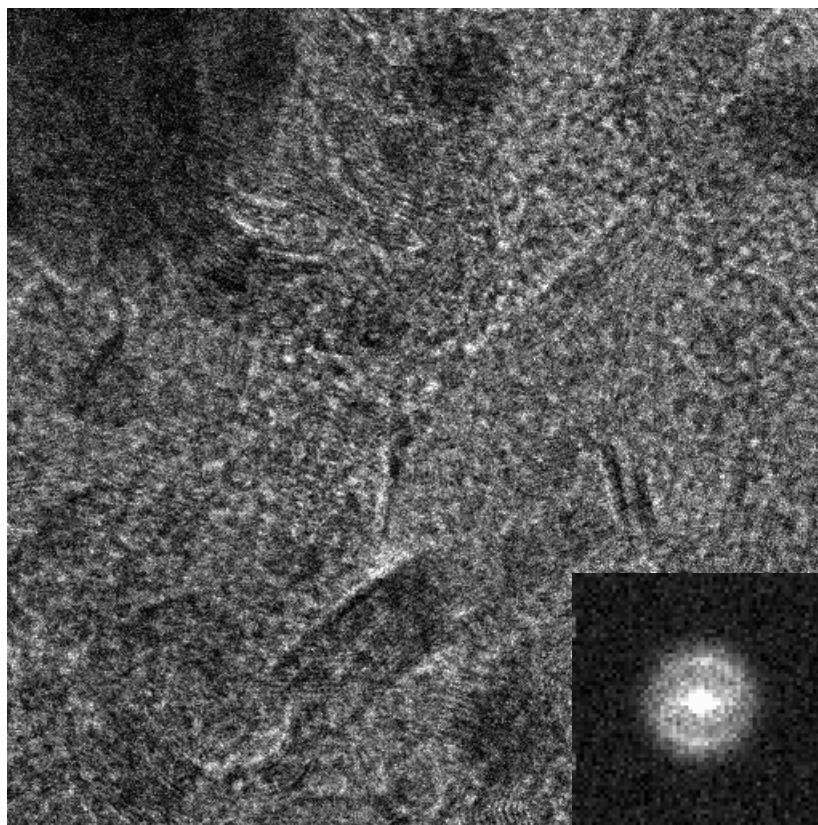
The principal aim of C_s -correction is to improve the resolution of the electron microscope. To examine the performance of the corrector, a standard test specimen was used. This consists of a thin amorphous carbon film suitable for the automatic techniques used to measure the aberration function, suspended on a copper grid. This also has patches of graphitised carbon and small gold particles (typically around 10 nm diameter) to provide a range of lattice spacings to test the achievable resolution.

7.4.1. Uncorrected Images

To obtain a high-resolution image in a STEM, after the initial alignment, it is necessary to balance the underfocus against spherical aberration. The effect of this on bright-field imaging was described in Chapter 5. The best images are obtained when the PCTF has a large flat region spanning the range of spatial frequencies that are required. Higher resolution images can be obtained by increasing the amount of underfocus used, although this reduces the range of spatial frequencies that is successfully transferred. This process is described by Spence (1977), and is ultimately restricted by the information limit of the microscope. The first test of resolution that was used was to obtain fringes from the graphitized carbon (lattice spacing 3.4 Å) in bright field. The objective aperture used was approximately 10 mrad and the collector aperture was 0.5 mrad.

An image of the standard test specimen is shown in Fig. 7.12(A). In this case the defocus has been optimized so that the graphitized fringes are most visible. The range of other spatial frequencies that are transferred to the image is restricted, and the fine detail in the amorphous carbon background is lost. Fig. 7.12(B) shows the same region of the sample after the defocus has been changed to optimize the visibility of the fine detail from the amorphous carbon film. The fringes from the graphitized carbon are less visible because of the oscillations in the contrast transfer function caused by the spherical aberration. (Using the measured value of 3.9 mm for C_s , the BF resolution from Table 5.1 is predicted to be 4.4 Å, so it is only by carefully optimizing the defocus that the 3.4 Å spacing can be resolved.)

A



B

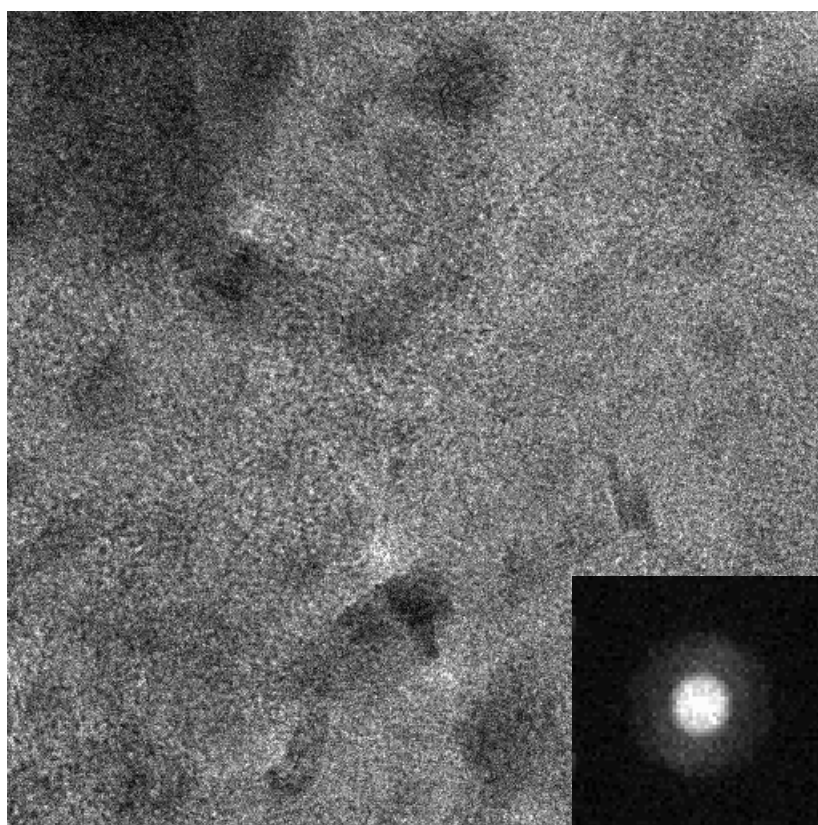


Figure 7.12. Bright field image of the standard test sample at 50000 times magnification from the uncorrected HB5 (with diffractograms). The defocus was optimized to maximize the visibility of:

(A) The graphitized carbon fringes. The fine detail of the background is lost due to C_s .

(B) The fine detail of amorphous carbon background. The graphite fringes are no longer visible.

7.4.2. Mark 1 Corrector

The mark 1 corrector and the HB5 STEM have been described in Chapter 2. Following the alignment of the corrector and the correction of the parasitic aberrations, this was used to record images with zero spherical aberration.

The spherical aberration of the corrected HB5 was measured using the methods described in Chapter 3. The output from the TIS method is illustrated in Fig. 7.13. Repeated measurements suggested that the spherical aberration was set to zero with an error of less than 100 μm . The TIS method was found to be the most suitable of the methods based on the bright field images for measuring the aberration function. This was because the diffractogram analysis method required higher quality images to give suitable diffractograms. The main problems with the TIS method were caused by the specimen drift over the time in which the images were acquired. High-resolution images in particular required rather long acquisition times. This problem was reduced by acquiring a line from each of the images in turn, so that they are recorded almost in parallel. Unfortunately with the existing control software, this led to acquisition times of several minutes. Therefore it was not possible to investigate the time dependence of the corrected state in great detail. The Ronchigram analysis method is expected to improve this in future.

A bright field image taken with zero spherical aberration is shown in Fig. 7.14(A). The defocus has been adjusted to maximise the visibility of the fringes from the graphitised carbon. The speckle from the amorphous carbon is more detailed than for the uncorrected case. Fig. 7.14(B) shows the same area with the defocus slightly reduced to maximise the fine detail from the amorphous carbon. Unlike the uncorrected microscope, the fringes from the graphitised carbon are still visible. Another benefit of C_s -correction is that in the bright field images, the graphitised carbon fringes are still visible in overfocus. This indicates that the aberration function is much flatter, and so there is no longer the need to use defocus to balance C_s to observe this spacing.

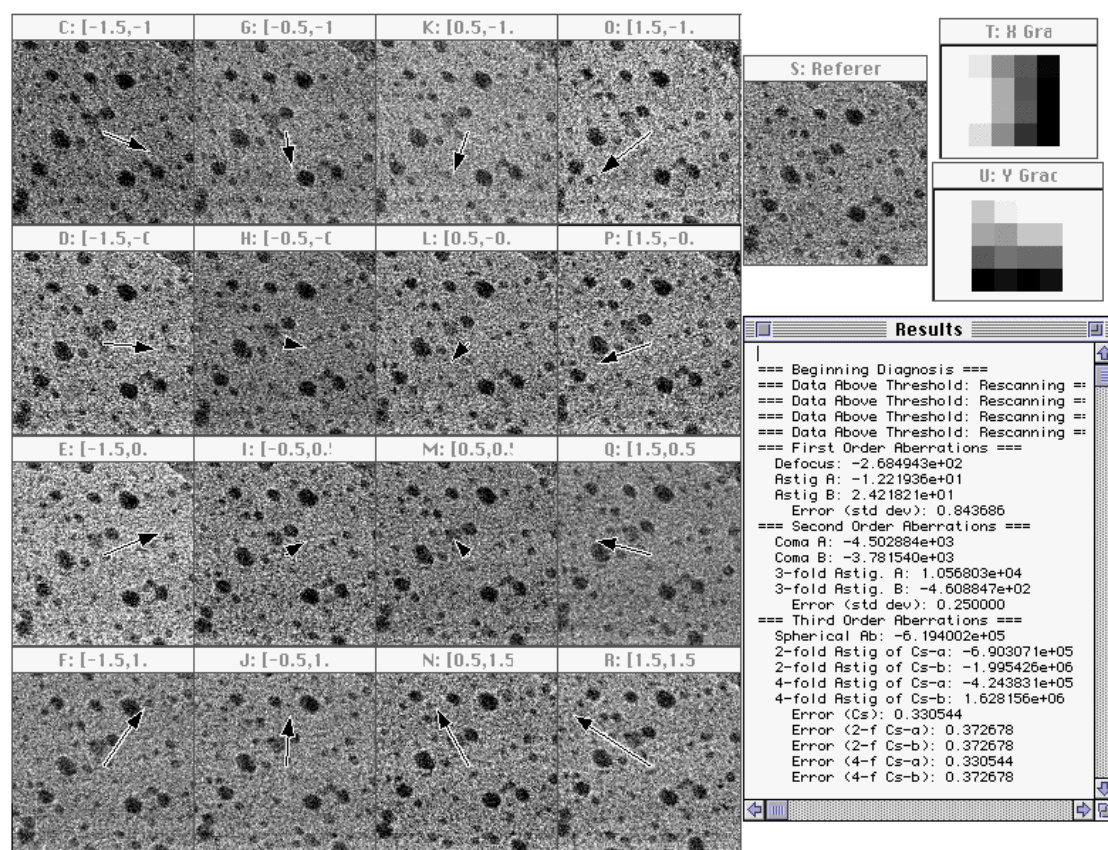
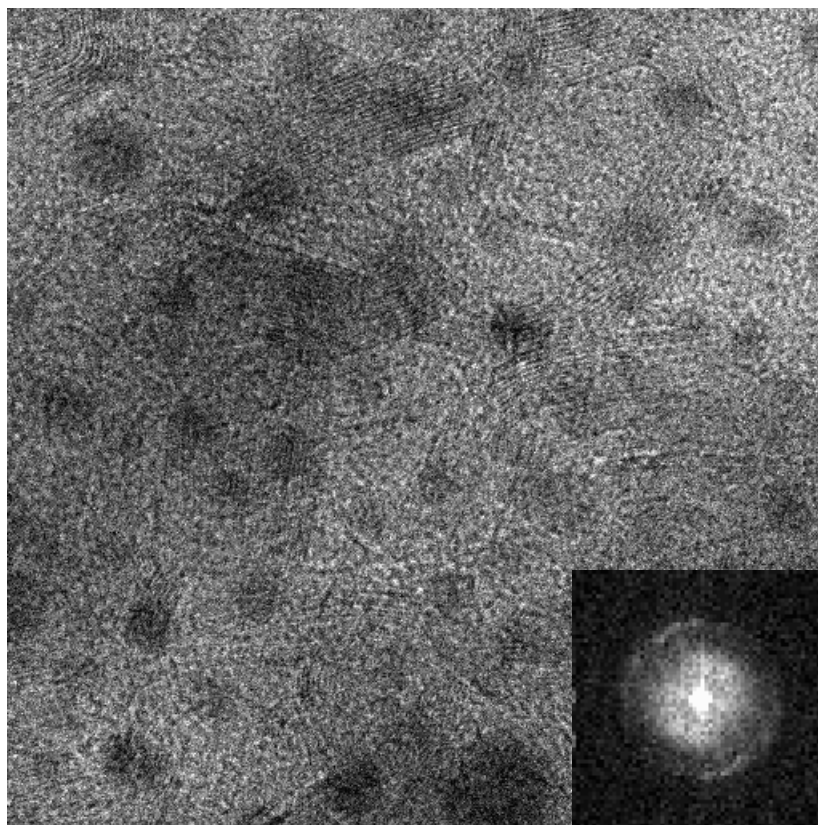


Figure 7.13. Example of the TIS autotuning procedure on the HB5. The reference image is the normal bright field image, and the other images are recorded at different settings of the Grigson coils. The arrows indicate the shift directions, which are used to give the gradient of the aberration function.

Images at an increased magnification before and after correction are shown in Fig. 7.15. These have been taken at the same magnification with the defocus optimised to maximise the visibility of the graphitised carbon fringes. Fig. 7.16 shows the integrated intensity in the diffractograms of these images. This is obtained by integrating all of the intensity at a particular radius and is plotted against the radius in the diffractogram. The peak corresponding to the 3.4 Å lattice spacing is shown, and both plots have been normalised to the same intensity scale. It is apparent from this figure, that a larger range of spatial frequencies is being more strongly transferred to the C_s -corrected image, even though in this case there is no evidence for an increase in resolution. Either instabilities or chromatic effects then limit the resolution and so the ADF imaging mode, which is less sensitive to C_c , was then investigated.

A



B

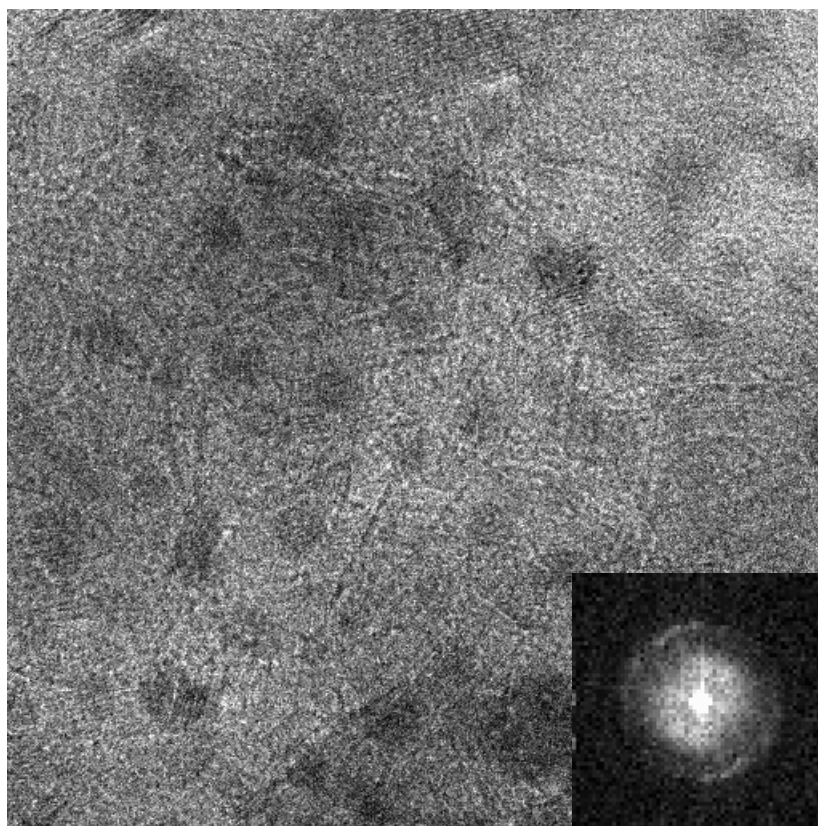


Figure 7.14. C_s -corrected bright field image taken on the VG HB5 taken at a magnification of 50000 times (with diffractograms). The defocus was optimized to maximize the visibility of:
(A) The graphitized carbon fringes. The fine detail of the background is still transferred.

(B) The fine detail of amorphous carbon background. The graphite fringes are still visible.

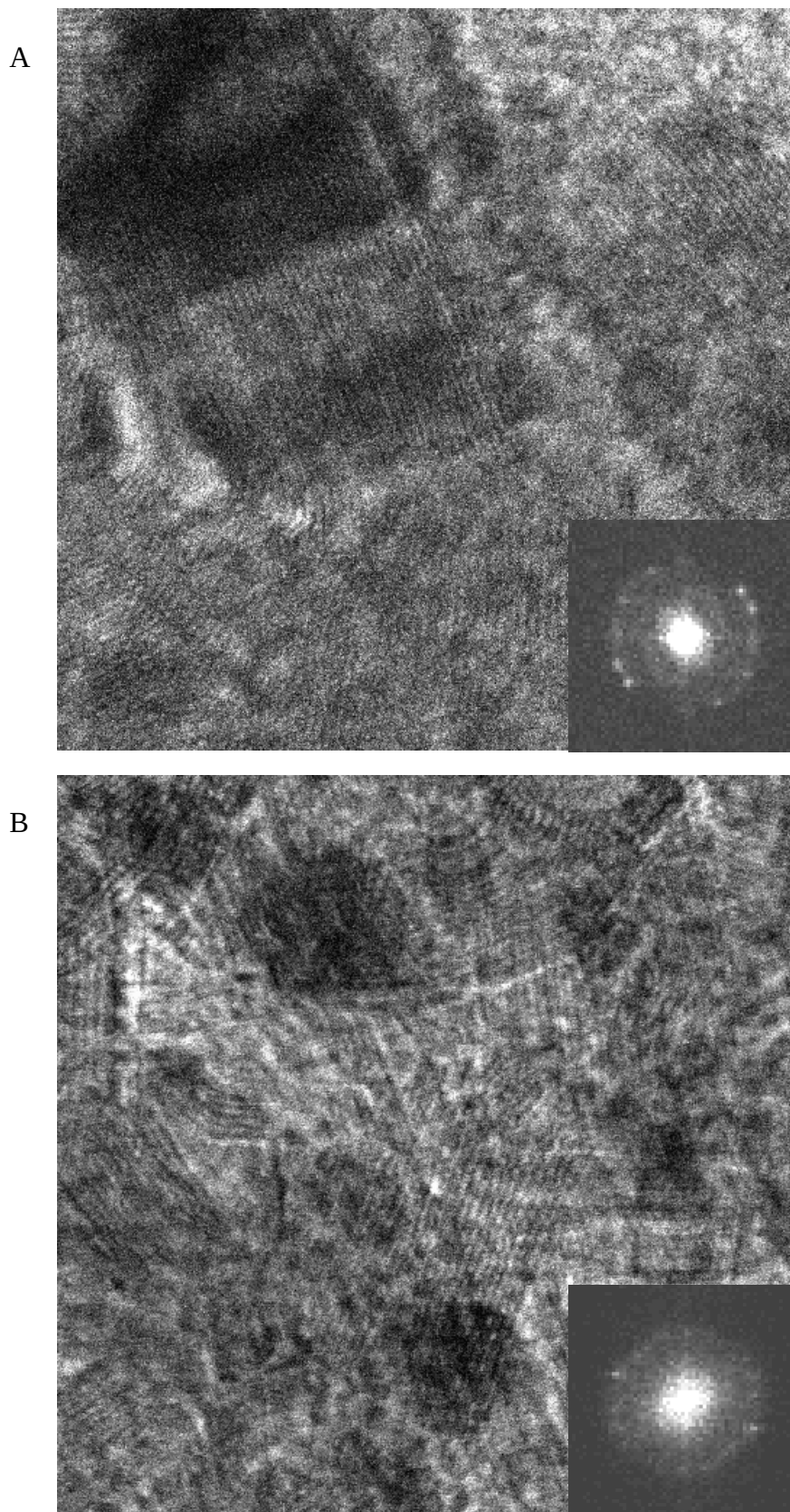


Figure 7.15. Bright field images taken at 10^5 times magnification, showing graphitised carbon.

(A) Before correction, although the fringes are visible, the background detail is lost.

(B) After correction, a greater range of spatial frequencies is transferred.

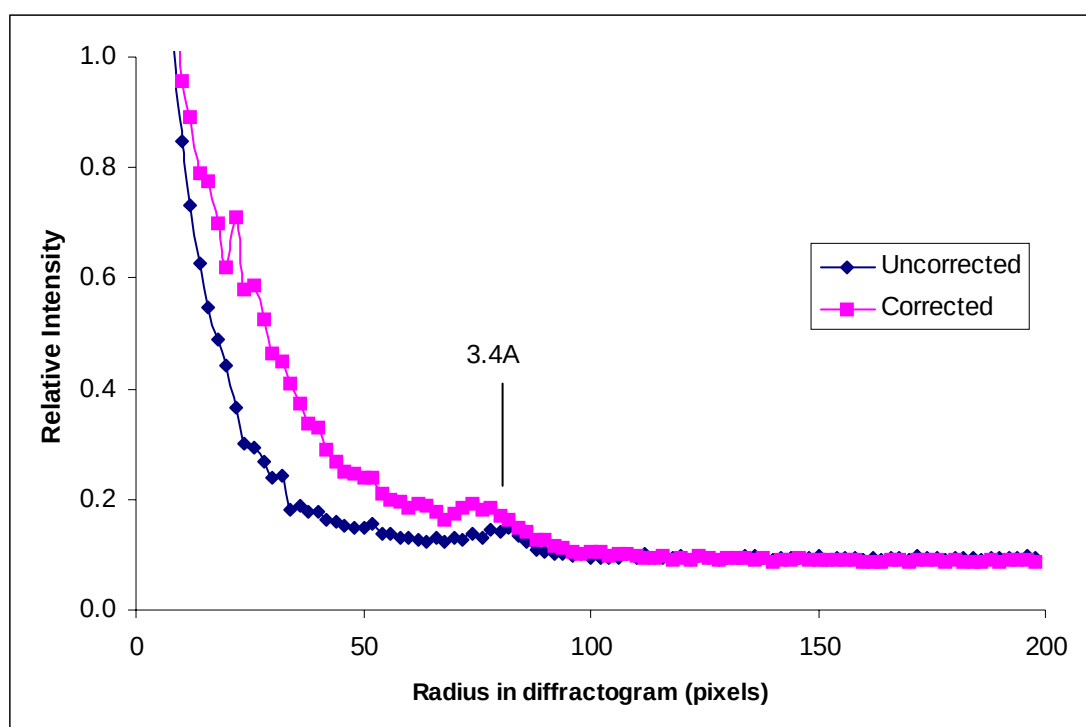


Figure 7.16. Integrated intensity as a function of radius in the diffractograms of the combined test specimen from regions with graphitised carbon fringes. The corrected and uncorrected images have been normalised to the same scale and the peak corresponding to the graphitised carbon spacing of 3.4 Å is shown.

Order	1 st	2 nd	3 rd
Average coefficient	-950 nm	9.2 µm	0.42 mm
Standard Deviation	200 nm	2.0 µm	0.24 mm

Table 7.3. Standard deviation from a series of measurements of the aberration coefficients from the TIS method. These were measured on the HB5 in a nearly corrected state. The first row is the average value of the coefficient measured, the second row is the standard deviation of the measurements.

The ADF image in Fig. 7.17 shows some of the initial problems with the first corrector. The image is distorted by the instabilities, which are particularly evident at the edges of the particles. The instabilities were reduced in several ways: The electrical supplies were adjusted to provide lower power outputs, reducing the effects of instabilities in the corrector power supply (it was not desirable to excite the coils too strongly because the pole-pieces would saturate). The mechanical coupling to the surroundings was reduced by suspending the entire microscope on elastic cabling. Potential dividers were fitted on the microscope column to reduce the effect of interference picked up by the cables leading to the scan coils from the console. The image quality was further enhanced when nearby electrical equipment was deactivated, because the corrector remained sensitive to stray fields.

A close-up of part of a C_s -corrected image is shown in Fig. 7.18, where fringes from the 2.35 Å lattice spacings of a gold particle are visible. The predicted resolution in Table 5.1 for the uncorrected system is 2.7 Å (using the measured C_s value of 3.9 mm for this pole-piece). This indicates that the resolution has been improved by the C_s -correction. However, there was a directional dependence to the fringes, and they are still distorted, showing that there were still problems with instabilities. The analysis in Chapter 4 suggested that this was caused by the relatively large beam diameter in the corrector. The original microscope power supplies also drifted and contributed to the instabilities.

Repeated measurements of the aberration function using the TIS method, on the corrected HB5 (here in a not fully corrected state) showed that the aberration coefficients could be measured with the precision given in Table 7.3. These do not achieve the precisions needed for 1 Å, which were given in Table 7.1. The values in Table 7.3 are based on a series of ten measurements around -1000 nm underfocus. The standard deviation was also affected by the amount of contrast in the image, because this affected the accuracy of the cross-correlations. The errors were not improved much by taking more measurements, because the measured values seemed to be stable only for several measurements, after which the values seemed to jump. Examination of the Ronchigram showed that this was due to a real change in defocus

rather than just spurious measurements. This was caused by instabilities in the microscope power supplies, particularly in the condenser lenses and the high voltage power supply. Small fluctuations in the power supplies to these elements were clearly reflected in defocus changes in the images.

The mark 1 corrector design has been shown to be capable of correcting the spherical aberration. This was clearly demonstrated in the appearance of the electron Ronchigram. The graphitised carbon fringes in the bright field images indicate that the stabilities were kept below 3.4 Å in all directions. The directional dependence of the fringes in the ADF images of the gold particles show that the instabilities were not reduced below 2.4 Å in all directions. Measurements of the aberration function and the uniform magnification in the Ronchigram confirmed that this was not simply caused by uncorrected aberrations. The problems with the mark 1 corrector were thus the increase in C_c (to approximately 7 mm), and the instabilities. The speed of measurement of the aberration coefficients was a problem, as it became difficult to differentiate between inaccuracies in the measurement process and real drifts in the coefficients. Overall, it appeared that the aberration corrector was beginning to be limited by the performance of the HB5, which is far less stable than current microscopes. Hence to improve the resolution further, a modified corrector on a more modern instrument was needed.

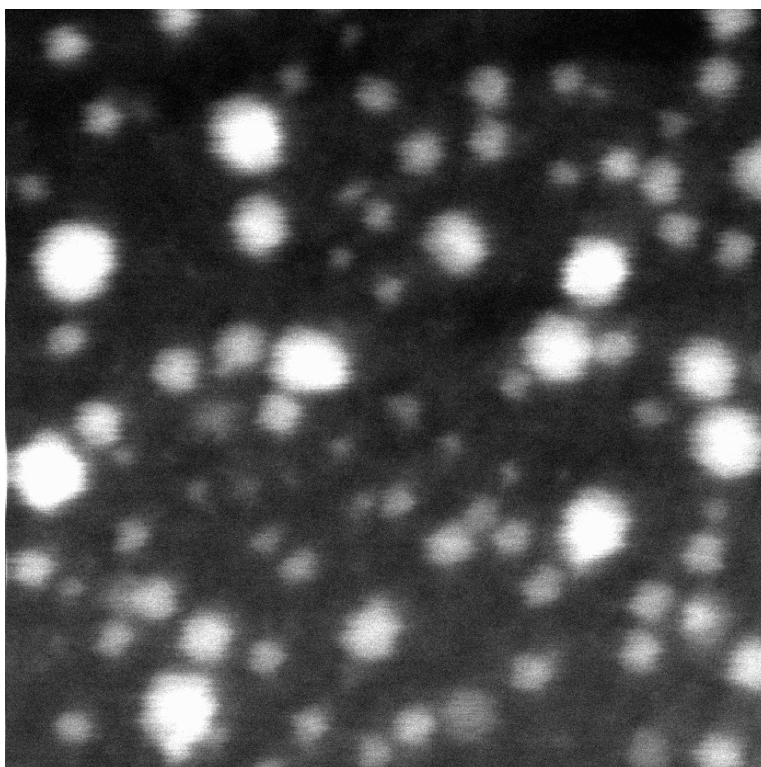


Figure 7.17. Aberration corrected ADF image not synchronized to mains frequency. Instabilities and stray fields from nearby electrical equipment limit the performance in this mode. The fast scan direction is horizontal in this image.

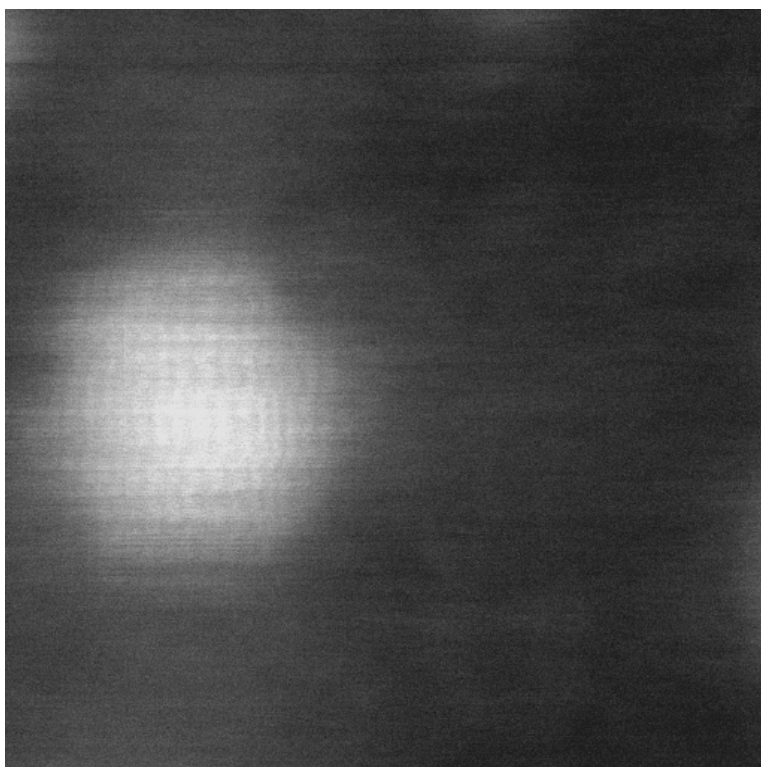


Figure 7.18. Aberration corrected ADF image synchronized to mains frequency. The lattice fringes from the 2.4Å spacing in the gold are just visible for correctly orientated gold particles.

7.4.3. Mark 2 Corrector

Based on the experience with the mark 1 corrector and the calculations described in this report, a second aberration corrector was constructed. This was fitted to a VG HB501 owned by Nion co. This microscope represents an updated version of the HB5 used in the rest of this report, with many small improvements. One important difference was the upgraded performance of the objective lens. This provided a higher excitation, with subsequently lower aberration coefficients. The increased post-sample compression is immediately obvious in the enlarged angular range present in the Ronchigram. The corrector design was also improved, as described in Chapter 4. The most significant improvements were the separation of the quadrupole and octupole elements, and the increased strength of the octupole elements. This meant that the beam diameter in the corrector was reduced by a factor of almost 10 relative to the mark 1 design, giving reduced sensitivity to stray fields, and a far lower chromatic aberration coefficient.

To address the problems of the speed of measurement and the need to measure fifth order aberrations, the Ronchigram analysis technique, that is presented here, was developed at the same time as this second corrector. This enabled a quantitative adjustment of the controls in the aberration corrector rather than by manual examination of the Ronchigram. Following the introduction of this routine, the next goal for future work is the completely automated adjustment of all corrector controls, which is presently being implemented (Dellby et al, 2001).

The images in this section were recorded in conjunction with O.L. Krivanek and N. Dellby of Nion Co. These were recorded on the Nion HB501 using the mark 2 corrector, aligned using the Ronchigram analysis software. This microscope uses the high-resolution pole-piece with $C_s=1.3$ mm, $f=1.7$ mm. The probe current is about 160 pA and the aperture half-angle 27 mrad.

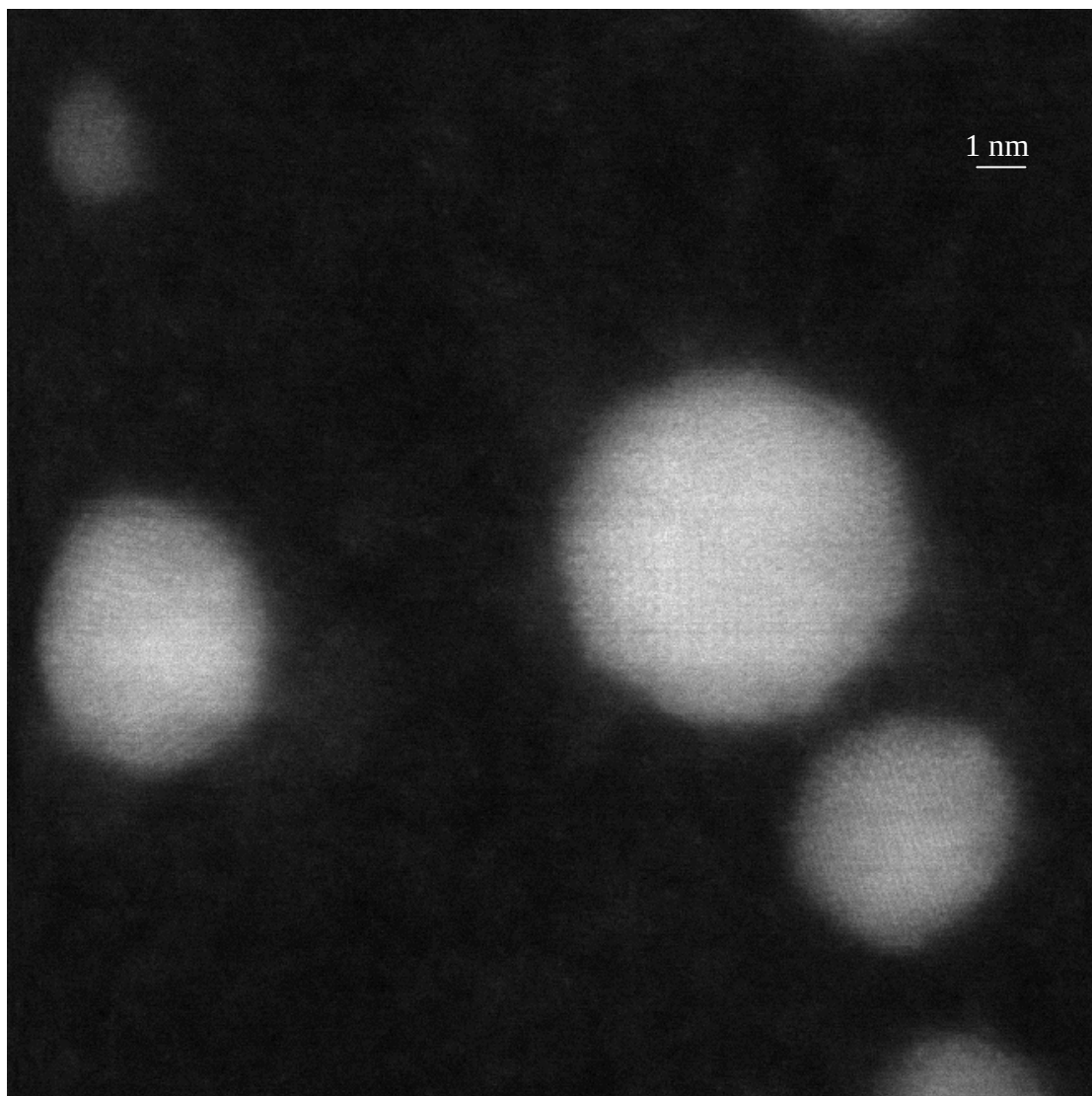


Figure 7.19. ADF image from the mark 2 corrector. Taken on the Nion HB501 with the high-resolution pole-piece and about 160 pA probe current at 32 μ S per pixel.

A high-angle ADF image from the HB501 with the mark 2 aberration corrector is shown in Fig. 7.19. The ADF mode is preferred for high-resolution imaging because of the reduced sensitivity to chromatic aberration. A close-up view of one of the particles is shown in Fig. 7.20, along with a diffractogram of that particle. This shows that spacings of 2.35 and 1.44 Å have been resolved. One final possibility from Fig. 7.19 is that some of the background detail between the gold particles might represent some single gold atoms on the carbon background, although further work would be necessary to confirm this.

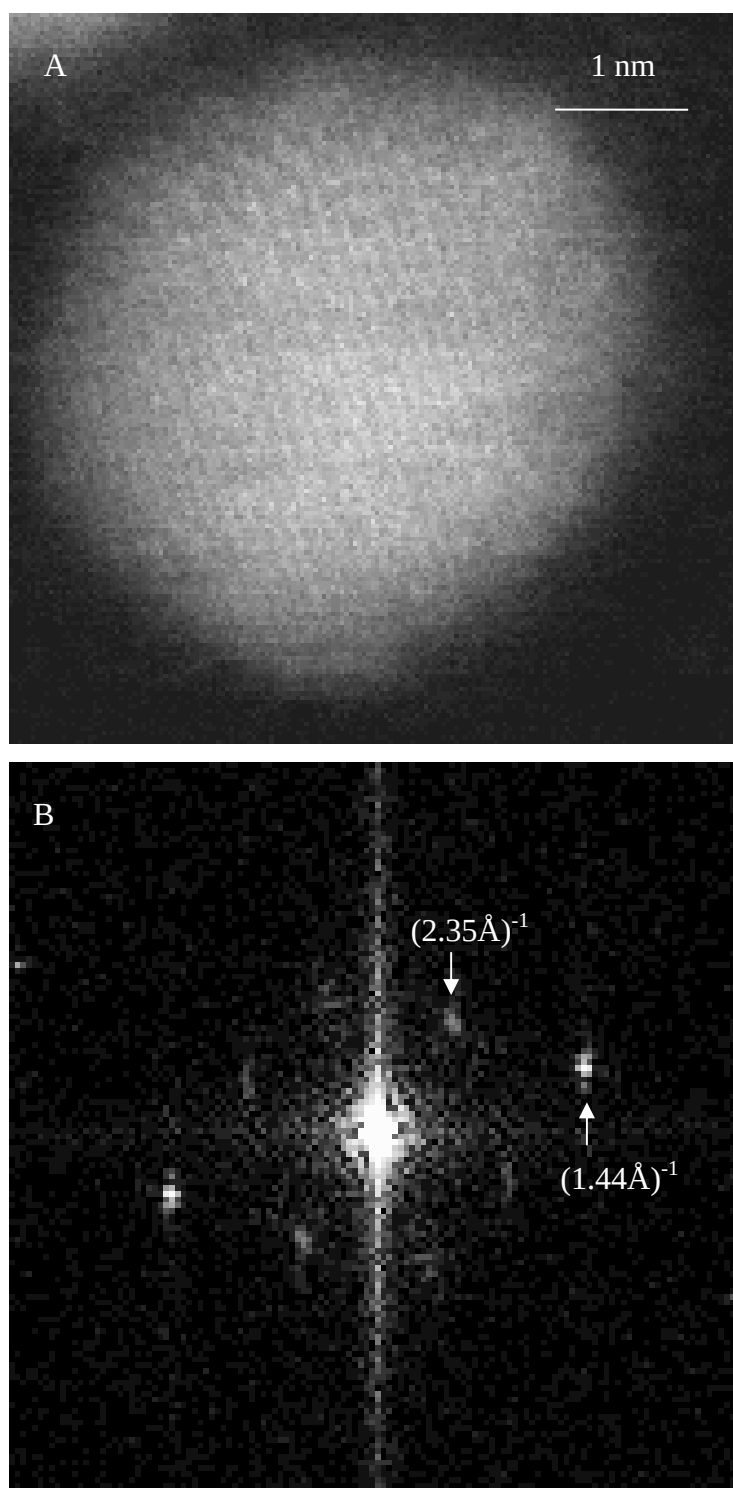


Figure 7.20. ADF image from the mark 2 corrector.

(A) Close up of one of the gold particles from Fig. 7.19.

(B) The diffractogram of the gold particle shows the 2.35 and 1.44 Å spacings.

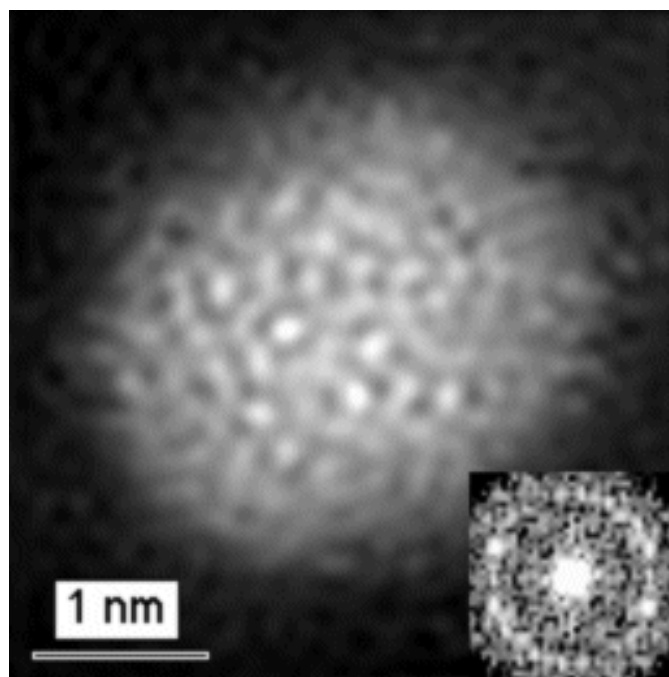


Figure 7.21. Gold particle (with inset diffractogram). Taken on the Nion HB501 with the high-resolution pole-piece. This appears to demonstrate a different structure from the larger size particles.

Figure 7.21 shows the high-angle ADF image of a small gold particle. This appears to show a different structure to the larger particles. The structure of such nano-particles is difficult to determine by other means, as discussed by Jefferson and Tilley (1999). Figure 7.22 shows an image of a larger gold particle with a stacking fault. In this case, spacings down to $1.23 \text{ \AA}$ are resolved, as revealed by carefully indexing the points in the diffractogram. This demonstrates that C_s -corrected ADF imaging will be able to produce unique images of structures such as dislocations and amorphous materials. Future developments will include applying the aberration corrector to different structures and problems.

These results show that the instability problems have been reduced in the second corrector. As predicted in the simulations, this offers improved performance over the mark 1 corrector. This is at least partly due to the improvements in the HB501 over the HB5. Other remaining experimental instabilities seem to be caused by the water-cooling of the objective lens and are being remedied (Dellby et al, 2001). After the instabilities, the chromatic aberration will be the next resolution limit.

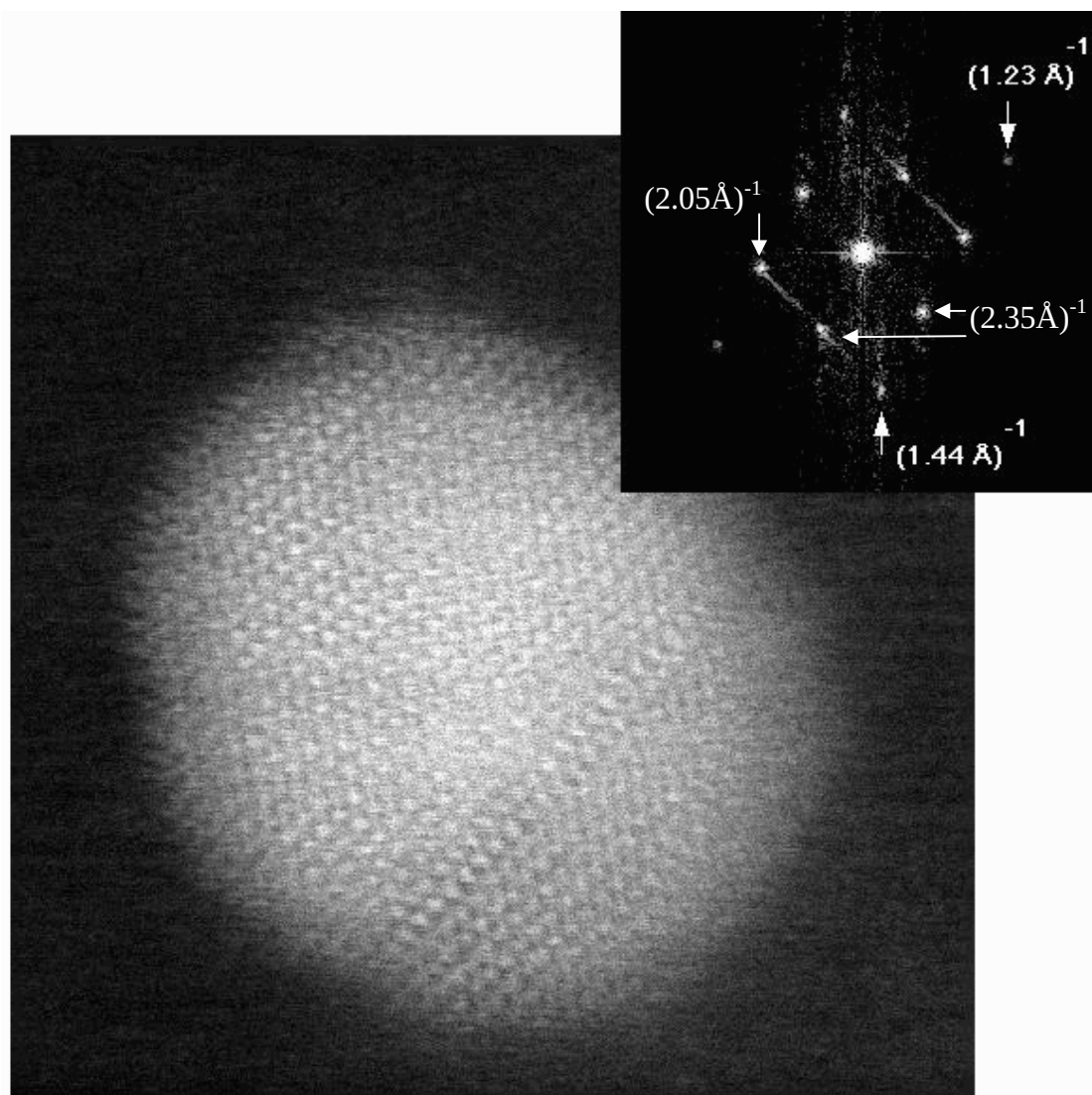


Figure 7.22. HAADF image of gold particle with a stacking fault. In this case, the particle is close to the $[110]$ orientation. The (111) , (200) , (220) and (311) lattice planes are resolved, the diffraction spots are shown in the diffractogram, corresponding to spacings of 2.35, 2.04, 1.44 and 1.23 $\text{\AA}$, respectively.

7.5. Conclusions

This Chapter has contained experimental tests of the Ronchigram analysis method. It has been shown to be a useful tool for general STEM work and forms an essential part of the aberration correction process. The techniques used to align the aberration corrector have been set out. Finally and most importantly the first images showing the successful correction of spherical aberration in STEM have been presented. The structure evident in the gold particles demonstrates that a C_s -corrected STEM will provide a uniquely useful tool for materials science. Future applications, particularly the combination of a C_s -corrector and an electron spectrometer are certain to provide information that can be obtained by no other route.

The results from the mark 1 corrector confirmed that it is possible to correct the spherical aberration in a high-resolution STEM. In the bright-field image this did not provide a large improvement in resolution, but has been shown to improve the range of frequencies transferred to the image. In the annular dark field image, the lattice spacing of gold planes just becomes visible, demonstrating a practical improvement in resolution. The ADF images showed convincingly that the practical performance of the mark 1 corrector was limited by the instrumental instabilities. This is not surprising, because the mark 1 corrector was fitted to the second HB5 ever made, which has performance far worse than current microscopes. However, this corrector succeeded in the main aim of proving for the first time that it is indeed possible for C_s -correction to increase the resolution of a STEM.

The improved performance of the mark 2 corrector confirms that the C_s -corrector can provide a practical improvement in the resolution of a modern high-resolution STEM.

The resolution in the C_s -corrected images is now limited primarily by the instabilities in the rest of the microscope. Chromatic aberration also acts as a limiting factor, but as was discussed in Chapter 5, this is less relevant to ADF STEM imaging than in the bright field case. It now appears that the next stage of instrumental development must concentrate on improving the rest of the microscope to the same level.

Chapter 8

Conclusions

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8.1. Summary of dissertation

This thesis has described the implementation and use of an aberration corrector to improve the resolution in a STEM. Experimental results from the mark 1 C_s -corrector and some theoretical analysis have been given. These led to the production of a second aberration corrector. This was the first ever such device to improve the resolution beyond that of any other STEM operating at the same voltage. The first results from the mark 2 corrector have been presented, and these confirm that a C_s -corrected STEM will provide a useful tool for materials analysis.

The system used to describe the aberration function was documented. Using this systematic description, summaries of the different imaging modes and the effects of the aberrations were given. This allowed the subsequent calculation of the probe size and expected resolution limits. By considering aberrations of all orders and symmetries, this represents an advance over previous work.

Some properties of aberration correctors were presented, and these calculations formed the basis for the design of a second aberration corrector. The resolution of a C_s -corrected microscope was shown to be limited by the fifth order and chromatic aberrations. The fifth order aberrations in a corrected system are primarily due to the combination aberrations. As has been proposed here, these can be adjusted and ultimately eliminated by relatively simple improvements to the present corrector design. The resolution will then be limited by the chromatic aberration and instabilities, which are significantly reduced in the mark 2 corrector.

One of the most significant advances in this work is a new method for measuring the aberration function of the microscope. This was based on the improved systematic description of the electron Ronchigram that was developed. This is one of the first experimental implementations of an automatic method to measure aberrations of all symmetries up to fifth order. This forms an essential part of the C_s -corrector, and is a necessary requirement for future automatic control of the microscope.

Finally, some of the first results from a C_s -corrected STEM have been presented. These show that the aim of improving the resolution of an electron microscope by aberration correction has finally been achieved, approximately 50 years after Scherzer's original proposal.

8.2. The future of C_s -correction

The correction of spherical aberration will become one of the most significant advances in transmission electron microscopy over the next few years. The aberration correctors that have described here are beginning to revolutionise the design and use of STEM instruments. Further correctors of the mark 2 design will soon be installed on several instruments. Future microscopes are already being designed from the start to include such devices.

The principal benefit of C_s -correction is that it increases the resolution beyond that of any other microscope operating at the same voltage. The resolution improvement has been demonstrated experimentally in Chapter 7. The target resolution of 1 Angstrom that has been suggested by several authors (for example Rose 1999, and Van Dyck et al, 1996) has been closely approached, and is expected to be achieved soon at 100 keV. This will allow materials analysis at higher resolution than ever before.

The aberration corrector will have further benefits, because the current density in the probe is increased. At the highest resolution, the total probe current is not increased because it is also necessary to increase the geometrical demagnification of the source. However, if the increased resolution is not required, the source demagnification can be kept constant and a larger aperture used. This results in the similarly beneficial situation of an increased current in the same size probe. Thus C_s -correction will offer improved current for EELS analysis of small areas.

In Chapter 5 it was demonstrated that the maximum current density was improved and the current in the probe tails was reduced. The background in ADF imaging is largely caused by these probe tails. Thus, this reduction in the fraction of the current outside the central maximum is expected to increase the visibility of weak contrast features, such as atoms of low atomic number (Krivanek et al, 2000a).

As a result of the work in this dissertation, the automatic control of the microscope has been significantly improved. Computer control has been shown to be essential for successful aberration correction. At the moment, this is not totally automated, although significant advances have been made. Instrumental improvements, such as improved stabilities, and total computer control of the microscope will allow further development.

The other essential requirement for automatic control is a convenient method of measuring the aberration function. One of the most exciting developments to result from this work was the description of the Ronchigram presented in Chapters 3 and 6. Ignoring dynamical effects, it was shown that the Ronchigram is equivalent to a distorted bright field image with a magnification that depends on the aberration function. This allowed the implementation of the Ronchigram analysis method for measuring the aberration surface. Further methods of analyzing the Ronchigram were proposed in Chapter 6. The use of the Ronchigram during this work, as well as for holography (Wang and Cowley, 1995) and super-resolution (Nellist and Rodenburg, 1994), suggests that this mode will become more widely used in future.

ADF STEM has already shown several benefits over bright field CTEM (Nellist and Pennycook, 1998), and this will continue in the aberration corrected instrument. If the fifth order aberrations can be kept sufficiently low, the chromatic aberration will limit the resolution in all modes. Unfortunately it appears that chromatic aberration correctors are not yet practical, because of the increased stability requirements. Thus monochromation would be needed in order to obtain the optimal resolution in bright field CTEM (Krivanek et al 2000a, Rose 1999), although this would reduce the available current. It has been shown by Brown (1981) that a reduction in beam current can reduce the achievable resolution, because of the degraded signal to noise ratio. The relative insensitivity to chromatic aberration of

ADF STEM reduces the need for monochromation, which therefore suggests that this mode will become more widely used than at present.

In a C_s -corrected microscope the fifth order aberrations become significant. From the calculations presented in Chapter 4, it is predicted that the present aberration corrector design can be changed to eliminate the fifth order effects. Consequently higher order aberrations, specifically sixth and seventh, will then have to be considered. Chromatic aberration again provides the next resolution limit.

The future for aberration correction in electron microscopy looks extremely bright. In the near future, the increased resolution that has been demonstrated will be applied to more practical problems in materials science. The next exciting development will be the combination of this corrector with EELS analysis, allowing higher resolution investigation of electronic structure. Work is already underway to improve the remaining components of the microscope, so that these do not limit the performance (Dellby et al, 2001). This will result in a new generation of STEM instruments, designed to include aberration correctors and offering significantly improved performance. C_s -corrected STEM will therefore become one of the most useful tools in many more areas of future materials research.

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Appendix 1

Ronchigram Analysis

Because of the volume of code produced, only a summary is presented here. The Ronchigram Analysis software was primarily written in the Digital Micrograph script language. The software was developed both on the PC and the Macintosh, and is further described by Krivanek et al (2000b). In both versions the least squares fitting and the final determination of aberration coefficients is performed in a series of functions developed in the C programming language. The least squares fitting algorithm was taken from the book *Numerical Recipes in C* (Press et al, 1988). The PC version featured a C++ wrapper to allow incorporation into the Digital Micrograph program as a plug-in (see Software References). To speed up the Macintosh version, more C-code was included, containing nearly all of the image processing.

The automated Ronchigram analysis was divided into the following sections, as separate Digital Micrograph script files:

Set-Up – The script that stores all of the calibrations and parameters needed for the analysis. As an example of the required values, a script that was used during the testing of the analysis has been included in this appendix.

Main – The main script that controls the analysis. This first calls another script, which handles the acquisition of the tableau of Ronchigrams and the movement of the probe. This was separated into a different script so that only that section would have to be modified in order to run on different hardware. The analysis then proceeds according to the values initialized in the set-up script, and finally the measured aberration coefficients are output to the user.

Second Pass – The script to do only the second pass of the analysis. This reacquires new Ronchigrams using an external script as before. Another script is called to perform the unwarping of the images. The analysis is then repeated, with the resulting shifts used to give corrections to the coefficient values from the first pass.

The remainder of this appendix consists of a calibration script that was used to test the Ronchigram analysis. This is included as a practical summary of all of the parameters that the full analysis requires. It contains comments explaining the non-obvious settings. Combining all of this in a single file was a convenient way of saving and editing entire sets of values without having to write a custom user-interface. The example included shows a set of values that were successfully used experimentally and in simulations. It is divided into a section of values that affect the processing of the cross-correlations and the values that are actually sent out to the microscope hardware. The main calibrations necessary are described in Chapters 6 and 7.

Initialisation Script for Ronchigram Analysis

```
// Setup tags for Ronchigram Analysis
// ARL 8/3/00
//*****
//*****
// Cross-correlation values:
//*****
// Whether to Re-acquire images or use previously acquired set: 1==Re-acquire, 0==Use
old set
SetpersistentNumberNote("Ronchigram:ReAcquire",1);
// Highest order of aberration coefficients to try fitting for:
SetpersistentNumberNote("Ronchigram:Order",5); // If set to 3 = Up to Cs
//Number of patches to use in cross-correlation:
SetpersistentNumberNote("Ronchigram:NumXcImages",10);
// How big is each patch:
SetpersistentNumberNote("Ronchigram:PatchSize",128); // Must be a power of 2
// Gap around edge of cross-correlation images (should be bigger than biggest shift!):
SetpersistentNumberNote("Ronchigram:GapSize",8);
// Spacing between centres of patches:
SetpersistentNumberNote("Ronchigram:StepSize",40); // Divide this by 2 for the radial
fits

// Faster version of analysis: 0 = Slow, 1 = No errors [ignored], 2 = No drift
subtraction
SetpersistentNumberNote("Ronchigram:Speed",0);
// Rotate coefficients between Camera and Corrector axis (radians)
```

```

SetpersistentNumberNote("Ronchigram:Rotation",0*3.1415927);
// Should our patches be in a concentric circles (0) or squares (1):
SetpersistentNumberNote("Ronchigram:Pattern",1);

// Add arrow annotations: 0 = No arrows, 1 = Average arrow only, 2 = Show all arrows
SetpersistentNumberNote("Ronchigram:UseArrows",0);
// Lengths of arrows
SetpersistentNumberNote("Ronchigram:ArrowMag",18);

// Image size:
SetpersistentNumberNote("Ronchigram:SizeX",640);
SetpersistentNumberNote("Ronchigram:SizeY",480);
// Coordinates of voltage centre:
SetpersistentNumberNote("Ronchigram:CentreX",320);
SetpersistentNumberNote("Ronchigram:CentreY",240);

// How many levels of undo -- also useful as it means you have a record of previous
runs
SetpersistentNumberNote("Ronchigram:Memory",16);
// Path for script files:
SetPersistentStringNote("Ronchigram:Path","C:/Gatan/autotuningscripts/");

//*****
// Values that go to hardware
//*****
// Value to shift EHT by in DACunits:
SetpersistentNumberNote("Ronchigram:DACunits",100);
// How long to allow EHT to settle for in 60'ths of a second
SetpersistentNumberNote("Ronchigram:Delay",60);

// Mis-rotation between the DigiScan axis and Camera axis (radians)
Number theta=(180-155.09)/180*3.141592;
// Other axis:
Number theta1=theta-3.141592/2;
// Aspect ratio to allow for DigiScan shift strengths varying with direction
Number aspect =0.936;
// How far in pixels to shift probe:  NOW IN DIGISCAN UNITS (-32767 <-> +32767)
Number r=4000;
// Scan coil strengths: This is in Angstroms/pixel in your DigiScan image "Signal 0"
Number DSScale=4.54e-3;
// Angular range visible in Ronchigram from centre to edge of image in the X direction
SetpersistentNumberNote("Ronchigram:MaxHalfAngle",53.8e-3); // From diffraction
rings...
SetpersistentNumberNote("Ronchigram:Shift1X",r*DSScale*1e-10);
SetpersistentNumberNote("Ronchigram:Shift1Y",0);
SetpersistentNumberNote("Ronchigram:Shift2X",0)
SetpersistentNumberNote("Ronchigram:Shift2Y",r*DSScale*1e-10);
SetpersistentNumberNote("Ronchigram:DSShift1X",-r/aspect*cos(theta));
SetpersistentNumberNote("Ronchigram:DSShift1Y",-r/aspect*sin(theta));
SetpersistentNumberNote("Ronchigram:DSShift2X",-r*aspect*cos(theta1));
SetpersistentNumberNote("Ronchigram:DSShift2Y",-r*aspect*sin(theta1));

```

```
result("Over-written tags\n");  
// End
```

Appendix 2

Converting Aberration Coefficients

It was necessary to convert experimental data and different forms of the aberration coefficients to the notation described in Chapter 2. The values of aberration coefficients also depend on the coordinate system, as was discussed in Chapter 4. Therefore, all coordinates should be referred to the same plane with any change in angular magnification included.

Here the conversion will be considered for the situation in a STEM, at the sample plane. The different styles of aberration coefficient, that have been introduced previously, fall into 2 classes:

- (1) The transfer matrix format of COSY or TRANSPORT (Chapter 4).
- (2) The $\chi(u, v)$ notation (Chapter 2, as used in the probe simulations).

It will be assumed that the transfer matrices have positions, (x, y) , and inclinations (tangents) to axis, (u, v) . It is also possible to convert between this and the variant of (1) used by COSY, involving sines of angles. However sines and tangents are equivalent to first order, so only higher order corrections are necessary.

To convert from (2) to (1) we equate terms in the gradient of the aberration function (Chapter 2) and the transfer matrices (Chapter 4). This is used to construct a series of matrices, $\underline{\mathbf{M}}_N$, representing the relationship between terms of similar orders, such that:

$$\begin{aligned} (R_{12}, R_{14}, R_{32}, R_{34}) &= \underline{\mathbf{M}}_1 (C_{12a}, C_{12b}, C_1) \\ (T_{122}, T_{124}, T_{144}, T_{322}, T_{324}, T_{344}) &= \underline{\mathbf{M}}_2 (C_{23a}, C_{23b}, C_{21a}, C_{21b}) \\ (U_{1222}, U_{1224}, U_{1244}, U_{1444}, U_{3222}, U_{3224}, U_{3244}, U_{3444}) &= \underline{\mathbf{M}}_3 (C_{34a}, C_{34b}, C_{32a}, C_{32b}, C_3) \end{aligned} \quad (\text{II.1})$$

As an example of this: R_{12} gives the first order dependence of the X-coordinate of a ray on the initial slope (to the X-axis). In the aberration coefficient notation, this dependence comes from the sum of the defocus (C_1) and astigmatism (C_{12a}) terms. Obviously to describe the same system, these two descriptions must be equivalent, giving:

$$R_{12} = 1C_{12a} + 0C_{12b} + 1C_1 \quad (\text{II.2})$$

The first row in $\underline{\mathbf{M}}_1$ is therefore (1,0,1). Thus the entries in $\underline{\mathbf{M}}_N$ are just the constants in the expressions for the gradient of the aberration function $\chi(u,v)$. This equation is defined for both position and angles at the Gaussian focal plane. Since the TRANSPORT coefficients are normally referred to the initial angles, the matrix elements including angular magnification must also be included.

To convert from (1) to (2) each inverse transform, $\underline{\mathbf{M}}_N^{-1}$, is defined such that:

$$(C_{12a}, C_{12b}, C_1) = \underline{\mathbf{M}}_1^{-1} (R_{12}, R_{14}, R_{32}, R_{34}) \quad (\text{II.3})$$

To find these transforms, we have to invert each $\underline{\mathbf{M}}_N$, which is not strictly possible because (2) is less general than (1) and so the matrices are not invertible. For example, (2) does not include rotations or spiral distortions, so some of the conversions are over or under determined. In under-determined situations we set the unknown coefficients to zero, and in the over-determined case we impose extra conditions based on symmetry considerations. The condition imposed was that if the terms relating two coefficients in $\underline{\mathbf{M}}_N$ are equal, then the corresponding inverse terms in $\underline{\mathbf{M}}_N^{-1}$ must be equal. Conveniently many of the entries will be zero and can be eliminated. Thus we can obtain *suitable* solutions for converting coefficients, but it is **not** strictly the inverse transformation.

$$\text{Thus } \underline{\mathbf{M}}_1 \text{ is: } \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \\ \bar{1} & 0 & 1 \end{pmatrix} \quad \text{And } \underline{\mathbf{M}}_1^{-1} \text{ is taken to be: } \frac{1}{2} \begin{pmatrix} 1 & 0 & 0 & \bar{1} \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix}.$$

Where bars have used to denote negative values. Examples of further conversion matrices are given in Table A2.1.

The output from simulations or experimental measurements can be in two additional formats:

- (3) From the numerical trajectory traces and the image shifts in the TIS aberration analysis method we obtain a table of ray positions (equivalent to first derivatives of the aberration function) against angles to the axis.
- (4) From the ADA and Ronchigram analysis method we obtain tables of second derivatives of the aberration function against angles.

The first stage in the processing in both cases is to perform a least squares fit, taken from the book *Numerical Recipes in C* (Press et al, 1988). For (3) the fit is over the polynomials:

$$\begin{aligned} x &= x_0 + R_{13}u + R_{14}v + T_{133}u^2 + T_{134}uv + T_{144}v^2 + U_{1333}u^3 + \dots \\ y &= y_0 + R_{23}u + R_{24}v + T_{233}u^2 + T_{234}uv + T_{244}v^2 + U_{2333}u^3 + \dots \end{aligned} \quad (\text{II.4})$$

The results of which can then be converted to our notation using the matrices defined by equation (II.1).

For (4) the fit will be over the polynomials:

$$\begin{aligned} \frac{\partial^2 \chi}{\partial u^2} &= a_0 + a_1u + a_2v + a_3u^2 + a_4uv + a_5u^2 + a_6u^3 + \dots \\ \frac{\partial^2 \chi}{\partial u \partial v} &= b_0 + b_1u + b_2v + b_3u^2 + b_4uv + b_5u^2 + b_6u^3 + \dots \\ \frac{\partial^2 \chi}{\partial v \partial u} &= c_0 + c_1u + c_2v + c_3u^2 + c_4uv + c_5u^2 + c_6u^3 + \dots \\ \frac{\partial^2 \chi}{\partial v^2} &= d_0 + d_1u + d_2v + d_3u^2 + d_4uv + d_5u^2 + d_6u^3 + \dots \end{aligned} \quad (\text{II.5})$$

Where we are fitting the $\partial^2 \chi / \partial u \partial v$ and $\partial^2 \chi / \partial v \partial u$ terms separately, although they are in principle equal. This is because these come from different measurements and so any inequalities will contain information about rotation and errors.

We then construct matrices relating equivalent terms in (II.5) to the second derivatives of the aberration function, such that:

$$\begin{aligned} (a_0, b_0, c_0, d_0) &= \underline{\mathbf{L}}_1 (C_{12a}, C_{12b}, C_1) \\ (a_1, a_2, b_1, b_2, c_1, c_2, d_1, d_2) &= \underline{\mathbf{L}}_2 (C_{23a}, C_{23b}, C_{21a}, C_{21b}) \\ (a_3, a_4, a_5, b_3, b_4, b_5, c_3, c_4, c_5, d_3, d_4, d_5) &= \underline{\mathbf{L}}_3 (C_{34a}, C_{34b}, C_{32a}, C_{32b}, C_3) \end{aligned} \quad (\text{II.6})$$

Where the entries in the matrices $\underline{\mathbf{L}}_N$ are then just the constants involved in the expressions for the second derivatives in the same way that $\underline{\mathbf{M}}_N$ were for the first.

The aberration coefficients can then be found by inverting the matrices, $\underline{\mathbf{L}}_N$ in exactly the same way as for $\underline{\mathbf{M}}_N$. In the current software, the coefficients were solved for using a least squares fit to find the most probable coefficients. This implicitly relies on a matrix inversion and so the principle is exactly the same (see Press et al, 1988).

$$\text{Thus } \underline{\mathbf{L}}_1 \text{ is: } \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \\ \bar{1} & 0 & 1 \end{pmatrix} \quad \text{And } \underline{\mathbf{L}}_1^{-1} \text{ is taken to be: } \frac{1}{2} \begin{pmatrix} 1 & 0 & 0 & \bar{1} \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix}.$$

The similarity between $\underline{\mathbf{L}}_1$ and $\underline{\mathbf{M}}_1$ (and their inverses) is not a general result for other values of N .

Most algebraic manipulation and generation of matrices was performed using MAPLE. It is likely that some sort of Fourier technique could provide alternatives to the method described here. However this method was chosen because all that is required to extend to arbitrary orders is to derive the next matrices, so this provided a simple method of analysis for any order.

An important point is that the data should cover an adequate range of angles – because of limited precision, there is no point in trying to fit fifth order coefficients at small angles. Similarly the low angles are dominated by the first order coefficients. The least squares fitting would ideally weight the data appropriately to minimize errors.

$$\underline{\mathbf{M}}_2 = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 2 & 0 & \frac{2}{3} \\ \bar{1} & 0 & \frac{1}{3} & 0 \\ 0 & 1 & 0 & \frac{1}{3} \\ \bar{2} & 0 & \frac{2}{3} & 0 \\ 0 & \bar{1} & 0 & 1 \end{pmatrix}, \underline{\mathbf{M}}_2^{-1} = \frac{1}{4} \begin{pmatrix} 1 & 0 & \bar{1} & 0 & \bar{1} & 0 \\ 0 & 1 & 0 & 1 & 0 & \bar{1} \\ 3 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 3 \end{pmatrix}$$

$$\underline{\mathbf{M}}_3 = \begin{pmatrix} 1 & 0 & 1 & 0 & 1 \\ 0 & 3 & 0 & \frac{3}{2} & 0 \\ \bar{3} & 0 & 0 & 0 & 1 \\ 0 & \bar{1} & 0 & \frac{1}{2} & 0 \\ 0 & 1 & 0 & \frac{1}{2} & 0 \\ \bar{3} & 0 & 0 & 0 & 1 \\ 0 & \bar{3} & 0 & \frac{3}{2} & 0 \\ 1 & 0 & \bar{1} & 0 & 1 \end{pmatrix}, \underline{\mathbf{M}}_3^{-1} = \frac{1}{8} \begin{pmatrix} 1 & 0 & \bar{1} & 0 & 0 & \bar{1} & 0 & 1 \\ 0 & 1 & 0 & \bar{1} & 1 & 0 & \bar{1} & 0 \\ 4 & 0 & 0 & 0 & 0 & 0 & 0 & \bar{4} \\ 0 & 2 & 0 & 2 & 2 & 0 & 2 & 0 \\ 3 & 0 & 1 & 0 & 0 & 1 & 0 & 3 \end{pmatrix}$$

$$\underline{\mathbf{L}}_2 = \begin{pmatrix} 2 & 0 & 2 & 0 \\ 0 & 2 & 0 & \frac{2}{3} \\ 0 & 2 & 0 & \frac{2}{3} \\ \bar{2} & 0 & \frac{2}{3} & 0 \\ 0 & 2 & 0 & \frac{2}{3} \\ \bar{2} & 0 & \frac{2}{3} & 0 \\ \bar{2} & 0 & \frac{2}{3} & 0 \\ 0 & \bar{2} & 0 & 2 \end{pmatrix}, \underline{\mathbf{L}}_3 = \begin{pmatrix} 3 & 0 & 3 & 0 & 3 \\ 0 & 6 & 0 & 3 & 0 \\ \bar{3} & 0 & 0 & 0 & 1 \\ 0 & 3 & 0 & \frac{3}{2} & 0 \\ \bar{6} & 0 & 0 & 0 & 2 \\ 0 & \bar{3} & 0 & \frac{3}{2} & 0 \\ 0 & 3 & 0 & \frac{3}{2} & 0 \\ \bar{6} & 0 & 0 & 0 & 2 \\ 0 & \bar{3} & 0 & \frac{3}{2} & 0 \\ \bar{3} & 0 & 0 & 0 & 1 \\ 0 & \bar{6} & 0 & 3 & 0 \\ 3 & 0 & \bar{3} & 0 & 3 \end{pmatrix}$$

Table A2.1. Matrices used to transform between coefficient notations. Higher orders would follow in exactly the same way. $\underline{\mathbf{L}}_2$ and $\underline{\mathbf{L}}_3$ are never explicitly inverted in the present system.

Appendix 3

Ronchigram Simulations

To test the software developed to analyze Ronchigrams and to study the effects of aberrations on images it was necessary to develop a basic method to simulate images, particularly Ronchigrams. This was performed using the wave-optical approach outlined in Chapter 3, which is consistent with other descriptions of Ronchigrams given in the literature (for example Nellist and Rodenburg, 1994). The Ronchigram is considered to be (the intensity of) the convolution of the aperture function and the sample diffraction pattern. To a similar level of approximation, the bright-field image would be (the intensity of) the convolution of the sample and the probe. As in the wave-optical theory, Fourier transforms are used to represent propagation from the objective aperture to the sample plane (and from there to the detector plane). All convolutions were performed as multiplications in the Fourier domain.

The steps involved in the wave-optical simulations were:

- (1) Generate a sample transmission function and its diffraction pattern.
- (2) Use the aberration coefficients to generate the aperture function.
- (3) Generate the probe by Fourier transforming the aperture function.
- (4) Do the appropriate convolution and take the intensities to give the image.

Tableaus of Ronchigrams were produced by shifting the generated samples before step (4). This used the same shift patterns as for the probe movement in the experiments, with linear interpolation for non-integer pixel shifts. These tableaus were then used to test the Ronchigram analysis method using known coefficients.

This method also includes the calculation of the probe intensity. Because the method is fully 2-dimensional all non-round axial aberrations can be included. Chromatic aberration was included by incoherently summing images simulated over a range of energies, using Gaussian weighting of intensities (Chapter 5).

Ronchigrams could also be simulated using a geometrical optics argument method based on that in Chapter 6. The geometrical optics simulation of the Ronchigram involves using the aperture function in (2) to give the gradient of the aberration function for each pixel in the image. This is used as a mapping to distort a bright field image of the sample. This was primarily used to verify that the magnification was the same in both the wave and geometrical optics descriptions. The loci of infinite magnification were identically situated in both approaches.

The method used to generate random amorphous samples was:

- (1) Fill an image with random gray levels.
- (2) Fourier transform and use a Gaussian filter in frequency space to give a characteristic feature size.
- (3) Reverse Fourier transform to generate the sample.

To make a phase object, these values are scaled and used as the phase of each point with a constant intensity. To make an amplitude object the values are just used as the amplitude of the transmission function. This ignores all dynamical effects and is intended to provide an illustration of the appearance of the different imaging modes rather than a quantitative analysis.